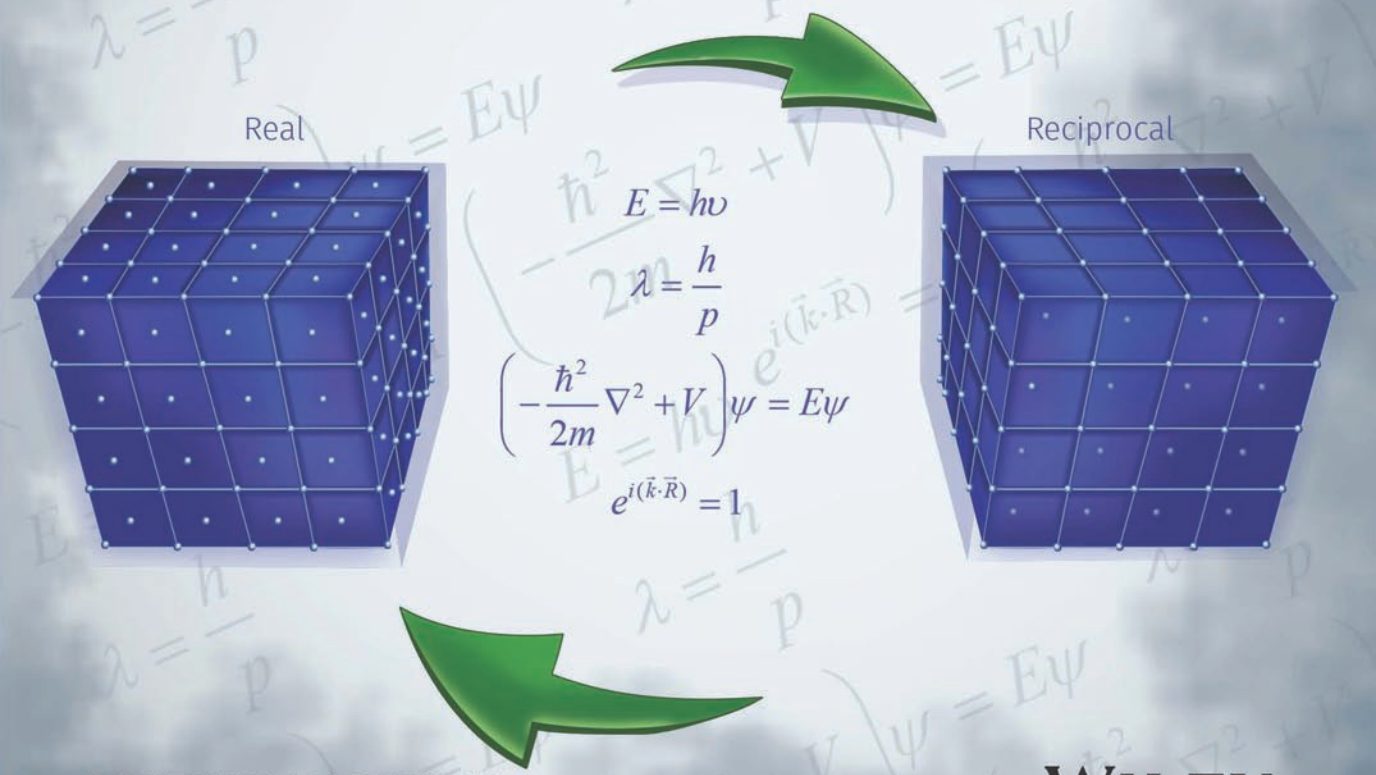
Physics of MATERIALS



Essential Concepts of Solid-State Physics



PRATHAP HARIDOSS

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Professor, Department of Metallurgical and Materials Engineering
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To my mother,
Mrs. Nithya Haridoss
and my father,
Mr. V. Haridoss

my first teachers.



Preface

This book "*Physics of Materials*" and the associated DVDs is a result of over 10 years of experience in teaching an undergraduate core course with the same title. The course was delivered to a collection of highly motivated and enthusiastic undergraduate students – my students in the Department of Metallurgical and Materials Engineering, at the Indian Institute of Technology, Madras. While teaching this course, I felt the absence of a book that presented the topics at the right level of detail for undergraduate students in engineering to grasp and benefit from. During my days as an undergraduate student and later as a graduate student too, I had felt this lack of a "good book" very strongly. Indeed there are excellent and well-established texts in Solid-State Physics. However, I found them to be very challenging to learn from, as a first-time learner of those topics, especially with an engineering background. I felt I was making insufficient headway into the topics being presented, despite considerable efforts. So as a student I felt the need for a book such as this, very strongly.

When I became a faculty at IIT Madras and got the opportunity to teach this course, I found the situation to be similar. There was no single source that I could turn to that could support my teaching process completely. I read parts of several books and looked up several sources of information. There were many gaps in understanding that had to be bridged to present the content coherently, effectively and satisfactorily to students. Books and resources varied very significantly in the level of complexity in their presentation. Some were appropriate for high school students and others already required you to be a "physicist". In the middle was a wide gap that had to be bridged which needed enormous effort and time.

In higher education sometimes too much new information is presented too quickly, and the students only gain vague familiarity of buzz words and not sufficient understanding of the concepts – an aspect that greatly affects their ability to benefit from subsequent courses as well. Therefore, in this course a conscious effort was made to ensure that the pace of the course was appropriate to enable students to stay with the course for the entire duration. In addition, expecting a student to fill in too many gaps in the information provided is also not realistic in view of their tight schedules. This is particularly true in basic foundational courses such as this. Therefore, in the book as well as in the lectures in the DVDs, derivations are made as complete as possible so that the student has the satisfaction of seeing in a single reading/viewing, how the concept is developed. The student does not have to break the reading process repeatedly just to figure out how one line in the book implies the next line.

During these 10 years, I also got the opportunity to develop this course into a video course for the National Programme on Technology Enhanced Learning (NPTEL), funded by the Ministry of Human Resource Development (MHRD), Govt. of India. Videos of classes are available for free through NPTEL and have been accessed by an international audience. Video lectures of "*Physics of Materials*" have been viewed over 100,000 times (adding the views across all the classes). The viewership is international and includes students as well as professionals from all over the world. Feedback received from users of NPTEL further convinced me that efforts to put together this content have been well directed.

Having putting over 10 years in this effort, I now feel I have put together a significant collection of concepts that I have been able to explain what I feel is the "right level of detail". I, therefore, feel that the time has arrived for this to be presented in the form of a book that students can use and benefit from.

The purpose of the book is to present essential concepts in *Solid-State Physics*; to make students comfortable and confident about important concepts that will make reading other advanced and established texts easy and much more effective. Along with the book come two DVDs which have 40 lectures of approximately an hour each, which provide the student with the classroom experience associated with this book. At the start of a section the icon  indicates the video lectures associated with that topic. Several animations are also included in the accompanying DVDs which enable a more visual experience to understanding the concepts discussed. Along with the caption of specific figures, the readers are directed to the associated animations.

The book begins with a general introduction, examines properties of materials, and then looks at thermal expansion to showcase the approach used to study properties. Given the focus on electronic properties, the book then looks at measuring electrical conductivity. The classical Drude model is developed and its limitations identified. Statistical mechanics and its use in understanding behaviour of large collections of particles such as electrons in a solid is examined. Maxwell–Boltzmann, Fermi–Dirac, and Bose–Einstein statistical distributions are derived and used in appropriate circumstances. In view of the limitations of the classical approach, after a brief look at the history and challenges in understanding quantum mechanics, the quantum mechanical approach of the Drude–Sommerfeld model is introduced and examined. The link between confinement and quantization is highlighted. The features introduced by the quantum mechanical approach, and how it addresses the limitations of the classical approach, are discussed. Reciprocal space is developed both from a materials science perspective as well as a physics perspective and the link between these approaches is brought out in detail – an important aspect that most other treatments pay insufficient attention to. The explanation for the origin of bands, as a consequence of the theories developed, is examined. Several material phenomena are then explained based on the theories developed. The book closes with a look at superconductivity and the physics of nano-scaled materials. In general, subsequent chapters do build on the theory of earlier chapters; however, specific topics such as Reciprocal space can be read independently as well.

In preparing this book, I am grateful to the students at the Department of Metallurgical and Materials Engineering, IIT Madras, who always inspired me to aim for greater clarity. There are numerous finer details in several chapters that are a direct result of questions that students raised in classes. My thanks also to the IIT system and IIT Madras, in particular, that give faculty abundant freedom to develop their courses. My gratitude to NPTEL for the opportunity to connect with an international audience, and my thanks to the Centre for Continuing Education (CCE), IIT Madras, for financially supporting the writing of this book.

I thank my teachers over the years, both in school as well as in college, who showed me the value of explaining something clearly. At a personal level, I thank my grandfather, the late Prof. K. Rajagopalan, who taught me mathematics at an important juncture; my parents and my sister who taught me in the early years; and my wife and my son who encouraged me while putting up with the intense efforts for this book, at the cost of time which clearly belonged to them.

In summary, I believe that this book responds to a strong and long felt need in the student community, and in view of the topics discussed as well as the manner of coverage of these topics, a very wide cross-section of students can benefit from this book and the associated DVDs.

Prathap Haridoss
September 2015

About the Author

Prathap Haridoss is a Professor in the Department of Metallurgical and Materials Engineering at the Indian Institute of Technology (IIT), Madras, India. He has been a faculty in the Department for 14 years. He received his B.Tech. in Metallurgical Engineering from IIT Madras is 1992. He completed his PhD in Materials Science from the University of Wisconsin-Madison, USA in 1999. During his PhD, he was a graduate research assistant at the Los Alamos National Laboratory in Los Alamos, New Mexico, USA. After his PhD, he worked as a Senior Scientist at Plug Power, a Fuel Cell company in Latham, New York, before joining IIT Madras as Faculty.

His research interests include PEM Fuel Cells, Carbon Nanomaterials including Carbon Nanotubes, Semiconducting Nanomaterials, and recycling printed circuit boards. He has developed a communication device for the speech impaired and a footwear-based device to enable gait analysis by timed mapping of foot contact points. He is also working on developing an exoskeleton to assist differently-abled as well as elderly people.

He has three US patents, and is the author of a copyrighted software. He has nearly 30 International Journal publications and has served as the advisor for 5 PhD students and more than 20 Masters students.

He is also the author of several articles to popularize science amongst school students as well as general-purpose articles aimed at the student community.

Dr Haridoss' video course on the topic covered in this book "*Physics of Materials*" has been viewed over 100,000 times (adding the views across all of the classes). The viewership is international and includes students as well as professionals from all over the world. These classes as well as comments received from users across the world can be viewed at the website of the National Programme on Technology Enhanced Learning (NPTEL), an initiative funded by the Ministry of Human Resource Development (MHRD), Govt. of India. They can also be viewed in the associated YouTube channel.

Contents

Preface **vii**

Chapter 1 Introduction – Use and Study of Materials 1

Learning Objectives 1
1.1 Introduction 1
1.2 Materials and the Engineer 1
1.3 Materials and a Scientist 2
1.4 Modeling a Material 2
1.5 Approach Used in This Book 3
Summary 3
Practice Questions 3
Answers 4

Chapter 2 Properties of Materials 5

Learning Objectives 5
2.1 Introduction 5
2.2 Mechanical Properties 6
2.3 Chemical Properties 7
2.4 Electrical Properties 8
2.5 Thermal Properties 9
2.6 Magnetic Properties 10
2.7 Optical Properties 10
2.8 Understanding Material Properties 11
Summary 12
Practice Questions 12
Answers 13

Chapter 3 Thermal Expansion 15

Learning Objectives 15
3.1 Introduction 15
3.2 Use of Thermal Expansion/Contraction 15
3.3 Model for Thermal Expansion 17
Summary 21
Practice Questions 21
Answers 22

Chapter 4 Electrical Conductivity 23

Learning Objectives 23
4.1 Introduction 23
4.2 Charge Carriers 24
4.3 Direct Current Conductivity Measurement 25
4.4 Alternating Current Conductivity Measurement 28
4.5 Short Note on Superconductivity 34
Summary 34
Practice Questions 34
Answers 35

Chapter 5 Free Electron Gas and The Ideal Gas 37

Learning Objectives 37
5.1 Introduction 37
5.2 The Free Electron Gas 37
5.3 Packing Fraction in Solids 38
5.4 Reasons for Caution with the Free Electron Model 40
5.5 The Kinetic Theory of Gases 41
Summary 44
Practice Questions 45
Answers 46

Chapter 6 The Drude Model 49

Learning Objectives 49
6.1 Introduction 49
6.2 Electrical Conductivity 49
6.3 Thermal Conductivity 55
6.4 The Wiedemann–Franz Law 61
6.5 Shortcomings of the Drude Model 62
Summary 67
Practice Questions 68
Answers 70

Chapter 7 Large Systems, Statistical Mechanics, and The Maxwell–Boltzmann Statistics 73

Learning Objectives 73
7.1 Introduction 73
7.2 Systems with Large Collections of Particles 73
7.3 Statistical Mechanics 75

7.3.1 System Consisting of Two Particles 75
7.3.2 System Consisting of Three Particles 76
7.3.3 System Consisting of Four Particles 77
7.4 The Maxwell–Boltzmann Statistics 79
Summary 82
Practice Questions 82
Answers 84

Chapter 8 A Brief History of Quantum Mechanics; Its Use in the Drude–Sommerfeld Model 85

Learning Objectives 85
8.1 Introduction 85
8.2 Classical Particles and Quantum Mechanical Particles 86
8.3 A Brief History of Quantum Mechanics 90
8.4 The Drude–Sommerfeld Model 100
Summary 102
Practice Questions 103
Answers 105

Chapter 9 Fermi–Dirac Statistics 107

Learning Objectives 107
9.1 Introduction 107
9.2 The Fermi–Dirac Statistics 107
9.3 Features of the Fermi–Dirac Distribution 114
9.4 Comparing Maxwell–Boltzmann and Fermi–Dirac Distributions 117
Summary 120
Practice Questions 120
Answers 123

Chapter 10 Anisotropy, Periodic Potential, Confinement, and Quantization 125

Learning Objectives 125
10.1 Introduction 125
10.2 Anisotropy in Crystalline Solids 126
10.3 Periodic Potential in a Crystalline Solid 128
10.4 Confinement and Quantization – Waves on a String Analogy 132
10.5 Confinement and Quantization – Quantum Mechanical Approach 134
Summary 138
Practice Questions 139
Answers 141

Chapter 11 Density of States, Fermi Energy, and The Electronic Contribution to Specific Heat at Constant Volume 143

Learning Objectives 143
11.1 Introduction 143
11.2 Density of States 143
11.3 Aspects Associated with Electrons Filling Energy Levels 147
11.4 Fermi Energy 149
11.5 Fermi Surface 149
11.6 Fermi Temperature 150
11.7 Estimating the Electronic Contribution to Specific Heat at Constant Volume 151
11.8 Further Improving the Model of the Solid 153
Summary 154
Practice Questions 155
Answers 157

Chapter 12 The Reciprocal Space 159

Learning Objectives 159
12.1 Introduction 159
12.2 Defining Reciprocal Space 160
12.3 Properties of Reciprocal Space 160
12.4 Diffraction in Reciprocal Space 163
12.5 The Ewald Sphere 166
12.6 Crystal Lattices in Reciprocal Space 167
12.7 Reciprocal Lattice as the Fourier Transform of the Real Lattice 170
12.7.1 Mathematical Representation of a 1D Lattice 170
12.7.2 Fourier Coefficients Corresponding to the 1D Lattice 171
12.7.3 Poisson's Summation Formula 171
12.7.4 Fourier Transform of a 1D Lattice 172
12.8 Another Way to Define or Describe Reciprocal Space 173
Summary 174
Practice Questions 175
Answers 177

Chapter 13 Wigner–Seitz Cell, Brillouin Zones, and The Origin of Bands 179

Learning Objectives 179
13.1 Introduction 179
13.2 The Wigner–Seitz Cell 180
13.3 The Brillouin Zone 181
13.4 Significance of Bragg Planes 185
13.5 Interaction of Electron Waves with Brillouin Zones 186
13.6 The Origin of Bands 188

Summary 193
Practice Questions 194
Answers 195

Chapter 14 Bands, Band Gaps, Free Electron Approximation, and Tight Binding Approximation 197

Learning Objectives 197
14.1 Introduction 197
14.2 Analytical Approach to The Band Structure – Free Electron Approximation 197
14.3 Band Structure and Types of Materials 202
14.4 Band Structure – The Tight Binding Approximation 203
14.5 Effect of Pressure on Band Structure 205
14.6 Comparison of Free Electron and Tight Binding Approximations 206
14.7 Conventions Regarding Representing Band Diagrams 207
Summary 209
Practice Questions 210
Answers 211

Chapter 15 Material Phenomena Explained using Theories Developed 213

Learning Objectives 213
15.1 Introduction 213
15.2 Semiconductors 214
15.3 Types of Semiconductors 214
15.3.1 Intrinsic Semiconductors 214
15.3.2 Extrinsic Semiconductors 215
15.4 Effect of Temperature on Semiconductors 217
15.5 Optical Properties of Semiconductors 218
15.6 The Utility of E versus k Diagrams 219
15.7 Phonons 219
15.8 Magnetism 221
15.9 Electron Compounds 224
Summary 226
Practice Questions 228
Answers 229

Chapter 16 Superconductivity and The Bose–Einstein Statistics 231

Learning Objectives 231
16.1 Introduction 231
16.2 Superconductivity 231

16.3 The Meissner Effect 233
16.4 The BCS Theory 233
16.5 The Bose–Einstein Statistics 234
16.6 The Bose–Einstein Condensate 237
Summary 238
Practice Questions 238
Answers 239

Chapter 17 Physics of Nano-Scale Materials 241

Learning Objectives 241
17.1 Introduction 241
17.2 The Exciton 242
17.3 Confining the Exciton 243
Summary 244
Practice Questions 245
Answers 245

Bibliography 247
Index 249

1 Introduction – Use and Study of Materials

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- State how materials science is utilized in several industries and compare that with its use in university research labs.
- State what is meant when we try to "model" a material.
- Indicate the hierarchy between theoretical models and experimental data.
- Indicate the implications of finding limits to a model.
- Indicate the use and limitations of empirical models.



1.1 Introduction

This book is aimed at undergraduate students majoring in Metallurgical and Materials Engineering. An attempt is made to ensure that the contents presented are quite complete so that a typical undergraduate student is able to read it in a continuous manner, without interrupting the process to read additional background information.

Although the book is primarily aimed at students in the Metallurgical and Materials Engineering stream, the content presented in this book will largely be accessible to most of the undergraduate engineering students.

While this is a bit of oversimplification, pursuing a career as a materials engineer or materials scientist broadly results in two types of jobs – an engineer in an industry focused on a single product or a scientist in a university/laboratory setting.

1.2 Materials and the Engineer

An engineer in an industry setting usually faces several technical troubleshooting challenges. Such challenges with stiff timelines require substantial familiarity with information databases and characterization techniques, to be successfully addressed. Addressing troubleshooting challenges are often based on the selection and use of the best available resources and substitutes. The types of questions that the engineer faces include the following:

1. Is there a lighter material readily available which can replace the existing material in the product?
2. Why is a specific batch of the product not as good as the other batch?
3. Will it be possible for this product to function with lesser parts than it currently has?

Answers to such questions enable the industry to meet the demands of their customers at lower costs and with greater reliability.

1.3 Materials and a Scientist

A scientist in a university or laboratory setting faces a different kind of challenge. The questions the scientist wishes to answer are as follows:

1. Why does a material display a specific value for a specific property from the range of values possible for that property?
2. What is the fundamental science behind the property?
3. What are the limits, if any, to this property?

Answers to such questions enable one to design new materials and push the capabilities of existing materials.

Based on the professional setting, a material specialist ends up in, he/she will likely get pulled into, one of the above types of activities. However, having both a good feel for the fundamental sciences as well as an understanding of the engineering approach to relate to the real world, can greatly enhance the value of a materials scientist/engineer for his or her organization.

At the same time, it usually requires considerable experience to fully understand, appreciate and make use of the linkages between fundamental science and the world of engineering. Experience and systematic studies provide us with the insight to make the connection between the real-world inventions and science behind them. Such an insight helps us to truly appreciate the contributions of the scientist as well as the engineer in shaping our interactions with the world around us. Such an insight also enables us to take our technical pursuits to greater heights and wider reaches, and hence is desirable.

1.4 Modeling a Material

In this book, we shall focus on trying to understand the science behind material properties. One of the approaches used for this is to model materials. The idea behind this approach can be described as follows: The constituents of the material, atoms, electrons, etc. are assumed to follow specific rules. The set of rules we assume for the constituents together constitute the "model" we propose for the material. Based on our best guess of what is reasonable to assume for these constituents, these rules are selected by us. Assuming that the constituents follow these rules, the material is then "built up" on the basis of these constituents. Based on the rules that we have assumed, predictions are then made on what the material properties are likely to be and how these properties will respond to external influences. These predictions are then compared with the experimental data. If the experimental data and trends in experimental data match the predictions, we conclude that our "model" is correct. In case we have incorrect predictions, the model needs to be changed or corrected until we get predictions to match the experimental data.

In this activity, we accept the hierarchy that experimental data are supreme. A model, no matter how sophisticated it may appear at the first glance, is of no value if it is unable to predict experimental data. At the same time, it must also be noted that when a failure is noted for a model, it identifies limits to be placed on the model's capabilities. For example, if a model correctly predicts experimental data between two temperatures T_1 and T_2, but makes incorrect predictions at temperatures outside this range, it is not correct to dismiss the model as "useless." A more appropriate reaction is to say that the model may need further refinement, and specifically, as long as the temperatures of interest lie between T_1 and T_2, the model is still very usable. Further refining the model may require us to add additional details into the model or perhaps recognize and incorporate new phenomena and processes that come into operation when specific threshold values of temperature, or other experimental parameters, are crossed.

In the literature as well as in many industries, models that are entirely empirical are used. This often means that there is no firm understanding of the science behind the response of a system, maybe because the system is too complicated, yet there is an empirical formula or curve fit that matches the experimental data from the system very well. Such models are of practical value since they enable some predictions of what is

likely to happen when specific parameters change in value, even though there may be a little understanding of why the response is changing the way it is.

1.5 Approach Used in This Book

In this book, we will broadly look at the following:

1. Properties of materials.
2. Features associated with some of the properties of materials.
3. Classical theories for the behavior of metallic systems and their limitations.
4. Quantum mechanics.
5. Description of material properties based on quantum mechanical approach for the behavior of electrons in crystalline structures.

The major focus of this book will be on electronic properties of materials, and relatively less on the other properties, although initially an overview of the properties and some relationships between them will be considered and discussed.

This book contains sections that are descriptive, and other sections that rely heavily on theory and derivations. It is also noteworthy that many of the topics discussed in this book as well as the corresponding equations and insights, resulted in the associated researchers receiving Nobel prizes. It is, therefore, not entirely surprising to face some initial difficulties in understanding the topics covered.

Summary

1. Studying materials science can broadly result in careers in industry or scientific organizations.
2. An industry often focuses on using available materials, finding material substitutes and identifying reasons for differences between batches of materials.
3. Scientific organizations are often interested in understanding why a material displays a given property.
4. One approach to understand the science behind material properties is to build models for materials.
5. Models for materials are rules imposed on constituents of the materials, which then enable prediction of material properties.
6. Models may have limits and are usable within these limits.
7. Experimental data are supreme. A model, no matter how sophisticated it may appear at first glance, is of no value if it is unable to predict experimental data.
8. Empirical models also exist and are widely used. They are based on correlations noticed in experimental data, even if the science behind the correlation is not yet fully understood.

Practice Questions

The questions in this chapter are intended to stimulate general awareness and curiosity. They are not directly addressed in the chapter, but are related to the ideas discussed.

Multiple Choice Questions

1. From among the materials below, in general the material type with the least density is
   (a) Metal **(b)** Polymer **(c)** Ceramic **(d)** Glass

2. Modern airplanes are being made lighter by the use of the following:
   (a) Composite materials
   (b) Aluminum
   (c) Steel
   (d) Copper
3. From among the materials below, in general which has the highest electronic conductivity at room temperature
   (a) Metal **(b)** Polymer
   (c) Ceramic **(d)** Glass
4. A ceramic sample can be
   (a) Opaque **(b)** Transparent
   (c) Translucent **(d)** All of the above
5. A metal sample can be
   (a) Opaque **(b)** Transparent
   (c) Translucent **(d)** All of the above

Short Answer Type Questions

1. Give one example of the type of problem an engineer in an industry is likely to be focused on.
2. Give one example of the type of problem a scientist in a university or laboratory is likely to be focused on.
3. Give one example of a model for material properties that is valid only within a range of conditions.
4. Give an example of a theory that many have tried to violate but has withstood the scrutiny it has been subject to.
5. Indicate one example of a theory that made predictions that could be verified only much later due to experimental limitations.

Answers

Multiple Choice Questions

1. (b)
2. (a)
3. (a)
4. (d)
5. (a)

Short Answer Type Questions

1. Here is an example from the power generation industry. In this industry, steam generation is often an important step. Much of the hardware therefore is in contact with water and steam. Sometimes specific batches of hardware may corrode much faster than expected. An engineer may be expected to figure out why the corrosion has occurred sooner in some cases and to design a fix for the problem.
2. Here is an example. Carbon nanotubes are allotropes of carbon that are available in various lengths and diameters. Typical synthesis techniques are unable to produce exactly one type of nanotube. What modifications can be made to the synthesis techniques to generate a specified type of carbon nanotube?
3. The Dulong–Petit empirical law predicts that the molar heat capacities of solid elements will be $3R$, where R is the universal gas constant. This law holds good for a wide range of temperatures but fails to make correct predictions both at very low temperatures as well as at very high temperatures.
4. The second law of thermodynamics. Many have tried to build and demonstrate a perpetual motion machine, which violates the second law of thermodynamics. No one has succeeded to do so.
5. There are a few examples of such theories, including one that will be discussed in greater detail later in this book. One example is Einstein's theory of relativity, some of whose predictions were verified using an opportune eclipse that enabled scientists to observe the movement of the planet mercury with great precision.

2 Properties of Materials

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Explain the process of classifying materials.
- List the properties of materials.
- List some of the important ways in each major category of material property, in which the property manifests itself.
- State examples of instances where a particular property is used or observed.
- Indicate the level of interaction (electronic, atomic, crystal structure, etc.) that impacts the display of a specific property.
- Indicate the different terminology used in Physics and Chemistry books to explain certain material properties.

2.1 Introduction



While we are all familiar with materials in our day-to-day usage. However, it is actually quite surprising to note that we often know very little about these materials. For example, metals are extremely commonplace. Yet ask yourself the following question:

"Define a metal."

You will find that the answer is not as commonplace as the material itself! The answer is "Metals are materials with a positive thermal coefficient of resistivity." This means that when the temperature of the metal is raised, its resistance increases. Other materials that can also conduct charge, such as ionic conductors and semiconductors; actually they lower their resistance when the temperature increases. Why do these materials perform differently when we raise the temperature? We will examine this a little later in this book. Now let us examine our knowledge of materials a little more and identify how we typically understand and classify materials.

In reality, it turns out that it is difficult to strictly define any type of material. In general, we look at the properties displayed by a material, and if the values of those properties fall in certain ranges, we classify it as a certain type of material. This approach to classify materials is affected by the fact that properties displayed by the materials are often significantly influenced by the environment that they are placed in. Materials that are insulating at room temperature and atmospheric pressure can become metallic under very high pressures. It is therefore important to keep in mind the conditions under which the material is being tested, to classify the materials within the framework of those conditions. In common usage, the conditions under which materials are being tested and classified are typically room temperature and atmospheric pressure. These test conditions are not explicitly stated at times, but we should be alert to these regardless.

When we examine material properties, we find that there are interesting correlations between properties. For example, good conductors of electricity are also often good conductors of heat. Such a correlation implies that something is fundamentally common to the process of conduction of electricity as well as heat.

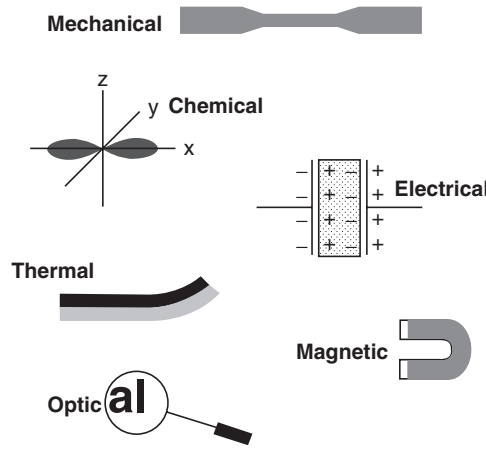


Figure 2.1 Schematic showing the properties commonly encountered in materials.

More than simply being aware of material properties, it is of interest to understand properties of materials and recognize relationships between these properties. Such an understanding enables us to see how material properties evolve from the behavior of the constituents of the materials such as atoms and electrons. Figure 2.1 schematically shows the properties we commonly encounter in materials.

The material properties that we commonly encounter and some examples of the instances where we encounter them are as follows:

1. **Mechanical properties** make their presence felt in a variety of objects we use in terms of the dimensional stability of the object as we use it and the response of the object to the physical forces it is subject to during use or testing.
2. **Chemical properties** can be encountered in our daily life, for example in the form of cleaning agents we use. Cleaning agents typically have instructions on what they can be used for and what they should not be used for. Such instructions reflect the chemical reactivity of the agents.
3. **Electrical properties** are used extensively in many household entertainment devices.
4. **Thermal properties** are used in commonplace equipment such as bimetallic strip in iron boxes, insulation in thermal wear, etc.
5. **Magnetic properties** are used in fans, motors and audio speakers.
6. **Optical properties** are used for creating sun control films and lenses for various applications.

A more elaborate, although not exhaustive, look at these properties is presented in the following sections.

2.2 Mechanical Properties

The major mechanical properties are shown in Fig. 2.2. A tensile test, during which a dog-bone shaped sample is pulled till it fractures, enables us to obtain a stress–strain curve for the material. From the stress–strain curve, the *modulus of elasticity* is obtained as the slope of the linear region of the curve. The *yield strength* is the stress at which the stress–strain curve just begins to lose its linearity.

Tensile strength is the maximum stress supported by the sample and is the highest stress recorded in the stress–strain curve. *Ductility* is the elongation that can be sustained by the sample before it fractures.

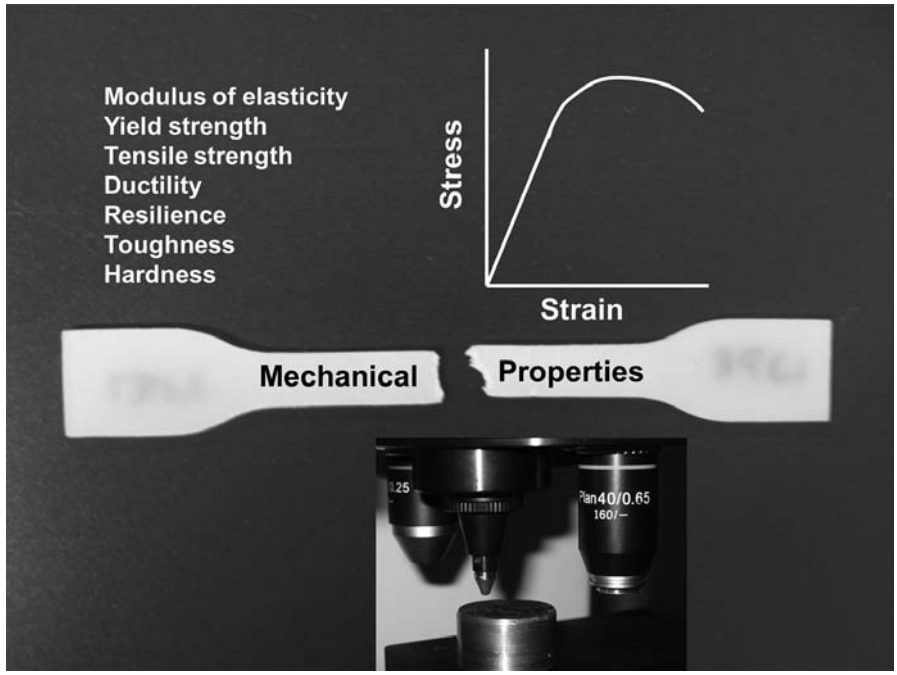


Figure 2.2 A list of the major mechanical properties of a material. Also seen is a "stress–strain" curve from which most of the mechanical properties can be obtained. A dog-bone shaped sample that has been tensile tested to fracture is shown. At the bottom of the figure is a hardness tester.

Resilience is the area under the stress–strain curve corresponding to only the linear region of the curve. It represents the amount of energy that the material can absorb and still not deform plastically. *Toughness* is the total energy absorbed by the material till it fractures and is obtained as the total area under the stress–strain curve till fracture. *Hardness* is the ability of the material to resist deformation when subject to a local compressive stress.

Mechanical properties are largely a result of material details at the microscopic level. The bonding present between the atoms in the material and crystal structure adopted by the material, and hence its slip systems directly impact the mechanical properties. While mechanical properties are measured on a macroscopic level, they originate from the atomic and crystal structure levels.

2.3 Chemical Properties

We are generally aware that chemical properties are related to the details at the atomic level. Figure 2.3 indicates a few important chemical properties.

Bonding energy, *ionization energy* and *electron affinity* represent quantities associated with the exchange of subatomic particles, specifically electrons, between atoms. Taken together they represent the chemical reactivity of the material and indicate, for example, the deteriorative characteristics of the material such as rusting.

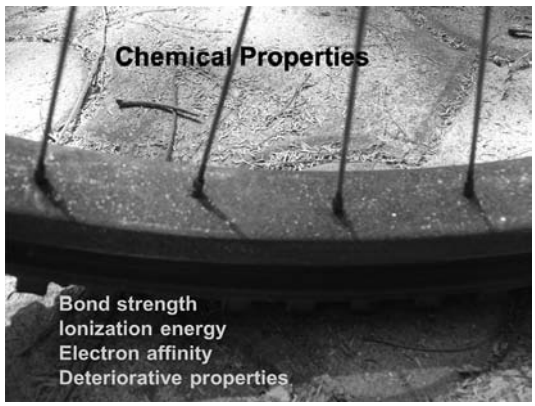


Figure 2.3 A list of the important chemical properties. In the background is a picture of a rusting bicycle wheel, a result of chemical properties of the material used for the wheel.

2.4 Electrical Properties

Figure 2.4 indicates the major electrical properties of materials. *Electrical conductivity*, or more generally conductivity, represents the transport of charge. Different charge carrier species can be present, such as electron, holes or ions, and hence we can obtain conductivity with respect to each of these species. It is important to note that simply connecting two different materials that have different charge carriers will not complete a circuit even if the conductivities of the two materials are the same, because the conductivities are defined with respect to the specific charge carrier in that material. At the interface between the two materials, charge carrier of one material will not be able to crossover if it is not the conducting species in the other material.

Wires connecting devices have electrons as charge carriers, while oxygen sensors in automobile exhausts usually contain oxygen ion conducting materials. Some materials may conduct more than one species.

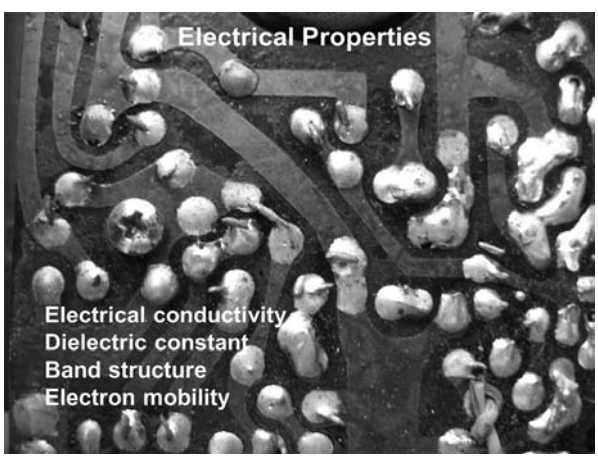


Figure 2.4 A list of the major electrical properties. In the background is the part of an electronic circuit.

It is also important to understand the origin of other electrical properties such as *dielectric constant* and *band structure*. It is of interest to see how constraints placed on the constituents of the materials, such as atoms and electrons, result in these properties for the materials.

Electron mobility represents the ease with which an electron moves in the presence of a field, and indicates the velocity that the electron will attain in the presence of a unit field. *Mobility* is also used for describing the movement of other species, such as atoms, in the presence of an appropriate driving force, and is hence used for diffusion-related phenomena as well. In other words, phenomena at electronic as well as atomic level show similarities to each other.

The circuit in Fig. 2.4 also shows signs of corrosion, which is an electrochemical phenomenon. Corrosion can degrade a circuit and limit its useful lifetime. It represents an interaction of electronic and chemical properties.

2.5 Thermal Properties

Good conductors of heat are also usually good conductors of electricity. Figure 2.5 lists the major thermal properties of materials. The figure shows a machine that is covered with snow. Such a machinery may be required to start in cold weather. The temperature outside could be well below 0°C, while the temperature inside the engine could be around 1000°C. The materials involved should be able to handle such temperature gradients.

Why do some materials expand significantly while heating and some others do not? Thermal expansion is considered in great detail in Chapter 3, and answers this question. *Thermal expansion* is associated with phenomena at the atomic level.

Thermal conductivity as well as *specific heat* are phenomena that have contributions at the atomic as well as electronic levels. Based on the material and circumstances, either the atomic or the electronic contribution to these properties can dominate.

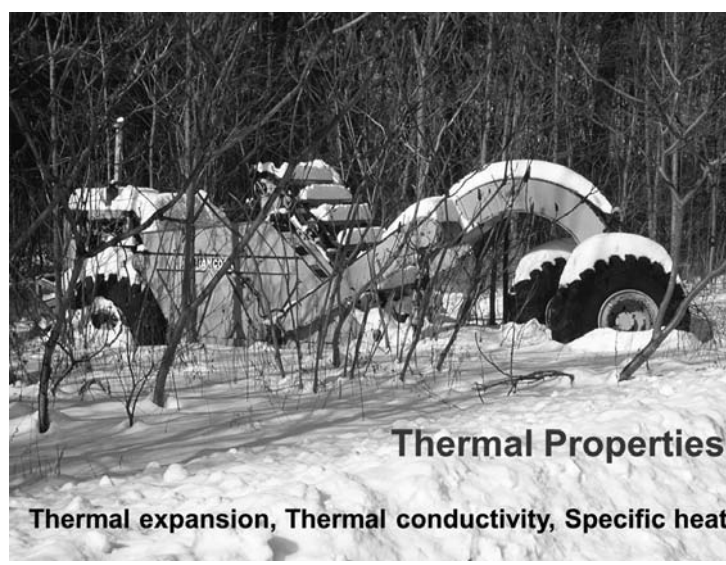


Figure 2.5 A list of the major thermal properties of materials. In the background is a machine that is covered with snow and may be expected to start and operate in cold weather.

2.6 Magnetic Properties

A variety of parameters describe the magnetic properties displayed by materials. Figure 2.6 lists the major magnetic properties displayed by materials. The *Curie temperature* is the temperature above which a ferromagnetic material displays paramagnetic behavior.

Materials respond in different ways to the presence of an externally applied magnetic field. This response leads to the display of properties such as *remanence* and *coercivity*. It also leads to the display of hysteresis. Based on the extent of the hysteresis, the materials are classified as *soft magnetic materials*, which show little hysteresis, or *hard magnetic materials*, which show large amounts of hysteresis.

The theories describing magnetism are quite involved and explored to some degree later in this book. Although the theories are quite involved, the use of magnetism is relatively commonplace. In addition to the compass shown in Fig. 2.6, audio speakers, fans and electric motors contain magnets, which are essential to their functioning. In fact, magnets in some of these equipment can affect the performance of other equipment near them, and hence some equipment come with magnetic shielding.

In the medical field, strong magnets are used in specific diagnosis equipment such as MRI (magnetic resonance imaging) equipment. For such strong magnets, superconducting wires are required, an aspect that is explored in greater detail later in this book.

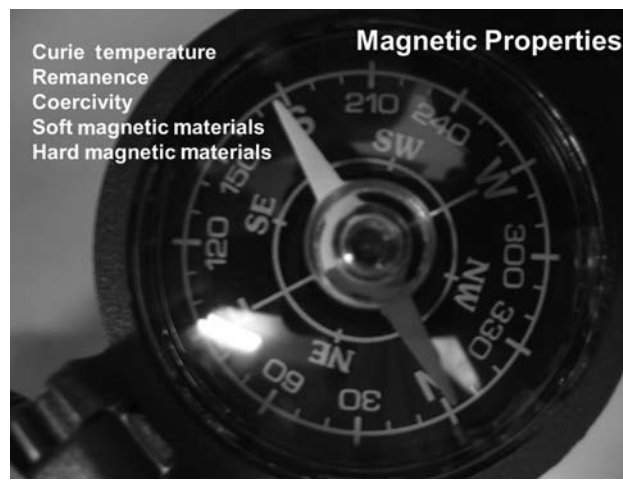


Figure 2.6 A list of major magnetic properties displayed by materials. In the background is a compass that uses magnetic properties to carry out its function.

2.7 Optical Properties

Sun control films that are used in the automobiles and construction industry rely on the *reflectivity*, *absorptivity* and *transmissivity* of materials, which are optical properties of those materials. Figure 2.7 lists the major optical properties of materials.

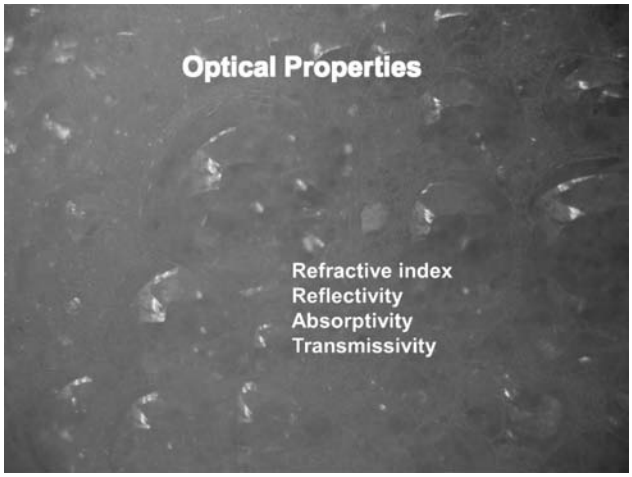


Figure 2.7 List of the major optical properties of materials. The background shows soap bubbles that display a range of optical properties.

The *refractive index* is a very important and fundamental optical property. Materials usage for optical purposes is often based on its refractive index.

Carbon in the form of graphite is opaque, but it is transparent in the form of diamond. Optical properties, just like electronic properties, depend significantly on the electronic structure of the materials. Photoelectric effect is a result of electronic as well as optical properties of the material, and helps us relate the two properties to fundamental phenomena in the material.

It is interesting to note that while lenses can be obtained very cheaply, and can even be made by filling curved surfaces with a transparent liquid such as water, high-end photographic equipment can be very expensive. The high cost of such equipment is partly due to the quality of glass used, and also in part to the special surface films that are applied to those glasses to ensure minimal to no additional reflections from the interfaces between the lenses and atmosphere.

2.8 Understanding Material Properties

Having discussed material properties in some detail in this chapter, a brief note on the approaches used for understanding the origin of these properties is presented in this section.

In physics books, discussion of properties involves reference to the band structure of the solids and understanding the origin of bands.

In chemistry books, the same topics are often described using atomic and molecular orbitals. Attention is drawn to terms such as the highest occupied molecular orbital and lowest unoccupied molecular orbital.

While the terminology used may seem different, they are simply different approaches to describe the same phenomena. The choice of approach is often based on convenience and familiarity.

Summary

1. Materials display many properties and are classified on the basis of these properties.
2. These properties are often interrelated.
3. The major properties displayed include mechanical, chemical, electrical, thermal, magnetic and optical.
4. These major properties manifest themselves in different ways during the study and use of these materials such as modulus of elasticity, yield strength, tensile strength, ductility, resilience, toughness, hardness; bond strength, ionization energy, electron affinity, deteriorative properties; electrical conductivity, dielectric constant, band structure, electron mobility; thermal expansion, thermal conductivity, specific heat; Curie temperature, remanence, coercivity, hard magnetic materials, soft magnetic materials; refractive index, reflectivity, absorptivity and transmissivity.
5. Physics books often utilize band structure to discuss material properties, while chemistry books often use bonds and orbitals to describe material properties.

Practice Questions

Multiple Choice Questions

1. The tire of an automobile operates using the following properties:
   (a) Magnetic
   (b) Thermal
   (c) Optical
   (d) Mechanical
2. The engine of an automobile does not use the following property:
   (a) Magnetic
   (b) Thermal
   (c) Chemical
   (d) Mechanical
3. The windshield of an automobile does not utilize the following property:
   (a) Thermal
   (b) Electrical
   (c) Optical
   (d) Chemical
4. The safe and sustained functioning of brakes of an automobile depend significantly on the following property:
   (a) Chemical
   (b) Electrical
   (c) Thermal
   (d) Optical
5. During its functioning, the paint on an automobile does not depend on the following property:
   (a) Optical
   (b) Chemical
   (c) Mechanical
   (d) Electrical

Short Answer Type Questions

1. List values of thermal coefficient of resistivity for a few metals and semiconductors.
2. What is the refractive index of glass and high-density glass?

3. Compare the mechanical properties of steel and aluminum from the perspective of usage in aeronautical industry.
4. Why are some materials chemically inert, for example Teflon?
5. What makes stainless steel "stainless"?

Answers

Multiple Choice Questions

1. (d)
2. (a)
3. (b)
4. (c)
5. (d)

Short Answer Type Questions

1.

Material	*Thermal Coefficient of Resistivity (K^{-1})*
Silver	0.0038
Copper	0.0039
Germanium	–0.048
Silicon	–0.075

2. The refractive index of glass is approximately 1.5, while of high-density glass is approximately 1.75. The higher refractive index enables lenses of the same "power" to be made of thinner, or with lesser amount of glass. This makes the lenses lighter.
3. As shown in the table below, the tensile strength of steel is higher than aluminum. However, the specific strength of aluminum, which is its strength per unit mass, is much higher than steel. This is because of the low specific gravity of aluminum.

Specitic Strength: Ratio of tensile strength to specific gravity for a material

	Tensile Strength (MPa)	*Specific Gravity (unitless)*	*Specific Strength (MPa)*
Steel (1020)	380	7.85	48.41
Aluminum (2024)	185	2.71	68.27

Similarly, steel has a much higher modulus of elasticity compared to aluminum; however, their specific stiffness values are nearly the same, as shown in the table below.

Specific Stiffness: Ratio of modulus of elasticity to specific gravity for a material

	Modulus of Elasticity (GPa)	*Specific Gravity (unitless)*	*Specific Modulus (GPa)*
Steel (1020)	207	7.85	26.37
Aluminum (2024)	72.4	2.71	26.72

Therefore, due to its higher specific strength and equal specific stiffness, aluminum is preferred over steel for aeronautical applications.

4. Chemically inert materials usually have strong bonds between the constituent atoms. In Teflon, for example, the C—F bonds are strong and it is difficult to break the bond to enable the material to react with any other species.

5. Stainless steel gets its superior deteriorative properties by virtue of a protective chromium oxide coating, which is stable and adherent to the base material. The steel has adequate chromium in it which ensures that the oxide layer formed, which is the protective layer, is continuous. However, if stainless steel is exposed to abrasive conditions, the protective coating is damaged and corrosion can occur.

3 Thermal Expansion

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Give examples where thermal expansion is useful, and examples where thermal expansion causes problems.
- Explain the operation of heat shrink materials.
- State the primary reason why many materials expand on heating.
- Draw an "*E* versus *r*" curve for a material.
- Differentiate between materials on the basis of the *E* versus *r* curve.



3.1 Introduction

In this chapter, we will look at a particular material property – thermal expansion. One of the reasons we will discuss thermal expansion in this chapter, early in this book, is that it lends itself to considerable analysis and understanding in a rather concise manner. Therefore, by looking at how a material is modeled to explain thermal expansion, we get a preview of the type of approach we will follow in the rest of this book. Unlike thermal expansion, some of the other properties, which are discussed later in this book, require considerably more involved analysis and more extensive development of the related background material. While we will be able to address thermal expansion in a single chapter, some of the other properties will require us to examine multiple models and develop the related background material over several chapters.

3.2 Use of Thermal Expansion/Contraction

Several day-to-day products use thermal expansion. Based on the temperature, a bimetallic strip, which consists of two metals with different coefficients of thermal expansion, will bend or straighten. Bimetallic strips are used in iron boxes, electric kettles and refrigerators. A schematic of the functioning of the bimetallic strip is shown in Fig. 3.1.

While thermal expansion is used for our advantage in many places, there are several situations in which thermal expansion is not a good phenomenon. In any application where dimensional stability of components is important, thermal expansion has to be avoided. An example where this is evident is shown in Fig. 3.2. Here two pipes are connected and the joint between them is sealed using a gasket. When the temperature of this setup is lowered, the two pipes as well as the gasket can shrink, which can result in the seal between pipes being broken.

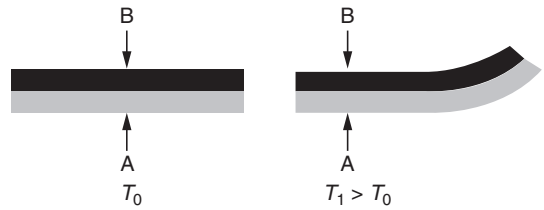


Figure 3.1 Schematic of a bimetallic strip made of metals "A" and "B." At a lower temperature T_0, the two metals have the same length. At a higher temperature T_1, metal "A" has expanded more than metal "B," and this results in the strip bending to accommodate the different lengths.

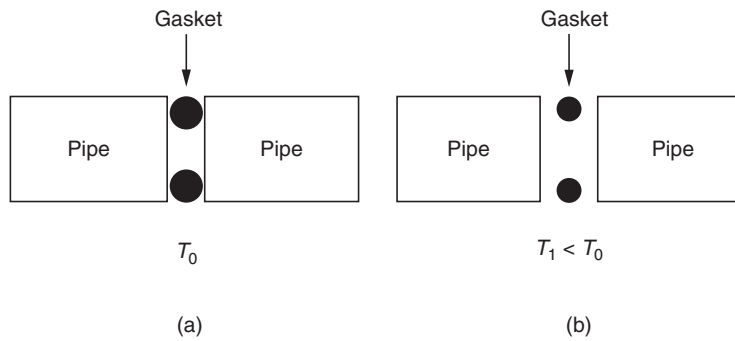


Figure 3.2 (a) Schematic showing the side view of two pipes connected using a gasket for sealing purposes, at temperature T_0. (b) As the temperature of the joint is lowered to T_1, the seal is broken because the pipes as well as the gasket have shrunk.

Incidentally, while most materials expand upon heating, there are some materials that are designed to contract or shrink upon heating. These materials are commercially referred as "heat shrinks." They are polymer-based materials that operate in one of two ways. In some cases, the cross-linking process is deliberately stopped at early stages and the material is supplied in that state. When such a material is heated, the cross-linking process proceeds to completion. During cross-linking, the molecules are pulled closer to each other, and as a result, the overall material shrinks. Another method to accomplish the same is to supply the material with additives that leave the material on heating. When the additives leave the material, the remaining material shrinks.

One of the commercial uses for this type of material is to enable electrical insulation of joints of exposed wires. A schematic of the application of a heat shrink material to enable electrical insulation is shown in Fig. 3.3. Heat shrink is available in the form of hollow tubes, and during its application one of the exposed wires is slid through an appropriate length of the tube. The length of the tube used should be adequate to cover the length of the exposed wires as well as to enable a limited overlap with the insulated parts of the wires. The exposed wires are then knotted to each other. The heat shrink tube is then slid on top of the knotted joint and is heated, typically using hot air. The tube shrinks and forms a good insulating cover over the knotted joint.

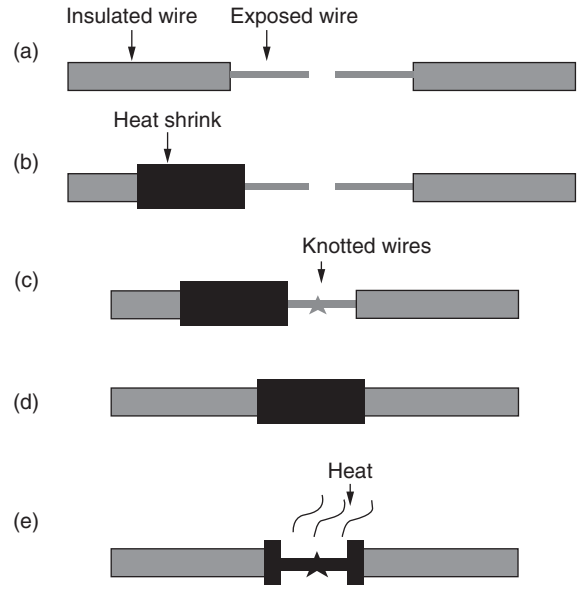


Figure 3.3 Usage of a heat shrink material: (a) Two exposed wires that need to be joined and insulated. (b) Heat shrink, in the form of a hollow tube, is slid over one of the wires. (c) The wires are knotted together. (d) The heat shrink is slid on top of the knotted wires. (e) Heat is applied and the material shrinks to form an insulating cover over the exposed wires and the knot. Since the material shrinks till it is physically stopped, it takes the shape of the knot.



Animation: "Use of heat shrink material" in the DVD.

3.3 Model for Thermal Expansion

Although, we have briefly talked about heat shrink materials in the discussion above, as indicated earlier, most materials expand on heating. In general, if the material has length L_0 at temperature T_0 and length L_1 at temperature T_1, the lengths can be related using the following expression:

$$L_1 = L_0 + L_0\alpha(T_1 - T_0)$$

where α is the linear coefficient of thermal expansion. The question is, why does thermal expansion occur?

Common perception is that materials expand on heating because atoms in a solid material vibrate more, or have larger amplitudes of vibration, at higher temperatures. While it is true that atoms in a solid material do vibrate with larger amplitudes at higher temperatures, this by itself cannot result in thermal expansion. The reason is that if the mean position of the vibration of the atoms does not change, then it does not really matter if the amplitude of the vibration has gone up – on average, the inter-atomic distance will be the same and the material will not expand. We, therefore, need to look deeper to understand why thermal expansion occurs.

To understand thermal expansion, let us take a simplified example of a one-dimensional, ionically bonded material, and see how it comes together starting from individual atoms. By examining this process and its implications, we will get a clearer understanding on why thermal expansion occurs.

Note: NaCl has a "rock salt" crystal structure that has the larger Cl^- ions occupying FCC sites, and the smaller Na^+ ions occupying the octahedral sites in the FCC structure. In the analysis below, we simplify the structure and look at it from a one-dimensional perspective.

Let us consider a one-dimensional crystal of NaCl. Since it is an ionic crystal, it consists of Na^+ and Cl^- ions. To begin with, we will look at Na and Cl separately and then put them together to form the crystal.

When an independent atom of Na becomes a univalent ion, the reaction can be written as follows:

$$Na \rightarrow Na^+ + e^- \qquad +5.1 \text{ eV}$$

Ionization energy of +5.1 eV has to be provided per atom for ion transition, and this raises the energy of the system.

Independently, let us consider the formation of a Cl^- ion:

$$Cl + e^- \rightarrow Cl^- \qquad -3.7 \text{ eV}$$

Electron affinity of 3.7 eV per atom to ion transition is released in this process.

The net energy required to create an Na^+ ion and a Cl^- ion can be obtained from the sum of the above two reactions, and is +1.4 eV per pair of ions produced as indicated in the following equation:

$$E = 1.4 \times 1.6 \times 10^{-19} \text{ J}$$

This energy does not depend on the inter-ionic distance between the two ions, and is therefore shown as the dotted line in Fig. 3.4(a). Since this is always a positive number, this alone will not enable the stable formation of the NaCl ionic crystal.

To consider the energy that results due to the electrostatic interaction between the ions, let us consider the possibility that they are infinitely separated from each other initially, and are therefore not interacting with each other, and then steadily brought closer. The energy due to the coulombic attraction between the ions is given by the following equation:

$$E = \frac{-1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

where r is the inter-ionic distance, e is the charge of an electron, and ε_0 is the permittivity of free space. This interaction occurs at an atomic or ionic level. The energy associated with this interaction is plotted as a function of inter-ionic distance in Fig. 3.4(b). This energy is negative, and if only this were to exist, the ions would collapse into each other.

As the ions get significantly close to each other, the electron clouds – especially the outer electron shells of the ions – begin to overlap. This forces electrons to move to higher energy levels in keeping with Pauli's exclusion principle. This interaction is of a sub-atomic level and causes the energy of the system to increase. The dependence of the energy on inter-ionic distance, due to this interaction, is given by the equation below, and is shown in Fig. 3.4(c).

$$E = \frac{B}{r^n}$$

This energy is positive and rises sharply as r decreases since its dependence on r is as r^{10}, that is, n is usually of the order of 10. The equation below gives the sum of the energies involved.

$$E = \frac{-1}{4\pi\varepsilon_0}\frac{e^2}{r} + \frac{B}{r^{10}} + 1.4 \times 1.6 \times 10^{-19}$$

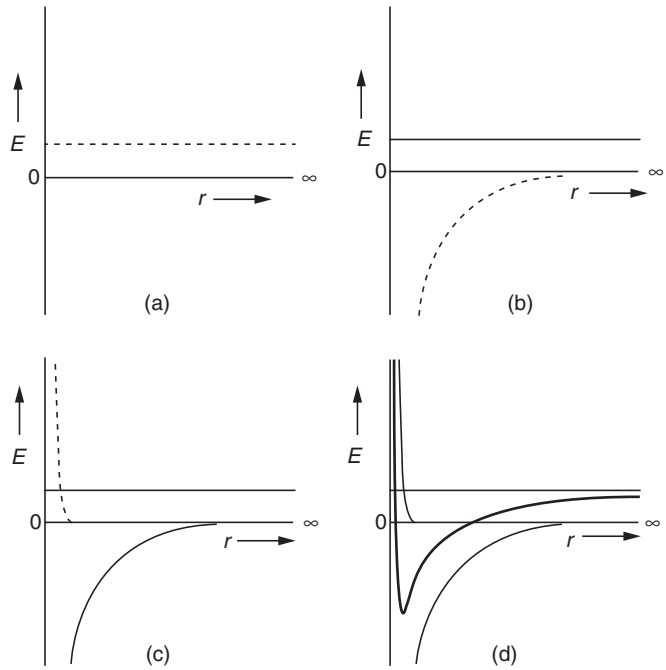


Figure 3.4 Schematic showing the contributions to the energy involved in the formation of a one-dimensional NaCl crystal: (a) Dotted line shows the sum of ionization energy and electron affinity. (b) Dotted curve results from the coulombic attraction between the ions. (c) Dotted curve shows the repulsion due to the overlap of electron clouds as the ions get closer. (d) The thick solid line shows the resultant energy as a function of inter-ionic distance.

Animation: "Energy for formation of 1D NaCl crystal – impacts thermal expansion" in DVD.

Figure 3.4(d) shows the sum of all of the energies as a function of r in the form of the solid curve.

The sum of the energies involved passes through a minimum E_0, at which point the inter-ionic separation is a single fixed value of r_0, as shown in Fig. 3.5.

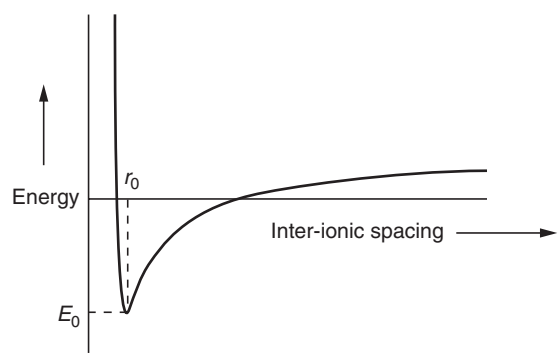


Figure 3.5 The minimum energy E_0 and equilibrium inter-ionic spacing r_0 at 0 K.

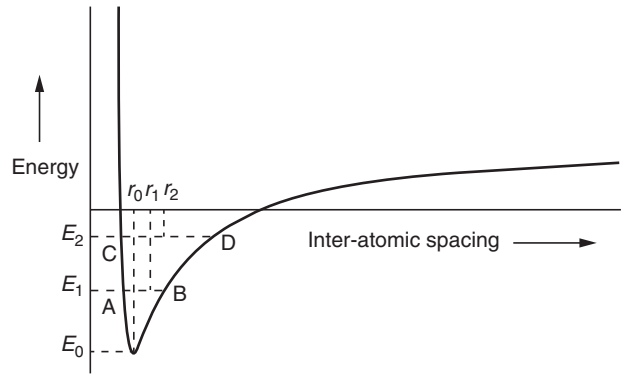


Figure 3.6 Effect of increasing temperature on the vibration of the ions and hence on the inter-ionic separation. The system is raised to temperature T_1 and then further to T_2. At T_1, when the energy is E_1, the ions are able to vibrate between the positions "A" and "B." Similarly, at T_2 when the energy is E_2, the ions are able to vibrate between the positions "C" and "D."

If the inter-ionic separation is fixed, it means the atoms are stationary, and this can occur only at 0 K. Therefore, the atoms are at their minimum energy E_0 and inter-ionic separation r_0 at 0 K.

As the temperature of the system is raised to T_1, and then further to T_2, the solid becomes consistent with the higher temperatures by having the ions gain energy, and vibrate with increased amplitudes. Therefore, as shown in Fig. 3.6, at T_1 when the energy is E_1, the ions are able to vibrate between the positions "A" and "B." Similarly, at T_2, when the energy is E_2, the ions are able to vibrate between the positions "C" and "D."

It is important to note in Fig. 3.6 that the midpoint between "A" and "B," which represents the mean inter-ionic distance at T_1, is r_1 that is greater than r_0. This is a direct result of the fact that the solid curve in Fig. 3.6 is asymmetric. The solid curve is asymmetric as a direct result of the form of the three terms (or curves) that add up to result in the solid curve. The manner in which the three terms in the equation depend on r and therefore the manner in which they display their influence in the sum, results in the solid curve being asymmetric. It is therefore important to note that thermal expansion occurs as a direct result of the fact that the "E versus r" curve is asymmetric.

A hypothetical solid, in which the E versus r curve is exactly symmetric, is shown in Fig. 3.7. In this case, when the temperature is raised, E increases, and the amplitude of vibration of the ions increases, but there is no thermal expansion since the mean inter-ionic distance is exactly the same at all temperatures.

It is also important to note that while the general approach used here can be extended to many systems, the exact shape of the resultant curve obtained will depend on the details of the specific system. In some cases, such as ceramic materials, the resultant curve will have a very deep and narrow trough – such materials will have a very low coefficient of thermal expansion. In other cases, the resultant curve will have a shallow and wide trough, which will cause the material to display a high coefficient of thermal expansion.

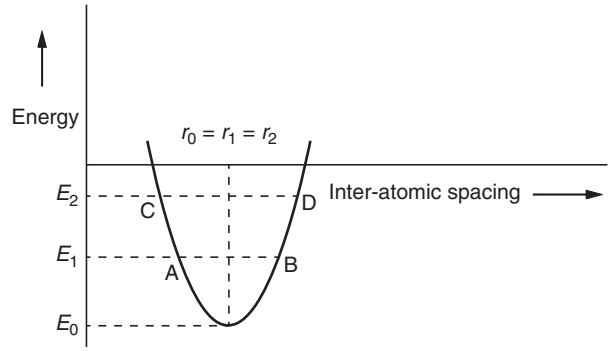


Figure 3.7 A hypothetical solid in which the *E* versus *r* curve is exactly symmetric. In this case when the temperature is raised, *E* increases, and the amplitude of vibration of the ions increases, but there is no thermal expansion since the mean inter-ionic distance is exactly the same at all temperatures.

Summary

1. Thermal expansion is useful in some situations such as the functioning of a bimetallic strip, and harmful in other situations such as sealing between tubes.
2. Some polymeric materials can be designed to shrink on heating. This shrinkage can be due to the polymerization process or the release of deliberately added additives.
3. Increase in atomic vibration with increase in temperature alone is insufficient to explain thermal expansion.
4. A model has been built to describe an ionic solid and thermal expansion, has been explained on the basis of this model.
5. This model uses interactions at the atomic and sub-atomic levels to explain a macroscopic phenomenon.
6. Thermal expansion occurs as a direct result of the asymmetry of the *E* versus *r* curve.
7. The exact shape of the *E* versus *r* curve decides the extent to which thermal expansion occurs upon heating.

Practice Questions

Multiple Choice Questions

1. Thermal expansion is useful in which one of the following situations:
   - **(a)** In the use of long continuous sections in bridges
   - **(b)** In the use of ceramic tiles as heat shields on a spacecraft
   - **(c)** In the storage of a liquid or gas in a container as the temperature increases
   - **(d)** In the attachment of a metal rim to a wooden wheel.
2. A solid is made up of atoms that display van der Waals bonding with each other. Upon heating, the following is true about the solid:
   - **(a)** It will expand more than a covalently bonded solid
   - **(b)** It will expand less than an ionically bonded solid
   - **(c)** It will first expand and then contract
   - **(d)** It will first contract and then expand
3. Upon heating a solid, the atoms of the solid
   - **(a)** Increase their amplitude of vibration
   - **(b)** Maintain the same mean position of vibration
   - **(c)** Decrease the mean position of vibration
   - **(d)** Vibrate unconstrained by their neighboring atoms

4. A rod of length l is heated at one end and cooled at the other. The location along the length of the rod where it will maintain its original cross section,
   (a) Will occur at $l/2$
   (b) Will be closer to the hot end
   (c) Will depend on the rate of heating and cooling at the two ends.
   (d) Will be independent of the rate of heating and cooling at the two ends.
5. At the equilibrium distance between atoms in an ionic solid held at temperature T,
   (a) The net force between the atoms is attractive.
   (b) The net force between the atoms is repulsive.
   (c) The net force is zero.
   (d) Net force cannot be determined, only net energy can be determined.

Short Answer Type Questions

1. Give examples of metals with high coefficient of thermal expansion, and low coefficient of thermal expansion.
2. Which metallic alloy has a very low coefficient of thermal expansion over a range of temperatures?
3. If a material undergoes a phase transformation within a range of temperature of interest, how will the model discussed in the chapter have to be modified to account for the phase change?
4. How is the boiling point of the material represented in the E versus r curve?
5. When a material is confined between two constraints, it is not allowed to expand with respect to heating, how can this be represented in the E versus r curve?

Answers

Multiple Choice Questions

1. (d)
2. (a)
3. (a)
4. (c)
5. (c)

Short Answer Type Questions

1. Among pure metals, lead and zinc have high coefficients of thermal expansion, while tungsten and molybdenum have low coefficients of thermal expansion.
2. Certain iron–nickel alloys (Fe 64%, Ni 36%) have extremely low coefficient of thermal expansion and are referred as invars (short for "invariable," due to the lack of change in length). These iron–nickel alloys are solid solutions and a detailed explanation for their behavior is not yet established.
3. Separate E versus r curves have to be drawn for each phase. These may be drawn initially for the entire range of temperatures; however, they will be valid only within the range of temperatures within which the phase exists. These ranges of temperatures will correspond to specific E values. Therefore, the material will follow the E versus r curve of the lowest temperature phase first till the E corresponding to the temperature of phase transformation, and above that the material will follow the E versus r curve of the higher temperature phase.
4. The temperature corresponding to the energy of formation of the ions represents the boiling point of the material since at that energy the ions are infinitely apart.
5. The material is forced to follow an imaginary dotted line on the expansion side of the curve that is symmetric with respect to the contraction side of the curve. Therefore, there is no net expansion.

4 Electrical Conductivity

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Identify different charge carriers.
- Write and utilize Ohm's law.
- State the different approaches to measuring conductivity.
- State the advantages of the four-probe method over the two-probe method.
- Differentiate between resistance and impedance.
- State the advantages of the AC method over the DC method.
- Graphically indicate the response of *R*, *C*, and *L* to AC signals.
- State and identify Bode and Nyquist plots.
- Draw the Nyquist plot for simple RC circuits.
- State the common approaches to analyze AC impedance data.

4.1 Introduction



In this chapter, we will first examine why it is of interest to focus our learning process on electrical conductivity. We will recognize that there are different types of charge carrying or conducting species and these species may exhibit different behaviors, and finally we will understand how conductivity is measured.

If we look at various gadgets we use in our daily life, it is easy to recognize that several technologies depend on or make use of electronic properties of materials. Toys, household appliances, and automobiles are examples where several mechanical systems have been replaced or augmented by electronic systems. Whereas, previously it was possible to open and repair toys because they were based on mechanical systems, opening present day toys brings you face to face with an electronic chip and the repair process usually stops there. Most modern automobiles have sophisticated electronics that help them maximize the fuel efficiency, manage the braking process or control the traction of the wheels, etc. These are just a few examples of how pervasive electronics has become in our present day society.

Of the electronic properties that a material can display, conductivity is of particular interest for us because of a few different reasons. First, conductivity is possibly the "most commonly used" electronic property in a technological sense, both in terms of good conductors for wires and bad conductors to provide insulation for the wires. Second, it turns out that the best conductors and worst conductors of electricity vary in conductivity by over 24 orders of magnitude. For example, silver has a conductivity of approximately $10^{7}\ \Omega^{-1}m^{-1}$, whereas Teflon has a conductivity of less than $10^{-17}\ \Omega^{-1}m^{-1}$. There is almost no other property that displays such a significant variation in its manifestation in various materials. Therefore, from a technological perspective it is of interest to study electrical conductivity, due to its widespread use, and from a scientific perspective it is of interest to see if it is possible to identify the theories that can explain such a large variation in the property.

4.2 Charge Carriers

As we take a closer look at some of the aspects associated with conductivity, it is important to recognize that in the most general sense, electrical conductivity means transport of the charge. The charge that is being transported can be carried by different species, which can be termed as charge carriers. Different charge carriers may have different mass, different interaction with their surroundings and may respond differently to changes in external environments, and may face different limitations in terms of what they can do and cannot do. The charge carriers most commonly encountered in physics and engineering are

1. Electrons
2. Holes
3. Ions

In a single circuit, different sections of the circuit may have different charge carriers. For example, consider a circuit that consists of an electrochemical power source connected to an external load, as shown in Fig. 4.1.

In this circuit, the wires connected to the external load have electrons as the charge carriers; in the electrolyte, ions are the charge carriers; and in the electrodes (anode as well as cathode), both electrons and ions are the charge carriers. This type of circuit is quite common; virtually every battery-operated gadget is an example of this circuit.

Consider also a circuit of a p–n–p transistor as shown in Fig. 4.2.

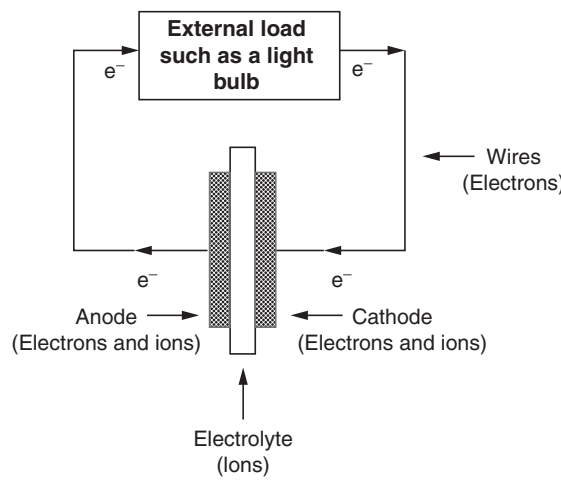


Figure 4.1 The charge carriers (indicated within brackets) in various sections of a circuit that connect an electrochemical power source, such as a battery to an external load such as a light bulb. The anode, electrolyte, and cathode together constitute the electrochemical power source.

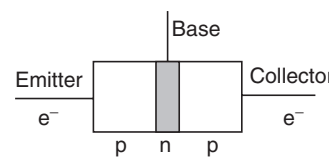


Figure 4.2 A p–n–p transistor. The charge carriers from left to right are electrons, holes, electrons, holes, and electrons.

The charge carriers in the wires that connect to the device are electrons. Within the device, the charge carriers are holes in the regions that show p-type semi-conductivity, and are electrons in the regions that show n-type semi-conductivity. In particular, in Fig. 4.2 going from left to right, the charge carriers are electrons, holes, electrons, holes, and electrons. Electronic circuits in most modern devices are built using several transistors; hence, this example is also very commonly encountered.

The two examples mentioned above highlight the fact that in our daily life we are now using devices, gadgets, and technologies that have multiple charge carriers. It is just that we are often not aware of this information, we simply think of "current" as being associated with "electrons."

As indicated earlier, in the most general sense, electrical conductivity means transport of charge. As we have already discussed that there can be different types of charge carriers, it is of interest to see how conductivity can be measured and how such a measurement handles the possibility of different types of charge carriers.

4.3 Direct Current Conductivity Measurement

This is the most familiar or common form of electrical conductivity measurement. In this form of conductivity measurement, there are two possible variations: Two-probe measurement and four-probe measurement. As the names suggest, in the two-probe measurement, the sample is simultaneously connected at two places and the conductivity is measured, whereas in the four-probe measurement, the sample is simultaneously connected at four places to measure its conductivity. While the difference in these two methods may seem trivial at first glance, there is a specific issue, discussed in this chapter, with the two-probe measurement that is effectively addressed by the four-probe measurement, and hence the latter is now the more commonly accepted technique for direct current (DC) conductivity measurement.

In a DC conductivity measurement process, current from a DC source should flow through the sample and the potential difference that develops across the sample must be measured. Using Ohm's law, the resistance of sample can be determined, and then using the dimensions of the sample, the conductivity can be determined. Ohm's law can be written as

$$V = IR$$

where V is the potential difference across the sample when the current I flows through it and R is the resistance of the sample. Once R is determined, the resistivity of the sample can be determined using the relationship:

$$R = \frac{\rho l}{A}$$

where ρ is the resistivity of the sample, l is the length of the sample, and A is the cross-sectional area of the sample. We will assume that the length and cross-sectional area of the sample are uniform and therefore each of a single value.

The conductivity of the sample is then simply the inverse of the resistivity. Conductivity σ is given by

$$\sigma = \frac{1}{\rho}$$

The units for conductivity are $\Omega^{-1}\text{m}^{-1}$.

Experimentally, the challenge, therefore, is to measure V and I values accurately and correctly, and also to determine the length and cross-sectional area of the sample.

The two-probe and four-probe methods differ in how well they determine the V value that is of relevance. As it turns out, the two-probe measurement results in a higher value for V. Hence it overestimates

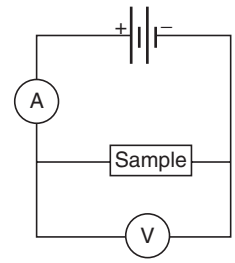


Figure 4.3 A two-probe measurement of the electrical conductivity of a sample. A battery serves as the source of DC power, an ammeter measures the current in the circuit and a voltmeter measures the potential drop across the sample.

the resistance of the sample, and therefore underestimates the conductivity of the sample. The four-probe method enables the measurement of V in a significantly more accurate manner, and is the preferred method for DC conductivity measurement.

Figure 4.3 shows the two-probe measurement technique to obtain conductivity of the sample. In the two-probe measurement of electrical conductivity of the sample, a battery may serve as the source of DC power, an ammeter can be used for measuring the current in the circuit, and a voltmeter can be used for measuring the potential drop across the sample.

When any two surfaces come in physical contact with each other, the contact is never perfect, nor are the surfaces themselves perfect. Typically, surfaces have thin non-conducting oxide layers on them, and may very well be rough at an atomic level even if polished and cleaned at a macroscopic level. As a result, when two wires come in contact with each other or when a wire is attached to any electrical component or device, the current experiences a significant resistance as it tries to flow from one contacting surface to the next. This resistance is simply referred to as *contact resistance*, which is denoted by R_c. In addition, the wires used for connecting the sample also have some finite resistance. If we ignore the resistance of the wires, we still find that the manner in which the voltmeter is connected in the two-probe measurement technique, results in a situation where the potential drop measured will be the result of the potential drop across the sample, as well as the potential drop caused by the contact resistances present where the wires from the external circuit contact the sample on the either side of the sample.

In other words, if the resistance of the sample is R_s and the current in the circuit is I, the potential drop measured by the voltmeter is given by

$$V = IR_c + IR_s + IR_c$$

The contact resistance term appears twice, because contacts have to be made on either sides of the sample. The potential as a function of position will therefore be as shown in Fig. 4.4. Since R_c can be a significant quantity, it is now evident from the equation as well as from Fig. 4.4 that the value of V is overestimated. The measurement of the potential drop attributed to the sample will be much more accurate if the positions at which the potentials are measured are changed, as indicated in Fig. 4.5.

It should also be noted that no matter what we do, we may not be able to reduce R_c beyond a point. Therefore, the trick to estimate V attributed to the sample more accurately, is to reduce the current going through the contact resistance without reducing the current going through the sample. This may seem impossible at first glance, but is actually achieved quite easily by a voltmeter. In a voltmeter, only an extremely small current flows through it and this fact is taken as an advantage in the four-probe measurement technique. In this technique also, the contact resistances cause a significant potential drop as shown in Fig. 4.5. However, the voltmeter leads contact the sample in the region where only the sample defines

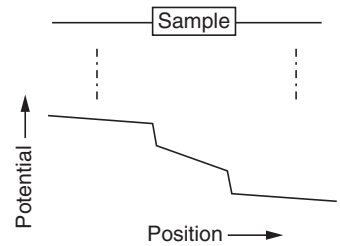


Figure 4.4 Potential as a function of position. Dotted lines indicate the positions at which a two-probe measurement technique will measure the potential drop across the sample.

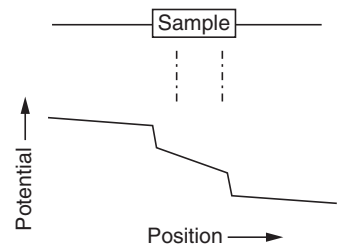


Figure 4.5 Potential as a function of position. Dotted lines indicate the positions at which the potentials are measured in a four-probe measurement. In this case, the potential drop measured can be more accurately attributed to the sample alone, and not to the contact resistance.

the potential drop. There will still be a contact resistance associated with the contact between the voltmeter leads and the sample; however, the current flowing through the voltmeter circuit will be several orders of magnitude smaller than I, the current flowing through the sample, and hence the errors caused are greatly reduced, and for all practical purposes virtually non-existent.

The manner in which connections are made to enable the four-probe conductivity measurement is shown in Fig. 4.6.

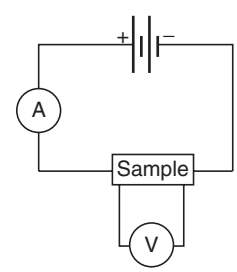


Figure 4.6 A four-probe measurement of the electrical conductivity of a sample. A battery serves as the source of DC power, an ammeter measures the current in the circuit, and a voltmeter measures the potential drop across the sample. The voltmeter leads contact the sample within the region defined by the sample and hence do not measure the potential drops due to the contact resistances on the either sides of the sample.

4.4 Alternating Current Conductivity Measurement

We have seen two examples where the charge carriers are different at different locations of a circuit. Suppose we wish to measure the conductivity of a sample that uses ions as charge carriers, for example oxygen ions, O^{2-}; the conductivity we measure is termed as *ionic conductivity*. Ionic conductivity is different from electronic conductivity that is more commonly encountered when we measure conductivity of wires. The challenge in measuring ionic conductivity is that most of our measuring instruments such as voltmeters and ammeters use electrons as charge carriers. These instruments are not designed to flow ions through them. Similarly, several (not all) ionic conductors are designed to ensure that they have minimal to no electronic conductivity in view of the end use they are aimed for. Therefore, a situation arises where we wish to determine the conductivity of a sample with a particular charge carrier, while the measuring instruments use a different charge carrier. When such a sample is connected in a typical DC conductivity measurement setup such as the one shown in Fig. 4.6, a problem arises at each of the sample–circuit interfaces, where the wires from the external circuit contact the sample. While the DC power source tries to send electrons into the sample, the sample is unable to conduct the electrons, so there is a buildup of charge within the sample in response to the potential applied, in just the manner that a capacitor responds to the application of a potential across its terminals. The buildup of charge inside the sample, by the movement of oxygen ions, opposes the buildup of charge at the electrodes contacting the sample due to the movement of electrons in the external circuit. The charge buildup in the circuit is shown in Fig. 4.7.

As a result of the charge buildup within the sample, the current in the circuit immediately drops to zero. Even during this brief interval, the current varies as a function of time though the voltage imposed is constant. Instruments with very high measuring speeds can measure the decay of current with time and this data can be analyzed to obtain information about the sample.

To measure conductivity in situations where there are different charge carriers, it is better to apply an alternating current (AC), where the direction of current flow changes with time and hence the charge in the sample is also forced to swing back and forth within the sample. This type of measurement is referred as AC conductivity measurement. AC conductivity can also be used for samples that use electrons as charge carriers, and it is a more versatile technique when compared to the DC conductivity measurement. However, the equipment required to measure AC conductivity is typically much more expensive than DC measurement instruments and, hence, where it is sufficient, DC conductivity is the preferred technique.

The schematic of the circuit used for an AC conductivity measurement is shown in Fig. 4.8 and is similar to the one used for DC measurement, except that it uses an AC power source. In case the sample is a pure resistor, Fig. 4.9 shows how current and voltage will vary with time in the circuit.

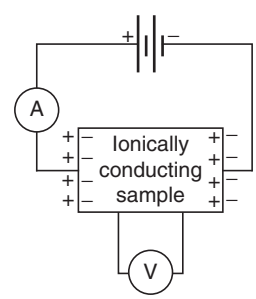


Figure 4.7 Buildup of charge when a sample that only conducts ions is connected to a DC power source.

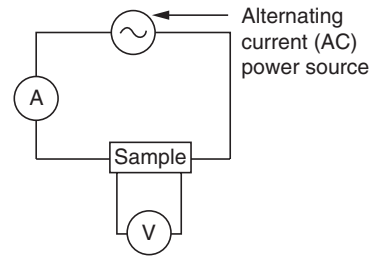


Figure 4.8 An AC four-probe measurement of the conductivity of a sample.

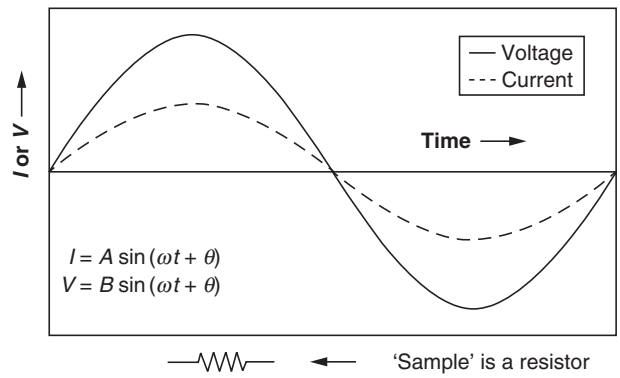


Figure 4.9 The response of a pure resistor to an applied AC signal. Here ω is the angular frequency and is equal to $2\pi\nu$, where ν is the frequency of the signal. A and B are the amplitudes of current and voltage, respectively, and θ is the phase angle.

In order to analyze the data obtained using AC signals and determine the conductivity of the samples, it is useful to understand some of the nomenclature and analysis techniques associated with AC signals.

AC signals vary with time and the current and voltage can be represented using equations of the form

$$I = A \sin(\omega t + \theta)$$

$$V = B \sin(\omega t + \theta)$$

where ω is the angular frequency and is equal to $2\pi\nu$, with ν being the frequency of the signal; A and B are the amplitudes of current and voltage, respectively; and θ is the phase angle.

In a DC measurement, the measurement process involves measuring a single data point. In a typical AC measurement, the frequency of the AC signal is an experimental variable and also a valuable tool in probing the properties of the sample. The electricity supplied in homes typically has a fixed frequency of 50 or 60 Hz around the world. However, this is just one of the possible frequencies that can be accomplished experimentally. In fact, for carrying out AC conductivity measurements, typically a very wide range of frequencies is employed – from mHz to several kHz. The sample is examined at several frequencies from within this range, and considerable information can be obtained about the sample behavior and its

fundamental properties in this process. Unlike the single-point DC measurement, the AC measurement, therefore, involves recording several data points – one each at a series of specified frequencies. The sample is subject to a relatively small amplitude AC signal (say an AC voltage) and the instruments record the AC current response to this imposed signal. The amplitude of the signal used has to be large enough to enable a recordable response from the system; however, it should also be small enough to ensure that the system responds linearly. During the experiment, variations in voltage as well as current are recorded as a function of time. The ratio of these quantities, appropriately determined, as described below indicates the response of the sample to the signals imposed on it.

While in the case of DC measurements on a pure resistor, we discuss the results in terms of resistivity – the more general phenomena as investigated using the AC technique is referred to as *impedance* and is denoted by Z. Impedance represents the tendency to obstruct the flow of current, and is the AC analog of DC resistance. In the case of a pure resistor, resistance and impedance are exactly the same. In view of the impedance offered to the flow of current, AC conductivity measurement technique is also referred to as the AC impedance technique.

To more conveniently analyze the AC data, complex number notation is used. Let us briefly look at how an AC signal is denoted using complex number notation and also consider the validity of such a representation.

As shown in Fig. 4.10, an AC signal can be thought of as having an x component and a y component at any given instant of time associated with the same modulus of the current I.

The AC signal or waveform can be thought of as a vector of fixed modulus I rotating about the origin such that the angle (and hence time) is obtained as $\tan^{-1}(I_i/I_r)$, and the amplitude $I = (I_i^2 + I_r^2)^{1/2}$. I_r and I_i are the real and imaginary components of I in the usual complex number notation, and j is $\sqrt{-1}$.

The simplicity that complex number notation offers is that multiplying any vector by j rotates the vector counter-clockwise by 90°. Therefore, for example, multiplying a vector by j twice rotates it by 180° and makes it opposite to the original vector. Therefore, by recognizing that the AC signal or waveform has varying x and y components associated with the same modulus of the current I and by denoting the waveform using complex number notation, it is possible to capture the details of the current and voltage quantities accurately, and use complex number mathematics to understand the interactions and implications of the quantities.

The response of a pure resistor to an AC signal, and the impedance offered by the resistor to the flow of current can be identified using the AC impedance technique as shown in Fig. 4.11.

In the case of a pure resistor, the impedance Z is not a function of the frequency. It is equal to the resistance R regardless of the frequency used in the measurement.

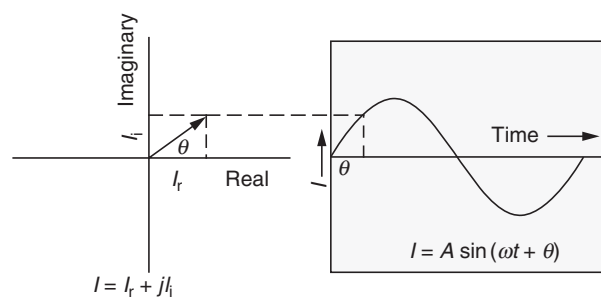


Figure 4.10 Representing an AC signal, in this case an AC current, using complex number notation. Here j is $\sqrt{-1}$, and I_r and I_i are the real and imaginary components of I in the usual complex number notation.

Animation: "Representing an AC signal using complex number notation" in DVD.

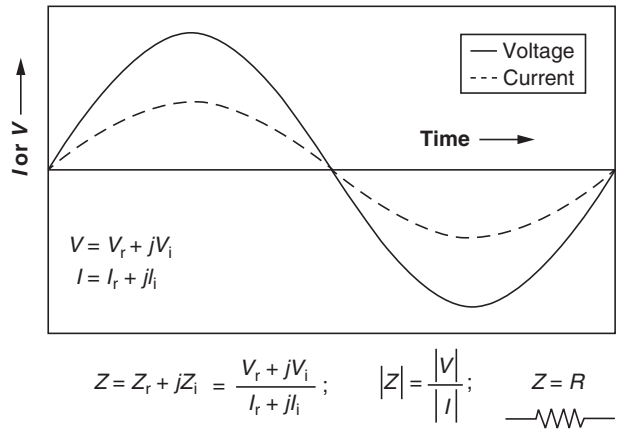


Figure 4.11 Response of a pure resistor to an AC signal. Current and voltage vary as a function of time but are exactly in phase. The impedance Z is equal to the DC resistance R.

When the sample is a pure capacitor, current and voltage are not in phase. Current leads voltage by 90°. This phase difference results in the impedance offered by a capacitor being a complex quantity.

Figure 4.12 shows the response of a pure capacitor to an AC signal. The impedance of a capacitor is observed to be dependent on the frequency used for the measurement. At very high frequencies (ω is high), the impedance Z drops to zero and the capacitor behaves as though it has been shorted internally. At very low frequencies, Z becomes a very high value and becomes infinity when ω drops to zero – or when a DC signal is employed, which is consistent with the behavior of a capacitor in a DC circuit.

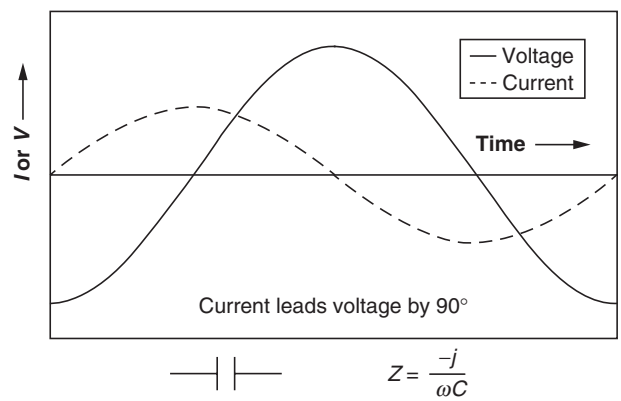


Figure 4.12 Response of a pure capacitor, of capacitance C, to an AC signal. Current and voltage vary as a function of time and current leads voltage by 90°. The impedance Z is a complex imaginary quantity and a function of the frequency ω.

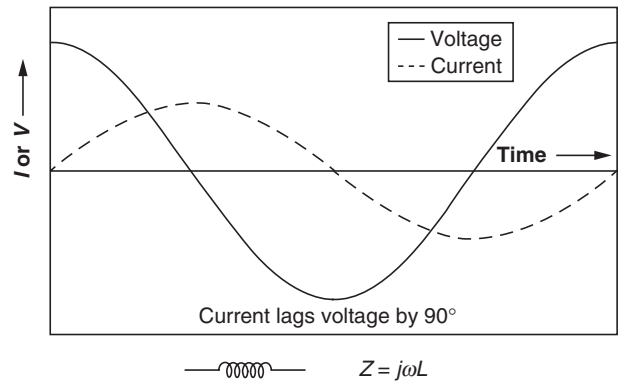


Figure 4.13 Response of a pure inductor, of inductance L, to an AC signal. Current and voltage vary as a function of time and current lags voltage by 90°. The impedance Z is a complex imaginary quantity and is a function of the frequency ω.

When an inductor is subject to an AC signal, its behavior is as shown in Fig. 4.13. The inductor shows a behavior that is descriptively inverse of the behavior shown by a capacitor, when subject to an AC signal. At high frequencies, its impedance is high, whereas it behaves as it is internally shorted when the frequency of the AC signal drops to zero or, in other words, when a DC signal is used.

The response of the three circuit elements discussed so far – a resistor, capacitor, and an inductor – as a function frequency of the AC signal used is summarized in Fig. 4.14. In each case, the data consist of a series of points, one each at specific frequencies, measured over a range of frequencies. In the case of a resistor, the points measured coincide within experimental error. In the case of capacitors and inductors, the points measured coincide with the y-axis. Please note that the negative of the imaginary impedance is plotted on the y-axis as a matter of convention since in many of the systems investigated capacitive responses are prominent. This type of plot of the real versus imaginary impedance values is called a Nyquist plot. It is also

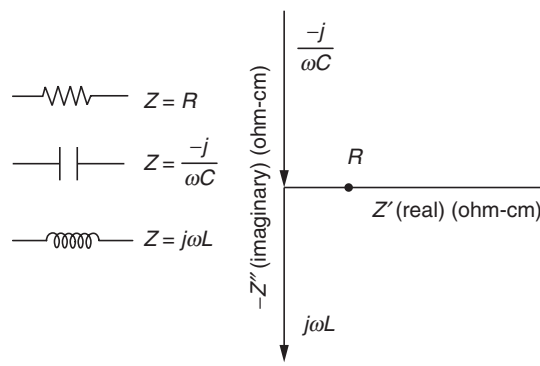


Figure 4.14 Impedance of a pure resistor R, a pure capacitor C, and a pure inductor L to an AC signal. The arrows indicate the direction of increasing frequency ω. The impedance of the resistor is unaffected by the value of ω. Z' is Z_r and Z'' is Z_i. As a matter of convention, $-Z''$ is plotted on the y-axis.

Animation: "Impedance as a function of frequency for *R*, *C*, and *L*" in DVD.

possible to plot the magnitude of the impedance as a function of logarithm of frequency or the phase angle as a function of logarithm of frequency, which are referred to as Bode plots.

While the discussion so far has looked at the individual circuit elements such as a pure resistor or a pure capacitor, real systems display characteristics that are equivalent to having a combination of resistors and capacitors. Figure 4.15 shows a possible combination of resistors and capacitors and the resultant impedance behavior as a function of frequency.

At high frequencies, the capacitor behaves as if it is internally shorted; therefore, the impedance is only R_0. At very low frequencies, the impedance of C_1 is almost infinite and the current flows through R_0 as well as R_1 and the impedance is $R_0 + R_1$. At intermediate frequencies, the impedances trace a semicircle as shown in Fig. 4.15.

AC impedance analysis is used for studying complex systems where several phenomena may be occurring in series or in parallel. By subjecting the system to an AC signal, the phenomena are forced to oscillate back and forth at each of the specific frequencies. The ease or difficulty with which the phenomena are able to follow the AC perturbation then decides the response of the system.

AC impedance data are analyzed by a few different approaches. In its simplest form, the AC impedance data are obtained from the sample or system under various experimental conditions or after the sample has been subjected to various experimental conditions, and the data are compared. Features such as the location of specific intercepts, size of any semicircles observed in the data, etc. are noted and inferences are made on what has happened to the system based on the prior experience with such systems.

A more sophisticated approach requires theoretically fitting the data to an equivalent circuit. Each element of the circuit is then associated with a physical phenomenon in the sample, and the changes in the value of the circuit element with experimentation is interpreted as changes in the parameters associated with the phenomenon. It is important to note that several circuits can simulate the same data. Therefore, the choice of circuit can greatly impact the effectiveness of the interpretation. Considerable experience is required to utilize this approach successfully.

An even more rigorous manner to handle AC impedance data is to compare it with theoretical models of the system. In this case, the theoretical models already incorporate the fundamental phenomena involved; therefore, when the theoretical curve matches the experimental data, interpretation is much easier than equivalent circuit fitting.

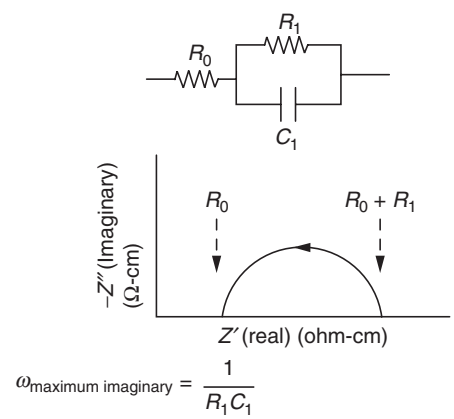


Figure 4.15 Impedance of a circuit containing pure resistors R_0 and R_1, and a pure capacitor C_1. The solid arrow indicates the direction of increasing frequency ω. The dotted arrows indicate the intercepts. **Animation: "Frequency dependence of impedance of an *RC* circuit" in DVD.**

In the curve fitting discussed above, there is one additional aspect that is important and different from typical data fitting encountered in engineering and science. In AC impedance analysis, each data point is obtained at a specific frequency. Even in the simulated data, each data point corresponds to a specific frequency. For a good fit, it is important that at each frequency the experimental and theoretical data points match. In other words, for example, the data point experimentally obtained at 35 Hz should match the data point obtained theoretically at 35 Hz. The fit is not considered acceptable if only the shape and size of the curves match, while the specific data points themselves do not match.

4.5 Short Note on Superconductivity

Superconductivity is a phenomenon that is displayed by some materials at very low temperatures. The present understanding of this phenomenon relates it significantly with magnetism and indicates a mechanism that is quite different from the mechanism displayed by materials at room temperature. Superconductivity is therefore described in greater detail in Chapter 16, later in this book.

Summary

1. Electrical conductivity is of interest both from an engineering perspective as well as scientific perspective due to its utility in many technologies as well as very large range of values of conductivity obtained when comparing excellent conductors of electricity with poor conductors of electricity.
2. Commonly encountered circuits can have different charge carriers in different parts of the circuit such as electrons, holes, and ions.
3. Conductivity can be measured using two-probe as well as four-probe methods.
4. The two-probe method gives erroneous results due to the inclusion of contact resistances in the measurement.
5. Four-probe approach significantly reduces errors from contact resistances and enables accurate measurement of conductivity.
6. Conductivity can be measured using DC methods as well as AC methods.
7. Impedance is the AC analogue to the DC resistance, and represents an obstruction to the flow of current.
8. AC impedance measurement enables handling of different charge carriers in the same circuit as well as in the evaluation of various phenomena that may be involved in the flow of charge in the circuit.
9. AC impedance data can be analyzed using Nyquist plots or Bode plots.
10. AC impedance data can be interpreted by comparing data obtained under specified experimental conditions using equivalent circuit approaches and theoretical models.

Practice Questions

Multiple Choice Questions

1. A typical laptop computer uses the following charge carriers:
   (a) Only electrons
   (b) Only holes
   (c) Only ions
   (d) Electrons and holes and ions
2. Contact resistance in a two-probe measurement can

(a) Be eliminated using silver paste at the contacts
(b) Be eliminated be soldering the contacts
(c) Be eliminated by welding the contacts
(d) Never be eliminated

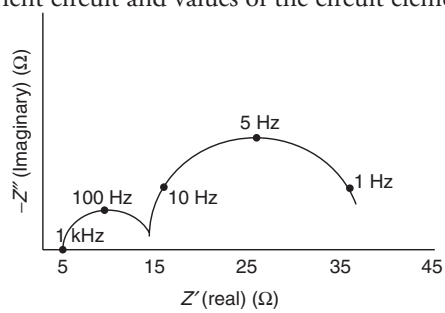
3. Contact resistance in a four-probe measurement
(a) Is twice that in a two-probe measurement
(b) Can be eliminated completely
(c) Is minimized relative to a two probe measurement
(d) Is zero

4. Impedance of a pure resistor is
(a) Lower than its resistance
(b) Equal to its resistance
(c) Higher than its resistance
(d) Unrelated to its resistance

5. Impedance of a capacitor is
(a) Low at low frequency
(b) High at low frequency
(c) High at high frequency
(d) Unrelated to frequency

Short Answer Type Questions

1. For the data given below, indicate the equivalent circuit and values of the circuit elements.



2. Assuming a sample length of 4 cm and cross-sectional area of 2 cm^2 for the data above, calculate the electrical conductivity

3. What is the percentage error in the DC two-probe measurement versus four-probe measurement, where the contact resistance is 0.1 Ω and the sample resistance is 0.01 Ω?

4. If a four-probe experimental setup were to be used for determining the conductivity of a single crystal by making contacts to its surface, what additional issues may affect the measurement?

5. Identify sample characteristics that can adversely affect the measurement of conductivity even while using a four-probe method.

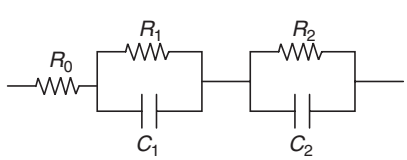
Answers

Multiple Choice Questions

1. (d)
2. (d)
3. (c)
4. (b)
5. (b)

Short Answer Type Questions

1. The equivalent circuit will be as follows:



Reading the intercepts and values of the frequency at the peaks of the graph, the values of the various circuit elements will be

$R_0 = 5\ \Omega$

$R_1 = 15 - 5 = 10\ \Omega$

$R_2 = 38 - 15 = 23\ \Omega$

$C_1 = 1/(10 \times 2 \times \pi \times 100) = 1.59 \times 10^{-4}$ F

$C_2 = 1/(23 \times 2 \times \pi \times 5) = 1.38 \times 10^{-3}$ F

2. $R = \rho l/A$

or

$\sigma = l/(AR)$

Therefore,

$\sigma = 4/(2 \times 5) = 0.4\ \Omega^{-1}\text{cm}^{-1}$

3. The four-probe measurement will sense only 0.01 Ω, whereas the two-probe measurement will measure two contact resistance and the sample resistance. Therefore, it will measure 0.21 Ω. The percentage error in the process is [(0.21 − 0.01)/0.01] × 100 = 2000% – A huge error!

4. The single crystal is contacted by the four probes at the top surface. Since electrons will choose the path of least resistance, different amounts of current will flow through different depths to minimize the potential drop. Such a variation of current as a function of depth can complicate the associated calculations to determine the conductivity of the sample.

5. Lack of homogeneity in the sample can lead to an inaccurate or erroneous estimate of the conductivity of the sample even while using the four-probe measurement technique.

5 Free Electron Gas and The Ideal Gas

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Describe features of a free electron gas.
- Calculate the fraction of empty space in a solid.
- Calculate and compare the density of a free electron gas with that of an ideal gas.
- State the justifications as well as reasons for concern with the free electron gas model.
- State the rules of the free electron theory of metals
- State the properties assumed for ideal gas atoms.
- Derive the expression for the average translational kinetic energy of atoms of an ideal gas.
- Derive the expression for the specific heat at constant volume for an ideal gas.

5.1 Introduction

Electronic conductivity, as discussed earlier, is of scientific as well as technological interest. It is of scientific interest since it varies over 24 orders of magnitude across various materials, and it is also of technological interest since many commonly used gadgets depend on electronic conductivity for their functioning. In the present as well as upcoming chapters, several models will be built to examine how the constituents of the materials interact with each other and how that results in the conductivity displayed by the material. In particular, we will focus on metals that display a positive thermal coefficient of resistivity.

It is noteworthy that in the immediate discussions, we will not include superconductors. Superconductors exhibit phenomena that are quite unique to them and are not seen in typical metallic conductors at room temperature. Superconductors are covered in a greater detail in Chapter 16.

Metals behave as there is a "free electron gas" inside them. In this chapter, we will examine the idea of a free electron gas in a considerable detail and specifically look at how reasonable or unreasonable such a "model" for a metal is.



5.2 The Free Electron Gas

The free electron gas picture is as follows: Atoms in metals occupy regular lattice sites. These atoms have a natural valence state; consistent with this valency, they release electrons and become positively charged ions. The lattice sites are therefore occupied by ionic cores. The released electrons are not confined to any single ionic core, but are free to roam within the extent of the solid. These electrons are, therefore, referred to as *free electrons* or the *free electron cloud*. The free electrons are, however, confined within the extent of the solid – they cannot escape the physical boundary of the solid. In this regard, they are said to be ***localized*** within the extent of the solid. A schematic of metallic solid containing the free electron gas is shown in Fig. 5.1.

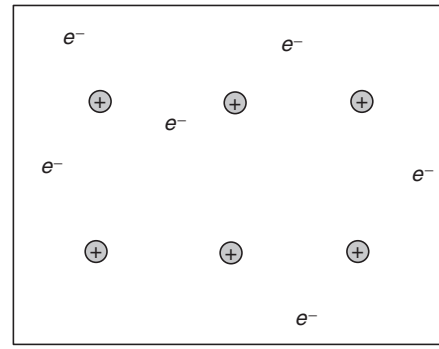


Figure 5.1 A schematic showing a metallic solid containing ionic cores at lattice sites and a free electron gas roaming through the solid.
Animation: "Free electron gas" in DVD.

For example, if there are 1000 atoms in a sample of a metal and if the natural valency of the atoms is +1, then each of the atoms releases one electron to form the free electron gas, which therefore consists 1000 electrons. These 1000 free electrons are free to roam through the extent of the metallic sample. The remaining electrons per atom remain in the vicinity of each individual ion and are not free to roam through.

The positively charged ionic cores will repel each other, and left to themselves, the solid will fall apart. The negatively charged free electron cloud provides the negatively charged atmosphere within which the positively charged ions can sit stably. Since the electrons are free to run within the extent of the solid, they can be thought of as atoms of a gas confined within a box that is the size of the solid.

Taking into account the model for the solid shown in Fig. 5.1, we will now put together equations and numbers relevant to the above picture and see if from these equations and numbers we are able to extract predictions about the conductivity of the solid. To do this, let us first focus on the phrase "free electron gas." In this phrase, the word that is important for our immediate discussion is "gas." When we discuss gases, we usually begin with the familiar phrase "ideal gas." We are familiar with the equations associated with the ideal gas and the behavior of the ideal gas. In the discussion in Chapter 6 we will impose the ideas, rules and concepts associated with an ideal gas on the free electron gas. The thought that the concepts associated with a gas can be extended to a solid, does not seem very well founded at first glance. In fact, as a state of matter, we treat a solid as being distinctly different from gas. Therefore, in this chapter we will examine the validity of the idea that ideal gas rules can be extended to one of the constituents of a metallic solid, namely the free electrons. Even at this stage, we must note that we are not attempting to treat the entire solid as being equivalent to the gas. Only the free electrons in the solid, which are running freely through the extent of the solid, are being compared with the atoms of a gas, which run freely through the extent of the container that holds them. There is, therefore, some similarities between the circumstances faced by the free electrons and ideal gas molecules. We are taking advantage of this similarity to extrapolate ideas from ideal gases to free electrons.

5.3 Packing Fraction in Solids

What aspects of this extrapolation are easily justifiable? What aspects of this extrapolation are causes for concern? We will now attempt to answer these questions.

Assume that the solid has a simple cubic crystal structure. There are atoms at the corners of the simple cube as shown in Fig. 5.2. The atoms have a radius r and the crystal structure has a lattice parameter a. The volume of the cube is a^3.

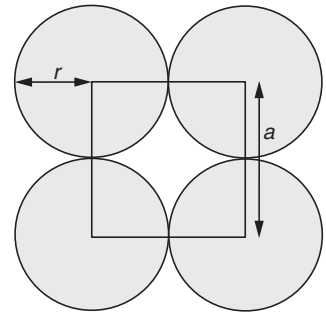


Figure 5.2 Side view of a simple cubic crystal.

There are eight corner atoms, each shared by eight adjacent unit cells, so these together contribute one atom to the unit cell on average. Therefore, the volume of the atom associated with the cube is

$$\frac{4}{3}\pi r^3$$

From Fig. 5.2, it can be observed that

$$r = \frac{a}{2}$$

The packing fraction is, therefore,

$$\frac{(4/3)\pi a^3/8}{a^3}$$

After simplifying, this fraction evaluates to approximately 50%. In other words, a metal having a simple cubic crystal structure is 50% empty!

A more closely packed structure, such as a face-centered cubic (FCC) structure, is shown in Fig. 5.3. There are eight corner atoms, each shared by eight adjacent unit cells; therefore, these together contribute one atom to the unit cell on average. Additionally, there are six face-centered atoms, each shared by two

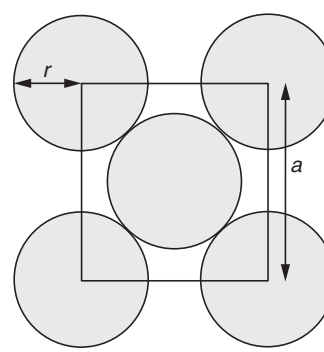


Figure 5.3 Side view of a face-centered cubic crystal.

adjacent unit cells. These altogether contribute a total of three atoms to the unit cell on average. Therefore, there are four atoms per unit cell. Here

$$4r = a\sqrt{2}$$

For FCC, there are four atoms per unit cell. Therefore, packing fraction is given by

$$\frac{4 \times \frac{4}{3}\pi\left(\frac{a\sqrt{2}}{4}\right)^3}{a^3}$$

which is approximately 74% BCC and has a packing fraction of approximately 68%. Therefore, metallic objects are approximately 25%–50% vacant. From the perspective of this available empty space within the solid and taking into account the extremely tiny size of electrons relative to atoms, it is not unreasonable to compare the state of free electrons in a solid with the atoms of an ideal gas in a relatively large vessel.

5.4 Reasons for Caution with the Free Electron Model

Consider the number density of particles in a solid and an ideal gas. At standard temperature and pressure (STP; 0°C and 1 atmosphere pressure), 1 mol of an ideal gas occupies 22.41. Now

$$1 \text{ mol} = 6.023 \times 10^{23} \text{ atoms}$$

Therefore, 6.023×10^{23} atoms are present in 22.41 at STP or in 22.4×10^{-3} m^3. On the basis of 1 m^3, a gas therefore has $(6.023 \times 10^{23})/(22.4 \times 10^{-3})$ atoms, which is approximately 3×10^{25} atoms/m^3.

Consider the metal silver, Ag, which has a density (ρ) of 10.5 g/cm^3 and atomic mass of 107 amu, and usually demonstrates a valency of +1. Now

$$\rho = 10.5 \times 10^6 \text{ g/m}^3$$

Therefore, the number of moles of Ag per cubic meter is given by

$$\frac{10.5 \times 10^6}{107} \text{ mol/m}^3$$

The number of atoms/m^3 is given by

$$\frac{6.023 \times 10^{23} \times 10.5 \times 10^6}{107} = 6 \times 10^{28} \text{ atoms/m}^3$$

Here we will assume that the atoms will be univalent, this is also the number of free electrons per cubic meter.

In summary, from the above calculations, we find that there are approximately 10^{25} particles per m^3 in an ideal gas, whereas there are 10^{28} particles per m^3 in the free electron gas. This means that there is a three orders of magnitude increase in the particles per cubic meter in a free electron gas with respect to that in an ideal gas. In other words, the free electrons are a thousand times more densely packed than atoms of an ideal gas. This is the reason we need to be cautious when we extend ideal gas behavior to the free electrons in a solid. In an ideal gas, we assume that the particles do not interact with each other between collisions. The more densely the particles are packed, the less reasonable it is to state that the particles do not interact between collisions.

Therefore, the vacant space within the solid makes the ideal gas–free electron gas comparison reasonable, while the significantly higher particle density in the free electron gas gives us reason for caution in this comparison.

With these ideas in mind, let us now list some rules that we will expect the free electron gas to obey. These rules are similar to the rules applicable to ideal gas atoms and develop what is called the "free electron theory for metals.

1. **Electrons undergo collisions with each other. These collisions are instantaneous and lead to scattering of the electrons**.
   As electrons run through the material, they collide with other electrons. We are assuming that the collisions occur without any significant time being associated with the act of the collision.
2. **Between collisions, details of the interaction of the electron with other electrons and ionic cores are neglected. However, an averaged resistive term is used to account for the interaction with the rest of the material**.
   In reality, each of the electrons will experience the negative charge of the rest of the electrons in the free electron cloud. However, the negative charge of the electron cloud is balanced by the positive charge of the ionic cores and therefore it is not entirely unreasonable to neglect interactions with the rest of the material in between collisions.
3. **The mean free time between collisions (τ) for the free electrons is independent of position and velocity of the electrons**.
   The electrons collide with other electrons as they move around. If an electron is moving fast, there is a greater chance that it will collide with another electron sooner compared to an electron moving slowly. However, it is impossible to keep track of every individual electron between every two collisions. Therefore, an average time between collisions is assumed and used for the entire collection of electrons. Since it is an average, we use it regardless of the position and velocity of any specific electron.
4. **Electrons attain equilibrium with their surroundings through collisions with other electrons**.
   When two electrons collide, there is an exchange of energy. Through this energy exchange, the collection of electrons over a period of time attains equilibrium with its surroundings. For example, when a block of cold metal is placed in a warm room, the temperature of the block reaches the temperature of the room. The reason behind it is that the free electrons close to the surface of the block experience high temperature and gain velocity and subsequently collide with free electrons in the interior of the solid and pass on the energy to those electrons as well. This process continues till all the free electrons in the solid move with velocities that are consistent with the increased temperature of the solid. In this context, it must be noted that there is always a distribution of velocities among the electrons and not a single velocity for all of the electrons. Taken in its entirety, the distribution will be consistent with the temperature of the solid. For ease of calculations, we can work with the mean of these velocities. (The ions also participate in the transfer of heat into the solid, but it is not a process that we will focus on in the immediate discussions.)

With the above rules, we will examine electronic conductivity and thermal conductivity of metallic solids. In this regard, the mathematical manner in which we employ the rules that we have stated above is based on the mathematical treatment we employ for ideal gases. Therefore, we will now briefly examine the mathematical treatment of ideal gases and highlight the results we will take from that analysis to be used with the free electron gas. We will then use these results to predict electronic conductivity and thermal conductivity of metals as dictated by the "free electron theory" of metals mentioned above.

5.5 The Kinetic Theory of Gases



The theory that describes the behavior of an ideal gas is termed as the kinetic theory of gases. Let us now briefly review the approach and some of the major predictions of the kinetic theory of gases.

An ideal gas consists of atoms considered to be point particles, which are in random motion and do not interact with each other except through collisions. These collisions are assumed to be elastic.

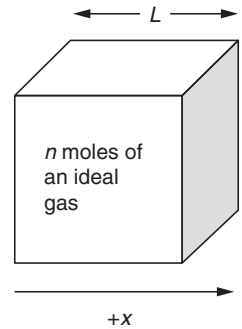


Figure 5.4 n moles of an ideal gas confined to a cubic box of length L.

Let us consider a closed volume V defined by a cubic box of length L, which contains n moles of an ideal gas, as shown in Fig. 5.4. Let a molecule of the ideal gas of mass m, traveling in the positive x direction, collide with the wall of the box and bounce back. If we assume the collision is elastic, then the change in momentum of the molecule is only in the x direction.

Initial momentum is $+mv_x$ (since it is moving in the positive x direction), final momentum is $-mv_x$ (since it moves in the negative x direction after collision with the wall of the cubic box) and change in momentum is final momentum – initial momentum $= -mv_x - (+mv_x) = -2mv_x$.

Given that the particle is traveling with $\pm v_x$ and the length of the box is L, the time between collisions is $2L/v_x$ (since the particle has to traverse the distance L in each direction once before it can collide with the same wall again).

Therefore, the rate at which momentum is delivered to the wall (which is opposite to the rate of change of momentum for the particle) is given by

$$\frac{-(-2mv_x)}{2L/v_x} = \frac{mv_x^2}{L}$$

which is then the force exerted by the particle on the wall. Considering all of the particles, the total force exerted on the wall is given by

$$F = \frac{mv_{x_1}^2}{L} + \frac{mv_{x_2}^2}{L} + \cdots + \frac{mv_{x_N}^2}{L}$$

The pressure exerted on the wall is given by

$$P = \frac{F}{L^2}$$

Therefore,

$$P = \frac{\sum_{i=1}^{i=N} mv_{x_i}^2}{L^3}$$

If we assume that the molecules have a mean-square velocity in the x direction denoted by $\langle v_x^2 \rangle$, the pressure exerted on the wall can be written as

$$P = \frac{\sum_{i=1}^{i=N} m\langle v_x^2 \rangle}{L^3} = \frac{mN\langle v_x^2 \rangle}{L^3}$$

Since there are n moles of gas in the system and N is the total number of molecules present, then $N = nN_A$, where N_A is the Avogadro number. Therefore, pressure can be written as

$$P = \frac{mnN_A\langle v_x^2 \rangle}{L^3}$$

If the overall velocity of a particle is v, then

$$v^2 = v_x^2 + v_y^2 + v_z^2$$

Since the gas molecules are in random motion, there is no preferred direction of motion; therefore we can reasonably assume that

$$v_x^2 = v_y^2 = v_z^2$$

Therefore,

$$v_x^2 = \frac{v^2}{3} = \frac{v_{\text{rms}}^2}{3}$$

Further, since the volume of the box (L^3) is equal to V, the equation for pressure becomes

$$P = \frac{mnN_A v_{\text{rms}}^2}{3V}$$

Rearranging and comparing with the equation for an ideal gas, we obtain

$$PV = \frac{nmN_A v_{\text{rms}}^2}{3} = nRT$$

where R is the universal gas constant and T is the absolute temperature.

If M is the molar mass of the gas, then $M = mN_A$. Hence,

$$v_{\text{rms}}^2 = \frac{3RT}{M}$$

or
$$\frac{Mv_{\text{rms}}^2}{3} = RT$$

Dividing by N_A on both sides, we obtain on a per molecule basis:

$$\frac{mv_{\text{rms}}^2}{3} = k_b T$$

where k_b is the Boltzmann constant and is related to the universal gas constant R through $k_b = R/N$, N_A being the Avogadro number. Therefore,

$$\frac{1}{2}mv_{\text{rms}}^2 = \frac{3}{2}k_b T$$

The above expression gives the average translational kinetic energy per ideal gas – atom/molecule in a system at equilibrium at temperature T and pressure P.

If there are N_f atoms per unit volume, the thermal energy per unit volume – which is stored as the kinetic energy of the atoms – is given by

$$E = \frac{3}{2}N_f k_b T$$

Therefore, the specific heat at constant volume (C_V^e) is given by

$$C_V^e = \frac{\partial E}{\partial T} = \frac{3}{2}N_f k_b$$

where the superscript e is introduced to indicate that the same expression will be extended to a collection of electrons in a solid.

The expressions obtained above for the average translational kinetic energy of atoms of an ideal gas and the specific heat at constant volume for an ideal gas, will be used in subsequent chapters.

Summary

1. Metals behave as though there is a free electron gas within them.
2. Atoms occupy lattice sites, while the free electrons are free to roam through the extent of the solid.
3. Only the valence electrons contribute to the free electron gas, the rest of the electrons are localized to the individual atoms.
4. The free electrons do not escape the solid.
5. There is plenty of vacant space within a solid and the electrons are extremely small. This justifies and supports the theory of a free electron gas and its comparison with an ideal gas.
6. The number density of free electrons is three orders of magnitude higher than atoms of an ideal gas, which suggests the need for caution in making the comparison.
7. The free electron theory for metals extends the laws of ideal gas behavior to the free electrons in the solid.
8. An ideal gas consists of atoms that are considered to be point particles in random motion

which do not interact with each other except through collisions. These collisions are assumed to be elastic.

9. The average translational kinetic energy of atoms of an ideal gas is given by the following expression:

$$\frac{1}{2}mv_{\text{rms}}^2 = \frac{3}{2}k_{\text{b}}T$$

10. The specific heat at constant volume for an ideal gas is given by the following expression:

$$C_v^e = \frac{\partial E}{\partial T} = \frac{3}{2}N_{\text{f}}k_{\text{b}}$$

Practice Questions

Multiple Choice Questions

1. The atoms in a metal behave as though
- **(a)** They form an ideal gas
- **(b)** They are fixed in position but can move about in a limited manner based on conditions
- **(c)** They can move like a liquid
- **(d)** They are rigidly fixed and can never be moved from their positions

2. Relative to the volume of a single electron the free electron gas
- **(a)** Occupies a large volume
- **(b)** Occupies a comparable volume
- **(c)** Occupies a small volume
- **(d)** Occupies the volume of a single atom

3. The number of free electrons in a metallic solid is
- **(a)** Equal to the total number of electrons in the solid
- **(b)** A small fraction of the total number of atoms in the solid
- **(c)** A small fraction of the total number electrons in the solid
- **(d)** Is independent of the number of atoms in the solid

4. Relative to the volume of a metallic solid, the amount of vacant space within the solid is of the order of
- **(a)** 1%
- **(b)** 10%
- **(c)** 40%
- **(d)** 90%

5. "Free electrons" are
- **(a)** Electrons that are attached to individual atoms
- **(b)** Electrons that are shared between adjacent atoms
- **(c)** Electrons that are released by one atom as part of an ionic bond
- **(d)** Electrons released by individual atoms but confined within the extent of the solid

6. Ideal gas atoms have the following properties
- **(a)** They attach to the atoms of the container they are present in
- **(b)** They form strong bonds with each other
- **(c)** They are reactive
- **(d)** They are non-reactive

7. The kinetic energy of atoms of an ideal gas
- **(a)** Depends on the temperature of the gas
- **(b)** Is independent of the temperature of the gas
- **(c)** Depends on the volume occupied by the gas at STP
- **(d)** Depends on the material used to make the container of the gas

8. The specific heat at constant volume of an ideal gas
- **(a)** Depends on the temperature of the gas
- **(b)** Is independent of the temperature of the gas
- **(c)** Depends on the volume occupied by the gas at STP
- **(d)** Depends on the material used to make the container of the gas

9. On heating a container containing an ideal gas
   (a) Average velocity of the gas atoms stays the same
   (b) Pressure exerted by the gas atoms decreases
   (c) Average velocity of the gas atoms increases
   (d) Average velocity of the gas atoms decreases
10. The specific heat at constant volume of an ideal gas
   (a) Depends on the number of atoms of the gas in the container
   (b) Depends on the number of atoms per unit volume of the gas
   (c) Depends on the number of electrons of the atoms in the gas
   (d) Depends on the number of free electrons in the gas

Short Answer Type Questions

1. Calculate the percentage of empty space in α-Fe, which has a body-centered cubic structure? Compare it with that of δ-Fe, which also has a body-centered cubic structure?
2. Calculate the percentage of empty space in graphite?
3. Metallic glasses are known to have high packing fractions and low free volume, how is this achieved?
4. What is the percentage of empty space in liquid Hg at room temperature?
5. What is the percentage of empty space in liquid nitrogen at its boiling point?
6. Indicate a few "real" gases that behave very similar to ideal gases.
7. Assuming Ar behaves as an ideal gas at STP, what is its specific heat at constant volume at STP?
8. What is the thermal energy per unit volume for Ar atoms at STP?
9. What is the mean square velocity of the Ar atoms at STP?
10. Assuming a radius for Ar atoms, what is the true "free space" available for the Ar atoms at STP?

Answers

Multiple Choice Questions

1. (b)
2. (a)
3. (c)
4. (c)
5. (d)
6. (d)
7. (a)
8. (b)
9. (c)
10. (b)

Short Answer Type Question

1. We shall make an assumption to answer this question. Let us assume that the atomic size of Fe does not change in these two structures. The lattice parameter of δ-Fe is 2.5% larger than that of α-Fe.

   Since the α-Fe has a BCC structure, and it is the lower temperature phase, we can assume that its packing fraction is the same as expected for the structure in general. There are two atoms in the unit cell. Assuming the radius of the atoms is r, the lattice parameter $a = 4r/\sqrt{3}$ Therefore, the packing fraction is given by

$$\frac{2\times(4/3)\pi r^3}{(64/3\sqrt{3})r^3} = \frac{\pi}{8/\sqrt{3}} = 0.68$$

   Therefore, the percentage of empty space in α-Fe is 32%.

In δ-Fe, the lattice parameter is about 2.5% larger than that of α-Fe. Therefore, the denominator is changed in the above calculation by a factor of $(1.025)^3$.

Therefore, packing fraction of δ-Fe is $0.68/(1.025)^3 = 0.63$. Therefore, the percentage of empty space in δ-Fe is 37%.

2. Graphite has a layered structure where alternate layers are exactly on top of each other in an *ABAB* configuration. The carbon atoms are arranged hexagonally in each layer. If the carbon–carbon bond length is a (which is 1.42 Å), the area of the base of the hexagon is

$$6 \times \frac{a^2}{2} \sin 60°$$

The height of the unit cell is twice the *AB* spacing of 3.35 Å = 6.7 Å. Therefore, the volume of the unit cell is

$$6.7 \times 6 \times \left(\frac{a^2}{2} \right) \sin 60° = 35 \text{Å}^3$$

Each atom at the corner of the hexagon is shared by six unit cells. Therefore, the base provides one atom per unit cell, and so does the top hexagon. There is one atom in between the two hexagons from *B* layer that belongs entirely to the unit cell. Along the vertical edges, there are three atoms of *B* layer that are shared by three unit cells each, therefore this provides one additional atom to the unit cell under consideration. Thus, there are four atoms per unit cell.

Since the carbon–carbon bond length is 1.42 Å, if r is the radius of each carbon atom then

$$2r = 1.42 \text{ Å or } r = 0.71 \text{ Å}$$

Therefore, the volume of the atoms in the unit cell is

$$4 \times \frac{4}{3} \pi \times 0.71^3 = 6 \text{Å}^3$$

Therefore, the packing fraction is 6/35 = 0.17 or 17%, which implies that the empty space in the graphite is 83%. In view of most of the volume of graphite being empty and its layered structure, it serves as a host for intercalation of various other species including lithium, an aspect that is used in some lithium ion batteries.

3. The high packing fractions of metallic glasses is achieved by mixing elements of significantly different atomic radii, which more effectively uses the space between atoms.

4. The density of Hg is 13.5 g/cm^3. The atomic weight of Hg is 200 amu. Therefore, 200 g of Hg will contain 1 mol of Hg. The volume corresponding to 200 g is 200/13.5 = 14.81 cm^3. The atomic radius of Hg is 1.51 Å. Therefore, actual volume of 1 mol of Hg atoms is

$$\begin{aligned} &6.023 \times 10^{23} \times \frac{4}{3} \pi (1.51 \times 10^{-10})^3 \\ &= 87.2 \times 10^{-7} \text{ m}^3 \\ &= 87.2 \times 10^{-1} \text{ cm}^3 \end{aligned}$$

Therefore, the percentage of empty space is

$$\frac{14.81 - 8.72}{14.81} \times 100 = 41.12\%$$

5. The density of liquid nitrogen is 0.807 g/cm^3. The molecular weight of N_2 is 28 amu. Therefore, 28 g of N_2 will contain 1 mol. The volume corresponding to 28 g is 28/0.807 = 34.69 cm^3. The atomic radius of N is 1.55 Å. The actual volume of 1 mol of N_2 molecules can be approximated to twice the volume of 1 mol of N atoms, and is given by

$$\begin{aligned} &2 \times 6.023 \times 10^{23} \times \frac{4}{3} \pi (1.55 \times 10^{-10})^3 \\ &= 188 \times 10^{-7} \text{ m}^3 \\ &= 188 \times 10^{-1} \text{ cm}^3 \end{aligned}$$

Therefore, the percentage of empty space is

$$\frac{34.69 - 18.8}{34.69} \times 100 = 45.81\%$$

6. Most inert gases such as helium, neon and argon behave very nearly like ideal gases. In particular when the pressure is low, inert gases will display behavior closer to that of ideal gases.

7. $C_v^e = \frac{3}{2} N_f k_b$

$= \frac{3}{2} \times \frac{6.023 \times 10^{23}}{22.4} \times 1.38 \times 10^{-23}$

$= 0.55$ J/lK

8. The thermal energy per unit volume is given by

$$E = \frac{3}{2} N_f k_b T$$

Therefore,

$$E = \frac{3}{2} \times \frac{6.023 \times 10^{23}}{22.4} \times 1.38 \times 10^{-23} \times 273$$

$= 151$ J/l

9. For ideal gases,

$$\frac{1}{2} m v_{rms}^2 = \frac{3}{2} k_b T$$

$k_b = 1.38 \times 10^{-23}$ J/atom-K and at STP, $T = 273$ K.

Atomic weight is 40 amu. Therefore 40 g will contain 6.023×10^{23} atoms of Ar. Therefore,

Mass of one atom $= \frac{40 \times 10^{-3}}{6.023 \times 10^{23}}$

$= 6.64 \times 10^{-26}$ kg.

Therefore,

$$v_{rms}^2 = \frac{3 k_b T}{m} = 170 \times 10^7 = 1.7 \times 10^5 \text{ m}^2/\text{s}^2$$

10. The density of Ar at STP is 1.784×10^{-3} g/cm^3. The atomic weight of Ar is 40 amu. Therefore, 40 g of Ar will contain 1 mol of Ar. The volume corresponding to 40 g is $40/(1.784 \times 10^{-3}) = 22.4 \times 10^3$ cm^3. The atomic radius of Ar is 1.88 Å. Therefore, the actual volume of 1 mol of Ar is given by

$$6.023 \times 10^{23} \times \frac{4}{3} \pi (1.88 \times 10^{-10})^3$$

$$= 167 \times 10^{-7} \text{ m}^3$$

$$= 167 \times 10^{-1} \text{ cm}^3$$

Therefore, the percentage of empty space is

$$\frac{22,400 - 16.7}{22400} \times 100 = 99.93\%$$

6 The Drude Model

LEARNING OBJECTIVES

After reading this chapter, you will be able to

- Derive an expression for the drift velocity of an electron.
- Derive the expression for electrical conductivity based on the Drude model.
- Calculate the electrical conductivity of a typical metal.
- State the different forms of heat transfer commonly encountered.
- Indicate the restriction on the thermal conductivity described by the Drude model.
- Write the Fourier's law of heat conduction.
- Derive the expression for thermal conductivity based on the Drude model.
- Calculate the thermal conductivity of a typical metal.
- Indicate specific instances where the Drude model has successfully predicted experimental observations.
- Write the Wiedemann–Franz law.
- Show that the Drude model is consistent with the Wiedemann–Franz law.
- Indicate specific instances where the Drude model is unsuccessful in predicting experimental observations.
- Estimate the error in Drude model's prediction of the electronic contribution to the specific heat at constant volume.
- Estimate the error in Drude model's prediction of the mean square velocity of the electrons.
- Estimate the drift velocity of electrons in Ag.
- Identify the fundamental cause for the shortcomings of the Drude model.

6.1 Introduction

The Drude model assumes that free electrons in a metallic solid behave like atoms of an ideal gas. In this chapter, we will derive expressions for the electrical conductivity and thermal conductivity of a metal based on the Drude model. We will examine how successful the model is in predicting experimental observations. Based on the comparison with experimental data, we will determine the cause for the shortcomings of the Drude Model.



6.2 Electrical Conductivity

We are familiar with Ohm's law written as an equation in the form:

$$V = IR$$

Since resistance experienced by electrons arises due to collisions with other electrons, it is not surprising that the resistance increases as the length of the conductor increases as increased length also increases the number of collisions that the electrons will experience. Similarly, the resistance is observed to decrease as the cross-sectional area of the conductor increases, since the increased area provides parallel paths for the electron to travel.

Therefore, if the resistivity of the sample is ρ, the resistance of the sample is given by

$$R = \frac{\rho L}{A}$$

Substituting into Ohm's law, we get

$$V = I\frac{\rho L}{A}$$

By rearranging, we get

$$\frac{V}{L} = \frac{I}{A}\rho \tag{6.1}$$

Since

$$\frac{V}{L} = \varepsilon$$

where ε is the electric field and

$$\frac{I}{A} = J$$

where J is the current density, Eq. (6.1) becomes

$$\varepsilon = J\rho$$

Since the resistivity ρ is the inverse of conductivity σ, we can rewrite the above as

$$J = \varepsilon\sigma$$

The above equation is simply another way to state Ohm's law. We will use the above expression in the discussion that follows since we will be more readily able to relate it to the phenomena in the material.

We will now examine how electrons move when they experience an electric field. In this process, we will get a relation between J and ε, and this will have another quantity that we will then associate with the conductivity σ.

In this exercise, we will use the free electron theory to arrive at an expression for σ of a metal. This approach is credited to Drude and Lorenz and treats the free electrons in a solid in a manner similar to atoms of an ideal gas.

Assume a one-dimensional sample of length L across which we apply a potential difference V. The field generated due to this is given by

$$\varepsilon = \frac{V}{L}$$

The force F experienced by the electrons is given by

$$F = \varepsilon e$$

where e is the charge of an electron.

Dimensionally, we can check that the above is correct since the RHS has the dimensions Coulomb Volt per meter, which is Joules per meter, which is equal to Newtons. The force F is mass times acceleration; therefore,

$$F = m\frac{dv}{dt} = \varepsilon e$$

Since the applied field, charge of the electron, and mass of the electron are constant, the above equation implies that the electrons will experience constant acceleration. The constant acceleration will result in a continuously increasing velocity of the electrons and hence a continuously increasing current. In reality this is not observed – the current reaches a steady-state value.

As the electrons move within the conductor, they collide with other electrons present. The faster they move, the more likely it is that they will undergo such a collision. Therefore, there is some form of a general resistive term that becomes more prevalent as the velocity of the electron increases.

In the assumptions that we listed in Chapter 5, in order to extend the ideal gas theory to free electrons, we mentioned the need to introduce an "approximate term" that accounts for the interaction between the electrons, and between the electrons and the ionic cores as a whole. We will now introduce this approximate term. We will list the resistive term as γv such that as v increases, the resistive force increases. Therefore, the net force experienced by the electron can now be written as

$$m\frac{dv}{dt} = \varepsilon e - \gamma v$$

In the above equation, the accelerating force is constant, while the resistive force increases with the velocity of the electrons. Therefore, with the passage of time, as the velocity of the electrons increases, the net force on the electrons drops to zero and the electrons attain a constant final velocity v_f known as the *drift velocity*:

$$\varepsilon e - \gamma v_f = 0$$

By rearranging, we get

$$\gamma = \frac{\varepsilon e}{v_f}$$

Substituting back, we get

$$m\frac{dv}{dt} = \varepsilon e - \frac{\varepsilon e}{v_f} v$$

Here, the velocity v is a variable, while v_f is a constant. It is relevant to note here that even in the absence of an applied electric field, the free electrons in a metal are in constant random motion at any temperature above absolute zero. However, in the absence of an applied field, the net velocity is zero. This is consistent with the fact that we do not see current in the absence of an applied potential or electric field. When an electric field is applied, a net velocity consistent with the direction of the applied field is observed. Therefore, the net velocity is a quantity that is initially zero, and then increases in a given direction in response to an applied field in that direction.

Incidentally, the ratio $v_f/\mathcal{E}$ represents the velocity attained by the species per unit driving force and is referred to as mobility (μ) of the species. Mobility is a more general concept and appears in the treatment of other phenomena such as diffusion.

We can now rearrange the above equation as follows:

$$m\frac{dv}{dt} = \mathcal{E}e\left[1 - \frac{v}{v_f}\right]$$

or

$$\frac{dv}{[1-(v/v_f)]} = \frac{\mathcal{E}e}{m}dt$$

The above can be integrated with v varying from 0 to v, while t varies from 0 to t:

$$\int_0^v \frac{dv}{[1-(v/v_f)]} = \int_0^t \frac{\mathcal{E}e}{m}dt$$

Therefore,

$$-v_f \ln\left[1 - \frac{v}{v_f}\right] = \frac{\mathcal{E}e}{m}t + \text{constant}$$

LHS of the equation above evaluates to zero when $v = 0$, and this value of $v = 0$ exists when $t = 0$, therefore the constant in the equation above is also zero.

By rearranging, we get

$$\ln\left[1 - \frac{v}{v_f}\right] = -\frac{\mathcal{E}e}{mv_f}t$$

The term $\mathcal{E}e/mv_f$ has the dimensions of (time)$^{-1}$, since it is acceleration (force/mass) divided by velocity. Let us denote the time indicated by this term as τ and examine what τ can be associated with. Assuming an electron starts with a velocity zero and is subject to an acceleration $\mathcal{E}e/m$, then from Newton's first law, the velocity it will attain after time τ is as follows:

$$v = 0 + \frac{\mathcal{E}e}{m}\tau$$

By the definition above, we see that for time = τ, the velocity $v = v_f$. Therefore

$$v_f = 0 + \frac{\mathcal{E}e}{m}\tau$$

Therefore, we can associate τ with the time taken to attain v_f if there is no change in the acceleration within this time, implying that there are no collisions within this time. We can therefore consider the time τ as the mean time between collisions. In reality, the typical electron may encounter a few collisions before it reaches v_f, we are looking at one possible scenario in the discussion above. Therefore, we can now write the equation as

$$\ln\left[1 - \frac{v}{v_f}\right] = -\frac{t}{\tau}$$

This equation can be rearranged as follows:

$$1 - \frac{v}{v_f} = \exp\left(-\frac{t}{\tau}\right)$$

$$\frac{v}{v_f} = 1 - \exp\left(-\frac{t}{\tau}\right)$$

$$v = v_f\left[1 - \exp\left(-\frac{t}{\tau}\right)\right]$$

This implies that at $t = 0$, $v = 0$ and at $t = \infty$, $v = v_f$.

A plot of this equation is shown in Fig. 6.1. τ is of the order of 10^{-14} s. Therefore, when we switch on a circuit, it takes approximately 5×10^{-14} s for the current to stabilize, which is virtually instantaneous from our perspective.

Considering the system from a different perspective, it is of interest to identify the charge carriers and the number of charge carriers that are moving, since this is what causes the current. In the system we are

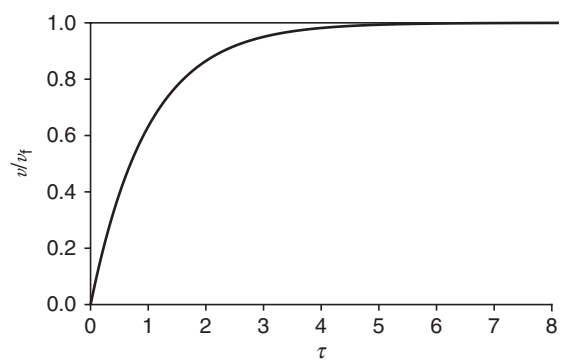


Figure 6.1 Variation of v/v_f as a function of τ.

examining, electrons are the charge carriers and the number of free electrons per unit volume N_f is the quantity of interest with respect to the current that is flowing in the system. The speed with which the electrons move is the drift velocity v_f.

Therefore, the current density J is given by

$$J = N_f v_f e$$

The RHS of the equation has the unit Coulombs per square meter per second, which is the unit of the current density. As indicated earlier,

$$\frac{\varepsilon e}{m v_f} = \frac{1}{\tau}$$

Therefore,

$$\frac{\tau \varepsilon e}{m} = v_f$$

Substituting in the expression for current density, we have

$$J = N_f \frac{\tau \varepsilon e}{m} e$$

or

$$J = \frac{N_f e^2 \tau}{m} \varepsilon$$

Comparing with

$$J = \varepsilon \sigma$$

we have

$$\sigma = \frac{N_f e^2 \tau}{m}$$

In this equation, $N_f = 10^{28}\ \text{m}^{-3}, e = 1.6 \times 10^{-19}\ \text{C}, \tau = 10^{-14}\ \text{s}, m = 9.1 \times 10^{-31}\ \text{kg}$. Therefore,

$$\sigma = 3 \times 10^6\ (\Omega\,\text{m})^{-1}$$

The above is a reasonably good prediction since metallic systems display conductivity values of the order of $10^7\ (\Omega\ \text{m})^{-1}$. As indicated in our earlier discussions, we are more interested in getting the correct order of magnitude for the predictions, rather than the exact value.

Here we have assumed a unit valency for the metal. The valency we assume will affect the N_f value since higher the valency, greater will be the N_f value. At the same time, if N_f increases, we can reasonably expect τ to decrease since the electrons will now be more likely to collide with each other.

Correct prediction of the electronic conductivity of metallic systems is therefore a success for the free electron theory of metals or the Drude model.

We will now use the same set of assumptions and try to predict the thermal conductivity of metallic systems. We will also examine how well the Drude model predicts the relationship between electrical and thermal conductivities of metals.

6.3 Thermal Conductivity



There are several instances in engineering and technology where heat transfer is a very important aspect for the successful functioning of a product. There are three modes for the transfer of heat. These are:

1. Conduction
2. Convection
3. Radiation

Let us examine these a little closer.

1. **Conduction:** This is the process of transferring heat through lattice vibrations, from atom to atom and through electrons in the solid. During this process, atoms are largely fixed to their respective lattice locations and there is virtually no large-scale movement of atoms. We typically consider conduction as the primary mode of transfer of heat within a solid.
2. **Convection:** This is the process of heat transfer that we associate with liquids and gases. This process of heat transfer involves significant atomic movement. Consider, for example, the beaker shown in Fig. 6.2.

   As the heat is introduced from the bottom into the beaker, the temperature of the liquid close to the bottom increases. In general as the temperature increases, the density of the liquid decreases. Therefore, the density of the liquid closer to the bottom of the beaker decreases, which is relative to the density of the same liquid at the top in the same beaker. This variation in density causes the liquid in the bottom to rise up and the liquid at the top to sink. The liquid and atoms of the liquid therefore get significantly mixed up in this process and heat gets transferred from the bottom to the top.

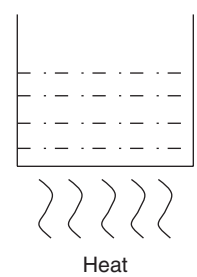


Figure 6.2 Beaker containing a liquid being heated slowly from the bottom.

This process of mixing up of the liquid is called convection. If it occurs in the manner described above, it is called natural convection. We can also speed up the process by adding stirrers or blowers to mix up the liquid faster – and hence transfer the heat faster. Such a process is called forced convection.

A day-to-day example where we see forced convection is in the functioning of air conditioners. In principle, we can have the air conditioner that provides a cold surface in one corner of the room and expect natural convection to eventually cool the room. However, this is likely to be very slow. Therefore, blowers are used for forcing air in the room to make contact with the cold surfaces provided by the air conditioner and the room gets cooled in the process. This cold air is forced to mix with the rest of the air in the room and therefore helps to cool the room quickly.

3. **Radiation:** This form of heat transfer involves the usage of electromagnetic waves. Consider the transfer of heat from the sun to the earth. There is no direct physical contact between the two, therefore, conduction is not possible. Except for the relatively thin layer of atmosphere around the earth, there is no fluid in the space between the sun and the earth, therefore convection is not possible. Energy carried through electromagnetic waves is the only way in which heat arrives from the sun to the earth.

   Incidentally, once the heat arrives on earth, the atmosphere around the earth uses convection to distribute the heat throughout the planet. The atmosphere around the earth ensures that the temperature difference between the side lit by the sun and the side that is facing away from the sun is not significantly different. On the other hand, since there is no atmosphere around a spacecraft, there is usually a very significant temperature difference between the side of the spacecraft facing the sun and the side facing away from the sun.

With this background on the different modes of heat transfer, let us now look at thermal conduction using the Drude model. While we carry out this analysis, we must keep one additional detail in mind – in metals heat conduction occurs both through the free electrons in the sample as well as the lattice vibrations present in the solid, which are called phonons. The lattice vibrations pass on the energy from atom to atom. In metallic systems, due to the presence of a large number of free electrons, the electronic contribution to thermal conductivity is the dominant contribution. In thermal insulators, the lattice contribution is significant. Since we are restricting our discussion to metals at the moment, we will only look at the electronic contribution to thermal conductivity.

So, there are different modes of heat transfer, and we are limiting our discussion to conduction. Further, conduction of heat can occur through the electrons in the sample as well as through lattice vibrations, and we are limiting our discussion to the conduction through electrons alone. For solid, metallic samples, these are reasonable limits to work within.

Thermal conductivity is denoted by K. Conductivity is the heat transferred per unit time, per unit cross-sectional area, per unit temperature gradient. If the heat energy transferred by conduction is Q, and this heat is transferred in time t through a cross-sectional area A, and in response to a temperature gradient $\Delta T/\Delta x$, the conductivity is given by

$$K = \frac{-[(Q/t) \times (1/A)]}{\Delta T/\Delta x} \tag{6.2}$$

The negative sign is present to indicate that the flow of heat occurs from the region of higher temperature to the region of lower temperature, or opposite the direction of increasing temperature – that is, opposite the direction of the positive temperature gradient – which is the term in the denominator.

The heat transferred per unit time per unit cross-sectional area is the flux of heat J_q. Therefore, in differential form, for a one-dimensional case, we can write Eq. (6.2) as follows:

$$K = \frac{-J_q}{dT/dx}$$

By rearranging, we get

$$J_q = -K\frac{dT}{dx}$$

The above equation is referred to as Fourier's law of heat conduction.

We are interested in determining K based on the assumptions of the Drude model. In order to determine K, let us consider the one-dimensional solid shown in Fig. 6.3, and the analysis that follows.

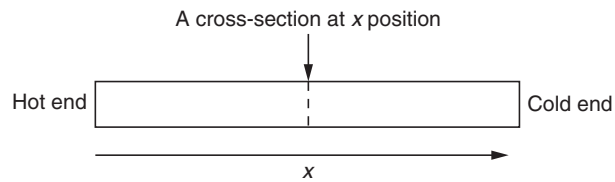


Figure 6.3 A one-dimensional sample, one end of which is hot and the other end is cold.

Let the left end of the sample be the hot end and the right end be the cold end, as shown in Fig. 6.3. In this case, the energy E is a function of the temperature T, and the temperature T itself is a function of the position x. These can be denoted as follows:

$$E[T] \quad \text{and} \quad T(x)$$

which can be combined and written as

$$E[T(x)]$$

Consider a cross-section at position x from the hot end of the sample. It is important to note that the atoms and electrons in the material do not "know" which are the hot end and the cold end in the sample. Heat energy moves from the hot end of the sample to the cold end across this cross-section, as well as from the cold end to the hot end – we perceive the net transfer of heat. In this context, the process is similar to diffusion of chemical species.

We are, therefore, interested in determining the energy of the electrons that arrive at and then cross the position x. As indicated earlier, electrons exchange energy only through collisions with other electrons. Therefore, we need to determine the last collision that the electron had before it arrived at x. The energy corresponding to that position will be the energy that the electron will possess as it arrives at x. We also noted as part of the Drude assumptions that there is a mean time between collisions τ, which is independent of position and velocity of the electrons.

So where did the electrons have their last collision before they arrived at position x? Since they are moving with the velocity v and have a mean time between collisions τ, the electrons coming from the hot end had

their last collision at $x - v\tau$ and the electrons coming from the cold end had their last collision at $x + v\tau$. The energies corresponding to these two positions are $E[T(x - v\tau)]$ and $E[T(x + v\tau)]$, respectively.

Since there are N_f free electrons in per unit volume at all positions within the sample and the electrons are assumed to move randomly, at each position we can assume that half the electrons are moving toward the hot end of the sample and half the electrons are moving toward the cold end of the sample. In other words, we are not imposing any directional preference for the movement of the electrons.

Since the velocity of the electrons is v, the flux of heat from the hot end to the cold end across the cross-section at x is given by

$$J_{\text{hot to cold}} = \frac{N_f}{2} vE[T(x - v\tau)]$$

Dimensionally this is

$$\text{m}^{-3} \times \text{m/s} \times \text{J} = \text{J} / \text{m}^2/\text{s}$$

which is consistent with the flux of heat.

Similarly, the flux of heat from the cold end to the hot end across the cross-section at x is given by

$$J_{\text{hot to cold}} = \frac{N_f}{2} vE[T(x + v\tau)]$$

Subtracting the two, we get the net transfer of heat from the hot end to the cold end of the sample across the cross-section at x.

$$J_{\text{net}} = J_{\text{hot to cold}} - J_{\text{cold to hot}}$$

Therefore,

$$J_{\text{net}} = \frac{N_f}{2} vE[T(x - v\tau)] - \frac{N_f}{2} vE[T(x + v\tau)]$$

For small changes in x, we can use the approximation

$$E[T(x - v\tau)] = E[T(x)] - v\tau \frac{dE}{dx}$$

Similarly,

$$E[T(x + v\tau)] = E[T(x)] + v\tau \frac{dE}{dx}$$

Substituting back in the expression for J_{net}, we get

$$J_{\text{net}} = \frac{N_f}{2} v \left\{ E[T(x)] - v\tau \frac{dE}{dx} - \left(E[T(x)] + v\tau \frac{dE}{dx} \right) \right\}$$

or

$$J_{\text{net}} = \frac{N_f}{2} v \left\{ E[T(x)] - v\tau \frac{dE}{dx} - E[T(x)] - v\tau \frac{dE}{dx} \right\}$$

Therefore,

$$J_{\text{net}} = \frac{N_{\text{f}}}{2} v \left\{ -2v\tau \frac{dE}{dx} \right\}$$

which leads to

$$J_{\text{net}} = -N_{\text{f}} v^2 \tau \frac{dE}{dx}$$

Since energy E is a function of the temperature T and the temperature T is a function of the position x, we can rewrite the above as

$$J_{\text{net}} = -N_{\text{f}} v^2 \tau \frac{dE}{dT} \frac{dT}{dx}$$

Since E is the energy of a single electron, dE/dT is the contribution to specific heat from the single electron, and since N_{f} is the number of free electrons per unit volume, $N_{\text{f}}(dE/dT)$ is the specific heat per unit volume associated with the electrons, and is therefore denoted using C_V^e. Therefore,

$$J_{\text{net}} = -v^2 \tau C_V^e \frac{dT}{dx}$$

Since the electrons are moving randomly, only one-third of the electrons can be expected to travel along the x direction. Therefore, in order to generalize from the one-dimensional case that we have considered to a realistic three-dimensional sample, we have to recognize that only one-third of the heat flux will occur in the x direction. Further, we have derived the above equation using the velocity of a single electron. In reality, a large collection of electrons in a solid will have a distribution of velocities associated with them. Therefore, as we average over the collection of electrons, we have to use the mean of the square of velocities of the electrons $\langle v^2 \rangle$ rather than the square of the velocity of a single electron v^2. Taking into account the above two observations, we now rewrite the equation as follows:

$$J_{\text{net}} = -\frac{1}{3} \langle v^2 \rangle \tau C_V^e \frac{dT}{dx}$$

Comparing this expression with Fourier's law of heat conduction:

$$J_{\text{q}} = -K \frac{dT}{dx}$$

We have the thermal conductivity of a metallic sample given by

$$K = \frac{1}{3} \langle v^2 \rangle \tau C_V^e$$

The dimensions of K work out as follows:

$$\frac{\text{m}^2}{\text{s}^2} \ \text{s} \frac{\text{J}}{\text{m}^3 \text{K}}$$

which works out to

$$\mathrm{J s^{-1} m^{-1} K^{-1}}$$

or

$$\mathrm{W m^{-1} K^{-1}}$$

In terms of the numerical value of the thermal conductivity, we can substitute the following into the expression for thermal conductivity:

$$\langle v^2 \rangle \approx 10^{10} \ \frac{\mathrm{m}^2}{\mathrm{s}^2}$$

$$\tau \approx 10^{-14} \ \mathrm{s}$$

$$C_V^e \approx 10^6 \ \mathrm{J/(m^3 K)}$$

Therefore,

$$K \approx 10^2 \ \mathrm{W m^{-1} K^{-1}}$$

The thermal conductivities of metallic conductors such as Ag, Cu and Au are in the same order of magnitude as the above prediction.

As discussed, we wish to understand from first principles why materials display the properties that they do. In particular, we wish to explain material properties based on our understanding of how constituents of the material behave, and how they interact with each other and with the surroundings.

Using the Drude model, we have obtained predictions for electronic and thermal conductivities of materials. They are given by the following expressions.

Electronic conductivity:

$$\sigma = \frac{N_f e^2 \tau}{m}$$

Thermal conductivity:

$$K = \frac{1}{3} \langle v^2 \rangle \tau C_V^e$$

With the expressions above, it has been possible to obtain an understanding of these two properties independently. The equations above match experimental data reasonably well, which is the most critical aspect of validation of any theory.

Having examined these two properties independently, let us now see if there is any inter-relationship between these properties. Several electrical appliances such as fans and motors use windings through which current flows in order to create electromagnets – more the current, more powerful is the electromagnet. At the same time, current flowing through any metallic conductor generates heat due to resistive heating. Therefore, metals chosen for preparing windings have to possess high electrical conductivity and consequently possess low electrical resistivity. The metal of choice for windings for electromagnets is copper because it is a good conductor of electricity.

One of the technologies that use thermal conductivity as a central aspect of its operation is *heat exchanger*. Heat exchangers extract heat from one location and pass the heat energy on to another

location. They are used in a variety of systems such as air conditioners, laptop computers, etc. The material of choice for this technology is also copper because it is also a good conductor of heat. Generalizing further, it is observed that materials that are good conductors of electricity are often also good conductors of heat.

In the Drude model for metallic systems, free electrons carry out the task of electrical conductivity as well as thermal conductivity. The model therefore creates a situation where factors impacting electrical conductivity also impact thermal conductivity, which results in the model being consistent with the observation that good conductors of electricity are also good conductors of heat.



6.4 The Wiedemann–Franz Law

Around the year 1850, Wiedemann and Franz experimentally investigated the relationship between electrical and thermal conductivities for several metals. They discovered that the ratio of the electrical to the thermal conductivity was constant for several metals at a given temperature. In particular, they found

$$\frac{K}{\sigma T} = \text{Constant} = 2.4 \times 10^{-8}\ \text{J}\Omega\text{s}^{-1}\text{K}^{-2}$$

Are the predictions of the Drude model consistent with the above experimental observation? Using the results of the Drude model, let us write an expression for $K/\sigma T$:

$$\frac{K}{\sigma T} = \frac{(1/3)\langle v^2\rangle \tau C_V^e}{(TN_\text{f} e^2 \tau)/m}$$

By simplifying, we get

$$\frac{K}{\sigma T} = \frac{m\langle v^2\rangle C_V^e}{3TN_\text{f} e^2}$$

The kinetic theory of gases, discussed in Chapter 5, gave us the following results:

$$\frac{1}{2} m\langle v^2\rangle = \frac{3}{2} k_\text{b} T$$

$$C_V^e = \frac{3}{2} N_\text{f} k_\text{b}$$

Assuming these results hold for free electrons as well and by substituting these results in the equation above we get

$$\frac{K}{\sigma T} = \frac{3}{2}\frac{k_\text{b}^2}{e^2}$$

This is then the prediction made by the Drude model. Substituting the values for Boltzmann's constant and the charge of an electron, we get

$$\frac{K}{\sigma T} = 1.1 \times 10^{-8}\ \text{J}\Omega\text{s}^{-1}\text{K}^{-2}$$

This is very close to the value obtained experimentally by Wiedemann and Franz, and certainly in the correct order of magnitude. Correctly predicting the Wiedemann–Franz law, in addition to correctly predicting the electrical and thermal conductivities independently, are the major successes of the Drude model.

6.5 Shortcomings of the Drude Model



As indicated earlier, the Drude model has extended ideal gas laws to the constituents of a solid, where the number density of particles is higher by three orders of magnitude. Therefore, there is a reason for concern. However as we have just seen, despite such concerns, the model displays significant success in the predictions it makes. As it turns out, the correct prediction of the thermal conductivity has occurred fortuitously.

The value of C_V^e predicted using the ideal gas laws is higher by two orders of magnitude than the experimental values obtained using low temperature measurements, where the electronic contributions are significant. Later in this chapter we will see that we can predict the value of τ with reasonable confidence. Therefore, the correct prediction of thermal conductivity implies that the prediction of $\langle v^2 \rangle$ is correspondingly lower by two orders of magnitude.

In addition to thermal and electrical conductivities, the Drude model is also successful in predicting some but not all of the instances of the Hall coefficient. The Hall effect was discovered around the year 1880. A schematic of the effect is shown in Fig. 6.4.

It is an important effect that enables us to determine the sign of the charge carrier in a conductor. Measuring a current alone does not tell us anything about the sign of the charge carrier in a conductor. It was noted that if a magnetic field is placed perpendicular to the direction of a flowing current, it deflects the charge carriers in a direction perpendicular to the magnetic field as well as the flowing current. A potential, therefore, develops perpendicular to the direction of flow of current. Buildup of charge occurs till the potential developed opposes any further movement of charge in the perpendicular direction. Depending on the sign of the charge carrier, the potential is either positive or negative. The Hall coefficient, R_H, which appears in the associated calculations, is negative if the charge carrier is negative and positive if the charge carrier is positive. The Drude model is consistent with a negative R_H, but is not able to predict a positive value for R_H.

While a vast majority of the elements in the periodic table are metallic in nature, any general theory for materials should also account for semi-conductors and insulators. While the Drude model uses N_f to distinguish between materials, this alone does not capture the differences between materials comprehensively. For example, the changes in material properties with changes in crystal structure and the existence of anisotropy

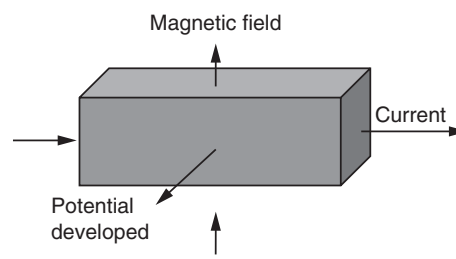


Figure 6.4 Appearance of the Hall effect in a conductor carrying current, which is subject to a magnetic field perpendicular to the current. A potential is developed, which is perpendicular to the magnetic field as well as the current.

in most crystalline solids cannot be explained simply on the basis of N_f. N_f is the same regardless of direction. While, in an ideal gas there is no preferred orientation, which is the reason we have

$$\langle v_x^2 \rangle = \frac{1}{3}\langle v^2 \rangle$$

In a crystalline solid, there is a distinct directionality in that the ionic cores are not randomly distributed. Therefore, to the extent that ionic cores impact material properties, the properties will also display directionality or anisotropy. In the Drude model, we have largely ignored the presence of the ionic cores except to introduce a general resistive term γv and it is therefore not surprising that the predictions demonstrate limitations. Therefore, the model needs to be further refined.

The Drude model successfully predicts electrical and thermal conductivities of metallic systems and the Wiedemann–Franz law but makes incorrect predictions of C_V^e, $\langle v^2 \rangle$ and R_H. It is now of interest to see how we can improve the model. In particular, we need to identify the specific fundamental assumptions of the Drude model, which need to be changed and to identify the appropriate manner to incorporate these changes. To understand the source of shortcomings in the Drude model, let us numerically estimate some of the following important quantities predicted by the Drude model:

$$\sigma = \frac{N_f e^2 \tau}{m}$$

$$K = \frac{1}{3}\langle v^2 \rangle \tau C_V^e$$

$$C_V^e = \frac{3}{2} N_f k_b$$

$$\frac{1}{2} m \langle v^2 \rangle = \frac{1}{2} k_b T$$

Consider the metal silver that is a good conductor of electricity and heat, and let us carry out calculations with respect to this element.

Element: Ag
Atomic weight: 108 amu
Density: 10.5 gm/cc

The density is therefore 10.5×10^6 g/m^3. The number of moles per cubic meter is given by

$$\frac{10.5 \times 10^6}{108} \sim 10^5 \text{ mol/m}^3$$

The corresponding number of atoms per cubic meter is given by $10^5 \times 6.023 \times 10^{23}$ atoms of Ag/m^3 $= 6.023 \times 10^{28}$ atoms of Ag/ m^3.

Assuming that the Ag atoms are univalent, the number of free electrons per cubic meter will be the same as the number of Ag atoms per cubic meter. Therefore, $N_f = 6.023 \times 10^{28}$. Since $k_b = 1.38 \times 10^{-23}$ J/atom/K, we have

$$C_V^e = \frac{3}{2} N_f k_b = \frac{3}{2} \times 6.023 \times 10^{28} \times 1.38 \times 10^{-23} \approx 12.5 \times 10^5 \text{ J/m}^3\text{/K}$$

Since there are approximately 10^5 mol/m^{3}of free electrons in solid Ag (same as the number of moles of Ag atoms per cubic meter),

$$C_V^e \approx 12.5 \text{ J/mol/K}$$

Data obtained at low temperatures in the 1960s indicate that the experimental value for C_V^e is ≈ 0.2 J/mol/K. Therefore, the theory overestimates the value of C_V^e by approximately two orders of magnitude. Since the estimation of thermal conductivity is correct and is given by the equation

$$K = \frac{1}{3}\langle v^2 \rangle \tau C_V^e$$

it appears that we are underestimating $\langle v^2 \rangle$ by two orders of magnitude – assuming that we are estimating τ correctly, which we will see shortly. The value of $\langle v^2 \rangle$ predicted by the Drude model may be calculated as follows:

$$\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_b T$$

Given that the mass of the electron is 9.1×10^{-31} kg at room temperature or approximately 300 K, we have

$$\langle v^2 \rangle = \frac{3 \times 1.38 \times 10^{-23} \times 300}{9.1 \times 10^{-31}} \text{ m}^2/\text{s}^2$$

Therefore, as predicted by the Drude model,

$$\langle v^2 \rangle \approx 1.38 \times 10^{10} \text{ m}^2/\text{s}^2$$

which is underestimated by two orders of magnitude.

Let us now estimate drift velocity of electrons in Ag that has been subject to a potential difference across its ends. We will need to make some assumptions about the dimensions of the wire and potential difference being applied across it. Let us assume that we have a silver wire that has a length $l = 1$ m and a diameter of 1 mm (10^{-3}), and we have applied 1 V across the length of the wire.

Given that the conductivity (σ) of Ag is 6.3×10^7 Ω^{-1} m^{-1}, we have the resistance R of the wire given by

$$R = \frac{\rho l}{A} = \frac{l}{\sigma A}$$

where ρ is the resistivity and A is the cross-sectional area of the wire. Therefore,

$$R = \frac{1}{6.3 \times 10^7 \times \pi \times 10^{-6}/4} \approx \frac{1}{50} \Omega$$

When we apply 1 V across the ends of this wire using Ohm's law we have

$$V = IR$$

Therefore,

$$I = 50 \text{ A}$$

which incidentally is a lot of current, but is to be expected since Ag is a good conductor of electricity.

Assuming that all of the free electrons are carrying the current and assuming that there is conservation of charge, that is, the charge entering the wire is equal to the charge leaving the 1-m length of wire, then in the time taken for electrons to travel the length of the wire, the total charge entering or leaving the length of wire is equal to the number of free electrons in the length of wire.

$$\text{The number of free electrons in the wire} = N_f \times l \times A$$
$$\text{Charge corresponding to these electrons} = N_f \times l \times A \times e$$

In the presence of the voltage, if we assume that the electrons have a drift velocity v_f, time taken by the electron to travel a distance l is l/v_f. If the current in the circuit is I, the total charge transported during this time is $I \times l/v_f$. Therefore,

$$I \times \frac{l}{v_f} = N_f \times l \times A \times e$$

Rearranging and simplifying, we get

$$v_f = \frac{I}{N_f \times A \times e} \approx \frac{50}{6 \times 10^{28} \times 0.8 \times 10^{-6} \times 1.6 \times 10^{-19}}$$

Therefore,

$$v_f = 6 \times 10^{-3} \text{ m/s}$$

or the drift velocity is approximately 6 mm/s.

It is quite interesting to discover that the value of drift velocity is so low. At this velocity, it will take approximately 150 s or 2.5 min for the current to travel the 1-m length of the wire. This drift velocity is much slower than normal human walking speed! Yet our day-to-day experience is contrary to this. As soon as we turn on the switch, the light bulb becomes on, no matter how far it is from the switch. How is this possible given the low drift velocity of the electrons? The answer is simple: Electrons close to the bulb start moving as well and light the bulb, electrons from the switch get there much later. The situation is the same as water in a pipe that is already full of water – just the way a wire is full of free electrons from end to end. When the tap is opened far away, water that is already close to the exit of the pipe begins to flow out of the pipe almost immediately. Water just exiting from the tap takes much longer time to reach the end of the same pipe.

While discussing the topic of electron velocity, we must note the difference between the drift velocity we have just calculated and the mean square velocity we calculated earlier. Drift velocity is the net velocity of the electrons in the direction of the applied field and turned out to be a relatively small number. The mean square velocity corresponds to the velocity with which electrons randomly move within the material – there is no net movement of electrons in any direction in this case. The mean square velocity turns out to be a large number, which itself we indicated had been underestimated by two orders of magnitude.

The drift velocity is a believable number since it is extracted directly from the current flowing in a conductor, assuming simply charge conservation. The meantime between collisions (τ) can be directly estimated from v_f, as we will see below. Therefore, we can place confidence in the estimate of τ and hence we are justified in doubting the estimate of $\langle v^2 \rangle$, as we indicated earlier.

Estimating τ:

$$\tau = \frac{mv_f}{\varepsilon e} = \frac{9.1\times10^{-31}\times6\times10^{-3}}{1.6\times10^{-19}\times1} = 3.5\times10^{-14}\ \text{s}$$

The relationship used for determining τ, as shown above, is derived from the experiments associated with movement of electrons in response to an electric field. However, we are "placing confidence" on its use in relation to the expression for thermal conductivity, where there is no electric field applied. Why is this so?

The justification is as follows: Electrons move very rapidly and in a random manner as a result of their thermal energy. This results in a high value for $\langle v^2 \rangle$, which is of the order of 10^{10} m^2/s^2 and can be loosely thought of as corresponding to a modulus of velocity of the order 10^5 m/s. As electrons move around with such high velocities within an enclosed region, *due to their thermal energy*, they collide with each other and this results in the meantime between collisions that is being identified as τ. The additional velocity that the applied electric field causes is relatively miniscule and is of the order of 10^{-3} m/s in the direction of the applied field. This drift velocity v_f in response to the electric field is limited by the collisions that the electrons encounter as a result of the random movement of the electrons due to their thermal energy. In other words, the collisions are dominated by the thermal state of the material. Therefore, even though we use the drift velocity (caused by the applied electric field) to gauge the meantime between collisions, the value obtained is dominated by the thermal state of the system, and therefore can be used for describing the thermal behavior with confidence. In other words, it is the v_f of the electrons that is limited by the existence of a τ in the system, rather than the τ being limited by the v_f of the electrons.

So we find that the Drude model overestimates the specific heat at constant volume of free electrons, and underestimates the mean square velocity or effectively the translational kinetic energy of the electrons. In other words, the Drude model is not able to accurately describe how energy is present within the system.

For the Drude model, the kinetic theory of gases has been used for describing how energy is held within the system of free electrons in a solid conductor. Given the very large number of free electrons in a conductor and the large number of atoms in a mole of an ideal gas, the kinetic theory does not try to determine the velocities of each electron or atom individually. Instead a statistical description is used for understanding how energy is held within the system. This statistical description used for ideal gases is called the Maxwell–Boltzmann distribution. It applies to a large collection of non-interacting particles, such as the atoms of an ideal gas, and estimates how many particles will occupy an energy level or possess a certain amount of energy, or in other words, how particles are distributed across energy levels. From this, the $\langle v^2 \rangle$ can be calculated.

To the extent that we have imposed the kinetic theory of gases on electrons in a solid, we have effectively imposed the Maxwell–Boltzmann statistics on the collection of free electrons.

The errors we see in the Drude model indicate that the assumption of Maxwell–Boltzmann statistics for the free electrons is incorrect at some fundamental level. This is the underlying message we have extracted from our analysis of the system so far. This is not surprising because we have imposed rules that pertain to non-interacting particles, on electrons that could interact with each other or with the ionic cores.

To arrive at improved models for electrons in a solid, in Chapter 7 we will look at the assumptions of the Maxwell–Boltzmann distribution and derive the distribution. We will examine the results we will obtain. We will recognize that only certain types of particles can be reasonably expected to follow the Maxwell–Boltzmann statistics. These are called classical particles. We will focus our attention on the definition of a classical particle and consider what other description will be reasonable for electrons. Once we arrive at an alternate description for electrons, we will again try and make predictions and examine how successful we have been in improving the model for electrons in a solid.

Summary

1. Ohm's law can be written as

$$V = IR$$

or alternately as

$$J = \varepsilon\sigma$$

2. The mean time between collisions is given by

$$\tau = \frac{mv_f}{\varepsilon e}$$

3. The current density in a metallic system is obtained from the following expression:

$$J = \frac{N_f e^2 \tau}{m}\varepsilon$$

4. The conductivity of a metallic is system is given by the expression:

$$\sigma = \frac{N_f e^2 \tau}{m}$$

5. There are three modes for the transfer of heat – conduction, convection and radiation.

6. In solids, conduction of heat can occur through the electrons in the sample as well as through lattice vibrations.

7. In metallic solids, conduction of heat through the electrons is the dominant mode of heat conduction.

8. Fourier's law of heat conduction is expressed as

$$J_q = -K\frac{dT}{dx}$$

9. The thermal conductivity of a metallic sample given by

$$K = \frac{1}{3}\langle v^2\rangle \tau C_V^e$$

10. Materials, which are good conductors of electricity, are often also good conductors of heat.

11. The Wiedemann–Franz law states

$$\frac{K}{\sigma T} = \text{constant} = 2.4\times10^{-8}\ \text{J}\Omega\text{s}^{-1}\text{K}^{-2}$$

12. The Drude model predicts

$$\frac{K}{\sigma T} = 1.1\times10^{-8}\ \text{J}\Omega\text{s}^{-1}\text{K}^{-2}$$

13. Correctly predicting the Wiedemann–Franz law, in addition to correctly predicting the electrical and thermal conductivities independently, is the major success of the Drude model.

14. The correct prediction of the thermal conductivity by the Drude model is fortuitous, since errors in predicting C_V^e and $\langle v^2\rangle$ cancels each other out.

15. The Drude model is also able to predict some aspects of the Hall effect though not all of it.

16. The Drude model predicts

$$C_V^e \approx 12.5\ \text{J/mol/K}$$

while the experimental data indicate

$$C_V^e \approx 0.2\ \text{J/mol/K}$$

17. The Drude model predicts:

$$\langle v^2\rangle \approx 1.38\times10^{10}\ \text{m}^2/\text{s}^2$$

The mean square velocity corresponds to the velocity with which electrons randomly move within the material – there is no net movement of electrons in any direction in this case.

18. The drift velocity of electrons in an Ag wire that can be estimated as

$$v_f = 6\times10^{-3}\ \text{m/s}$$

Drift velocity is the net velocity of the electrons in the direction of the applied field.

19. The meantime between collisions is estimated as

$$\tau = \frac{mv_f}{\varepsilon e} = 3.5 \times 10^{-14} \text{ s}$$

20. To the extent that we have imposed the kinetic theory of gases on electrons in a solid, we have effectively imposed the Maxwell–Boltzmann statistics on the collection of free electrons.

Practice Questions

Multiple Choice Questions

1. On the application of a constant potential across the ends of a conductor the current
- **(a)** Steadily increases
- **(b)** First increases and the stays constant
- **(c)** Increases then stays constant and then decreases
- **(d)** Stays constant and then decreases.

2. The mean time between collisions for an electron in a conductor
- **(a)** Decreases with increase in time
- **(b)** Increases with increase in temperature
- **(c)** Decreases with increase in temperature
- **(d)** Is a function of position

3. In comparison to a metallic conductor a ceramic material shows significantly lower conductivity because
- **(a)** It has lower N_f
- **(b)** It has lower τ
- **(c)** It has lower e
- **(d)** It has lower m

4. The acceleration experienced by an electron, as a function of time, when a conductor is subjected to an electric field
- **(a)** Is a constant.
- **(b)** First increases and then becomes a constant
- **(c)** First increases, then decreases and then becomes a constant
- **(d)** Starts from a positive value and then decreases and becomes zero

5. Current density J observed in a conductor, when subject to a constant electric field
- **(a)** Increases when the cross sectional area of the conductor is increased
- **(b)** Decreases when the length of the conductor is increased
- **(c)** Depends on the cross sectional area of the conductor but is independent of its length
- **(d)** Is independent of the length as well as cross sectional area of the conductor.

6. Heat from the top of the ocean travels into the interior through
- **(a)** Conduction only
- **(b)** Conduction and radiation
- **(c)** Conduction and convection
- **(d)** Convection and radiation

7. In a ceramic sample, heat transfer occurs through
- **(a)** Electrons
- **(b)** Lattice vibrations only
- **(c)** Electrons as well as lattice vibrations
- **(d)** Radiation

8. The predominant form of heat transfer inside molten metal is
- **(a)** Conduction
- **(b)** Convection
- **(c)** Radiation
- **(d)** Conduction and radiation

9. Heat conduction occurs
- **(a)** Against the direction of the positive temperature gradient
- **(b)** Along the direction of the positive temperature gradient
- **(c)** Independent of the temperature gradient
- **(d)** Only when the temperature gradient is zero

10. Thermal conductivity of a metal
- **(a)** Decreases with increase in mean time between collisions
- **(b)** Decreases with increase in mean square velocity of the electrons
- **(c)** Increases with increase in the temperature gradient
- **(d)** Increases with increase in mean time between collisions

11. The following is both a good conductor of electricity as well as heat

(a) Glass **(b)** Wood
(c) Diamond **(d)** Gold

12. The following is a good conductor of heat but not a good conductor of electricity
(a) Glass **(b)** Wood
(c) Diamond **(d)** Gold

13. The constant predicted by the Wiedemann–Franz law
(a) Increases with increase in temperature
(b) Is independent of temperature
(c) Decreases with increase in temperature
(d) Is defined only at room temperature

14. Drude model correctly predicts electrical conductivity of a metal
(a) Since the model is inherently correct with respect to electrical conductivity
(b) Since it has errors but the errors are insignificant
(c) Since it has errors but they cancel each other out
(d) Since electrical conductivity does not involve lattice vibrations

15. Drude model correctly predicts thermal conductivity of a metal
(a) Since the contribution of lattice vibrations is important
(b) Since errors in predicting mean time between collisions are cancelled by errors in predicting specific heat
(c) Since errors in predicting mean square velocity are cancelled by errors in predicting specific heat
(d) Since errors in predicting mean square velocity are cancelled by errors in predicting mean time between collisions

16. In a metallic conductor at room temperature, in the absence of any applied potential difference across its ends, the state of electrons in the conductor is
(a) They are stationary
(b) Moving slowly from top to bottom
(c) Moving fast, but in random directions
(d) Are all bound to the respective ionic cores

17. In a metallic conductor at room temperature, in the presence of an applied potential difference across its ends, the state of electrons in the conductor is
(a) Moving fast, but in random directions only
(b) Moving slowly but along a single straight line from one end of the wire to the other
(c) Mostly moving fast in random directions but also slowly moving from one end to the other of the wire
(d) Moving fast and along a single straight line from one end of the wire to the other

18. The drift velocity in glass when subject to 1 V across its ends is
(a) 10^{-3} m/s **(b)** 0 m/s
(c) 10^{5} m/s **(d)** 10^{10} m/s

19. Kinetic theory of gases assumes that the particles
(a) Follow Maxwell–Boltzmann statistics
(b) Follow Fermi–Dirac statistics
(c) Follow Bose–Einstein statistics
(d) Are all at the same state and all possess exactly the same values for their properties

20. Drude model predicts several properties correctly because
(a) There are no errors in the model
(b) Errors in the model cancel each other always
(c) Errors in the model cancel each other sometimes
(d) Thermal properties do not have significant impact on other properties

Short Answer Type Questions

1. List a few common metals and their electronic conductivity values.

2. What is the strength of the electric field that must be applied to get a current density of 1 A/cm^2 through a silver wire?

3. Under what conditions will electrons accelerate above the drift velocity values?

4. Compare the electrical conductivity of silver with that of stainless steel. What are the reasons for the differences in their values.

5. What can be done to improve the electrical conductivity of a metallic sample?
6. List a few common engineering metals and their thermal conductivity values.
7. What are the mechanisms that contribute to the thermal conductivity but not to the electrical conductivity of metals?
8. Indicate an example of the addition of a metal layer over another metal, specifically to improve its thermal conductivity.
9. How is the thermal conductivity of diamond different from that of silver?
10. What are the factors that can enable improved thermal conductivity in metallic systems?
11. How does the Drude model account for variation in properties with temperature?
12. In what kind of samples will the Drude model succeed significantly?
13. In what kind of samples will the Drude model fail significantly?
14. Indicate a few other correct predictions of the Drude model.
15. Indicate a few other limitations of the Drude model.
16. Indicate the correct value for the mean square velocity, $\langle v^2 \rangle$, of free electrons.
17. How is the root mean square velocity different from the drift velocity of electrons?
18. Describe the experimental approach to determine C_V^e.
19. What is the manner of interaction of the "free electrons" with the ionic cores?
20. How is the value of N_f determined in the Drude model?

Answers

Multiple Choice Questions

1. (b)	**6.** (c)	**11.** (d)	**16.** (c)
2. (c)	**7.** (b)	**12.** (c)	**17.** (c)
3. (a)	**8.** (b)	**13.** (b)	**18.** (b)
4. (d)	**9.** (a)	**14.** (a)	**19.** (a)
5. (d)	**10.** (d)	**15.** (c)	**20.** (c)

Short Answer Type Questions

1. The approximate value of conductivity of a few common metals is as

Element	*Conductivity* ($\Omega^{-1}\mathbf{m}^{-1}$)
Silver	6×10^7
Copper	5×10^7
Gold	4×10^7
Aluminum	3×10^7

2.

$$J = \varepsilon\sigma$$

The conductivity of silver is 6×10^7 ($\Omega^{-1}\text{m}^{-1}$). The current density is 1 A/cm^2 = 1×10^4 A/m^2. Therefore, the field required is given by

$$1 \times 10^4 = \varepsilon \times 6 \times 10^7$$

Therefore,

$$\varepsilon = 1.67 \times 10^{-4} \text{ V/m}$$

3. When an electron is ejected from a metal, such as in an electron microscope, it can accelerate to velocities well above that of its drift velocity in the source metal. The velocities attained can be high enough to require relativistic corrections.

4. Silver has a higher electrical conductivity, approximately $6 \times 10^7\ \Omega^{-1}m^{-1}$, than stainless steel, which has a conductivity of approximately $1 \times 10^6\ \Omega^{-1}m^{-1}$. The difference in conductivities of these two metallic samples is more than an order of magnitude. One major reason for this difference is that silver is a pure metal, whereas stainless steel is an alloy. In any alloy, the variations in the elements present in the material results in additional scattering events that the electron encounters as it travels through the material. This leads to a lowering of the mobility of the electrons in the alloy, and hence the lower value of electrical conductivity in stainless steel.

5. In general a metallic sample will show higher electrical conductivity if it is free of impurities and has little or no defects. Therefore, a pure metallic element, which is well annealed and hence has minimal dislocations, grain boundaries and porosity, will display the highest electrical conductivity that it can possibly display.

6.

Metal	*Thermal Conductivity* ($W\ m^{-1}\ K^{-1}$)
Ag	425
Cu	400
Au	300
Al	200

7. In metallic systems both free electrons as well as lattice vibrations, or phonons, can contribute to thermal conductivity. However, of these two, only free electons can contribute to electrical conductivity. Even for thermal conductivity, it is free electron contribution that dominates in the case of metals.

8. Several cooking utensils, which are made of stainless steel, usually have a copper bottom attached to them specifically to improve the thermal conductivity of the vessel during its use. The copper bottom is positioned to be in direct contact with the source of heat, to transmit the heat uniformly and quickly to the rest of the vessel.

9. In terms of values of thermal conductivity, the materials differ as indicated below:

Material	*Thermal Conductivity* ($W\ m^{-1}\ K^{-1}$)
Diamond	2200
Ag	425

In terms of mechanism, Ag depends on free electrons for most of its thermal conductivity, however diamond has no free electrons and depends on lattice vibrations (phonons) associated with its strong covalent bonds for its thermal conductivity.

10. Since metallic samples depend on the movement of free electrons for thermal conductivity, in general they will display higher thermal conductivity if they are free of impurities, and have little or no defects. Therefore, a pure metallic element, that is well annealed and hence has minimal dislocations, grain boundaries and porosity, will display the highest thermal conductivity that it can possibly display.

11. The Drude model varies the mean time between collisions to account for variation of properties with changes in temperature.

12. Specifically looking at electrical and thermal properties, the Drude model is successful with metals such as Ag and Cu.

13. For ceramic and polymeric samples, the Drude model is unlikely to be able to make any successful predictions.

14. The Drude model is able to predict specific optical properties of metallic systems correctly.

15. Anisotropy of electrical properties, for example in a single crystal, cannot be explained by the Drude model.

16. The correct value for the mean square velocity of free electrons is

$$\langle v^2 \rangle \approx 10^{12}\,\mathrm{m}^2/\mathrm{s}^2$$

17. Drift velocity indicates net movement in one direction, while root mean square velocity is the same in all directions and hence there is no net movement associated with it.

18. C_V^e is determined by carrying out calorimetric studies at low temperatures, where other contributions to the specific heat of the solid are minimized.

19. While the Drude model ignores the presence of the ionic cores, in reality the free electrons will experience strong Coulombic attraction whenever they pass close to any ionic core.

20. The Drude model assumes that the valence electrons contribute to the free electrons per unit volume. However, this assumption is not easy to work with since several metals can display multiple valencies.

7 Large Systems, Statistical Mechanics, and The Maxwell–Boltzmann Statistics

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Indicate the different approaches used in chemistry, chemical engineering, physics and metallurgy to deal with systems containing a large collection of particles and explain the reason for this difference.
- State the problem in determining the energy of a system with a large collection of particles.
- Distinguish between microstate and macrostate of system.
- Identify the most probable microstate of a few small systems.
- State the key principle of statistical mechanics.
- Write the expression for the number of ways to accomplish a microstate for a collection of classical particles.
- Indicate the restrictions placed with respect to the total energy of the system and the total number of particles.
- Indicate the significance of the Lagrange method of undetermined multipliers.
- Use the Lagrange method of undetermined multipliers to arrive at the Maxwell–Boltzmann statistical distribution.

7.1 Introduction

Our discussions so far have enabled us to recognize that we are dealing with a system that consists of a large collection of particles in the form of free electrons in a solid. We wish to predict the properties of such a system as a whole without knowledge of the details of every single particle that goes to make up the system – a feat that we can accomplish using an approach that is referred to as "statistical mechanics." In this chapter we will begin by looking at how systems consisting of large collections of particles are handled in different disciplines. We will then examine the basic approach used to predict the behavior of such systems using statistical mechanics. Finally, using the approach of statistical mechanics, we will develop the Maxwell–Boltzmann Statistics.



7.2 Systems with Large Collections of Particles

Before we examine and use the ideas of statistical mechanics, there is a detail which we will examine, because it will be essential for much of our subsequent discussion. The need for this detail will become clear as we return to the topic of statistical mechanics a little later in this chapter. Systems consisting of a large collection of particles are encountered in books and classes associated with chemistry, chemical engineering, metallurgy and physics. While the central ideas are essentially the same, there is a subtle difference in how the topics are handled in these different disciplines. The subtle difference in approach results in different aspects of the results being highlighted in various disciplines. In general, the difference is as follows.

1. In chemistry- and chemical engineering-based discussions, systems are examined under constant temperature and pressure.
2. In metallurgy-based discussions, systems are examined under constant temperature and pressure with the additional constraint that the pressure is usually 1 atm.
3. In physics-based discussions, systems are examined under constant temperature and volume.

We must immediately note that there is an element of generalization in the above statements and exceptions do exist. However, in general they are true and many standard texts will use these approaches. Since textbooks are often aimed at students of a specific stream, this variation in approach is often not elaborated but merely assumed as necessary for that specific stream. However, it is informative to understand the reasons behind and significance of such a variation in approach. We will now examine this variation.

In general, to study any system, certain parameters have to be held constant within the framework of which the system is examined. Usually it is sufficient if two parameters are held constant or controlled. The state of the system is then fixed and predictions can be made about the system.

In chemistry as well as in chemical engineering, liquids and gases are typically examined under experimentally controlled circumstances such as during the refining of oil. From the perspective of experimental control, temperature and pressure are much easier to control and manipulate. With a liquid, for example, constantly changing its volume in a controlled or predetermined manner cannot be accomplished easily experimentally. Experimental ease is, therefore, the reason for choosing temperature and pressure as variables. Therefore, phase diagrams in chemical engineering text books typically have pressure and temperature as two of the axes.

In metallurgical engineering too, emphasis is on experimentally controllable parameters because it pertains to ore extraction and steel production. However, due to the predominant need to handle solids and liquid melts, and large quantities of the materials involved, much of the processing is carried out under 1 atm pressure. Often the experimental work will be done in open air atmosphere itself. Phase diagrams in metallurgical engineering textbooks often plot composition versus temperature with the understanding that the diagram has been drawn under 1 atm.

In physics, to carry out theoretical studies, it is necessary to know the energy levels present within the system and then to look at how these energy levels are occupied by the particles in the system. As we will see in Chapter 10, it turns out that if the volume of the system is fixed, the allowed energy levels of the system get fixed. It is for this specific reason that physics-based approach, to discussing systems consisting of large particles, use constant volume and temperature as the setting within which the system is examined.

While the above discussion has highlighted the differences between the approaches used, this is not, by any means, intended to suggest that the results obtained using these different approaches cannot be compared. If a surface is used for plotting all possible allowed states of the system, the different approaches mentioned above merely refer to different points on the same surface and therefore are not inconsistent with each other. Therefore, the different approaches merely suggest different perspectives on the same system. There is no difference in a fundamental sense. It is similar to looking at the same mountain from different locations.

In the present textbook, we wish to examine the physics behind material properties, and therefore we will adopt the approach of constant temperature and constant volume for our calculations. As we proceed, this assumption will remain in the background but will impact our discussions accordingly. Presently, we will merely accept the impact of constant volume on the system. In Chapter 10, we will explore how and why constant volume enables us to fix the energy levels present in the system and, therefore, justify what we will merely accept for now.

The systems we are interested in are very large and may contain 10^{28} particles per cubic meter. Our discussions will revolve around the energy of such a system and its behavior, for example, how the energy goes up or down with changes in temperature as indicated by its specific heat.

The requirement that a system is at a fixed temperature is associated with the necessity that the constituents of the material have, on average, an energy corresponding to that temperature. Given the energy of a large system, it is not possible for us to know the energy of each of the 10^{28} atoms or free electrons that comprise the system. It is physically not possible for us to determine all the details of each of the particles and then arrive at an average for subsequent use. So we need an approach that enables us to make statements about the system as a whole without knowing all the details of every single particle in the system. As indicated earlier, such an approach is referred to as statistical mechanics.

7.3 Statistical Mechanics

We will now look at a few small systems and see if understanding them can help us in handling large systems.

Let us also introduce a couple of terms that we are more likely to encounter in a book on thermodynamics – "Macrostate" and "Microstate." Macrostate refers to the macroscopic details of the system such as specifying its temperature, volume, and total energy. Microstate refers to the additional microscopic detail of the system, such as how many particles are occupying which energy level that is consistent with the overall energy of the system. Since we can vary the number of particles occupying the various energy levels while still keeping the total energy constant, we can have several microstates corresponding to the same macrostate.

Let us now look at three examples of systems having a fixed macrostate, and see what generalization can be made about their microstates. In all the examples, let us assume that the system is at a constant temperature, has a constant volume, and its total energy is fixed at a value 3E. Since the system has a fixed volume, we will assume that it has fixed allowed energy levels – we will establish this in Chapter 10, but just assume it for now. Let the allowed energy levels in the system be 0, E, 2E and 3E. Let us look at the examples where the system consists of either two or three or four particles. Each arrangement of particles in the allowed energy levels, which makes the overall system consistent with the macrostate of 3E, is an allowed microstate of the system. The calculations below assume identical but distinguishable particles – an aspect that will be explored in greater detail in Chapter 8.

7.3.1 System Consisting of Two Particles

The total energy of 3E can be accomplished in two ways or using two microstates in this system. In one microstate, that is microstate-1, there is one particle at the energy level 3E and the other particle is at the energy level 0. In the other microstate, that is, microstate-2, there is one particle each at the energy levels E and 2E. An examination of the system will show that there is no other way to accomplish the same macrostate. Hence

1. Microstate-1 can be accomplished in $\dfrac{2!}{1! \times 1!}$ ways = 2 ways.

2. Microstate-2 can be accomplished in $\dfrac{2!}{1! \times 1!}$ ways = 2 ways.

Table 7.1 summarizes this information.

Table 7.1

	Number of Particles at Each Energy Level				*Number of Ways to Accomplish This Microstate*
	0	***E***	***2E***	***3E***	
Microstate-1	1	0	0	1	2
Microstate-2	0	1	1	0	2

Since there is no reason why any specific arrangement should be preferred over the rest, a random snapshot of the two-particle system will show the particles to be in either of the two microstates with equal probability.

7.3.2 System Consisting of Three Particles

The total energy of 3E can be accomplished through three different microstates. In microstate-1, there is one particle at the energy level 3E and the other two particles are at the energy level 0. In microstate-2, there is one particle each at the energy levels 0, E, and 2E, and in microstate-3, all the three particles are at the energy level E. An examination of the system will show that there is no other way to accomplish the same macrostate.

1. Microstate-1 can be accomplished in $\frac{3!}{2! \times 1!}$ ways = 3 ways.
2. Microstate-2 can be accomplished in $\frac{3!}{1! \times 1! \times 1!}$ ways = 6 ways.
3. Microstate-3 can be accomplished in $\frac{3!}{3!}$ ways = 1 way.

Table 7.2 summarizes this information. It indicates that of the 10 possible microstates to accomplish the same macrostate of total energy of 3E, microstate-2 occurs six times. Since there is no reason to assume that any one microstate is preferred over the others, if one takes a snapshot of the system, the probability of observing any given microstate is simply proportional to the number of microstates of that type. In the

Table 7.2

	Number of Particles at Each Energy Level				*Number of Ways to Accomplish This Microstate*
	0	***E***	***2E***	***3E***	
Microstate-1	2	0	0	1	3
Microstate-2	1	1	1	0	6
Microstate-3	0	3	0	0	1

above example, microstate-1 is likely to be observed 30% of the time, while microstate-2 60% of the time, and microstate-3 10% of the time. Therefore, in the present example, microstate-2 is the "most probable state" and occurs 60% of the time. In other words, the most probable state is more probable than all the other states combined – in this case, one and half times more probable than all of the other states combined, and is twice as probable as the next most probable state.

7.3.3 System Consisting of Four Particles

The total energy of 3E can be accomplished through three different microstates. In microstate-1, there is one particle at the energy level 3E and the other three particles are at the energy level 0. In microstate-2, there is one particle each at the energy levels E and 2E and two particles at the energy level 0. And in microstate-3, three particles are at the energy level E and one particle is at the energy level 0. An examination of the system will show that there is no other way to accomplish the same macrostate.

1. Microstate-1 can be accomplished in $\frac{4!}{3! \times 1!}$ ways = 4 ways.

2. Microstate-2 can be accomplished in $\frac{4!}{1! \times 1! \times 2!}$ ways = 12 ways.

3. Microstate-3 can be accomplished in $\frac{4!}{3! \times 1!}$ ways = 4 ways.

Table 7.3 summarizes this information. It indicates that of the 20 possible microstates to accomplish the same macrostate of total energy of 3E, microstate-2 occurs 12 times. Since there is no reason to assume that any one microstate is preferred over the others, if one takes a snapshot of the system, the probability of observing any given microstate is simply proportional to the number of microstates of that type. In the above example, microstate-1 is likely to be observed 20% of the time, while microstate-2 is likely to be observed 60% of the time, and microstate-3 is likely to be observed 20% of the time. Therefore, in the present example, microstate-2 is the "most probable state," and is more probable than all of the other microstates combined, and is three times more probable than the next most probable state.

From the examples above we can make the following observations:

1. As the number of particles increase, the microstates are not equally probable. One microstate becomes more probable. In the examples above, as the number of particles increases from two to three and then to four, the ratio of the occurrence of the most probable state to the next most probable state changes

Table 7.3

	Number of Particles at Each Energy Level				*Number of Ways to Accomplish This Microstate*
	0	***E***	***2E***	***3E***	
Microstate-1	3	0	0	1	4
Microstate-2	2	1	1	0	12
Microstate-3	1	3	0	0	4

from 1:1 to 2:1 to 3:1. This general trend will continue as we keep increasing the number of particles – the most probable state will continue to become more probable than the other states in the system combined.

2. The systems we are interested in have 10^{28} particles. The trend of the most probable state becoming more probable becomes significantly magnified in this case. In this type of a large system, the most probable state overwhelms the probability of occurrence of all other states combined; that is, the probability of occurrence of the most probable microstate is nearly 100%. Therefore, in large systems, the most probable microstate is effectively the equilibrium state of the system, the state in which it will be found almost all of the time, and hence the only microstate worth further consideration.

Statistical mechanics takes advantage of this situation, and while the mathematics to calculate all possible states may be quite laborious, statistical mechanics ignores all other states and focuses on identifying only the most probable microstate, which is significantly easier to do.

We can further understand the generalization with respect to large system as follows. If there are N_f free electrons per unit volume, the general form of the term giving the number of ways in which a microstate can be accomplished will be of the form

$$\frac{N_f!}{n_1!n_2!n_3!}$$

where n_1, n_2, n_3, etc. are the number of particles at various energy levels within the system. At very large numbers, the removal of even a single term from the denominator will change the value of the expression above by several orders of magnitude. Therefore, the most probable state of the system will occur many orders of magnitude more often than the next most probable state, and more often than all the other states combined. Statistical mechanics acknowledges that the system can possess other microstates, other than the most probable state, but argues that the chance that the system will be found in the other states is infinitesimally small and hence can be ignored. Statistical mechanics provides us with the tools to understand the most probable state. It enables us to determine properties of large systems without knowing the details of every single particle in the system. It will enable us to see where the error in C_V^e and $\langle v^2 \rangle$ came from. Therefore, statistical mechanics is the tool we will use to look at these large systems.

From the discussions in Chapter 6, we have noted that the Drude model needs to be improved. In particular, there appears to be an incorrect assumption of how energy is distributed among the free electrons in the Drude model, which led to incorrect estimations of C_V^e and $\langle v^2 \rangle$. The system we are dealing with is extremely large, as it contains 10^{28} particles per cubic meter. This number is too large to compute averages by determining the properties of each and every particle and we will have to use statistical approaches to handle these problems. Statistical approaches for these problems recognize that such systems have a "most probable state." With a very large system, the most probable state of the system is more probable than all other states of the system combined. In such large systems, the most probable state overwhelms the probability of occurrence of all other states combined, that is, the probability of occurrence of the most probable state is nearly 100%. Therefore, in large systems, the most probable state is effectively the equilibrium state – the state in which it will be found almost all of the time – and hence the only state worth further consideration. Statistical mechanics takes advantage of this situation and while the mathematics to calculate all possible states may be quite laborious, statistical mechanics ignores all other states and focuses on identifying only the most probable state, which is significantly easier to do.

For the particles we are dealing with – free electrons in a solid – the Drude model has made the assumption that their behavior is similar to the atoms of an ideal gas. Specifically, this means that the electrons behave as though they are "identical but distinguishable." The words "identical but distinguishable" may

convey something different in a conversational sense but they mean something specific in the context of statistics of particles. In this chapter, we will not dwell on the significance of "identical but distinguishable" but merely note that we have made such an assumption. In Chapter 8, we will examine these words – "identical but distinguishable" – and examine what they mean and consider what other possibilities and variations exist in this context.

The statistical distribution of properties of a large collection of identical but distinguishable particles has been calculated and is attributed to Maxwell and Boltzmann and hence bears their names. By assuming that the free electrons in a solid behave like atoms of an ideal gas, the Drude model has assumed that the free electrons follow the Maxwell–Boltzmann statistics.



7.4 The Maxwell–Boltzmann Statistics

In the analysis that follows, we will initially talk about all the possible states and initiate the mathematics toward that goal, but at an appropriate step we will narrow down to the most probable state and from there on focus only on the most probable state and its properties.

Consider a system that has a total energy U, a total volume V, and a total number of particles n, all of which are constant and together represent the macrostate of the system. Let the system has allowed energy levels ε_0, ε_1, ε_2, ε_3, ε_4, …, ε_r. We will now find how many particles – or what fraction of particles – are in a particular energy level. In other words, our goal is to identify the distribution of particles corresponding to the most probable microstate of the system.

In a given microstate corresponding to the above macrostate, let there be n_0 particles at ε_0, n_1 particles at ε_1, n_2 particles at ε_2, n_3 particles at ε_3, n_4 particles at ε_4, … and n_r particles at ε_r. The number of ways in which this microstate can be accomplished is denoted by Ω, which is given by

$$\Omega = \frac{n!}{n_1! \times n_2! \times n_3! \times n_4! \times \cdots \times n_r!}$$

$$\Omega = \frac{n!}{\prod_{i=0}^{i=r} n_i!}$$

where the symbol Π denotes the product of the factorials. We now take advantage of Sterling's approximation, which states that for large n,

$$\ln n! = n \ln n - n$$

Since the system is large, we assume each of the n_i is large enough to justify the usage of this approximation. Therefore,

$$\ln \Omega = n \ln n - n - \sum_{i=0}^{i=r} (n_i \ln n_i - n_i)$$

Our goal is to identify the specific distribution, that is, the specific values of n_i, which will maximize Ω. Please note that in the system we are examining, the values of ε_i are fixed. Only the values of n_i at each of the ε_i are varied. In other words, we wish to know what is the value of n_0 at ε_0, n_1 at ε_1, n_2 at ε_2, etc. such that the combination of n_0, n_1, n_2, etc. maximizes Ω.

To determine the combination that maximizes Ω, we make note of the fact that by the nature of the behavior of Ω and $\ln \Omega$, the combination that maximizes Ω is the same combination that also maximizes $\ln \Omega$.

If ln Ω has been maximized, its differential with respect to n_i will equal to zero. In other words, for a small rearrangement of particles between the different states,

$$\delta \ln \Omega = 0$$

Therefore differentiating the expression for ln Ω with respect to n_i and noting that the total number of particles (n) is a constant, we get

$$\sum_{i=0}^{i=r} \partial n_i \ln n_i + \partial n_i - \partial n_i = 0$$

or

$$\sum_{i=0}^{i=r} \partial n_i \ln n_i = 0 \tag{7.1}$$

In addition, since the total number of particles is fixed, any rearrangement of particles does not change the total number of particles. Therefore,

$$\sum_{i=0}^{i=r} \partial n_i = 0 \tag{7.2}$$

Finally, since the total energy of the system is fixed, rearrangement of particles between the different states cannot change the total energy of the system. Therefore,

$$\sum_{i=0}^{i=r} \varepsilon_i \partial n_i = 0 \tag{7.3}$$

Our problem can therefore be restated as follows: We wish to maximize ln Ω, as denoted by Eq. (7.1), subject to the constraints imposed by Eqs. (7.2) and (7.3).

Mathematically, this problem is identical to trying to find the maximum of a surface subject to the constraint that you have to move along a specific path. In other words, we are not finding the absolute maximum of the surface, but merely the highest point in the surface that can be reached while constrained to moving along the specific path.

The mathematical approach used for solving the problem is referred to as the Lagrange method of undetermined multipliers. The approach and idea behind the method is as follows: First, the three equations are rolled into one by multiplying Eq. (7.2) by an initially unknown multiplier α and Eq. (7.3) by a multiplier β, and adding the three equations:

$$\sum_{i=0}^{i=r} \partial n_i (\ln n_i + \alpha + \beta \varepsilon_i) = 0 \tag{7.4}$$

The method assumes that we are able to find α and β such that the various δn_i, become independent of each other. In other words, upon reaching a maximum, the result should not change due to minor variations of specific n_i and therefore should be independent of the various δn_i.

The implication of δn_i becoming independent of each other is that the term within the parentheses in Eq. (7.4) now equals zero. Therefore,

$$\ln n_i + \alpha + \beta \varepsilon_i = 0$$

which implies that

$$n_i = e^{-\alpha} e^{-\beta \varepsilon_i}$$

In other words, our result indicates that maximizing ln Ω occurs when there is an exponential decrease in n_i as ε_i increases.

Since $\sum_{i=0}^{i=r} n_i = n$, we have

$$e^{-\alpha} \sum_{i=0}^{i=r} e^{-\beta \varepsilon_i} = n$$

The term $\sum_{i=0}^{i=r} e^{-\beta \varepsilon_i}$ is called the partition function P. Therefore,

$$e^{-\alpha} = \frac{n}{P}$$

Further, the value of β works out to be $1/k_b T$, where k_b is the Boltzmann constant and T is the absolute temperature. Hence,

$$n_i = \frac{n}{P} e^{-\varepsilon_i / k_b T}$$

is an exponential decrease in n_i as ε_i increases. This expression is referred to as the Maxwell–Boltzmann distribution.

Figure 7.1 shows a plot of the Maxwell–Boltzmann statistics. The energy levels available in the system are plotted on the *y*-axis, while the number of allowed states at a given energy level is plotted on the *x*-axis.

As shown in the plot above, the higher the energy level, the lower is the number of particles present. Atoms of an ideal gas, and electrons in the free electron gas when described by the classical Drude model, are assumed to conform to the above distribution.

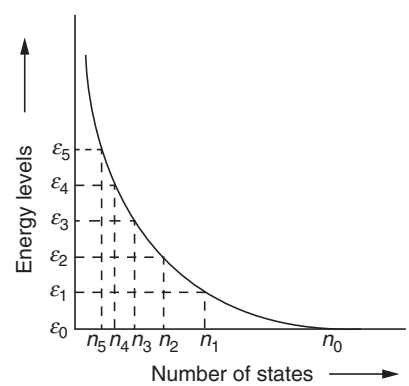


Figure 7.1 A plot showing the number of states as a function of energy level determined by the Maxwell–Boltzmann statistics.

Summary

1. In chemistry, chemical engineering and metallurgical engineering-based discussions, systems are typically examined under constant temperature and pressure. In metallurgy, the additional constraint is that the pressure is usually 1 atm.
2. In physics-based discussions, systems are typically examined under constant temperature and volume.
3. Constant temperature and pressure are easier to control experimentally.
4. Constant volume results in the identification of the allowed energy levels in a system, which is useful from a theoretical perspective.
5. An approach that enables us to make statements about the system as a whole without knowing the entire details of every single particle in the system is referred to as statistical mechanics.
6. Specifying the macroscopic details of the system such as its temperature, volume and total energy is referred to as the macrostate of the system.
7. Specifying microscopic detail of the system such as how many particles are occupying which energy level, consistent with the overall energy of the system, amounts to specifying the microstate of the system.
8. As the number of particles increase the different microstates corresponding to the same macrostate are not equally probable – one microstate becomes more probable.
9. In large systems, the most probable microstate overwhelms the probability of occurrence of all other states combined, that is, the probability of occurrence of the most probable microstate is nearly 100%. It is effectively the equilibrium state of the system, the state in which it will be found almost all the time, and hence the only microstate worth consideration.
10. The statistical distribution of properties of a large collection of identical but distinguishable particles has been calculated and is referred to as the Maxwell–Boltzmann statistics.
11. By assuming that the free electrons in a solid behave like atoms of an ideal gas, the Drude model has assumed that the free electrons follow the Maxwell–Boltzmann statistics.
12. The Lagrange method of undetermined multipliers addresses the mathematical problem of determining the highest point on a surface that can be reached while being constrained to moving along a specific path.
13. The Maxwell–Boltzmann distribution indicates that in the most probable state, there is an exponential decrease in the number of particles occupying higher energy levels relative to lower energy levels.

Practice Questions

Multiple Choice Questions

1. Experimentally the following is easier to control
   (a) Temperature and pressure
   (b) Volume and pressure
   (c) Volume and temperature
   (d) Temperature and specific heat
2. In the processing of crude oil, the parameters controlled are typically
   (a) Temperature and pressure
   (b) Volume and pressure
   (c) Volume and temperature
   (d) Temperature and specific heat
3. In the first principles calculation of the equilibrium state of a large collection of electrons, the parameters controlled are typically
   (a) Temperature and pressure

(b) Volume and pressure
(c) Volume and temperature
(d) Temperature and specific heat

4. Statistical mechanics is the appropriate approach in trying to determine the equilibrium state of
(a) A single stationary pendulum
(b) A collection of 10 atoms at high temperature
(c) One mole of a gas at room temperature
(d) A collection of 100 atoms at low temperature

5. In large systems at equilibrium
(a) There is only one microstate
(b) There are many microstates that are all equally probable
(c) There are microstates but one of these is as probable as the rest of them combined
(d) There are many microstates, but one of these is vastly more probable than the rest of them combined

6. Maxwell–Boltzmann statistics applies to a
(a) A small collection of identical and distinguishable particles
(b) A large collection of identical and indistinguishable particles
(c) A small collection of identical and indistinguishable particles
(d) A large collection of identical and distinguishable particles

7. Free electrons in a metallic system
(a) Are a small collection of identical particles
(b) Are a large collection of identical particles
(c) Are a small collection of individually unique particles
(d) Are a large collection of individually unique particles

8. The Lagrange method of undetermined multipliers
(a) Always determines the absolute maximum of a function
(b) Always determines the absolute minimum of a function
(c) Determines the maximum of a function subject to specific constraints
(d) Determines the constraints placed on the function

9. The Maxwell–Boltzmann distribution indicates that at equilibrium
(a) As energy of the state increases, the energy states are increasingly unoccupied
(b) As energy of the state increases, the energy states are increasingly occupied
(c) As energy of the state decreases, energy states are increasingly unoccupied
(d) As energy of the state decreases, there is no change in the level of occupancy

10. As per the Drude model
(a) Atoms follow Maxwell–Boltzmann statistics
(b) Ions follow Maxwell–Boltzmann statistics
(c) Free electrons follow Maxwell–Boltzmann statistics
(d) Bound electrons follow Maxwell–Boltzmann statistics

Short Answer Type Questions

1. Distinguish between a macrostate and microstate of a system with the help of an example.

2. Name one distribution used for describing commonplace large systems such as the population of a region and indicate a few of its key features.

3. Is it appropriate to use statistical mechanics for smaller systems?

4. What is the significance of a mean and a standard deviation?

5. What is the difference between Gibbs free energy and Helmholtz free energy?

6. Indicate real-life systems where Maxwell–Boltzmann statistics is followed.

7. Using the Maxwell–Boltzmann distribution, explain how to calculate the average speed of electrons in a solid.

8. What happens to the applicability of the Maxwell–Boltzmann statistics for a collection of electrons if the volume of the system is decreased?

9. Comment on the use of Lagrange method of undetermined multipliers and the concept of absolute maxima.

10. What will happen to the distribution of energy in a collection of electrons if the temperature of the system is raised?

Answers

Multiple Choice Questions

1. (a)
2. (a)
3. (c)
4. (c)
5. (d)
6. (d)
7. (b)
8. (c)
9. (a)
10. (c)

Short Answer Type Questions

1. Knowing the pressure, temperature and volume of an ideal gas at equilibrium, fixes its macrostate, while knowing the energies of each of its atoms defines its microstate.

2. The Gaussian or normal distribution is used for describing commonplace large systems such as the population of a region, and many other natural phenomena. Some of the key features of this distribution are that about two-thirds of the population is within one standard deviation of the mean, and 95% is within two standard deviations from the mean.

3. It is usually inappropriate and inaccurate to use statistical mechanics for small systems. The primary approximations made in statistical mechanics become reasonable only for very large systems.

4. Mean represents the average value of a property for a population, while standard deviation represents the extent to which the population is scattered about the mean. Therefore, if all the values of the property are nearly the same, then the standard deviation will be very small for a given mean, on the other hand, if the values are widely scattered, even though the mean may be the same, the standard deviation will be large.

5. Gibbs free energy is used for describing the state of the system under constant T and P, while Helmholtz free energy describes the state of the system under constant T and V.

6. Most inert gases follow the Maxwell–Boltzmann statistics at high temperatures and low pressures.

7. The average speed of electrons following the Maxwell–Boltzmann distribution is calculated as follows:

$$\langle v \rangle = \int_0^\infty v f_{\mathrm{MB}} dv$$

where f_{MB} is the normalized form of the Maxwell–Boltzmann distribution function such that

$$\int_0^\infty f_{\mathrm{MB}} dv = 1$$

8. If the volume of the system is decreased, the extent of interaction between the particles can increase, hence the applicability of the Maxwell–Boltzmann statistics decreases.

9. The concept of absolute maxima involves finding the highest value a function can possibly attain without any additional constraints placed on it. However, the Lagrange method of undetermined multipliers applies to situations where the function is faced with additional constraints, and the maximum is desired within the scope of the constraints on the function.

10. When the temperature of a system consisting of a collection of electrons is raised, the mean energy corresponding to the electrons goes up. All of the electrons gain energy, but do not necessarily attain the energy corresponding to the temperature. Electrons attain energies both higher and lower than that corresponding to the temperature, only the mean of the energy of the collection of electrons corresponds to the temperature of the system.

8 A Brief History of Quantum Mechanics; Its Use in the Drude–Sommerfeld Model

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- State the features and give examples of classical particles.
- State the features and give examples of quantum mechanical particles.
- Indicate the significance of particles being identical.
- Indicate the significance of particles being distinguishable.
- Indicate the size scales associated with various commonly studied particles and whether or not they display classical behavior.
- Indicate a few experiments whose results cannot be explained by classical physics.
- Describe a blackbody.
- Describe the form of blackbody radiation.
- Write and utilize Wien's displacement law and Steffan–Boltzmann equation.
- State the significance of radiant intensity.
- State the important aspects associated with Planck's approach to describe blackbody radiation.
- Write and utilize the Rayleigh–Jean law and Planck's law.
- Explain the significance of h.
- Explain the photoelectric effect.
- Explain the wave particle duality and de Broglie wavelength.
- Write the time-independent Schrödinger wave equation and explain the terms.
- Explain the idea of conjugate variables.
- Write and explain the Born interpretation.
- State the important features of Drude–Sommerfeld model.
- Identify the statistical distribution that is relevant for the system of electrons in a solid.
- Indicate the key features of fermions.
- Distinguish between the Classical Drude model and Drude–Sommerfeld model.

8.1 Introduction

The manner in which we view the world around us changed drastically around the year 1900, with the discovery of Quantum Mechanics. This discovery enabled new theoretical framework to be put in place and many phenomena were re-examined. Significantly improved predictions could now be made for several phenomena that up until that time had very unsatisfactory to no explanations. In this chapter we will briefly review the history of quantum mechanics, its key postulates, and how it can be used in the context of modeling material behavior.

8.2 Classical Particles and Quantum Mechanical Particles



In the Drude model, we assumed that the free electrons in a solid behaved in a manner similar to the atoms of an ideal gas. This meant that the electrons followed the Maxwell–Boltzmann statistics. We derived the Maxwell–Boltzmann statistics in Chapter 7. Let us now briefly consider how the distribution of energy in the system varies as a function of temperature. Figure 8.1 plots the variation of the Maxwell–Boltzmann distribution as a function of temperature.

Figure 8.1 shows that at a higher energy level such as ε_{10}, more states are occupied at the higher temperature T_2 than at the lower temperature T_1. Whereas, at the lower energy level ε_3, less states are occupied at the higher temperature T_2 than at the lower temperature T_1. This layout of energy in the system is consistent with the fact that the overall energy of the system has increased with an increase in the temperature.

Figure 8.1 also indicates how the layout of energy in the system changes with the temperature – which is essentially the information that is captured in the specific heat at constant volume of the system. The system, in this case, is the free electrons in the solid.

The analysis of the predictions of the Drude model has shown that it erroneously predicts the distribution of energy in the system and the specific heat of the system. Therefore, we conclude that the Maxwell–Boltzmann distribution is not appropriate for free electrons in a solid. It is very informative to understand why the Maxwell–Boltzmann distribution is inappropriate for electrons in a solid. Such an understanding will enable us to make better decisions on what will be a more appropriate assumption for electrons in a solid. In this chapter, we will therefore closely examine a very fundamental assumption of the Maxwell–Boltzmann distribution and understand the implications of the same as well as recognize the possibilities that exist to modify those assumptions.

The Maxwell–Boltzmann distribution assumes that the particles are what physicists refer to as "classical" or, in other words, are "identical but distinguishable" particles. It turns out that this is the central assumption that makes the Maxwell–Boltzmann distribution inappropriate for free electrons in a solid. We will therefore examine what is meant by "identical and distinguishable" and also identify other possibilities.

The words "identical but distinguishable" mean something specific when used in the context of physics. For a long time, all of the particles and objects we were aware of were assumed to be "classical" in nature and hence were called classical particles, and the associated physics was called "classical physics." Newton's laws apply to classical particles. Only around the year 1900 did the idea emerge that particles could be considered to behave in a manner other than classical.

Consider any two objects, say two balls – whether they are identical or not can be decided by comparing their attributes. In Fig. 8.2, a few different possibilities are considered.

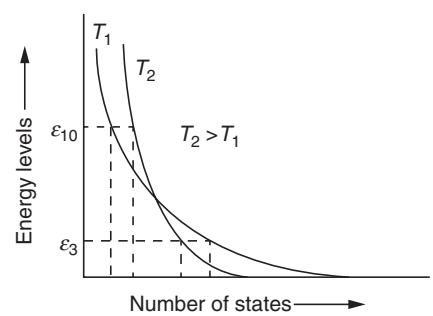


Figure 8.1 Variation of the Maxwell–Boltzmann distribution as a function of temperature.

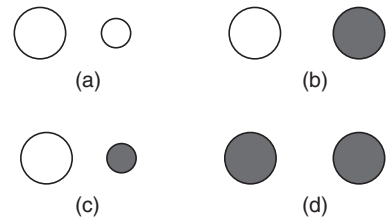


Figure 8.2 Two balls that are (a) not identical in size but identical in color, (b) not identical in color, but identical in size, (c) not identical in size as well as in color, and (d) identical in size as well as in color.

In the example chosen in Fig. 8.2, it is seen that when two balls are compared based on their attributes, they could differ in size or color, or both, or could be identical in size as well as color. If the material of manufacture of the balls is the same, then with the same size and color we will have two balls that we can reasonably consider as "identical." The question then is whether we can take two such identical balls and still distinguish between them. Figure 8.3 considers some possibilities.

In the macroscopic world that we are used to, we can look at the position of the two balls and state that a specific ball is on the right and a specific ball is on the left. As long as the balls are stationary, which we can easily verify, the ball on the right will remain on the right and the ball on the left will remain on the left for any length of time. In this manner, even though the balls are identical, we are able to distinguish between them. We could use the same sort of reasoning if instead of left and right, one ball was held at a higher position and another at a lower position – again the respective positions will remain undisturbed if there is no relative movement, and we can distinguish between the identical balls. Similarly, if the balls were to move toward each other, collide, and then move apart – as long as we know the initial conditions – we can confidently state which ball is where after the collision. Specifically, there is no possibility that the balls could have mysteriously interchanged their positions without our knowledge. This is the basis of the idea of "identical and distinguishable." The analysis above may not seem profound at this stage, but its significance will be clear when we consider other possibilities.

While the dimensions of the objects we have discussed above are large, balls that could be several tens of centimeters in diameter (10^{-1} m), atomic and sub-atomic particles are of much smaller dimensions as indicated in Table 8.1.

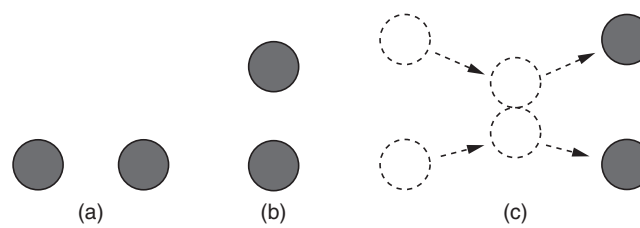


Figure 8.3 Two identical balls that are distinguished based on their position: (a) Left and right, (b) up and down and (c) distinguished even when they undergo a collision. (The dotted circles represent the position of the balls before and during collision, and the arrows indicate the path of each specific ball.)

Table 8.1 The size scale of some atomic and subatomic particles

Particle	*Size Scale*
Atoms	10^{-10} m (1 Å)
Protons	10^{-15} m
Neutrons	10^{-15} m
Electrons	10^{-18} m

The size scales of the particles listed in Table 8.1 are several orders of magnitude less than that of the balls discussed so far. The limit of material characterization techniques is only marginally better than the atomic level of 10^{-10} m. As it turns out, when the size scale decreases, the certainty with which we can simultaneously indicate the position of the particle as well as its velocity also begin to decrease. This is an idea that is central to the field in physics known as "quantum mechanics." It is important to note that this decrease in certainty is *not an experimental limitation* but a natural phenomenon – something that we will discuss more later in this chapter.

We can compare sub-atomic particles in a manner similar to balls in the earlier discussion. Here the attributes of significance are size and charge. Protons and neutrons are similar in size but different in their charge. Protons and electrons are different in their charge as well as their size. So these different particles are clearly not identical. The challenge that we face is when we compare two electrons with each other. Two electrons, by definition, have the same size as well as the same charge. They are therefore identical particles. The question is can we distinguish between two identical particles such as two electrons. In classical physics, we make the assumption that we can treat two electrons as no different from the two macroscopic balls that we have described earlier, and therefore assume that it will always be possible to distinguish between them. Specifically, in the example of the two balls colliding and moving away as shown in Fig. 8.3(c), if each of the balls was actually an electron, classical physics says that the electron at the top of the figure will, with certainty, remain at the top after the collision.

Since the particles are sub-atomic, and are in motion, the concept of interest is the trajectory of the particle, which is examined in Fig. 8.3(c). In classical physics, we say that we can keep track of each specific electron by simply keeping track of its trajectory.

According to quantum mechanics, there is only a probability that a particle's position is at a given location. This probability could be high or low. If we follow the trajectory of the particle, we can say that there is a high probability that the particle is where we think it is. There is a definite, and hence non-zero, probability that the particle could be elsewhere too. As particles approach closer to each other, there is an increasing probability that they could interchange positions without our knowledge. In other words, identical particles could swap positions at any time without our knowledge, and hence these identical particles cannot be distinguished from each other. Particles discussed in the context of quantum mechanics are therefore "identical but indistinguishable."

The central concepts of classical mechanics and quantum mechanics, as discussed above, are highlighted in Fig. 8.4. The distinction made above is very important because it changes the way we count the number of states of a system – an aspect that affects the predictions we make about the system. In classical physics – which uses Maxwell–Boltzmann statistics – when two identical particles that are occupying two different energy levels swap positions, the new arrangement is counted as an additional microstate available to the system. In quantum mechanics, when two identical particles that are occupying two different energy levels swap positions, it is not counted as a new microstate available to the system because the particles may have swapped back without our knowledge anyway.

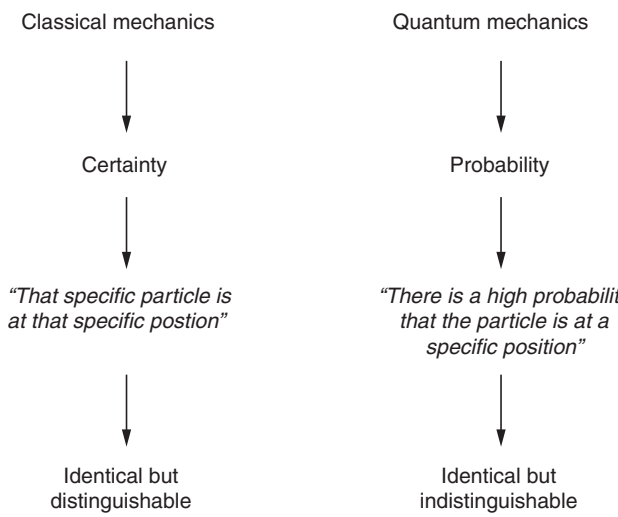


Figure 8.4 The concepts that form the basic ideas of classical mechanics and how they are compared with the basic ideas of quantum mechanics.

One of the comments we made earlier is that we will make assumptions about materials and their constituents and then see if the behavior predicted by those assumptions is validated by experimental data. At this time we find that assuming free electrons to be classical particles, that is, identical but distinguishable, has not led to a satisfactory concurrence with the experimental data. It is, therefore, of interest to see if the other available approach, which is to treat electrons as quantum mechanical particles, that is, identical but indistinguishable, leads to a better validation by the available experimental data.

Our present understanding of the way nature functions is that it seems to follow the rules of quantum mechanics. However, quantum mechanical effects become prominent only under specific conditions. We will, at present, just make the assertion that quantum mechanical effects are not prominent in the scheme of ideal gas atoms at room temperature, but are prominent in the scheme of free electrons in a solid. We therefore have to take into account quantum mechanical behavior of electrons and re-evaluate the predictions that will result.

We will now look at the history of quantum mechanics. In Chapter 9, we will derive the statistical behavior for a specific collection of quantum mechanical particles, which is referred to as the Fermi–Dirac statistics in honor of its authors. We will then employ these statistics to the free electrons in the solid and examine the predictions that result.

We have considered the possibility of treating electrons in a solid either as classical particles or as quantum mechanical particles. Upfront there is no reason to believe that one or the other assumption will be more appropriate. The approach taken is to try out each assumption and compare the predictions made with experimentally obtained data. The approach that results in a better match with experimental data is then validated due to the match.

This brings us to the general topic of quantum mechanics. We will now take a step back and examine the origin of quantum mechanics, history of quantum mechanics, and major concepts of quantum mechanics. While we can use these concepts directly, the present discussion is a very useful exercise to undertake since it will help put things in perspective. While there is overwhelming proof that nature follows the rules of quantum mechanics, our discussion will enable us to recognize the reasons for our difficulty in coming to terms with it. We will look at the people who opened the door of quantum mechanics for us and the circumstances under which they led us down this path.

8.3 A Brief History of Quantum Mechanics



As recently as 1997, a book was published titled *The End of Science* by John Horgan, which examined the idea that all the major phenomena that can be discovered in science have already been discovered. If this were true, many ideas we come across in science fiction will forever remain in the realm of fiction. Surprisingly, a similar thinking existed almost a hundred years earlier in the late 19th century. It was believed then that all major discoveries in science had been made and only minor details remained to be ironed out. The feeling was that there was no real future in physics for any aspiring young person. In hindsight, we note that this sense of the state of science at the end of the 19th century was far from the reality we have seen since. Even today, in the early part of the 21st century, while we know so much more, there is so much that is still to be discovered or understood. For example, scientists who study the nature and origins of the universe note that all of the science we know explains only about 20% of what goes to make up our universe. There is an overwhelming 80% of the universe we do not understand yet, which scientists call "dark energy" and "dark matter." So while there are some who understand relativity and quantum mechanics, it is with humility we note that there may be a lot more for us to figure out.

In the late 19th century, when Max Planck went to take up physics, it was generally believed that all major discoveries in physics had already been made and only minor details needed to be sorted out. Some of the minor details that needed to be sorted out are as follows:

1. Blackbody radiation.
2. Discrete nature of atomic and molecular spectra.
3. Compton-modified scattering.
4. Photoelectric effect.

We will spend most of this chapter focusing on blackbody radiation; let us therefore briefly consider the rest before we proceed.

It was known that when atoms and molecules absorb energy, they initially gain energy, or get excited and later release energy to go back to their original state. The energy released by the excited atoms or molecules appears only at specific wavelengths and not at all wavelengths. There was no explanation available at that time as to why the energy released is not continuous across all wavelengths but appears only at specific or discrete wavelengths.

When X-rays interacts with matter, it was observed that after the interaction, the wavelength of the X-ray increases. This phenomenon, known as Compton-modified scattering, could not be explained at that time, it was discovered – around the year 1923.

When light is incident on materials, in some cases electrons are ejected. This phenomenon is referred to as the photoelectric effect. It was observed that as long as the frequency of the incident radiation was less than a certain threshold value, which varied with materials, no electrons were ejected, no matter how intense the beam of light was. At the same time, once the threshold frequency was crossed, electrons were ejected from the material even when the intensity was very low. The impact of the frequency and the lack of impact of the intensity in initiating the photoelectric effect was unexplained.

Even though several experimentally observed phenomena such as the above were unexplained, the general belief remained that the knowledge prevalent in the late 19th century would only need to be extended marginally and explanations would emerge for these phenomena. There was no expectation that the study of one of these phenomena, the blackbody radiation, would result in the discovery of an entirely new science – quantum mechanics, a discovery that would explain other phenomena as well and fundamentally change our view of the workings of nature. Let us therefore revisit the journey that led to this discovery.

Any given body will give out radiation consistent with its temperature. For example, at room temperature humans give out infrared (IR) radiation. This is the reason that militaries use IR goggles to spot

people at night. At around 1000°C, bodies give out visible light, which is how conventional light bulbs function.

When electromagnetic radiation is incident on a body, some of it will be absorbed, some will be reflected, and some will be transmitted. A body can be imagined and constructed that absorbs all radiation incident on it as long as it is cooler than its surroundings. This body will also emit radiation if it is hotter than its surroundings. Such a body is referred to as a "blackbody." Graphite as a material comes close to satisfying this description. People tried to design a blackbody and in 1859, Kirchoff unveiled the design that has since been accepted as a good design for a blackbody. Figure 8.5 shows the schematic of the blackbody designed by Kirchoff.

In general, electromagnetic radiation emitted by a blackbody comes out over a range of wavelengths; however, it is not emitted with uniform intensity across all wavelengths. The maximum intensity of the radiation occurs at one wavelength and the intensity decreases for all other wavelengths. An example of the spectral distribution of the radiation emitted by a blackbody is shown in Fig. 8.6.

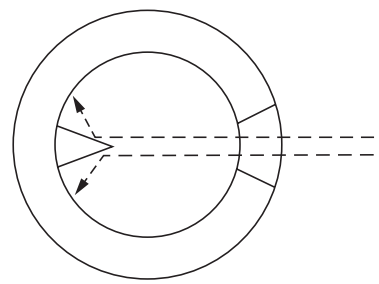


Figure 8.5 A blackbody as designed by Kirchoff. Arrows indicate how radiation entering the body will get absorbed by the internal surfaces of the body.

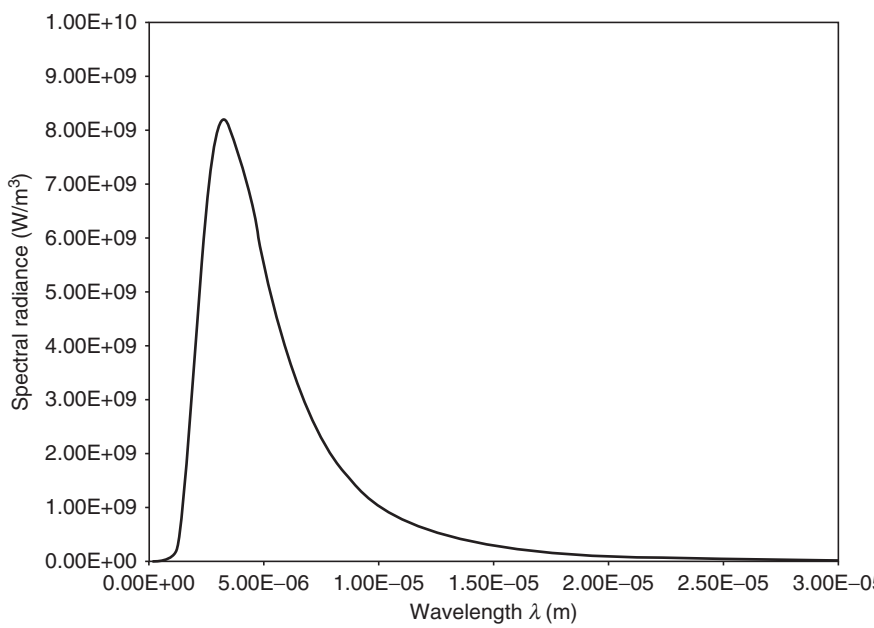


Figure 8.6 The spectral distribution of the radiation emitted by a blackbody.

Before proceeding we present a short note on the axes in the graph above. The *x*-axis plots the wavelength in meters. The spectral radiance plotted on the *y*-axis can be understood as follows: Energy is measured in Joules (J); energy per unit area is measured in Joules per square meter (J/m^2); power per unit area is represented by Joules per square meter per second ($J/m^2/s = W/m^2$); spectral radiance is power per unit area per unit wavelength and is therefore represented by Watt per square meter per meter ($W/m^2/m = W/m^3$), which is the unit shown on the *y*-axis.

Intensity, which is power per unit area, is therefore the area under the curve in Fig. 8.6. Mathematically,

$$I = \int_0^\infty E(\lambda)\, d\lambda$$

There are two observations that can be made about blackbody radiation:

1. As temperature T of the body increases, intensity of the radiation from the body also increases.
2. Higher the temperature, lower is the wavelength of the most intense part of the spectrum.

These two observations are indicated in the schematic in Fig. 8.7. These two trends in blackbody radiation were mathematically stated in the form of two laws:

1. **Stefan–Boltzmann law:**

$$I(T) = \sigma T^4$$

where $\sigma = 5.67 \times 10^{-8}\ Wm^{-2}K^{-4}$.

2. **Wein's displacement law:**

$$\lambda_{(max)} T = \text{constant} = 2898\ \mu mK$$

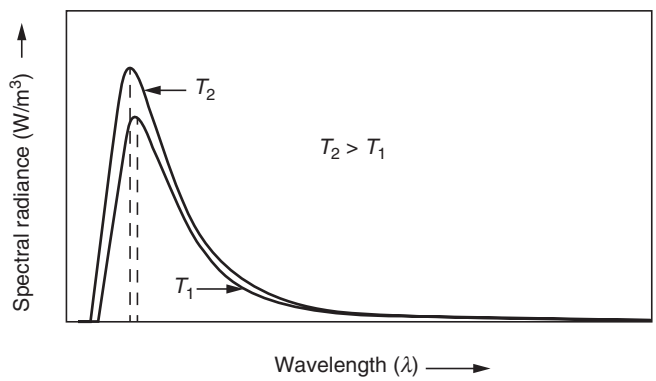


Figure 8.7 The variation of blackbody radiation with temperature. At the higher temperature T_2, the area under the curve, the intensity has increased relative to the curve at T_1. At the higher temperature T_2, the wavelength corresponding to the maximum intensity (identified using the dotted lines in the figure) has decreased relative to that at T_1.

The intensity predicted by the Stefan–Boltzmann law should match the expression for the area under the curve indicated earlier; therefore,

$$I(T) = \int_0^\infty E(\lambda) d\lambda$$

The scientific challenge that remained was to determine the exact form of the spectral radiance, or power per unit area at a particular wavelength, $E(\lambda)$. Obtaining the equation for $E(\lambda)$ was expected to result in a fundamental understanding of how matter interacted with radiation.

Several researchers worked to determine the form of $E(\lambda)$. One of the early attempts described the matter – radiation interaction in a classical manner – that is, assumed an equipartition of energy wherein all modes available to the solid through which it could absorb energy participated in the process equally. This led to the law known as the Rayleigh–Jean law, which provides an equation for the spectral radiance as follows:

$$E(\lambda, T) = \frac{8\pi kT}{\lambda^4}$$

At higher values of λ, this led to a good match between theory and experiment. However as λ decreases, the theory predicts an ever-increasing spectral radiance – a prospect dubbed as "ultraviolet catastrophe." A common experience shows that this does not occur as bodies do not spontaneously release infinite energy. Therefore, the Rayleigh–Jean law comprehensively fails at lower wavelengths. However, it was believed that only a minor correction was required to sort out this discrepancy. The mismatch between theory and experiment is shown in the schematic in Fig. 8.8.

Max Planck perceived the data differently and came up with a new expression. He made the assumption that the presence of intensity at any given frequency meant that an oscillator at that frequency was active in the blackbody. He then arbitrarily assumed that for any oscillator to become active in the blackbody, a certain minimum energy was required, although he did not know what that minimum energy was. He placed these assumptions on a mathematical framework saying, with no immediate basis at that time, that the minimum energy required to activate an oscillator of frequency υ was proportional to the frequency υ itself.

In other words, Energy = Constant × υ was required to activate a single oscillator with the frequency υ. For each additional oscillator at the same frequency υ – additional energy – of the same quantity as above will be required, as per this theory. It is important to note that this was an arbitrary assumption in order to

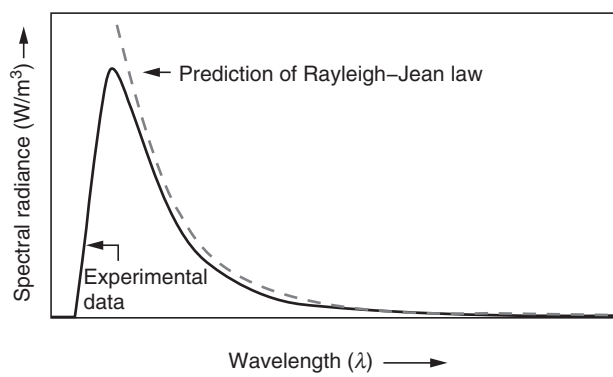


Figure 8.8 The overlay of blackbody radiation data with the prediction of the Rayleigh–Jean law. The theory and data match well at high wavelengths but diverge at lower wavelengths.

generate a better fit to the experimental data. Planck designated the constant as h, and hence $E = h\upsilon$ was required to activate a single oscillator with frequency υ.

This assumption of Planck, although arbitrary, was useful in generating a better curve fit for the experimental data. It created a situation where when the wavelength decreased, and hence frequency increased, the quantity of energy required to activate the oscillator kept increasing with $h\upsilon$. Based on the finite energy available in the system, with decreasing λ and increasing υ, it would become less likely to impossible to activate the corresponding oscillators, since the $h\upsilon$ would keep increasing to larger values with increasing υ. Therefore, the contributions of oscillators to the spectrum would decrease with increasing frequency. Planck's assumption created a situation that enabled higher frequencies to be "switched off."

At lower frequencies the step size or energy increment ($h\upsilon$) was small enough to be switched on with the energy available to the system – a possibility that declined and disappeared at higher frequencies. The energy increment, $h\upsilon$, was known as "quanta." At that time, there was no basis for this type of a model and its assumption of quanta. The model was merely put together to obtain an acceptable curve fit to experimental data.

While his assumptions were arbitrary at that time, Planck enforced the assumption to see what equation he would get for the spectral radiance of a blackbody. Planck did not know the value of the constant h, so when his equation contained h he had to vary the value of h till he got a good curve fit to the experimental data. The equation he obtained was as follows:

$$E(\lambda,T) = \frac{8\pi ch}{\lambda^5}\frac{1}{e^{hc/\lambda kT}-1} \tag{8.1}$$

Note: In Planck's paper, he uses θ to denote temperature in this equation. In addition, a similar expression can also be written in terms of frequency υ instead of the wavelength λ. The conversion between these two expressions is accomplished keeping in mind that we are trying to convert

$$E(\lambda)d\lambda$$

to

$$E(\upsilon)d\upsilon$$

Since

$$d\lambda = \frac{cd\upsilon}{\upsilon^2}$$

and since

$$c = \upsilon\lambda$$

Eq. (8.1) becomes

$$E(\upsilon,T) = \frac{8\pi\upsilon^3 h}{c^3}\frac{1}{e^{h\upsilon/kT}-1}$$

(We are ignoring the negative sign here.)

The model and its resultant equation fit the experimental data very well. Planck used h as an adjustable parameter to get the model to fit the data and found that the model matched the predictions with $h = 6.55 \times 10^{-34}$ Js. (The value of h that is accepted at present is 6.626×10^{-34} Js.)

Planck appears to have insisted that this was only a model and there was no reason to believe that the universe actually followed these rules with respect to the blackbody radiation. Inadvertently, Planck had stumbled upon the most fundamental rule of what has since evolved as an entirely new field of physics – quantum mechanics.

It is of interest to note that at high λ, we can make the following approximation:

$$e^{hc/\lambda kT} - 1 \approx \left[1 + \frac{hc}{\lambda kT} + \cdots\right] - 1 = \frac{hc}{\lambda kT}$$

which when substituted in the Planck equation makes it identical to the Rayleigh–Jean equation. Therefore, the Planck equation reduces to the Rayleigh–Jean equation at higher wavelengths, but at lower wavelengths or higher frequencies it provides additional detail that the Rayleigh–Jean equation does not provide.

While the conventional thinking at that time was matter and radiation exchanged energy in a continuous manner, Planck suggested that the transaction of energy between matter and radiation had a step size associated with it. It is believed that Planck himself was initially unconvinced that nature behaved in this manner; he had merely made the assumption of quanta to obtain a better fit to the experimental data.

The discovery of quantum mechanics is considered profound. Max Planck was awarded the Nobel Prize in 1918 for his discovery.

Blackbody radiation and its analysis is not a matter of esoteric curiosity of a few people. It captures a fundamental piece of information of how matter interacts with energy. Almost a hundred years after Max Planck's work on blackbody radiation, the Nobel Prize in physics for the year 2006 was awarded to Mather and Smoot for their discovery of the blackbody form of the cosmic microwave background radiation.

Based on the discovery of the blackbody form of the cosmic microwave background radiation, it has been possible to estimate the background temperature of the universe. It is now estimated at 2.75 K. The significance of this estimate is that we can conclude that this is the lowest temperature that can naturally exist anywhere in the universe.

Using the analysis of blackbody radiation, it has been possible to estimate the temperature of stars that are millions of light years away.

Max Planck's study of blackbody radiation represents the origin of the field of quantum mechanics. His unexpected discovery that matter exchanged energy with radiation in discrete increments of $h\upsilon$, which he called quanta, led to the discovery of quantum mechanics. This discovery represents the start of quantum mechanics as a field of science. At that time Max Planck was unsure what his discovery implied, or whether nature actually behaved this way, but it led to a good fit with the data obtained from blackbodies.

It is of interest to note that if at the end of Planck's analysis h had turned out to be zero, then it would have implied that matter exchanged energy with radiation in a continuous manner at all frequencies and quantum mechanics would not have existed. The small but non-zero value of h is the sign that nature follows quantum mechanics.

We will now look at other important relationships associated with quantum mechanical behavior. We will become familiar with these relationships and see how they relate to each other. Discussion of these relationships is important, because it is this body of relationships that we will take and utilize together when we examine electrons in a solid.

Max Planck's discovery helped in the explanation of not only the blackbody radiation but also other major experiments that had eluded satisfactory explanation up until that time.

The photoelectric effect displayed the following features:

1. No electrons are ejected regardless of intensity of the incident radiation, unless the frequency υ of the radiation exceeds a threshold value.

2. Kinetic energy of the ejected electrons increases with frequency υ of the incident radiation and is independent of the intensity of the radiation.
3. Even at low intensities, electrons are ejected if the threshold frequency υ is exceeded.

To explain these observations, Einstein proposed the following equation:

$$\frac{1}{2}m_e v^2 = h\upsilon - \phi$$

where $(1/2)m_e v^2$ is the kinetic energy of the ejected electrons and ϕ is the work function of the solid from which the electrons are being ejected.

Einstein suggested that the energy of the incident light at any given frequency υ was given by the expression $h\upsilon$ – an idea that he borrowed and extended from Planck's analysis of blackbody radiation. Einstein's equation, which successfully explained the data corresponding to the photoelectric effect, implied the following:

1. Electromagnetic radiation of frequency υ cannot possess any arbitrary amount of energy – it can only possess energies $h\upsilon$, $2h\upsilon$, $3h\upsilon$, …, $nh\upsilon$.
2. Electromagnetic radiation of frequency υ behaves as though it consists of 1, 2, 3, …, n particles each with energy $h\upsilon$.

Specifically the equation implied that at a frequency υ, light could not have possessed energy that was non-integral multiples of $h\upsilon$. For example, an energy of $1.25\,h\upsilon$ was prohibited at υ.

Light, which had been thought of as exhibiting wave-like behavior for a very long time, suddenly seemed to display a particle-like behavior. Einstein called these particles of light as "photons." It is interesting to note that nearly 200 years before Einstein's analysis, Newton suggested that light consisted of particles called "corpuscles." However, when experiments showed that light displayed diffraction and interference phenomena, Newton's idea of particles of light was abandoned and the wave nature of light was "established." With Einstein's analysis, the idea of light as particles resurfaced and the ideas came through a full circle. Although Newton preceded Einstein in the idea of light as particles, Einstein takes the credit for this idea, since he placed it on a stronger theoretical footing and successfully explained phenomena on this basis, while Newton had merely suggested the possibility. For his contribution to explaining the photoelectric effect, Einstein was awarded the Nobel Prize in 1921.

Einstein was also responsible for introducing the world to the ideas of relativity, general theory of relativity, and special theory of relativity. But Einstein was awarded the Nobel Prize for his work on the photoelectric effect – an indication perhaps of how significantly quantum mechanics changed the landscape of science.

While Einstein suggested that light – which had been thought of as waves – could also be thought of as particles, Louise de Broglie considered the opposite – particles could be thought of as waves.

In 1924, Louis de Broglie proposed the idea that any particle, traveling with a linear momentum "p," can be thought of as having a wavelength λ given by

$$\lambda = \frac{h}{p}$$

This possibility that particles could be thought of as waves was successfully explored by Davisson and Gremer. In 1927, they demonstrated that a beam of electrons could diffract. Electrons from a heated filament, incident on a Ni sample, demonstrated diffraction. There was an element of luck in their work because the wavelengths involved in their experiment worked out just right for them to observe the diffraction easily in reflection mode.

Louise de Broglie's theory suggested that even large, day-to-day objects could be thought of as having a wavelength associated with them. Using his theory, we find that indeed such a wavelength can be determined even for large day-to-day objects, but it is just that the wavelength turns out to be insignificant in the size scale of day-to-day objects and hence does not impact our physical observation and interactions with these objects.

Between Einstein, Louis de Broglie, and Davison and Gremer, the world was introduced to the ideas that waves could show particle-like behavior and all particles could show wave-like behavior.

Within its framework, quantum mechanics presented the idea of wave–particle duality. The approach presently adopted in physics is to treat matter either as waves or as particles entirely based on circumstances. Whenever convenient, particle-like description of matter is used and wherever convenient, wave-like description of matter is used. For his insights into the wave nature of particles, Louis de Broglie was awarded the Nobel Prize in 1929.

New tools and approaches were required to deal with the ideas of quantum mechanics. The trajectory of particles in classical physics needed to fit inside the description of a wave, a requirement that was successfully addressed by Erwin Schrödinger.

Quantum mechanics accepts the wave–particle duality by treating the trajectory of a particle as a wave, which is represented by a wave function ψ. The wave function ψ has the properties of the system and can be obtained by solving the Schrödinger wave equation:

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi = E\psi$$

The Schrödinger wave equation *cannot be derived* from more fundamental principles. It is itself considered the fundamental principle. It merely states that the total energy of the system consists of the kinetic energy and potential energy of the system.

In the studies involving quantum mechanics, we typically identify the constraints placed on a system and solve the Schrödinger wave equation consistent with these constraints to obtain the wave function ψ of the system. This wave function ψ then encapsulates the properties of the system. Schrödinger provided us with the tool using which we could extract an understanding of the system within the framework of quantum mechanics. For his contributions to quantum mechanics, Schrödinger was awarded the Nobel Prize in 1933. He shared that year's Nobel Prize with Paul Dirac.

The Schrödinger wave equation is easy to solve only for a few simple cases. In general, it can get quite complicated to solve and may require specific mathematical tools in some cases.

The Schrödinger wave equation indicated above is one form of the equation known as the time-independent Schrödinger wave equation. There is another form of the equation called the time-dependent Schrödinger wave equation. This later equation is in conflict with aspects of relativity and is yet another indication that there may be more to discover about the working of nature.

While the Schrödinger wave equation became accepted as a fundamental aspect of the quantum mechanical world, considerable confusion prevailed on the significance of the wave function ψ. Max Born provided the interpretation of the wave function ψ. If ψ^* is the complex conjugate of ψ, he stated that $\psi\psi^* dx$ or $|\psi|^2 dx$, is the probability of finding the electron between x and $x + dx$. For this contribution to quantum mechanics, called the Born interpretation, Max Born was awarded the Nobel Prize in 1954.

For a bound electron, for example, the wave function ψ will turn out such that $\psi\psi^* dx$ will have a high value in the vicinity of the atom and will be virtually zero everywhere else.

Niels Bohr considered atomic and molecular spectra and attempted to explain the discrete nature of these spectra. He proposed that electrons went around the nucleus in fixed orbits and the energy released when electrons jumped from one orbit to another was a fixed value depending on the orbits between which

the jump occurred. While this planetary model of the atom has not turned out to be exactly right, it did explain the experimental spectra observed. Schrödinger showed that the results obtained using his wave functions were consistent with the predictions of Niels Bohr.

For his contributions to explaining atomic and molecular spectra, Niels Bohr was awarded the Nobel Prize in 1922.

As indicated earlier, it is not easy to solve the Schrödinger wave equation in many cases. The wave function itself may end up being very complicated. If the wave function can be thought of as a sum of many waves, then each measurement of momentum of the particle will result in a value corresponding to any one of the waves that constitutes the wave function. This idea and its surprising implication was explored by Heisenberg and led to his identification of the "uncertainty principle."

The "uncertainty principle," attributed to Heisenberg, effectively states that if the location of a particle is identified very precisely then determining its momentum will be very difficult or will be very imprecise. High school texts, in order to simplify the complexity involved, explain the theory by saying that to determine the position of a particle very precisely we need to shine a light on the particle. In this process, the light impinges on the particle and disturbs it. As a result, the momentum of the particle changes and we are hence unable to determine the exact position as well as the exact momentum of the particle at the same time.

This suggests that the Heisenberg uncertainty principle is just an experimental limitation. If it were only an experimental limitation, the situation should have likely improved over the years – we should have much lower uncertainty now, than nearly a hundred years ago, in view of our improved experimental facilities. However, this "improvement" does not seem to have occurred. We do not hear news reports of lowered uncertainty with respect to Heisenberg's uncertainty principle. Such a lack of improvement is a hint that this principle is perhaps not one of experimental limitation.

A more advanced view of the uncertainty is as follows. Momentum and position of a particle are related to each other through a Fourier transform and hence are referred to as conjugate variables of each other. It turns out that when we carry out Fourier transforms, as a direct result of the Fourier transform process, the product of the variability of one variable and the variability of its conjugate variable is equal to or greater than a certain value.

For example, if A and B are variables related by a Fourier transform and ΔA and ΔB represent the variability associated with each of these variables, then

$$\Delta A \times \Delta B \geq C$$

where C is a constant.

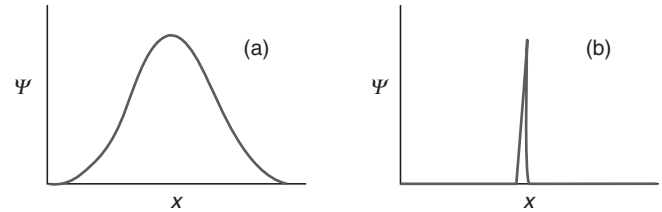
Any conjugate variable pair will therefore create the situation that when we try to decrease the variability in one variable, we will automatically increase the variability in the other. Therefore, this rule appears as a result of a mathematical necessity arising from the Fourier transform process. It has no direct link to experimental limitations.

The momentum p and the position x of a particle are conjugate variables. Therefore, Heisenberg proposed that if p is the momentum associated with a particle and Δp is the uncertainty in determining this momentum, and if x is the location of the particle and Δx the uncertainty in determining its location, then the "uncertainty principle" states:

$$\Delta p \times \Delta x \geq \frac{h}{4\pi}$$

One way to understand the situation is that many waves have to be added to get a description that ensures that the particle is confined to a small location. Therefore, the particle can now have many wavelengths and hence many possible momenta.

A **wave function** can be considered as a combination of several waves, each with a corresponding momentum. Such a wave function can indicate the different values of momentum during measurements – resulting in an "**uncertainty**" in the value of momentum. More the waves necessary, greater the uncertainty in momentum.



A "**Localized**" particle requires that its wave function is non-zero only in the region of its localization. Localization implies **less uncertainty** in the position of a paticle.

Compared to the wave function plotted in (a), the wave function plotted in (b) represents greater localization, but requires more waves to be added to ensure that they add up and constructively interfere only in a narrower region. More waves implies more values of momentum are possible. Therefore **lower the uncertainty** in position **greater the uncertainty** in momentum.

Stated mathematically: $\Delta p \Delta x \geq h/4\pi$

Figure 8.9 The concept of the uncertainty principle.

For his discovery of the "uncertainty principle," Heisenberg was awarded the Nobel Prize in 1932. The concept of the uncertainty principle is summarized in Fig. 8.9.

Specifically with respect to electrons, Wolfgang Pauli proposed that no two electrons can have all of their quantum numbers being identical. This is referred to as Pauli's exclusion principle and resulted in a Nobel Prize for Pauli in the year 1945.

The key concepts of quantum mechanics and the people credited with their discovery are summarized in Fig. 8.10.

It is relevant to note that several of the ideas associated with quantum mechanics resulted in Nobel Prizes for their proponents. Such is the significance of these discoveries. At the same time, although several major unexplained phenomena were successfully explained using the idea of quantum mechanics, there was considerable discomfort with the idea. There was a school of thought that perhaps there was a deeper theory

$$E = h\upsilon \qquad \text{Max Planck}$$

$$\frac{1}{2} m_e v^2 = h\upsilon - \phi \qquad \text{Einstein}$$

$$\lambda = \frac{h}{p} \qquad \text{Louis de Broglie}$$

$$\left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi = E\psi \qquad \text{Schrödinger}$$

$$\psi\psi^* dx \quad \text{or} \quad |\psi|^2 dx \qquad \text{Born}$$

$$\Delta p \times \Delta x \geq \frac{h}{4\pi} \qquad \text{Heisenberg}$$

Figure 8.10 The key concepts of quantum mechanics and the people credited with their discovery.

that did not require quantization but which would still explain the phenomena observed. Such thoughts did not materialize in all the investigations since, and quantum mechanics has persisted. So, when we sometimes face difficulties in understanding and accepting the ideas of quantum mechanics and feel that these ideas are not intuitive, we can take comfort from the fact that some of the greatest minds in the history of science also struggled to accept it and worried about its significance.

Our intention is to identify theories that match experimental data on material properties. The Drude model was only partially successful in this process. In fact, it was partially successful only due to a stroke of luck that certain errors in the model canceled each other in specific instances. After recognizing the Drude model's limitations, the search for the cause of its limitations led us to the idea that we had to treat electrons as quantum mechanical particles rather than classical particles. It is in this context that we looked at the history of quantum mechanics, the key concepts of quantum mechanics and how the key concepts are related to each other.

Quantum mechanical effects are relevant in the context of the systems we are studying. At the same time, we must note that quantum mechanics is not inconsistent with the macroscopic world. It is merely that at large-size scales, quantum mechanical effects become insignificant and hence can be reasonably ignored.

Now let us introduce the scientist who made the contribution that significantly improved the Drude model – Arnold Sommerfeld. To understand how significant his contributions to science in general have been, let us briefly consider a few other people and their accomplishments.

Heisenberg, Debye, Pauli, and Bethe are four Nobel laureates. Their awards span a time period of 35 years and involve work related to subatomic particles, quantum mechanics, and nuclear reactions and energy in stars – three Nobel Prizes in physics and one in chemistry. What do these four scientists have in common, other than being Nobel laureates? Interestingly, they all had the same Ph.D. guide or advisor – Arnold Sommerfeld.

Arnold Sommerfeld never won the Nobel Prize himself, but made significant contributions to the fields of mathematics and physics. Of immediate interest to us is the Drude–Sommerfeld model that takes some features of the Drude model and modifies some others, and makes an overall improvement to the Drude model.

8.4 The Drude–Sommerfeld Model



The important features of the Drude–Sommerfeld model are listed in Fig. 8.11. The Drude–Sommerfeld model is a free electron model similar to the classical Drude model. This means that the electrons responsible for conduction are not bound to any particular atom and are free to roam within the extent of the solid. The potential within the solid is assumed to be uniform and its implication is that the electrons do not have any preferred site that they are likely to aggregate toward.

In view of the application of quantum mechanical principles, the electrons are assumed to be identical and indistinguishable. This assumption impacts the manner in which the statistics corresponding to the electrons, and their distribution across energy levels, is developed. In addition, there is an assumption that the electrons obey Pauli's exclusion principle.

Identical and indistinguishable particles that obey Pauli's exclusion principle follow the statistical description developed by Fermi and Dirac, which is referred to as the Fermi–Dirac statistics. Particles that follow the Fermi–Dirac statistics are called fermions just as the particles following Maxwell–Boltzmann statistics are called classical particles. Fermions have the additional characteristic of possessing half-integer spins, which electrons do.

In dealing with fermions, we must recognize that there is a concept of a fixed number of states at any given energy level, which sets an upper limit to the number of electrons that can occupy that energy level. This was not a restriction in the classical Drude model that we discussed earlier. While deriving the Maxwell–Boltzmann statistics, we began by stating that let there be n_0 particles at $\mathcal{E}_0$, n_1 particles at $\mathcal{E}_1$, n_2 particles at $\mathcal{E}_2$, n_3 particles at $\mathcal{E}_3$, n_4 particles at $\mathcal{E}_4$, … and n_r particles at $\mathcal{E}_r$. While deriving the Fermi–Dirac statistics, we will have to modify that statement and say instead that let there be n_0 particles in s_0 states at $\mathcal{E}_0$, n_1 particles in s_1 states at $\mathcal{E}_1$, n_2 particles in s_2 states at $\mathcal{E}_2$, n_3 particles in s_3 states at $\mathcal{E}_3$, n_4 particles in s_4

The Drude–Sommerfeld model

- A free electron model
- Applies quantum mechanical principles to the Drude model
- Applies the Pauli's exclusion principle
- Potential is constant within the metal
- Electrons are identical and indistinguishable
- Behavior subject to Fermi–Dirac statistics

Figure 8.11 Important features of the Drude–Sommerfeld model.

states at ε_4, … and n_r particles in s_r states at ε_r. If the manner in which we define the states includes all of the quantum numbers, then we can have a maximum of only one particle per state.

In view of the particles being identical and indistinguishable, if the total number of particles at two energy levels remains the same and a few particles from one energy level are simply swapped for the same number of particles from the other energy level, it does not count as a new microstate for the system.

The limits on the number of particles at a given energy level and the change in the manner of defining a new microstate for the system together significantly change the resulting statistics of the system. The Fermi–Dirac statistics, therefore, provides us with results and predictions that are vastly different from those obtained using the Maxwell–Boltzmann statistics. The important differences between the classical Drude model and Drude–Sommerfeld model are summarized in Fig. 8.12.

In Chapter 9, we will derive the Fermi–Dirac statistics and the results of the derivation will tell us if the estimation of the behavior of electrons in a solid has changed. Specifically, we will see if the anomalies in the estimation of C_V^e and v^2 have been addressed.

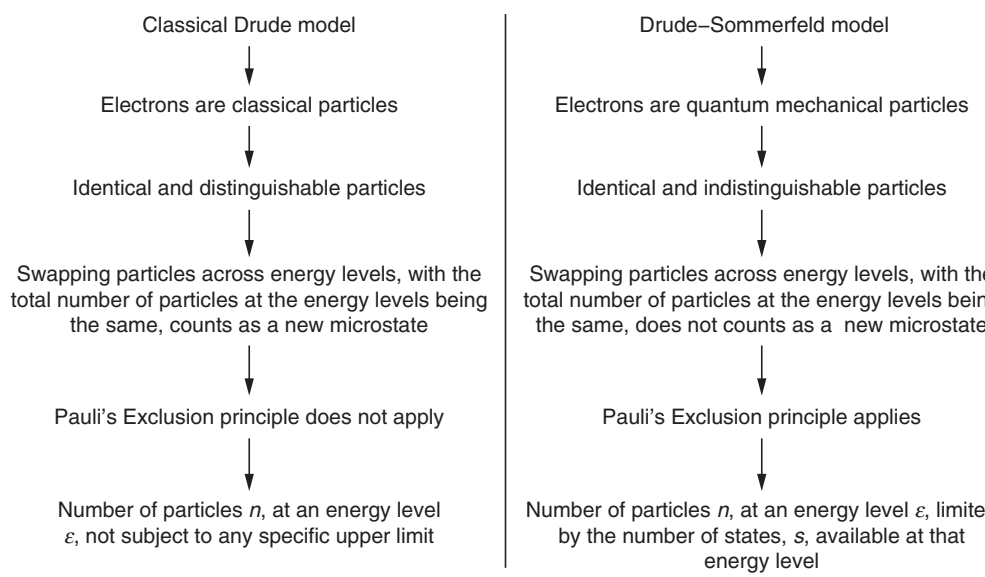


Figure 8.12 Differences between the classical Drude model and the Drude–Sommerfeld model.

Summary

1. The Maxwell–Boltzmann distributions indicate that at higher temperatures, more states at higher energy levels are occupied, whereas at the lower temperatures, lesser number of states at higher energy levels are occupied.
2. Classical particles can be identical and yet be distinguishable.
3. Quantum mechanical particles, if identical, are not distinguishable.
4. In the late 19th century, it was generally believed that all major discoveries in physics had already been made. Only minor details needed to be sorted out, which included: blackbody radiation, discrete nature of atomic and molecular spectra, Compton-modified scattering and photoelectric effect.
5. For blackbody radiation, as the temperature T of the body increases intensity of the radiation from the body increases, and the wavelength of the most intense part of the spectrum decreases.
6. Stefan–Boltzmann law:

$$I(T) = \sigma T^4$$

where $\sigma = 5.67 \times 10^{-8}$ Wm^{-2} K^{-4}.

7. Wein's displacement law:

$$\lambda_{max} T = \text{constant} = 2898\ \mu\text{mK}$$

8. The intensity predicted by the Stefan–Boltzmann law should match the expression for the area under the curve indicated earlier, therefore,

$$I(T) = \int_0^\infty E(\lambda) d\lambda$$

9. Rayleigh–Jean law for the spectral radiance:

$$E(\lambda, T) = \frac{8\pi kT}{\lambda^4}$$

10. Planck's expression for spectral radiance:

$$E(\lambda, T) = \frac{8\pi ch}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

h was an adjustable parameter, which he determined to be 6.55×10^{-34} Js (the value of h that is accepted at present is 6.626×10^{-34} Js).

11. Planck's equation reduces to the Rayleigh–Jean equation at higher wavelengths.
12. Einstein suggested that light showed particle-like behavior and called these particles of light as "photons."
13. Louise de Broglie considered the opposite – particles could be thought of as waves.
14. The possibility that particles could be thought of as waves was successfully explored experimentally by Davisson and Gremer.
15. Between Einstein, Louis de Broglie, and Davison and Gremer, the world was introduced to the ideas that waves could show particle-like behavior and all particles could show wave-like behavior.
16. Quantum mechanics accepts the wave–particle duality by treating the trajectory of a particle as a wave, represented by a wave function ψ. The wave function ψ has the properties of the system and can be obtained by solving the Schrödinger wave equation.
17. Max Born provided the interpretation of the wave function ψ. If ψ^* is the complex conjugate of ψ, he stated that $\psi\psi^* dx$ or $|\psi|^2 dx$ is the probability of finding the electron between x and $x + dx$.
18. Niels Bohr explained the discrete nature of atomic and molecular spectra.
19. Heisenberg's uncertainty principle states that if the location of a particle is identified very precisely, then determining its momentum will be very difficult or very imprecise.
20. Heisenberg's uncertainty principle is not an experimental limitation that can be overcome with advances in experimental techniques. It is a mathematical necessity associated with any conjugate variable pair arising from the Fourier transform process.
21. Wolfgang Pauli proposed that no two electrons can have all of their quantum numbers being identical.

22. The Drude–Sommerfeld model is a free electron model that applies quantum mechanical principles to electrons.

23. The electrons are assumed to be identical and indistinguishable.

24. Electrons follow Pauli's exclusion principle.

25. Identical and indistinguishable particles, which obey Pauli's exclusion principle, follow Fermi–Dirac statistics and are called fermions.

26. Fermions possess half-integer spins.

Practice Questions

Multiple Choice Questions

1. The Maxwell–Boltzmann distribution indicates that as temperature increases
- **(a)** Higher energy states are increasingly unoccupied
- **(b)** Higher energy states are increasingly occupied
- **(c)** Lower energy states are increasingly occupied
- **(d)** For lower energy states, there is no change in the level of occupancy

2. A pair of cricket balls can be considered as
- **(a)** Identical but distinguishable
- **(b)** Identical but indistinguishable
- **(c)** Different but indistinguishable
- **(d)** Different and only therefore distinguishable

3. A pair of electrons can be considered as
- **(a)** Identical but distinguishable
- **(b)** Identical but indistinguishable
- **(c)** Different but indistinguishable
- **(d)** Different and only therefore distinguishable

4. Quantum mechanical behavior is prominent when
- **(a)** Size scale of objects involved is huge
- **(b)** Size scale of objects is in the visible to unaided eye regime
- **(c)** The objects are hidden from view
- **(d)** Size scale of the objects is in the subatomic particle range

5. For day-to-day objects visible to the unaided eye
- **(a)** Quantum mechanical aspects are absent
- **(b)** Quantum mechanical behavior is present but is insignificant
- **(c)** Quantum mechanical behavior is present and significant
- **(d)** Quantum mechanical behavior is significant when temperature is low

6. The state of science today is that
- **(a)** All major discoveries have been made, nothing remains to be discovered
- **(b)** A huge fraction of what makes up the universe is unknown
- **(c)** All subatomic particles have been discovered and identified
- **(d)** Only engineering challenges remain

7. A body at a million degrees centigrade releases
- **(a)** Only infrared radiation
- **(b)** Only visible radiation
- **(c)** Only ultraviolet radiation
- **(d)** X-rays

8. Planck's equation reduces to the Rayleigh–Jean equation
- **(a)** At high frequencies
- **(b)** At low frequencies
- **(c)** At low wavelengths
- **(d)** At room temperature

9. Compton modified scattering is explained using
- **(a)** Particle nature of X-rays
- **(b)** Wave nature of X-rays
- **(c)** High frequency of X-rays
- **(d)** Low wavelength of X-rays

10. The temperature of a distant star is typically determined by
- **(a)** Wein's displacement law
- **(b)** Measuring X-rays from the star
- **(c)** Measuring infrared radiation from the star
- **(d)** The value of the Planck's constant

11. Particles of light are presently referred to as
- **(a)** Phonons **(b)** Photons
- **(c)** Corpuscles **(d)** Fermions

12. Quantum mechanics suggests that
    (a) Waves can behave as particles while the converse is not possible
    (b) Particles can behave as waves while the converse is not possible
    (c) Particles can behave as waves and waves can behave as particles
    (d) Neither waves nor particles can behave like each other
13. Heisenberg's uncertainty principle is
    (a) An experimental limitation that can be overcome
    (b) An experimental limitation that cannot be presently overcome
    (c) A theoretical limitation that can be overcome
    (d) A theoretical limitation that cannot be overcome
14. The photoelectric effect
    (a) Can be explained entirely using the wave nature of light
    (b) Can be explained using the particle nature of light
    (c) Is yet to be fully explained
    (d) Can be explained using the wave nature of electrons
15. Heisenberg's uncertainty principle is more prominently observed for
    (a) Day-to-day objects
    (b) Atoms
    (c) Electrons
    (d) Molecules
16. The Drude–Sommerfeld model assumes the following
    (a) All electron are bound to ions
    (b) Electrons are classical particles
    (c) Electrons are distinguishable
    (d) The solid is an equipotential region
17. Free electron approximation means
    (a) Electrons have escaped the solid
    (b) Electrons are unaffected by the presence of other electrons in the solid
    (c) Electrons are unaffected by the presence of ions in the solid
    (d) Electrons are unaffected by the surface of the solid
18. Independent electron approximation means
    (a) Electrons have escaped the solid
    (b) Electrons are unaffected by the presence of other electrons in the solid
    (c) Electrons are unaffected by the presence of ions in the solid
    (d) Electrons are unaffected by the surface of the solid
19. The following is true about fermions
    (a) They have integer spin
    (b) They have half-integer spin
    (c) They are distinguishable
    (d) Several of them can have all of their quantum numbers be the same
20. Fermi–Dirac statistics assumes
    (a) Unlimited electrons can be in an energy level with finite states
    (b) There are infinite states at an energy level
    (c) There are finite states at an energy level
    (d) Unlimited electrons can be in a state

Short Answer Type Questions

1. Give a real-life example where the quantum mechanical behavior of electrons enables a phenomenon that would not have been possible if electrons only displayed classical behavior.
2. By improving instrumentation, is it possible to improve the certainty with which subatomic particles can be identified and distinguished?
3. Is an ideal gas atom a classical particle?
4. If electrons are not classical particles, how are we able to use them for imaging in an electron microscope?
5. Other than electrons, indicate a few particles that display non-classical behavior.
6. Using the blackbody form of the cosmic microwave background radiation, estimate the background temperature of the universe as obtained by Mather and Smoot.
7. For the sun, with a surface temperature of 6000°C, calculate the wavelength at which the maximum intensity of radiation will be seen.
8. What is the intensity of the radiation from the Sun?

9. What is the current accepted explanation for the discrete nature of atomic and molecular spectra?
10. What is the current accepted explanation for Compton-modified scattering?
11. What is a Hamiltonian?
12. How were the discrete spectra of atoms explained by Niels Bohr?
13. Calculate the wavelength of an electron traveling at 10 m/s, a ball of mass 100 g also traveling at 10 m/s.
14. What is the wavelength of an electron in an electron microscope subject to an acceleration of 200 kV?
15. If the work function of a material is 2.5 eV, what is the minimum frequency of light that must be incident on the surface to eject a photo electron?
16. What are the quantum numbers?
17. What do quantum numbers represent?
18. How do we arrive at the quantum numbers?
19. What is Pauli's exclusion principle?
20. From the literature, indicate a few particles that are considered as fermions.

Answers

Multiple Choice Questions

1. (b)	**6.** (b)	**11.** (b)	**16.** (d)
2. (a)	**7.** (d)	**12.** (c)	**17.** (c)
3. (b)	**8.** (b)	**13.** (d)	**18.** (b)
4. (d)	**9.** (a)	**14.** (b)	**19.** (b)
5. (b)	**10.** (a)	**15.** (c)	**20.** (c)

Short Answer Type Questions

1. The tunneling phenomenon, which enables the operation of the scanning tunneling microscope, is a quantum mechanical phenomenon that would not have been possible if electrons had displayed only classical behavior.
2. No. The inability to distinguish between identical subatomic particles, such as electrons, is not an experimental limitation.
3. Ideal gas atoms display behavior that is consistent with them being classical particles.
4. In an electron microscope, the image is built by a large number of electrons striking the screen. While individual electrons may be difficult to control due to the uncertainty principle, large collections of electrons will display a predictable behavior overall.
5. Photons and neutrinos are a few particles that display non-classical behavior.
6. Based on the blackbody form of the cosmic microwave background radiation, the background temperature of the universe has been estimated to be about 2.75 K.
7. From Wein's displacement law:

$$\lambda_{\max} T = \text{constant} = 2898\ \mu\text{mK}$$

Therefore at 6000°C or 6273 K, the wavelength at which the maximum intensity of radiation will be seen is given by

$$\lambda_{\max} = \frac{2898}{6273} = 0.462\ \mu\text{m}$$

8. From the Stefan–Boltzmann law

$$I(T) = \sigma T^4$$

where $\sigma = 5.67 \times 10^{-8}\ \text{Wm}^{-2}\ \text{K}^{-4}$. Therefore,

$$I(6273) = 5.67 \times 10^{-8} (6273)^4 = 8.8 \times 10^7\ \text{Wm}^{-2}$$

9. Quantum mechanics shows that confinement of electrons results in quantization of their energies, this results in the transition between these energy levels having only specific allowed values, and hence this results in discrete atomic and molecular spectra.

10. Due to inelastic scattering between X-ray photons and loosely bound electrons, the energy of the X-ray photons changes, which results in a change in their wavelength in keeping with the wave particle duality of quantum mechanics.

11. Hamiltonian is an operator that represents the total energy of the system in quantum mechanics.

12. Neils Bohr postulated the existence of "stationary orbits" possessing defined and fixed values of energies for electrons around an atomic nucleus. This postulation also stated that for transitions between these orbits, the difference in energy between these orbits will involve radiation that follows the Planck energy–wavelength relation

$$\Delta E = h\upsilon$$

This postulation successfully explained the discrete spectra of atoms and related it to ideas in quantum mechanics.

13. The mass of an electron is 9.1×10^{-31} kg. Therefore at 10 m/s, its momentum is given by 9.1×10^{-30} kg m/s.
Using the Louis de Broglie relation:

$$\lambda = \frac{h}{p}$$

we have

$$\lambda = \frac{6.626\times10^{-34}}{9.1\times10^{-30}} = 7.28\times10^{-3}\,\text{m}$$

For the ball, the momentum is $0.1 \times 10 = 1$ kg m/s. Therefore,

$$\lambda = \frac{6.626\times10^{-34}}{1} = 6.626\times10^{-34}\,\text{m}$$

which is an extremely small value compared to value of the electron.

14. The velocity of an electron subject to an acceleration of 200 kV is high enough that it requires significant relativistic corrections.
The wavelength of the electron is given by

$$\lambda = \frac{h}{\sqrt{2meV}} \times \frac{1}{\sqrt{1+(eV/2mc^2)}}$$

where V is the accelerating voltage and c is the speed of light. Therefore,

$$\lambda = \frac{6.626\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times1.6\times10^{-19}\times200\times10^{3}}} \times \frac{1}{\sqrt{1+\dfrac{1.6\times10^{-19}\times200\times10^{3}}{2\times9.1\times10^{-31}\times(3\times10^{8})^{2}}}}$$

$\lambda = 0.025$ Å

15. Based on Einstein's theory for the photoelectric effect, the minimum frequency, υ, required will be such that

$$h\upsilon = \phi$$

where ϕ is the work function of the material. Therefore,

$$6.26\times10^{-34}\times\upsilon = 2.5\times1.6\times10^{-19}$$

Simplifying,

$$\upsilon = 6.0\times10^{14}\,\text{s}^{-1}$$

16. Quantum numbers are a set of numbers associated with a system that describe states in the system.

17. In atoms, quantum numbers represent the size of the orbitals, shape of the orbitals, magnetic orientation of the orbital and spin of the electrons in the orbital.

18. Quantum numbers can be arrived at by solving the Schrodinger wave equation for the system.

19. Pauli's exclusion principle states that no two electrons can have all of their quantum numbers being exactly the same.

20. Electrons and neutrinos are examples of fermions.

9 Fermi–Dirac Statistics

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Indicate the general approach to be taken to arrive at the Fermi–Dirac distribution function.
- Indicate and justify the approximation being made in the approach.
- Write the expression for the number of ways in which a microstate can be accomplished given a macrostate.
- Mathematically indicate the other constraints on the system.
- Write the general equation to determine the probability of occupancy of a state at energy level ε_i, for a collection of fermions.
- Use the Lagrange method of undetermined multipliers to arrive at the Fermi–Dirac statistical distribution
- Plot the Fermi–Dirac distribution function at $T = 0$ K.
- Plot the Fermi–Dirac distribution function at $T > 0$ K.
- Define Fermi energy for $T = 0$ K and $T > 0$ K.
- Indicate important information about electrons in a metal that is not captured by the Fermi–Dirac distribution function.
- Indicate macroscopic analogies to electrons in a solid following the Fermi–Dirac distribution function.
- Pictorially compare the impact of the choice of the distribution function on the estimation of C_V^e.
- Identify further scope for refining the model for electrons in solids.

9.1 Introduction

As the name suggests, the Fermi–Dirac distribution is attributed to Enrico Fermi and Paul Dirac as a result of their work in 1926. Particles that display Fermi–Dirac statistical predictions are referred to as "fermions." Fermi–Dirac statistics describes the distribution of a collection of fermions across energy levels for a system in thermal equilibrium. Fermi–Dirac statistics is of relevance because electrons qualify as fermions. Using this statistical distribution will enable us to decide what the electrons can do, cannot do, and will tend to do in the solid. In this chapter we will derive the Fermi–Dirac distribution and examine some of its implications.

9.2 The Fermi–Dirac Statistics



Fermions follow quantum mechanical rules, are identical and indistinguishable, and specifically adhere to Pauli's exclusion principle that states that no two particles can occupy the same quantum state at the same time. One of the requirements that a particle needs to satisfy to be a fermion is that it should have a

half-integer spin. Electrons satisfy these requirements and are therefore termed as fermions. The criterion of being identical and indistinguishable implies that we are unable to distinguish between cases where electrons are swapped between energy levels, leaving the total number of electrons at each energy level unchanged. We are only able to distinguish between situations where the numbers of electrons occupying a state changes.

Within the framework of electrons being fermions and the total number of particles being constant, the total volume of the system being constant and the total energy of the system being constant, we wish to find out the probability that a state with energy ε is occupied.

Let us consider a system of particles that are allowed to occupy energy levels

$$\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4, \ldots, \varepsilon_i$$

Let the number of states at these energy levels be $s_1, s_2, s_3, s_4, \ldots s_i$, respectively. These energy levels as well as the number of states at these energy levels are fixed for the system. What values these are fixed at is a topic we will take up at a later stage. For the moment, we simply accept that they are fixed. The variable we have at our disposal is the number of particles that will occupy these states. Let the number of particles at the respective energy levels be $n_1, n_2, n_3, n_4, \ldots, n_i$.

A microstate of the system is then defined as a specific set of numerical values for $n_1, n_2, n_3, n_4, \ldots, n_i$. A different microstate will have a different set of values for $n_1, n_2, n_3, n_4, \ldots, n_i$. In the analysis that we will carry out, the values of s_i and ε_i are fixed for the system because we have defined or identified a system with those values. The value of n_i is, however, variable, and it is our intention to find out what will be the equilibrium values of n_i at the s_i states at each of the ε_i given the various constraints we are placing on the system. More than focusing on the exact numerical value of n_i, we will aim to determine the probability of occupancy of a state at energy level ε_i that will be based on a ratio of n_i to s_i, which will be a more useful parameter.

In our analysis to find the probability of occupancy of a state at energy level ε_i, we will approach the calculations as follows:

1. We will start from a narrow starting point that isolates the type of term we are interested in.
2. We will then build a general equation that appears to handle the system in its entirety.
3. On examining this general equation, we will note that only one term in the equation is of interest to us, so we will neglect the rest of the terms and focus only on that term from then on for the rest of the derivation.

Conventionally, the Fermi–Dirac distribution is denoted by $f(\varepsilon)$, which represents the probability that the energy level ε is occupied. Therefore, our goal is to determine an expression for $f(\varepsilon)$.

We note that each possible state of the system is characterized by a specific set of $n_1, n_2, n_3, n_4, \ldots, n_i$, which are the number of particles occupying energy levels $\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4, \ldots, \varepsilon_i$, respectively. We will denote an individual microstate of the system and its specific combination of $n_1, n_2, n_3, n_4, \ldots, n_i$ by $\bar{N} = (n_1, n_2, n_3, n_4, \ldots)$.

Our narrow starting point is that when the system is in the specific individual state $\bar{N}$, the probability that a state at energy level ε_i is occupied is referred to as the "conditional probability," because it is conditional on the system being in the individual microstate $\bar{N}$. Once an $\bar{N}$ is chosen, all the values ε_i, s_i and n_i are fixed for the system. Therefore, under this specific condition the probability that a state at energy level ε_i is occupied is simply n_i/s_i. This conditional probability is denoted by $f(\varepsilon_i \mid \bar{N})$. For example, if at a given energy level there are 10 states and there are four particles at that energy level due to a specific choice of $\bar{N}$ then, under those conditions, the conditional probability of occupancy of a state at that energy level is 40%.

If a different $\bar{N}$ were to be chosen which resulted in seven particles at that energy level, then under the conditions of the new $\bar{N}$, the conditional probability of occupancy of a state at that same energy level is 70%.

From this conditional probability that a state at a given energy level is occupied, how do we get the general probability that a state at that energy level is occupied? For this, we have to also determine the probability of occurrence of a specific $\bar{N}$ given that the system can exhibit a variety of different $\bar{N}$.

The probability that the system is displaying the individual microstate $\bar{N}$ – denoted by $p(\bar{N})$ – is given by the ratio of the number of ways in which such a microstate can be achieved – denoted by $\omega(\bar{N})$ – to the total number of ways in which all possible microstates of the system can be achieved – denoted by ω_{Total}, that is,

$$p(\overline{N}) = \frac{\omega(\overline{N})}{\omega_{\text{Total}}}$$

Therefore the total probability that a state at energy level ε_i is occupied, $f(\varepsilon_i)$, is given by

$$f(\varepsilon_i) = \sum_{\bar{N}} f(\varepsilon \,|\, \overline{N})\, p(\overline{N}) = \frac{1}{\omega_{\text{Total}}} \sum_{\bar{N}} f(\varepsilon \,|\, \overline{N}) \omega(\overline{N})$$

The above equation takes into account all the factors necessary to make the calculation of $f(\varepsilon_i)$ complete. This equation handles the situation in its entirety.

Our analysis is based on the idea that there is no inherent preference for any one individual arrangement over the other, that is, they are all equally probable. However the arrangements differ in the number of ways they can be arrived at, and therefore a snapshot of the system will show the arrangements that can be arrived at more often. Therefore, if one arrangement can be arrived at in 10 different ways, and another arrangement can be arrived at in 100 different ways, then the second arrangement is 10 times more probable than the first.

For large systems, the amount of information required, as well as the amount of computation required to fully calculate all the terms of the above expression, can be unacceptably large. Fortunately, large systems provide us with the simplification recognized in statistical mechanics, that the most probable microstate swamps the probability of all other microstates combined. In other words, we will find that one of the microstates or a specific $\bar{N}$ occurs vastly more often than the rest. This is the most probable microstate, which we will designate as $\bar{N}_0$.

The $\omega(\bar{N}_0)$ corresponding to this most probable microstate is larger than the sum of all the possible ways in which rest of the microstates can be arrived at, combined. Stated mathematically,

$$\omega(\bar{N}_0) \oplus \omega_{\text{Total}}$$

The ω_{Total} will, to be exact, contain contributions from all other possible microstates, but these contributions will be insignificant. Therefore, from the equation above we find that when we identify $\bar{N}_0$,

$$f(\varepsilon) \oplus f(\varepsilon \,|\, \bar{N}_0)$$

Since, as indicated earlier, $f(\varepsilon \,|\, \bar{N}_0)$ is simply n_i / s_i when $\bar{N} = \bar{N}_0$, all other contributions can be neglected.

To better understand the different terms we have used so far in this discussion, the following numbers, for $\bar{N}$, $f(\varepsilon \,|\, \bar{N})$, and $p(\bar{N})$ have been arbitrarily selected to indicate a few different possibilities.

The numbers as well as the associated calculations below can be considered purely as an illustrative example.

$\bar{N}$	$f(\phi \mid \bar{N})$ (%)	$p(\bar{N})$
$\bar{N}_1$	90	0.00010
$\bar{N}_2$	10	0.00090
$\bar{N}_3$	40	0.99900

Since $p(\bar{N})$ of $\bar{N}_3$ is 0.99900, it is designated as the most probable microstate $\bar{N}_0$. Therefore,

$$f(\varepsilon) = \sum_{\bar{N}} f(\varepsilon \mid \bar{N}) \times p(\bar{N}) \approx f(\varepsilon \mid \bar{N}_0) \approx 40\%$$

Even though for $\bar{N}_1$, the $f(\varepsilon \mid \bar{N})$ is 90%, it has virtually no impact on the final $f(\varepsilon)$ since the $p(\bar{N})$ corresponding to it is negligible. The final value of $f(\varepsilon)$ is dominated by the contribution of the most probable microstate, $\bar{N}_3$, which as indicated above, is therefore also designated as $\bar{N}_0$.

Therefore, even though we are faced with a computationally unwieldy expression, we can conveniently choose to ignore most of the computation involved and focus simply on the computation that maximizes $\omega(\bar{N})$. In the process of computing the maximum possible value for $\omega(\bar{N})$, we will be able to identify an expression for n_i/s_i, which corresponds to the maximum possible value of $\omega(\bar{N})$. This expression for n_i/s_i, for all practical purposes, is treated as the overall probability that a state at energy level ε_i is occupied. It is important to note that we have neglected a vast majority of the terms from our original expression for $f(\varepsilon_i)$. This is because we are accepting that all of those terms, taken together, have negligible impact when compared to the impact of the most probable state, therefore we focus entirely on the probability of occupancy of an energy level when the system is in its most probable state and have stated that this is the probability of occupancy of the energy level even when all of the other states are fully taken into account.

To reiterate, if we denote the most probable microstate of the system as $\bar{N}_0$, then mathematically the approximation can be stated as

$$f(\varepsilon) \oplus f(\varepsilon \mid \bar{N}_0)$$

Our task is therefore to determine the $\bar{N}_0$ which results in the maximum possible value for the corresponding $\omega(\bar{N}_0)$. The method for identifying $\omega(\bar{N}_0)$ is to write a general expression for $\omega(\bar{N})$ and to look for the conditions that maximize $\omega(\bar{N})$. A given $\bar{N}$ consists of:

1. Energy levels: $\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4, \ldots, \varepsilon_i$
2. Number of states: $s_1, s_2, s_3, s_4, \ldots, s_i$
3. Number of particles: $n_1, n_2, n_3, n_4, \ldots, n_i$

The number of ways to accomplish a given $\bar{N}$, denoted by $\omega(\bar{N})$, is given by the product of the number of ways in which n_1 particles can be arranged in s_1 states multiplied by the number of ways in which n_2 particles can be arranged in s_2 states … multiplied by the number of ways in which n_i particles can be arranged in s_i states.

The number of ways in which n_i particles can be arranged in s_i states is denoted by $\Omega(n_i)$. More elaborately, at a given energy level ε_i, the number of ways in which we can place n_i identical and indistinguishable particles in the available s_i states with no more than one particle per state (Pauli's exclusion principle) can be expressed mathematically as

$$\Omega(n_i) = \frac{s_i!}{n_i!(s_i - n_i)!}$$

The above expression can be thought of as the number of combinations possible with n_i identical and indistinguishable particles and $(s_i - n_i)$ identical and indistinguishable empty states in a total of s_i states. For a specific $\bar{N}$, therefore,

$$\omega(\bar{N}) = \Omega(n_1) \times \Omega(n_2) \times \Omega(n_3) \cdots = \prod_i \Omega(n_i) = \prod_i \left[\frac{s_i!}{n_i!(s_i - n_i)!} \right]$$

Our intent is to maximize $\omega(\bar{N})$ subject to the two constraints placed on the system, where the total energy of the system and the total number of particles in the system are constant, that is,

$$\sum \varepsilon_i n_i = \text{constant}$$

and

$$\sum n_i = \text{constant}$$

In this process of maximizing $\omega(\bar{N})$, subject to the above two constraints, we will find n_i / s_i in the most probable $\bar{N}_0$ – which is the $f(\varepsilon | \bar{N}_0)$ and hence the $f(\varepsilon)$ that we seek – since contributions from all other terms will be negligible. Therefore, for a collection of fermions, we have examined the process by which we expect to find the probability of occupancy of state at energy level ε_i. The general equation we arrived at is indicated in Fig. 9.1.

The equation corresponding to Fig. 9.1 is as follows:

$$f(\varepsilon) = \sum_{\bar{N}} f(\varepsilon | \bar{N}) \times p(\bar{N})$$

Written more elaborately, we note that

$$f(\varepsilon) = \frac{1}{\omega_{\text{Total}}} \sum_{\bar{N}} f(\varepsilon | \bar{N}) \times \omega(\bar{N})$$

Probability of occupancy of a state at energy level ε_i = $\sum$ (Sum over all possible microstates) [Probability of occupancy of a state at energy level ε_i, in a particular microstate $\bar{N}$ × Probability that the system is at that particular microstate $\bar{N}$]

Figure 9.1 For the collection of fermions, the general equation that enables us to determine the probability of occupancy of state at energy level ε_i.

Based on the approximation that statistical mechanics operates on in such systems, there exists a most probable microstate $\bar{N}_0$ which is more probable than all other microstates combined, we have

$$\omega_{\text{Total}} = \omega(\bar{N}_1) + \omega(\bar{N}_2) + \omega(\bar{N}_3) + \cdots \approx \omega(\bar{N}_0)$$

Therefore,

$$f(\varepsilon) \oplus f(\varepsilon \mid \bar{N}_0)$$

In other words, our problem reduces to determining n_i / s_i when $\bar{N} = \bar{N}_0$. We determine this n_i / s_i by *partially* going over the math to determine $\omega(\bar{N}_0)$ during which we obtain an expression for n_i / s_i when $\bar{N} = \bar{N}_0$. In the calculations below, we will proceed in this direction.

$$\omega(\bar{N}) = \frac{s_1!}{n_1!(s_1 - n_1!)!} \times \frac{s_2!}{n_2!(s_2 - n_2!)!} \times \cdots = \prod_i \frac{s_i!}{n_i!(s_i - n_i!)!}$$

Since the functions $\omega(\bar{N})$ and $\ln \omega(\bar{N})$, either both increase or decrease, their maxima occur at the same point $\bar{N}_0$. Therefore, instead of maximizing $\omega(\bar{N})$, we can choose to maximize $\ln \omega(\bar{N})$, which is mathematically easier to handle and arrive at the same desired result $\bar{N}_0$.

$$\ln \omega(\bar{N}) = \sum_i \ln s_i! - \sum_i \ln n_i! - \sum_i \ln(s_i - n_i)!$$

We can now use Stirling's approximation that for large X

$$\ln X! = X \ln X - X$$

Therefore,

$$\ln \omega(\bar{N}) = \sum_i \{s_i \ln s_i - s_i - (n_i \ln n_i - n_i) - [(s_i - n_i)\ln(s_i - n_i) - (s_i - n_i)]\}$$

Simplifying we get

$$\ln \omega(\bar{N}) = \sum_i s_i \ln s_i - \sum_i n_i \ln n_i - \sum_i (s_i - n_i)\ln(s_i - n_i)$$

When $\ln \omega(\bar{N})$ reaches a maximum, $\partial \ln \omega(\bar{N}) = 0$. Since s_i are fixed, $\partial \ln \omega(\bar{N}) = 0$ implies

$$-\sum_i [\ln n_i - \ln(s_i - n_i)]\partial n_i = 0$$

or

$$-\sum_i \ln\left(\frac{n_i}{s_i - n_i}\right)\partial n_i = 0$$

In addition, the total number of particles present in the system is fixed. Therefore, even if we move a few particles from one energy level to another, the total sum of those exchanges must be zero. Hence,

$$\sum_i \partial n_i = 0$$

Similarly the total energy of the system is fixed. Therefore, even if we move a few particles from one energy level to another, the total energy change corresponding to those exchanges must be zero:

$$\sum_i \varepsilon_i \partial n_i = 0$$

Using the Lagrange method of undetermined multipliers discussed in greater detail in Chapter 7, we obtain

$$-\ln\left(\frac{s_i - n_i}{n_i}\right) + \alpha + \beta\varepsilon_i = 0$$

Note that the negative sign in the first term in the equation above, is inserted by convention to make the final results easier to interpret. Since the equations leading up to the above are all equal to zero, introduction of a negative sign before summing up is acceptable.

Simplifying further, we get

$$\frac{s_i}{n_i} - 1 = e^{\alpha+\beta\varepsilon_i}$$

Therefore,

$$\frac{n_i}{s_i} = \frac{1}{1 + e^{\alpha+\beta\varepsilon_i}} = f(\varepsilon_i)$$

It is relevant to restate here that the above expression for $f(\varepsilon_i)$ or $f(\varepsilon)$ in general, has appeared during our efforts to maximize $(\bar{N})$. However, we take advantage of the underlying concept of statistical mechanics – that this expression for $f(\varepsilon)$ will be significantly greater in magnitude than all other possible contributions to n_i / s_i. Therefore, we consider it justifiable for a system subject to the constraints we have imposed – and as a very reasonable approximation – to treat this expression for $f(\varepsilon)$ as the overall probability of occupancy of a state at energy level ε. This expression for $f(\varepsilon)$ is then referred to as the Fermi–Dirac distribution. Only the values of the constants α and β remain to be specified. It is noted that the constants α and β work out in such a way that

$$f(\varepsilon) = \frac{1}{1 + e^{[(\varepsilon - E_f)/k_b T]}}$$

where E_f is referred to as the Fermi energy and will be elaborated later. T is the absolute temperature and k_b is the Boltzmann constant.

We have thus derived the Fermi–Dirac distribution that applies to a collection of fermions in thermal equilibrium. We will now examine the features of the Fermi–Dirac distribution and see what these features imply.

9.3 Features of Fermi–Dirac Distribution



The Fermi–Dirac distribution function is as follows:

$$f(\varepsilon_i) = \frac{1}{1 + e^{[(\varepsilon_i - E_f)/k_b T]}}$$

In the above expression, ε_i is a variable and we wish to evaluate $f(\varepsilon_i)$ for various values of ε_i. This evaluation can be carried out at various temperatures, so T is the other variable. k_b is Boltzmann's constant and E_f is a particular value of the energy level and a constant for a given system. We will consider the significance of E_f a little later in this chapter.

On examining the expression for $f(\varepsilon_i)$, we note that there are three different regimes of values of ε_i over which the response of the function can vary significantly. The regimes are as follows:

$$\varepsilon_i < E_f$$
$$\varepsilon_i = E_f$$
$$\varepsilon_i > E_f$$

Let us first consider these three regimes when the temperature tends to absolute zero Kelvin.

1. For all $\varepsilon_i < E_f$, $(\varepsilon_i - E_f)/k_b T$ tends toward $-\infty$ and hence $f(\varepsilon_i)$ tends to 1. In the limiting case of $T = 0$ K, $f(\varepsilon_i) = 1$.
2. Similarly, when the temperature tends to absolute 0 K, for all $\varepsilon_i > E_f$, $(\varepsilon_i - E_f)/k_b T$ tends toward $+\infty$ and hence $f(\varepsilon_i)$ tends to 0. In the limiting case of $T = 0$ K, $f(\varepsilon_i) = 0$.
3. When $\varepsilon_i = E_f$, in the limiting case of $T = 0$ K, $f(\varepsilon_i)$ is undefined and varies between the two limits 1 and 0. Figure 9.2 shows the behavior of $f(\varepsilon_i)$ as a function of energy at $T = 0$ K.

Let us now consider the Fermi–Dirac distribution at temperatures greater than absolute zero. The effect of increasing temperature on the Fermi–Dirac distribution is such that as the temperature increases, the change in $f(\varepsilon_i)$ from 1 to 0 occurs over a larger range of energies.

When $\varepsilon_i = E_f$ and the temperature $T > 0$ K, $(\varepsilon_i - E_f)/k_b T = 0$ and $f(\varepsilon_i) = 0.5$ regardless of the actual value of the temperature.

When the temperature is greater than 0 K, for all $\varepsilon_i < E_f$, $(\varepsilon_i - E_f)/k_b T$ is a negative number that becomes a larger negative number as ε_i decreases. Hence, $f(\varepsilon_i)$ starts from a value of 0.5 at $\varepsilon_i = E_f$ and tends toward 1 as ε_i decreases.

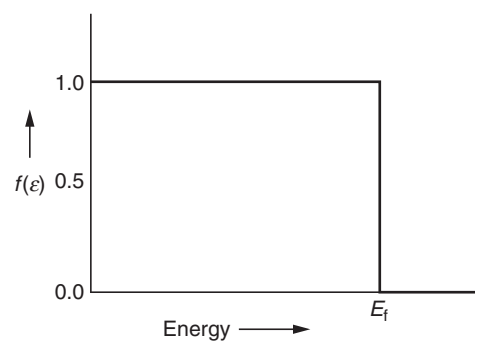


Figure 9.2 Variation of $f(\varepsilon_i)$ as a function of energy at $T = 0$ K.

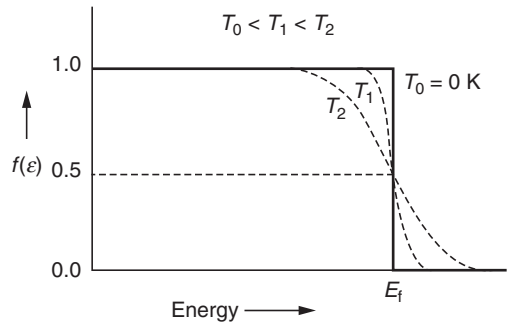


Figure 9.3 Variation of $f(\varepsilon_i)$ as a function of energy at various temperatures.

Similarly, when the temperature is greater than 0 K, for all $\varepsilon_i > E_f$, $(\varepsilon_i - E_f)/k_bT$ is a positive number which increases as ε_i increases. Hence, $f(\varepsilon_i)$ starts from a value of 0.5 at $\varepsilon_i = E_f$ and tends toward 0 as ε_i increases. This behavior of $f(\varepsilon_i)$ is summarized in Fig. 9.3.

While we have looked at the Fermi–Dirac distribution in a mathematical sense above, let us now consider the impact of this distribution in a descriptive sense.

We have noted earlier that there are a fixed and finite number of states at any given energy level, the framework within which we derived the Fermi–Dirac distribution. Therefore, because of Pauli's exclusion principle, there is a limit on the number of particles we can place on these states. At $T = 0$ K, if we arrange the energy levels of the system in order of increasing energy, nature will choose to fill the lowest energy level first. When all the states at the lowest energy level are full, if there are still particles that remain, they will start filling the states at the next higher energy level. This process will continue till we run out of particles to place in the energy levels. Therefore, even at absolute zero temperature, we are forced to fill energy levels that are higher than the lowest energy level available in the system – which is directly a result of the fact that the electrons are behaving in a manner consistent with the restrictions on fermions. The number of electrons available in the system is a large but a finite value, therefore as we continue the process of filling up states at increasing energy levels, we will eventually run out of electrons to fill further states. Even higher energy levels may be defined for the system, but will remain unfilled. In other words, as we fill the states at E_0 we run out of states, then at E_1 we run out states and so on and then at some higher energy level E_f we run out of electrons. This E_f at $T = 0$ K is called the Fermi energy and identified in Figs. 9.2 and 9.3. Since all states are getting filled up for all $E < E_f$, $f(\varepsilon_i) = 1$ for all energy levels less than the Fermi energy. At $E = E_f$, the $f(\varepsilon_i)$ transitions from 1 to 0 and then stays at 0 for all values of $E > E_f$. We are therefore able to see why the Fermi–Dirac distribution function looks the way it does at 0 K.

The Fermi energy for a metal is therefore defined as the highest energy level that is occupied at 0 K and it is of the order of 2.5 eV for a typical metal. There is an alternate definition for the Fermi energy, for temperature greater than 0 K, which we will describe shortly.

It is important to note that $f(\varepsilon_i)$ is only the probability of occupancy of a state at a given energy level ε_i. It does not give any information on the number of states actually available at that energy level. The number of available states will typically vary from one energy level to another energy level. There is another function which will give us that information and will be developed in Chapter 11. The $f(\varepsilon_i)$ merely tells us the probability of occupancy of states at a given energy level, regardless of the actual number of states at that energy level.

Purely for illustrative purposes, in order to understand the significance of $f(\varepsilon_i)$, let us use some hypothetical numbers of states at various energy levels as shown in Table 9.1.

Table 9.1 Number of states, probability of occupancy and actual occupancy

Energy Level States	*Number of States*	$f(\varepsilon_i)$	*Number of Occupied States*
ε_1	50	1.0	50
ε_2	70	1.0	70
ε_3	100	0.0	0

It is evident from the discussion that there is an additional and important piece of information that is not captured in the plots of the Fermi–Dirac distribution function that we have seen so far. So, for example, if there are some 3500 states at $E = E_f + 1$, all of these states will be empty at $T = 0$ K. We will consider this point in detail in Chapter 11. The Fermi–Dirac distribution is itself an important piece of information and hence we are focusing on it at this time.

As indicated above, when the temperature is greater than 0 K, some energy levels close to but below E_f have $f(\varepsilon_i) < 1$ and some energy levels close to but higher than E_f have $f(\varepsilon_i) > 1$. At higher temperatures, the range of energy values over which the $f(\varepsilon_i)$ varies from 1 to 0 increases, as shown in Fig. 9.3. Using Figs. 9.2 and 9.3, we are therefore able to understand the variation of $f(\varepsilon_i)$ as a function of energy as well as temperature.

From the equation for $f(\varepsilon_i)$, we find that when $T \neq 0$ K at $\varepsilon_i = E_f$, $f(\varepsilon_i) = 0.5$. This is then the other definition for the Fermi energy – it is the energy level at which the probability of occupancy is 1/2, when the temperature is greater than absolute zero.

To understand the concept of states and filling up of states consistent with the Fermi–Dirac statistics, we can draw an analogy to a vessel being filled with water or sand. The vessel will fill from bottom to top till we run out of water or sand. This analogy creates an energy level similar to E_f assuming that we are drawing this analogy at 0 K. The information captured by the Fermi–Dirac distribution is that the probability of occupancy is 100% for energy levels below E_f and 0% for energy levels above E_f. The information that has not been captured by the Fermi–Dirac distribution is the shape of the vessel and therefore the number of states at each of those energy levels. As shown in Fig. 9.4, the shapes of the vessels can be significantly different and hence they can get filled to significantly different levels with the same amount of sand or water.

The Fermi energy level signifies that it is the energy level at which the electrons are in a position to interact with energy levels above them. Therefore, only energy values close to E_f participate in the process of gaining temperature. Greater the change in temperature, more the number of states on either side of E_f that participate in the change in temperature. Much farther from E_f, the states are oblivious to the change in the temperature.

We have thus far looked at the plot of the Fermi–Dirac distribution and examined how the distribution function varies as a function of energy and how it varies as a function of temperature. We have also looked at the Fermi energy, which is the border across which all transactions with respect to energy occur. These are the major features of the Fermi–Dirac distribution. We have also noted that the important information that we have not addressed is the actual number of states at each energy level or the shape of the container in our analogy.

We will now compare the Fermi–Dirac distribution to the Maxwell–Boltzmann distribution and examine how the Drude–Sommerfeld model is better than the classical Drude model.

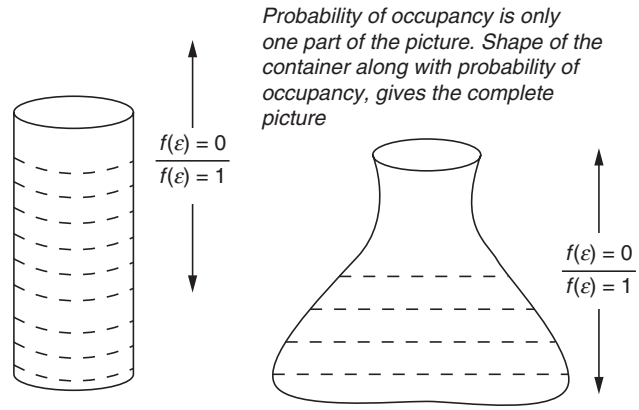


Figure 9.4 Analogy of sand or water being used to fill vessels of different shapes. The probability of occupancy of the container at any given height merely indicates whether or not sand or water is available at that height. To know the actual amount of sand or water held in the containers we also need to know detailed information about the shape of the containers, an aspect that is not captured by the Fermi–Dirac distribution.

9.4 Comparing Maxwell–Boltzmann and Fermi–Dirac Distributions



The Maxwell–Boltzmann distribution and its variation with temperature are shown in Fig. 9.5. At higher temperatures higher energy levels get increased numbers of electrons, while the lower energy levels get decreased numbers of electrons. The Maxwell–Boltzmann distribution function can be represented in a normalized manner as follows:

$$P(\varepsilon) = P_0 e^{(-\varepsilon)/k_b T}$$

such that

$$\int_0^\infty P(\varepsilon) d\varepsilon = 1$$

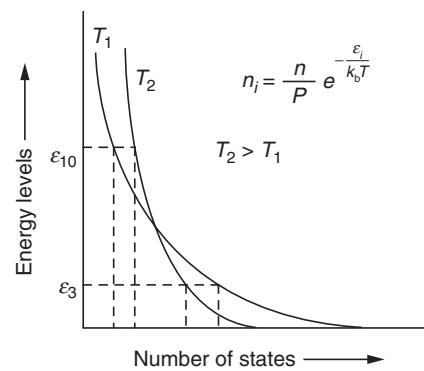


Figure 9.5 Maxwell–Boltzmann distribution function and its variation with temperature.

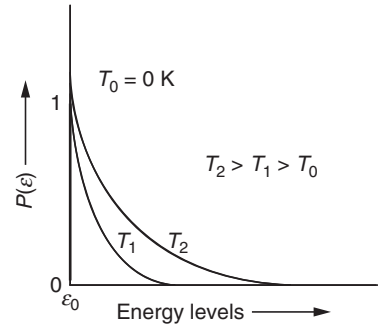


Figure 9.6 Plot of the normalized Maxwell–Boltzmann distribution function as a function of energy at different temperatures.

This normalized function is typically plotted with its axes interchanged with respect to the axes in Fig. 9.5, that is, with energy on the x-axis and $P(\varepsilon)$ on the y-axis. The normalized Maxwell–Boltzmann distribution function coincides with the y-axis at $T = 0$ K. At higher temperatures, higher energy levels are occupied as indicated in the curves in Fig. 9.6.

The Maxwell–Boltzmann distribution overestimates C_V^e . The reason for this is evident from the plot for the normalized distribution shown in Fig. 9.6. As the temperature increases, all the electrons can access empty states at energy levels higher than the energy level they occupy. Therefore all of them can, in principle, participate in the process of increasing the temperature by gaining energy. Since there are a large number of electrons in the system, the energy required to raise the temperature is large, and this is directly reflected in the value of the specific heat that is a measure of this energy.

The Fermi–Dirac distribution creates a situation wherein at lower energy levels all the states are full. Therefore, when an electron at a relatively low energy level tries to gain energy and move to a marginally higher energy level, it is unable to do so since all the states at that marginally higher energy level are already full. This situation is indicated illustratively in Fig. 9.7.

Only electrons close to the Fermi energy level are able to access empty states at energy levels marginally higher than those they already occupy. Therefore only electrons close to the Fermi energy participate in the gain in energy as the temperature increases. Since this is a very small fraction of the electrons in the system the energy required to raise the temperature is relatively small, and this is directly reflected in the smaller value of the specific heat that is a measure of this energy. The limitation in the number of electrons that can participate in the process of raising the energy of the system limits the C_V^e. This situation is analogous to water evaporating only from the surface of a container that is full of water.

We are therefore able to see how by applying the Fermi–Dirac distribution, we are able to correct one of the major anomalies of the classical Drude model. Figure 9.8 compares the important features of the Maxwell–Boltzmann distribution with those of the Fermi–Dirac distribution.

To further refine the model, we have to take a fresh look at the assertion that the potential within the solid is flat or uniform. In reality, the potential that an electron will experience close to an ionic core will be significantly different from the potential it will experience far from the ionic core. We therefore need to build a picture of the potential as a function of position in a solid. It will turn out that we will be able to approximate the potential as a function of position in a solid in a manner that makes it convenient for subsequent computations. We will also obtain an expression for the number of states at a given energy level in the upcoming chapters.

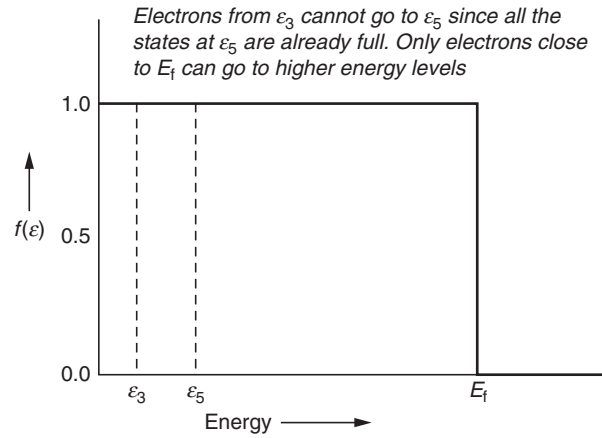


Figure 9.7 An illustrative example showing how lower energy levels are unable to participate in the process of gaining energy consistent with an increased temperature of the system. Electrons at energy level ε_3 are unable to gain energy and go to states at energy level ε_5, since those states are already full. Only electrons close to Fermi energy are in a position to gain energy and go to unoccupied states at higher energy levels.

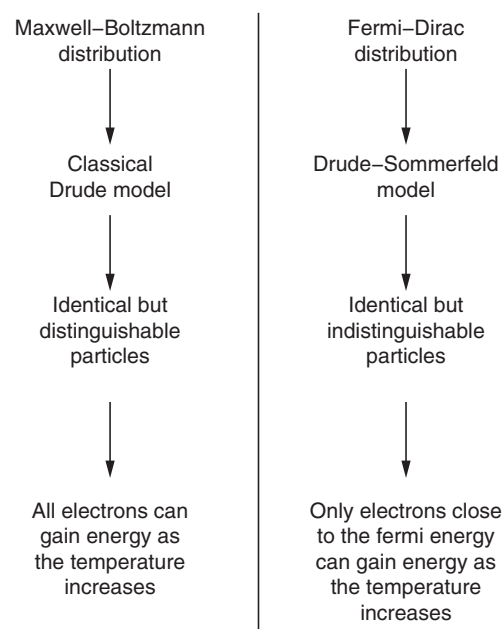


Figure 9.8 A comparison of some of the important features of the Maxwell–Boltzmann distribution with those of the Fermi–Dirac distribution.

Summary

1. Electrons have half-integer spin and are fermions.
2. The Fermi–Dirac distribution is denoted by $f(\varepsilon)$, which represents the probability that the energy level ε is occupied.
3. The total probability that a state at energy level ε_i is occupied, $f(\varepsilon_i)$ is given by

$$f(\varepsilon_i) = \sum_{\overline{N}} f(\varepsilon|\overline{N})p(\overline{N})$$

$$= \frac{1}{\omega_{\text{Total}}} \sum_{\overline{N}} f(\varepsilon|\overline{N})\omega(\overline{N})$$

4. The $\omega(\overline{N}_0)$ corresponding to the most probable microstate is so large that it is larger than the sum of all of the possible ways in which rest of the microstates can be arrived at, combined. This implies that

$$\omega(\overline{N}_0) \oplus \omega_{\text{Total}}$$

5. We intend to find n_i/s_i in the most probable $\overline{N}_0$ and ignore contributions from all other terms since they are negligible.
6. Based on the ideas of statistical mechanics, there exists a most probable microstate, $\overline{N}_0$ and hence it is necessary and sufficient if we can determine n_i/s_i when $\overline{N} = \overline{N}_0$.
7. Using appropriate mathematical simplifications, it is determined that

$$\frac{n_i}{s_i} = \frac{1}{1 + e^{\alpha + \beta\varepsilon_i}} = f(\varepsilon_i)$$

8. The Fermi–Dirac distribution works out to

$$f(\varepsilon) = \frac{1}{1 + e^{[(\varepsilon - E_f)/k_b T]}}$$

where E_f is referred to as the Fermi energy.

9. When $T = 0$ K, $f(\varepsilon_i) = 1$ for all $\varepsilon_i < E_f$.
10. When $T = 0$ K, $f(\varepsilon_i) = 0$ for all $\varepsilon_i > E_f$ for all
11. When $T = 0$ K at $\varepsilon_i = E_f$, $f(\varepsilon_i)$ varies from 1 to 0.
12. The Fermi energy for a metal is defined as the highest energy level that is occupied at 0 K and it is of the order of 2.5 eV for a typical metal.
13. As the temperature increases, the change in $f(\varepsilon_i)$ from 1 to 0 occurs over a larger range of energies.
14. For a metal, when $\varepsilon_i = E_f$ and the temperature $T > 0$ K, $(\varepsilon_i - E_f)/k_b T = 0$ and $f(\varepsilon_i) = 0.5$ regardless of the actual value of the temperature.
15. When the temperature is greater than absolute zero, Fermi energy for a metal is defined as the energy level at which the probability of occupancy is 1/2.
16. $f(\varepsilon_i)$ merely tells us the probability of occupancy of states at a given energy level, regardless of the actual number of states at that energy level.
17. One important significance of the Fermi energy level is that it is the energy level at which the electrons are in a position to interact with energy levels above them.
18. If electrons are assumed to follow the Maxwell–Boltzmann distribution, as the temperature increases all the electrons can access empty states at energy levels higher than the energy level that they occupy. This results in the overestimation of C_V^e.
19. If electrons are assumed to follow the Fermi–Dirac distribution, only the electrons close to the Fermi energy level are able to access empty states at energy levels marginally higher than those they already occupy. This results in the correct estimation of C_V^e.

Practice Questions

Multiple Choice Questions

1. Consider a collection of electrons subject to Fermi–Dirac statistics. The following is false about them:
   (a) Number of allowed states can be higher than the number of energy levels
   (b) Number of energy levels can be higher than the number of electrons

(c) Number of electrons can be higher than the number of allowed states
(d) Number of electrons can be higher than the number if energy levels occupied

2. The conditional probability that a state is occupied and the probability that the system displays a particular microstate are related as follows:
(a) Conditional probability of occupancy of the state is always the larger of the two
(b) Probability the system displays a microstate is always the larger of the two
(c) The two probabilities are the same
(d) The two probabilities are unrelated

3. The most probable microstate occurs nearly
(a) 50% of the time
(b) 90% of the time
(c) 95% of the time
(d) 100% of the time

4. In calculating the Fermi–Dirac statistics it is important to determine
(a) The two most probable microstates
(b) All states of the system
(c) Only the most probable microstate
(d) The most probable as well as the least probable microstate

5. If $f(\varepsilon|\bar{N})$ is 99%, and $p(\bar{N})$ corresponding to it is 1% (0.01), the following is true:
(a) This probability of occupancy of the state is 99%
(b) The probability of occupancy of the state cannot be determined from this information
(c) The probability of occupancy of the state is 1%
(d) The probability of occupancy of the state is 0.99%

6. For a system following Fermi–Dirac statistics, with increase in temperature, the probability of occupancy of a state at an energy level less than the Fermi energy
(a) Increases **(b)** Stays the same
(c) Decreases **(d)** Is undefined

7. Consider two systems at room temperature, both following Fermi–Dirac statistics but with different Fermi energies. The probability of occupancy of a state at an energy level less than the Fermi energy
(a) Is higher for the system with higher Fermi energy
(b) Is higher for the system with lower Fermi energy
(c) Is the same for both systems
(d) Is independent of the Fermi energy

8. For a system at room temperature, following Fermi–Dirac statistics, the probability of occupancy of a state at an energy level significantly lower than the Fermi energy is
(a) Nearly 1 **(b)** Nearly 0.5
(c) Nearly 0.1 **(d)** Nearly 0

9. For a system at room temperature, following Fermi–Dirac statistics, the probability of occupancy of a state at an energy level significantly higher than the Fermi energy is
(a) Nearly 1 **(b)** Nearly 0.5
(c) Nearly 0.1 **(d)** Nearly 0

10. For a system at room temperature, following Fermi–Dirac statistics, the probability of occupancy of a state at an energy level marginally less than the Fermi energy is
(a) Nearly 1
(b) Slightly higher than 0.5
(c) Slightly lower than 0.5
(d) Nearly 0

11. At the Fermi energy, for a metal at $T = 0$ K, the probability of occupancy is
(a) 1/2 **(b)** Less than 1/2
(c) More than 1/2 **(d)** Undefined

12. If $f(\varepsilon_i) = 0.9$, it implies that
(a) 90% of all occupied states exist at that energy level
(b) 90% of all states at that energy level are occupied
(c) The number of states at that energy level are 90% of all available states
(d) 90% of the energy in the system is available in that energy level.

13. At room temperature, in the process of exchange of energy with the surroundings, the fermions that participate in the process are
(a) Fermions close to the Fermi energy
(b) All the fermions
(c) Only fermions above the Fermi energy
(d) Only Fermions below the Fermi energy

14. At $T > 0$ K, Fermi energy is defined as the energy level at which the probability of occupancy is (1/2) for

(a) A metal **(b)** A semiconductor
(c) A ceramic **(d)** All of the above

15. Higher the Fermi energy
(a) Higher is the number of states with that energy
(b) Higher is the temperature required to change the occupancy levels of states near the Fermi energy to the same extent
(c) Higher is the energy with which electrons are bound to the atoms
(d) Lower is the temperature required to change the occupancy levels of states near the Fermi energy to the same extent

16. If electrons are assumed to follow the Maxwell–Boltzmann distribution
(a) There is a limit on the number of electrons that can occupy a given energy level independent of the temperature
(b) There is a fixed number of electrons that can occupy a given energy level and that fixed number is dependent on the temperature
(c) There is no limit on the number of electrons that can occupy a given energy level at a given temperature
(d) The number of electrons that can occupy any given energy level is a constant

17. If electrons are assumed to follow the Maxwell–Boltzmann distribution as opposed to Fermi–Dirac distribution
(a) More energy is required to raise the temperature by the same degree
(b) The amount of energy required to raise the temperature by the same degree is the same
(c) Less energy is required to raise the temperature by the same degree
(d) Energy required to raise the temperature is independent of the distribution assumed

18. If electrons are assumed to follow the Fermi–Dirac distribution
(a) There is a limit on the number of electrons that can occupy a given energy level independent of the temperature
(b) There is a fixed number of electrons that can occupy a given energy level and that fixed number is dependent only on the temperature
(c) There is no limit on the number of electrons that can occupy a given energy level at a given temperature
(d) The number of electrons that can occupy any given energy level is a constant

19. Differences between the Maxwell–Boltzmann distribution and the Fermi–Dirac distribution arise because
(a) Particles are identical
(b) In Maxwell–Boltzmann distribution, particles are indistinguishable
(c) In Fermi–Dirac distribution, particles are indistinguishable
(d) In Fermi–Dirac distribution, particles are distinguishable

20. In both Maxwell–Boltzmann distribution and the Fermi–Dirac distribution, as temperature is increased
(a) Occupancy of all lower energy states is unchanged
(b) Occupancy all of higher energy states is unchanged
(c) Occupancy of some lower energy states is decreased
(d) Occupancy of some lower energy states is increased

Short Answer Type Questions

1. In the context of deriving the Fermi–Dirac distribution, what is the difference between conditional probability and total probability?

2. Why is the value of $p(\bar{N})$ important?

3. Given the data that follows:

$\bar{N}$	$f(\phi \mid \bar{N})\%$	$p(\bar{N})$
$\bar{N}_1$	70	0.00010
$\bar{N}_2$	35	0.00090
$\bar{N}_3$	48	0.99900

which is the most probable microstate of the system?

4. From the data above, which is the microstate with lowest probability of occupancy?
5. What is the overall probability of occupancy of a state at $\mathcal{E}$?
6. What is the impact of the Fermi energy of a system on the variation of the Fermi–Dirac distribution as a function of temperature?
7. What is the value of $f(\varepsilon_i)$ at $\varepsilon_i = E_f$ when $T = 0$ K?
8. In the above example, what is the significance of $f(\varepsilon_i)$ being undefined?
9. What is the significance of the change in shape of the Fermi–Dirac distribution as a function of increasing temperature?
10. Why are lower energy states not initially affected by the increase in temperature?
11. If the Fermi energy of a metal is 2.5 eV at 1000°C, at what energy is the probability of occupancy 80%?
12. For the example above, at what energy is the probability of occupancy 20%?
13. If the material is heated to 2000°C, at what energy is the probability of occupancy 80%?
14. At 2000°C, what is the probability of occupancy at the Fermi energy?
15. How is probability of occupancy different from actual occupancy?
16. Is there any relation between the Fermi–Dirac distribution and Maxwell–Boltzmann distribution?
17. Indicate examples of particles that follow the Fermi–Dirac distribution.
18. Show that filling states with identical and indistinguishable particles results in different number of microstates compared to filling states with identical and distinguishable particles.
19. When the temperature of a collection of fermions is increased, what happens to the energy of the lowest energy fermions?
20. When are the lowest energy fermions affected by changes in temperature?

Answers

Multiple Choice Questions

1. (c)	**6.** (c)	**11.** (d)	**16.** (b)
2. (d)	**7.** (a)	**12.** (b)	**17.** (a)
3. (d)	**8.** (a)	**13.** (a)	**18.** (a)
4. (c)	**9.** (d)	**14.** (a)	**19.** (c)
5. (b)	**10.** (b)	**15.** (b)	**20.** (c)

Short Answers Questions

1. Conditional probability is the probability that a state at a specific energy level is occupied and is subject to the system being in a specific microstate. Total probability takes into account all possible microstates of the system.
2. The value of $p(\bar{N})$ is the parameter that decides the identification of the most probable state of the system and hence is instrumental in deciding the equilibrium state of the system and is therefore important.
3. $\bar{N}_3$ is the most probable microstate of the system.
4. $\bar{N}_1$ is the least probable microstate of the system.
5. The overall probability of occupancy is dominated by the most probable microstate $\bar{N}_3$ and is $48 \times 0.999 = 47.952\%$.
6. For systems with higher Fermi energy, even higher temperatures must be attained to see the same type of distribution as that obtained at a lower temperature with a system with lower Fermi energy.
7. The value of $f(\varepsilon_i)$ at $\varepsilon_i = E_f$ when $T = 0$ K is undefined.

8. At $T = 0$ K and $\varepsilon_i = E_f$, the value of $f(\varepsilon_i)$ varies from 1 to 0, as a result it is undefined as a single value under these conditions.

9. As the temperature of the system increases, more energy levels are involved in the energy absorption process. This is reflected in the change in shape of the Fermi–Dirac distribution as a function of increasing temperature. The distribution deviates from that at 0 K at even lower energies as the temperature of the system increases.

10. The lower energy states of a system following Fermi–Dirac statistics are initially unaffected by the increase in temperature since there are no vacant states of higher energy available in the vicinity of the energy they already possess.

11.
$$f(\varepsilon_i) = \frac{1}{1 + e^{[(\varepsilon_i - E_f)/k_b T]}}$$

Therefore,

$$0.80 = \frac{1}{1 + e^{[((\varepsilon_i - 2.5)\times 1.6\times 10^{-19})/1.38\times 10^{-23}\times 1273]}}$$

Simplifying, we obtain $\varepsilon_i = 2.34$ eV which is the energy at which the probability of occupancy is 80%.

12.
$$f(\varepsilon_i) = \frac{1}{1 + e^{[(\varepsilon_i - E_f)/k_b T]}}$$

Therefore,

$$0.20 = \frac{1}{1 + e^{[((\varepsilon_i - 2.5)\times 1.6\times 10^{-19})/1.38\times 10^{-23}\times 1273]}}$$

Simplifying, we obtain $\varepsilon_i = 2.65$ eV is the energy at which the probability of occupancy is 20%.

13.
$$f(\varepsilon_i) = \frac{1}{1 + e^{[(\varepsilon_i - E_f)/k_b T]}}$$

Therefore,

$$0.80 = \frac{1}{1 + e^{[((\varepsilon_i - 2.5)\times 1.6\times 10^{-19})/1.38\times 10^{-23}\times 2273]}}$$

Simplifying, we obtain $\varepsilon_i = 2.23$ eV is the energy at which the probability of occupancy is 80%. Comparing with answer for question 11, we find that as the temperature of the system increases, we have to access even lower energy levels, compared to the Fermi energy, for the same probability of occupancy. This is consistent with Fig. 9.3.

14.
$$f(\varepsilon_i) = \frac{1}{1 + e^{[(\varepsilon_i - E_f)/k_b T]}}$$

At $\varepsilon_i = E_f$ the exponent term reduces to 0 and since $e_0 = 1$, $f(\varepsilon_i) = 0.5$ regardless of the temperature (except at 0 K when it is undefined).

15. The actual occupancy requires information on the number of states available, however the probability of occupancy is independent of such a restriction. Actual occupancy is the product of the probability of occupancy and the number of states available.

16. At high temperatures most particles behave closer to classical particles, therefore at high temperatures the Fermi–Dirac distribution reduces to the Maxwell–Boltzmann distribution.

17. Electrons and neutrinos follow the Fermi–Dirac distribution.

18. If there are two particles designated as '1' and '2' that are identical and distinguishable, and there are two states each of which can hold only one particle, then the possible microstates are as follows:

	***State* 1**	***State* 2**
Microstate-1	1	2
Microstate-2	2	1

If the particles are indentical and indistinguishable, both particles can be designatd as '1'. Then the only possible microstate is as follows:

	***State* 1**	***State* 2**
Microstate-1	1	1

Therefore, the number of microstates changes significantly depending on the nature of the particles and drastically changes the statistics corresponding to them.

19. When the temperature of a collection of fermions is increased, the energy of the lowest energy fermions is largely unaffected.

20. The energy of the lowest energy fermions is affected by changes in the temperature when the temperature is very high, in particular at temperatures close to the Fermi temperature.

10 Anisotropy, Periodic Potential, Confinement, and Quantization

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Identify the specific assumption of the Drude–Sommerfeld model that is not realistic.
- Distinguish between properties of a material and properties of a sample.
- Plot the energy of an electron as a function of its distance from an ionic core.
- Plot the potential experienced by an electron as a function of position with respect to a one-dimensional lattice of uniformly spaced ionic cores.
- Describe the concept of a "potential well."
- Plot an approximation of the potential experienced by an electron as a function of position with respect to a one-dimensional lattice of uniformly spaced ionic cores.
- Differentiate between "free electrons," "nearly free electrons," and "bound electrons."
- Identify the extent of space within which "bound electrons" reside and the extent of space within which "nearly free electrons" reside.
- Make a plot showing the relative values of the energy of free electrons, nearly free electrons and bound electrons and also indicate the Fermi energy and work function on the same plot.
- Indicate the wavelengths that can be supported by a string tied at one end only.
- Calculate the wavelengths that can be supported by a string tied at both ends.
- Identify the link between confinement and quantization.
- Solve the Schrödinger wave equation in one dimension to identify the impact of confinement on the allowed k vectors and hence allowed values of energy of an electron.
- State the impact of the extent of confinement on the spacing between adjacent allowed energy levels.

10.1 Introduction

In the progress made so far in trying to build a model for the properties displayed by materials, an underlying theme is that we wish to build the simplest model. This is the reason we examined the Drude model, which is very simple, but still makes good predictions. However, we have been able to identify its limitations as well. It is worth noting that in this approach, we typically start by assuming that the system follows specific rules and then build equations consistent with these rules. In this process, the equations are typically "correct" within the context of the rules assumed, and hence when the predictions do not turn out to be accurate, it is not the equations, but the rules themselves that need to be re-examined. In other words, the rules chosen for the system are themselves not appropriate for the system, and hence need to be modified. This is the context in which the Drude–Sommerfeld model differs from the classical Drude model. The Drude–Sommerfeld model relinquishes the assumption of classical behavior of the electrons and imposes

the assumption of quantum mechanical behavior of electrons, thereby changing the rules for the system. The new rules result in improved predictions.

In the further refinements that we will make as we proceed, the overall approach will be the same. Limitations of models that we will try to overcome will appear as changes or modifications in the rules followed by the system, and hence the resulting equations will differ.

10.2 Anisotropy in Crystalline Solids



We note that the Drude–Sommerfeld model is still a "free electron model." There is no variation in the potential experienced by electrons in the solid, as they traverse the extent of the solid. The justification for this assumption is that all the negative charges in the solid – from the free electron cloud – will be neutralized by all the positive charges in the solid – from the ionic cores – since the solid is charge neutral. Therefore, the free electron model is not unreasonable at first glance. However, it is also true that while the overall material is charge neutral, the positive and negative charges are not distributed uniformly across the extent of the solid when examined at an atomic level. The ionic cores occupy specific locations within the solid and their influence is more strongly prevalent in the vicinity of their locations. Therefore, when an electron moves through the solid, although the solid is charge neutral from an overall perspective, the potential experienced by the electron is a very strong function of its position – that is, a flat uniform potential is far from reality.

Having recognized that the flat potential model is too simplistic, it is also of interest to identify specific aspects of material properties that the Drude–Sommerfeld model may not be in a position to address. Perhaps the most prominent aspect of material properties that cannot be addressed by the Drude–Sommerfeld model in its present form is the anisotropy or dependence on direction in properties displayed by most materials. Let us briefly examine how anisotropy manifests itself in material properties, and why we may not always notice it, even though the material is inherently anisotropic.

Consider the samples shown in Fig. 10.1. In a single crystal, shown in Fig. 10.1(a), the atoms are in perfect crystallographic order from one end of the sample to the other. Therefore, each direction in the sample that is a single crystal represents a specific order of atoms and atomic planes. So a single crystal sample typically shows strong directionality or anisotropy in its properties. The sample properties represent inherent material properties.

In a polycrystal with large crystal sizes as shown in Fig. 10.1(b), there is an increased chance that the sample properties are averaged versions of the material properties across a few different directions, assuming that the crystals are randomly oriented. The same sample direction will likely be different crystallographic directions in each of the crystals. Therefore, the overall property displayed in a given direction by the sample will have contributions from different crystallographic directions from each of the crystals making up the sample.

In a polycrystal with very small crystal sizes as shown in Fig. 10.1(c), assuming that the crystals are randomly oriented, the sample properties will necessarily be averaged versions of the material properties across many different crystallographic directions. This averaging is a direct result of the fact that a given direction in the sample will have contributions from a very large number of crystallographic directions corresponding to the many randomly oriented crystals making up the sample.

So while Fig. 10.1(a) results in the display of "material properties" during measurements, Figs. 10.1(b) and (c) progressively result in the display of "sample properties" rather than the inherent "material properties" during measurements. While the sample properties do originate from the inherent material properties, averaging of the properties due to the polycrystalline nature of the samples masks details of the directionality of the inherent material properties. The sample may be isotropic while the material itself is anisotropic.

A classic example of a material that shows significant anisotropy is graphite. Figure 10.2 shows the layered structure of graphite.

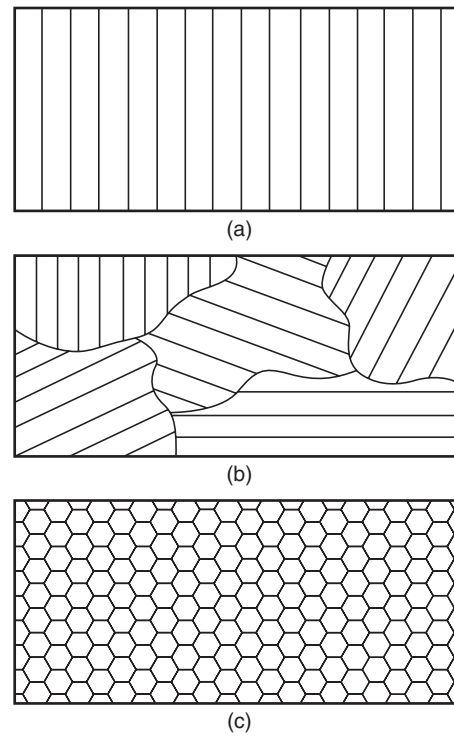


Figure 10.1 (a) A single crystal. (b) A polycrystal with large crystal sizes. (c) A polycrystal with very small crystal sizes. The lines and shading within the crystals are representatives of crystal planes. The direction of the lines is representative of the orientation of the crystal planes.

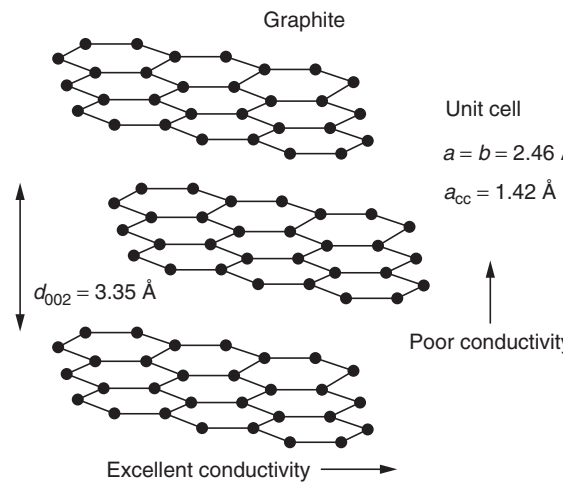


Figure 10.2 The layered structure of graphite. Graphite displays poor electronic conductivity perpendicular to the sheets of carbon atoms, and excellent electronic conductivity parallel to the sheets of carbon atoms.

Graphite has excellent electronic conductivity parallel to the sheets of graphite and very poor conductivity perpendicular to the sheets.

There is nothing in the Drude–Sommerfeld model to suggest anisotropy. The model assumes a uniform field within the material and ignores the ionic arrangements. The model has therefore effectively enforced isotropy. The model assumes isotropy, therefore the results are isotropic.

Thus, there is a need to develop a better picture of the potential an electron experiences as it travels through the solid. In other words, we need to incorporate into the model the dependence of potential on the position of the electron.

10.3 Periodic Potential in a Crystalline Solid



Let us first consider a single ionic core and examine the potential the electron experiences as it moves from infinity toward the ionic core. The energy, or potential, due to the coulombic force of attraction between the negatively charged electron and positively charged ionic core is given by the expression:

$$E = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

where ε_0 is the permittivity of free space, e is the charge of the electron, and r is the distance between the electron and ionic core. The electron will experience the above potential as a function of position regardless of the direction from which it approaches the ionic core. For the sake of simplicity, a one-dimensional example is considered and the potential experienced by the electron as a function of position, as it approaches the ionic core from the positive x direction and the negative x direction is shown in Fig. 10.3.

In reality, if the electron gets extremely close to the ionic core, it will start interacting with the electron cloud around the ionic core and a repulsive force will result. For the purpose of the present discussion, this repulsive interaction is being ignored based on the idea that only "free" electrons are being examined and hence they can be expected not to get very close to any particular ionic core.

If we extend the above analysis to a one-dimensional lattice, with regularly spaced ionic cores, we can expect the interaction of the electron with each of the ionic cores to be identical to that described above for the single ionic core. Figure 10.4(a) shows the interaction of the electron with each individual ionic core, which neglects the influence of the remaining ionic cores on the electron.

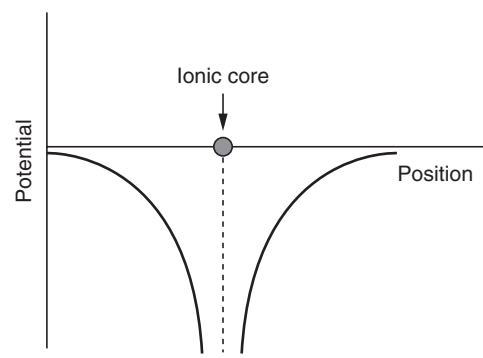


Figure 10.3 A one-dimensional example of potential experienced by an electron, as a function of position with respect to an ionic core.

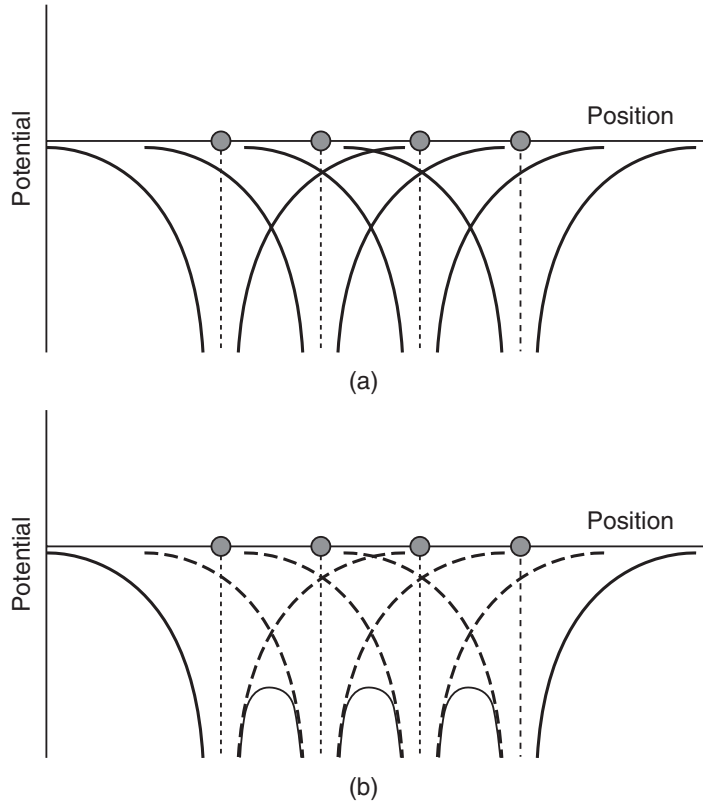


Figure 10.4 The potential experienced by an electron as a function of position with respect to a one-dimensional lattice of uniformly spaced ionic cores. (a) Interaction with each ionic core, ignoring the impact of the other ionic cores. (b) Interaction with the lattice as a whole.

In reality, since the ionic cores are relatively close to each other, as the electron moves from left to right in Fig. 10.4, it will begin to experience the presence of each succeeding ionic core before it has fully escaped from the influence of the preceding ionic core. The interaction of the electron with the one-dimensional lattice as a whole is shown in Fig. 10.4(b).

The steep fall in potential close to each ionic core is called a "potential well." As shown in Fig. 10.4, the potential inside a solid is not constant or "flat." It is therefore necessary to build enough detail into the model to incorporate the potential experienced by an electron as a function of position in the one-dimensional lattice.

However, it turns out that while it is important to capture the reality of potential wells in the model, it is not necessary to immediately focus on the exact shape of the curve associated with the formation of the potential well. It will be adequate if we can make a reasonable approximation of the potential landscape experienced by an electron in the solid. The approximation we make should capture enough detail of the reality of the situation while minimizing the associated complexity.

Figure 10.5 represents an approximation of the potential landscape experienced by an electron in a one-dimensional solid.

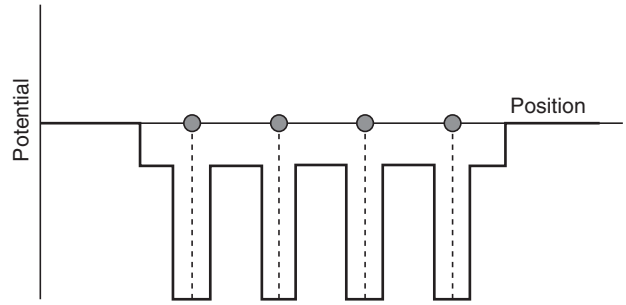


Figure 10.5 An approximation of the potential landscape experienced by an electron moving along the *x*-direction due to the presence of a one-dimensional solid, which is also aligned along the *x*-direction.

The primary features of the above approximation are as follows: When the electron is very far from the one-dimensional solid, it experiences zero potential due to the lack of interaction with the solid. This is true on either side of this hypothetical one-dimensional solid. This situation represents the electron having completely escaped from the solid, such as in the case of a photo electron. Such an electron is a truly free electron.

As the electron gets closer to the solid, it experiences a slight drop in the potential due to its interaction with the overall solid. This situation represents the electron being confined to the solid, but not attached to any specific ionic core. Such electrons are "nearly free" electrons, and these electrons participate in the electronic conductivity process since they are free to roam through the solid. The "free electron gas" of the Drude model also referred to these nearly free electrons only. When the electron gets very close to an ionic core, it experiences a very sharp drop in potential.

The electrons trapped in this deep potential well represent "bound electrons." These electrons are confined to each individual ionic core. In a typical metallic solid, these electrons represent the majority of the electrons, but they do not contribute to the electronic conductivity of the solid since they are unable to move through the solid.

In the approximation above, the curves corresponding to the actual potential experienced by the electron have been approximated to straight lines. This approximation results in the potential wells becoming "square potential wells" (in view of their rectangular shape) and makes subsequent calculations simpler. It will be necessary to judge at a later stage if this approximation is reasonable.

It is of interest to examine more features of the above picture and see how it relates to parameters discussed so far, such as the Fermi energy and work function of the solid.

We have noted that when an electron travels through a solid, it is not realistic to assume that there is no interaction with the solid, or that the interaction is featureless. We have also identified an approximation of the potential landscape that the electron experiences in the solid.

Just to get an idea of the numbers involved, let us assume that we have solid that is entirely made up of atoms of an element having an atomic number 50. Further, let us assume that each of the atoms in the solid releases one electron to the free electron gas. In terms of the approximate potential landscape we have just identified, this means that 49 of the 50 electrons associated with each ion are confined to the nearly 1 Å space adjacent to the location of the ionic core, or are "bound" to the location of the ionic core. The remaining one electron per ion is free to go around the solid but is confined to the physical extent of the solid. The extent of the solid can be of the order of several meters. Therefore, while all of the electrons within the

solid are confined, the bound electrons are confined to a very small region, which is 10 orders of magnitude smaller than the region over which the nearly free electrons are confined.

In the calculations so far, we have neglected the 49 electrons that are bound to the ionic cores and only focused our attention on the one electron per ionic core that has been released to be part of the free electron gas. We have adopted this approach because the properties we are examining are affected only by the nearly free electrons and not by the bound electrons. As indicated in our discussion regarding the Drude–Sommerfeld model, when the electrons fill up the states available to the nearly free electrons starting from the lowest available energy level, the highest energy level that gets filled at 0 K is called the Fermi energy level. In terms of the energy levels shown in Fig. 10.6, when an electron is at infinity, it has zero potential energy due to the solid. As it gets closer to the solid, the energy becomes increasingly negative in a relative sense. The nearly free electrons exist in the energy window between zero potential and a somewhat negative potential. In this potential window, the nearly free electrons fill the lowest energy level available first and then progressively fill higher and higher energy levels till the system runs out of free electrons. The highest energy level occupied within the levels available to the nearly free electrons is then the Fermi energy level.

When external energy is supplied to the sample in the form of light and a photo electron is ejected from the sample, the least amount of energy required will be to pull the highest energy electron available in the sample and take it to infinity with respect to the sample. The highest energy electrons available in the metal are the electrons at the Fermi energy. The energy that will therefore be required to remove this electron will be the difference between the Fermi energy and zero potential corresponding to the escaped electron. This energy required is defined as the work function which is also indicated in Fig. 10.6.

Based on the above discussion, we now have an understanding of the confinement experienced by the electrons in a solid and the relationships between quantities such as the Fermi energy and work function of a solid. Let us now use Fig. 10.6 to understand the behavior of electrons in a solid.

In general, here are two types of electrons:

1. Truly free electrons.
2. Confined electrons that can either be the nearly free electrons or the bound electrons.

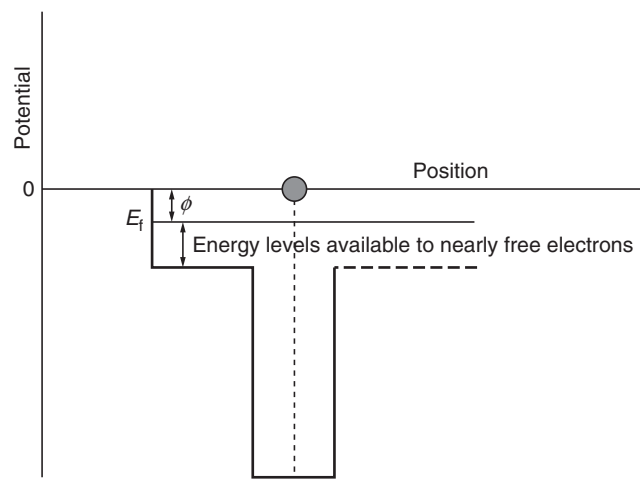


Figure 10.6 The energy levels available to the nearly free electrons and the Fermi energy level E_f, as well as work function ϕ, in a metallic sample.

To understand how electrons behave, we have to employ quantum mechanical tools to the problem. However, we will initially use a much simpler approach that will give us an idea of the key concepts involved and enable us to understand the quantum mechanical approach and its results more easily. To initiate the simpler approach, let us look at how the quantum mechanical approach will proceed and then make the simplification at the appropriate stage.

Quantum mechanical description acknowledges that there exists a wave particle duality. For the moment, let us look at electrons as waves. Based on the de Broglie equation, associated with a particle having a momentum p, there exists a wavelength λ such that

$$p\lambda = h$$

where h is Planck's constant. Rearranging, we have

$$p = \frac{h}{\lambda}$$

Defining $\hbar = h/2\pi$ and a vector called the wave vector $k = 2\pi/\lambda$, we can write the equation for the momentum as follows:

$$p = \frac{h}{2\pi} \times \frac{2\pi}{\lambda} = \hbar k$$

Since energy and momentum are related as

$$E = \frac{p^2}{2m}$$

We have

$$E = \frac{\hbar^2 k^2}{2m}$$

Therefore, the energy, momentum, and wave vector corresponding to a particle are related. Any restriction on one of these quantities will automatically restrict the other two quantities as well.

We have previously assumed that electrons in a solid have specific allowed or discrete energy levels ε_1, ε_2, ε_3, …. Now it is time to examine why the energy levels of electrons in a solid cannot be continuous and what are these discrete energy values. To understand what energy values and corresponding wavelengths are allowed for electrons in a solid, let us begin by examining a simpler analogy.

10.4 Confinement and Quantization – Waves on a String Analogy

Consider a string that is tied to a wall on one end and with the other end free as shown in Fig. 10.7. The question we wish to ask is what are the waves that can be supported by this string?

Since one end is free, any and all wavelengths can be supported by the string, or in other words, the string can adopt a shape consistent with any λ. Now consider a string that is tied at both ends as shown in Fig. 10.8.

The significance of the fact that both ends of the string are tied is that the string can only support zero displacement at those two ends. In other words, the waves supported by the string must have nodes at those two end points or the waves must be such that the distance between the ends of the string must correspond to integral multiples of half wavelengths.

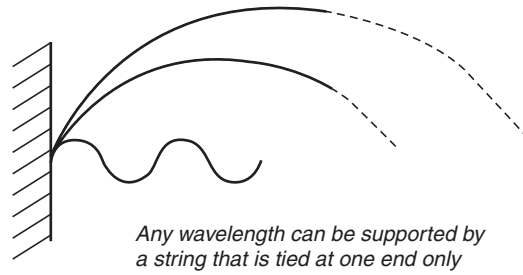


Figure 10.7 Some examples of waves that can be supported by a string tied at only one end.
Animation: "Waves on a string tied on one end only" in DVD.

If the distance between the walls where the string is tied is a, then the wavelengths supported by the string can only be of the kind that satisfies one of the following relationships:

$$a = \frac{\lambda}{2}$$

or

$$a = \frac{2\lambda}{2}$$

or

$$a = \frac{3\lambda}{2}$$

and so on. The general form being $a = n\lambda/2$, where n is an integer. These are the only values of wavelength that are allowed for a string tied at two ends.

Therefore, for a string that is free or tied at only one end, there are no restrictions on the wavelengths that can be supported. Hence, values of energy corresponding to these wavelengths can also be supported by the string. The continuous values of allowed wavelengths permit continuous values of energy to be supported by the string as well. For a string tied at both ends, not all values of wavelength can be supported due to the constraint that the ends of the string must be nodes. Therefore, only specific values of energy corresponding to these specific values of wavelength can be supported by the string. For every discrete value

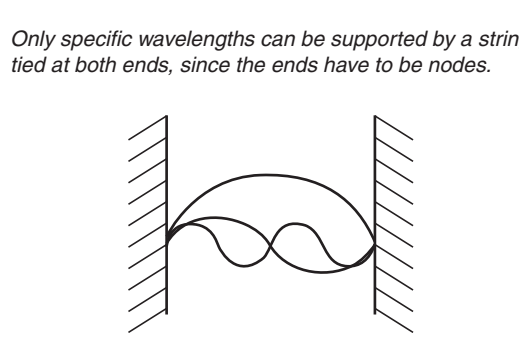


Figure 10.8 Some examples of waves that can be supported by a string tied at both ends.
Animation: "Waves on a string tied at both ends" in DVD.

of λ permitted (or k permitted), there is a corresponding discrete value of energy that is permitted. When $a = \lambda/2$, a corresponding energy value is permitted to be displayed by the string. The next nearest value of energy that can be displayed by the string will correspond to the situation where $a = 2\lambda/2$. Between these two energy values, no other energy value can be displayed by the string. By confining the string at both ends, we have prevented it from supporting a continuous set of energy values. This discretization of energy values is called "quantization." In other words, "confinement" of the string (tying both ends) has directly resulted in "quantization" (discretization) of allowed energy levels.

10.5 Confinement and Quantization – Quantum Mechanical Approach



We will now look at the quantum mechanical approach to handle this problem of confinement and examine if the results obtained by using the analogy of waves on a string is consistent with the results obtained from quantum mechanics.

So far, we have:

1. Examined the potential experienced by an electron as a function of position in a solid.
2. Developed a reasonable approximation of the same, such that curves become square shaped potential wells.
3. Recognized the idea of confinement of an electron. In particular, noted that some electrons are confined to the extent of the solid which could be of the order of a few meters, and are called the nearly free electrons. Some electrons are confined to the vicinity of individual ionic cores and are called bound electrons. Only if the electron has escaped the solid it becomes a truly free electron.
4. Examined the analogy of waves on a string to understand the effect of confinement on waves.
5. Noted that confinement leads to quantization of energy.

When an electron is trapped in a potential well, its wave-like behavior interacts with the confinement it faces and responds by adopting only specific wavelengths consistent with the extent of confinement. These wavelengths correspond to the fixed values of energy resulting in the quantization of energy of the confined electron.

Confinement within an extent a places the restriction on wavelengths that electron can adopt such that

$$\frac{n\lambda}{2} = a$$

where n has to be an integer. This implies that the allowed values of k are such that

$$k = \frac{2\pi}{\lambda} = \frac{n\pi}{a}$$

Since allowed energy values are related to allowed k values as follows

$$E = \frac{\hbar^2 k^2}{2m}$$

the only values of energy permitted to a confined electron are

$$E = \frac{\hbar^2 n^2 \pi^2}{2ma^2}$$

We arrived at the above result using the analogy of waves on a string. Let us now examine the problem of a confined electron using quantum mechanical principles. To use quantum mechanics, we have to obtain the wave function corresponding to the system. To obtain the wave function, we need to solve the Schrödinger wave equation with the conditions placed on the system. The wave function obtained will contain the properties of the system. The Schrödinger wave equation in one dimension is as follows:

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+V(x)\right]\psi(x)=E\psi(x)$$

A free electron does not experience any potential; therefore, $V(x) = 0$. Hence, for a free electron,

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\right)\psi(x)=E\psi(x)$$

Rearranging, we have

$$\frac{d^2\psi(x)}{dx^2}=\frac{-2mE}{\hbar^2}\psi(x)$$

In other words, a function of x that is differentiated twice results in a constant times the original function. Simply by observation, we can conclude that trigonometric functions of the type

$$\psi(x)=A\cos(kx)+B\sin(kx)$$

will solve the differential equation above, where we can identify k with the wave vector we have discussed earlier – it has the correct dimensions since kx should be dimensionless. Substituting for $\psi(x)$ in the left-hand side of the equation above, we have

$$\frac{d^2\psi(x)}{dx^2}=-k^2[A\cos(kx)+B\sin(kx)]$$

Therefore,

$$k^2=\frac{2mE}{\hbar^2}\Rightarrow E=\frac{\hbar^2k^2}{2m}$$

The k being a one-dimensional wave vector can adopt positive as well as negative values. Since the above analysis does not place any restrictions on the value of k, there are no restrictions on the values that E can adopt. This result is consistent with the results obtained through the analogy of waves on a string that is tied on one end only.

Since E is proportional to k^2 and the other quantities in the equation are constants, the plot of E versus k results in a parabola as shown in Fig. 10.9.

Let us now consider the case of an electron trapped in a potential well. Figure 10.10 shows an electron trapped in a potential well.

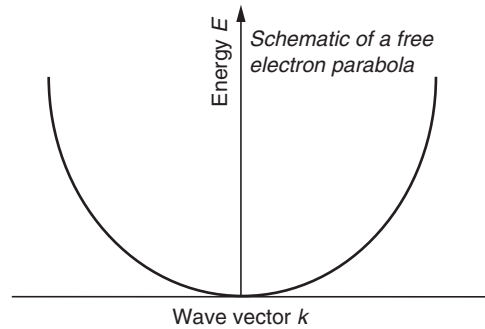


Figure 10.9 Plot of E versus k for a free electron. All values of k, and hence all values of E, are permitted. The parabola is continuous and E can adopt any value along the y-axis.

Therefore, within the well $V(x) = 0$, the form of the solution is the same as before:

$$\psi(x) = A\cos(kx) + B\sin(kx)$$

While this looks similar to what we obtained earlier for a free electron, it differs in one important detail. Due to the infinite potentials at the walls, the probability of finding the electron outside the walls is zero. This means the wave function is zero outside the walls. To avoid a discontinuity at the walls, the wave function inside the potential well must also drop to zero at the walls. Therefore, $\psi(0) = 0$ and $\psi(a) = 0$.

Since $\psi(0) = A$ therefore, $A = 0$.

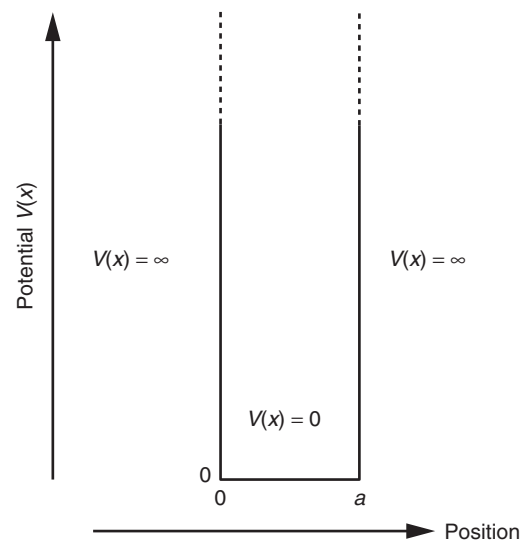


Figure 10.10 A potential well in which an electron is trapped. The potential is zero between the walls of the potential well and becomes infinity at and beyond the walls.

This leads us to

$$\psi(a) = B\sin(ka) = 0$$

This is possible when

$$ka = n\pi$$

Therefore, for an electron trapped in a potential well, the only values of wave vector k that are permitted are

$$k = \frac{n\pi}{a}$$

a result that is identical to the one we obtained when we examined a string tied at both ends. Since k can only have specific values, only corresponding values of energy are permitted. In other words, energy of the electron is now quantized. Since energy is related to the k vector through the same relationship as before,

$$E = \frac{\hbar^2 k^2}{2m}$$

the plot of the E versus k is still a parabola. It is just that not all points on the parabola are permitted. Only specific values of the k vector are permitted on the x-axis, and hence only the corresponding values of E are permitted on the y-axis. The allowed values are shown in the schematic in Fig. 10.11.

There is one additional point to note, which is regarding the level of confinement faced by the different electrons in the solid. In the equation for the wave vector:

$$k = \frac{n\pi}{a}$$

Here a represents the extent of confinement. The extent of confinement of nearly free electrons and bound electrons is shown in Fig. 10.12.

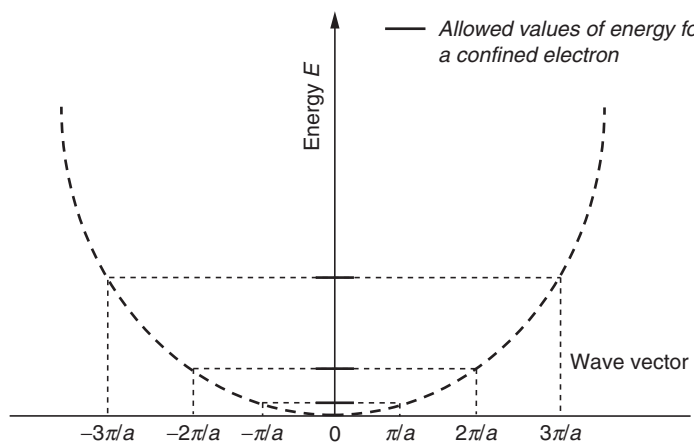


Figure 10.11 Allowed values of k vector and the corresponding values of E for an electron trapped in a potential well.

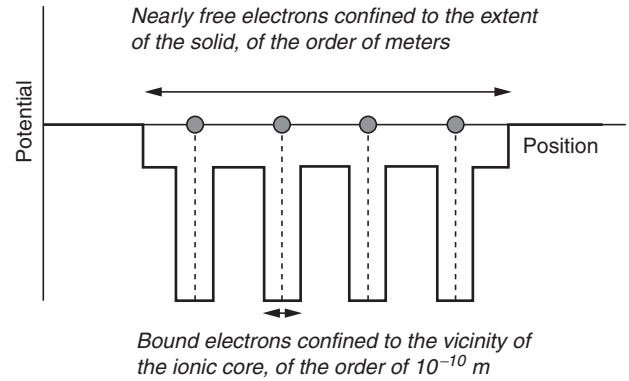


Figure 10.12 The difference in the extent of confinement of nearly free electrons and bound electrons.

For nearly free electrons, the extent of confinement is the extent of the solid and is in the order of meters. Therefore, the value of a is relatively large in this case and since it is in the denominator, it causes the adjacent values of k to be closely spaced. This in turn causes the allowed values of energy to also be relatively closely spaced as well.

For bound electrons, the extent of confinement is the length scale of an ionic core and is in the order of 1 Å or 10^{-10} m. Therefore, the value of a is extremely small in this case and since it is in the denominator, it causes the adjacent values of k to be spaced extremely widely apart. This in turn causes the allowed values of energy to also be relatively widely spaced as well.

Summary

1. While the overall solid is charge neutral, the positive and negative charges are not distributed uniformly across the extent of the solid when examined at an atomic level. Therefore, the potential experienced by the electron, as it moves through the solid, is a very strong function of its position.
2. While properties of single crystalline solids are inherently anisotropic, polycrystalline samples tend to show averaged properties masking the material properties. Graphite is an example of a highly anisotropic material.
3. The Drude–Sommerfeld model assumes a uniform field within the material and ignores the ionic arrangements. The model assumes isotropy, therefore the results are isotropic.
4. The energy due to the coulombic force of attraction between a negatively charged electron and the positively charged ionic core is given by the expression:

$$E = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

5. The potential landscape inside a solid can be approximated using square potential wells. This captures significant aspects of the real picture while minimizing the complexities of associated calculations.
6. Electrons far from the solid, which have escaped from the solid, are "free electrons." Electrons that are held within the confines of the solid but are not associated with any specific ionic core and can roam the extent of the solid, are "nearly free electrons." Electrons bound to each ionic core are "bound electrons."

7. Bound electrons associated with each ion, are confined to the nearly 1 Å space adjacent to the location of the ionic core, or are "bound" to the location of the ionic core. These form the majority of the electrons in the solid. The remaining one electron per ion (if the ion is univalent) is free to move around the solid but is confined to the physical extent of the solid, which can be of the order of several meters.
8. The highest energy level occupied within the levels available to the nearly free electrons is the Fermi energy level.
9. The energy required to remove an electron from the Fermi energy level and move it to the zero potential corresponding to the escaped electron is defined as the work function.
10. Wave vector k is defined as: $k = 2\pi/\lambda$.
11. The energy, momentum and wave vector corresponding to a particle are related by the equation:

$$E = \frac{\hbar^2 k^2}{2m}$$

12. A string tied at only one end can adopt a shape consistent with any λ and hence support all values of energy.
13. A string tied at two ends separated by a distance a can only adopt shapes consistent with the equation $a = n\lambda/2$, where n is an integer since the ends are nodes. Therefore, only specific wavelengths and energies corresponding to these wavelengths can be demonstrated by such a confined string.
14. "Confinement" of the string (tying both ends) therefore results in "quantization" (discretization) of allowed energy levels.
15. For free electrons:

$$k^2 = \frac{2mE}{\hbar^2} \Rightarrow E = \frac{\hbar^2 k^2}{2m}$$

where k being a one-dimensional wave vector can adopt positive as well as negative values. There are no restrictions on the value of k.
16. The plot of E versus k for a free electron is a continuous parabola.
17. For an electron confined to the extent a, plot of the E versus k is still a parabola. However, only specific values of k are given by

$$k = \frac{n\pi}{a}$$

and corresponding values of E are allowed.
18. For nearly free electrons, the extent of confinement a is large, therefore the allowed wavelengths and energy are closely spaced.
19. For bound electrons, the extent of confinement a is very small, therefore the allowed wavelengths and energy are very widely spaced.

Practice Questions

Multiple Choice Questions

1. State of matter that has higher tendency to be anisotropic is
   (a) Liquid **(b)** Solid
   (c) Gas **(d)** Plasma
2. For a poly crystalline sample, anisotropy is a strong function of
   (a) Sample size
   (b) Crystal size
   (c) Sample geometry
   (d) Sample orientation
3. Due to the presence of evenly spaced ionic cores, the potential within a single crystalline solid appears in the form of
   (a) A flat potential
   (b) A linear potential
   (c) A potential well
   (d) Zero potential
4. Electronic properties of a solid arise primarily through the contribution of
   (a) Nearly free electrons

(b) Free electrons that have escaped the solid
(c) Bound electrons
(d) Covalently bonded electrons

5. The potential experienced by an electron as it moves through a metallic solid is
(a) Uniform due to the equal number of positive and negative charges
(b) Non uniform due to the presence of the bound electrons
(c) Non uniform due to the presence of the ionic cores
(d) Non uniform due to the positions of the valence electrons

6. When the sample size increases
(a) The space available for bound electrons alone increases
(b) The space available for bound a well as nearly free electrons increases
(c) The space available for nearly free electrons alone increases
(d) The space for photo electrons increases

7. Larger the work function of a metallic solid
(a) Higher (closer to zero potential) is the value of the Fermi energy
(b) It has no impact on the Fermi energy
(c) Lower (further from zero potential) is the value of the Fermi energy
(d) Fermi energy has the same value as the work function of the solid

8. A string tied at only one end has
(a) Nodes at both ends
(b) A node at the tied end, always
(c) A node at the open end, always
(d) No nodes

9. A string tied at both ends has
(a) Nodes in the middle, always
(b) Nodes exactly in the midpoint, sometimes
(c) Nodes exactly in the midpoint, always
(d) Nodes in the end point, only sometimes

10. Energy values permitted for a string tied at both ends are
(a) Continuous
(b) Quantized and equally separated from each other
(c) Quantized but not equally spaced from each other
(d) Undefined

11. Free electrons, that have escaped the solid can
(a) Adopt any value of energy
(b) Can only adopt specific values of energies that are closely spaced
(c) Can only adopt specific values of energies that are widely spaced
(d) Possess no energy

12. Nearly free electrons, that are confined to the solid, but not confined to individual ionic cores can
(a) Adopt any value of energy
(b) Can only adopt specific values of energies that are closely spaced
(c) Can only adopt specific values of energies that are widely spaced
(d) Possess no energy

13. At the boundaries of a finite potential well, the wave function of the electron
(a) Drops to zero
(b) Is a finite non zero value
(c) Is infinite
(d) Is undefined

14. Outside the boundaries of an infinite potential well, the wave function of the electron
(a) Is zero
(b) Is a finite non zero value
(c) Is infinite
(d) Is undefined

15. Outside the boundaries of an infinite potential well, the probability of finding the electron is
(a) 0 **(b)** 0.2
(c) 0.5 **(d)** 1

Short Answer Type Questions

1. For a given metallic material, under what conditions of usage is having a large grain size beneficial?

2. For the same metallic material, under what conditions is having a small grain size beneficial?

3. What unusual effects can extremely small grain size cause?
4. What is the energy of interaction between an electron and univalent positively charged ion that are 2 Å apart?
5. Indicate a commonly available solid that is isotropic in its properties.
6. List the four longest allowed wavelengths of a string tied between two points of length 1 m.
7. Give examples of confinement of electrons.
8. Since confinement results in quantization, what energy values can be adopted by an electron confined to a dimension "*a*"?
9. What is the wavelength of an electron with velocity 2 m/s?
10. What is the energy corresponding to this electron?
11. What would happen to the *E* versus *k* curve for a confined electron if the extent of confinement were to decrease?
12. How can the discrete spectra of atoms be explained?
13. If the walls of the potential well did not have infinite potential, what would be its consequence on the electrons?
14. List the four lowest energies that an electron confined to 2 Å can possess.
15. In the expression for the allowed values of energy for a confined electron, what is the significance of *n*?

Answers

Multiple Choice Questions

1. (b)
2. (b)
3. (c)
4. (a)
5. (c)
6. (c)
7. (c)
8. (b)
9. (b)
10. (c)
11. (a)
12. (b)
13. (b)
14. (a)
15. (a)

Short Answer Type Questions

1. For typical high-temperature applications, having a large grain size is beneficial since it gives the material better creep properties.
2. Typically, room temperature strength of most materials is increased when the grain size decreases since the lower grain size significantly impedes the movement of dislocations, which manifests as the strength of the material. The equation that governs this strengthening of the material with decreasing grain size is called the Hall–Petch relation.
3. At extremely small grain sizes, materials can deform using other mechanisms such as grain boundary sliding. As a result, the material starts becoming weaker as the grain size decreases beyond a certain level. This behavior is referred to as the inverse Hall–Petch behavior.
4. The energy of interaction between an electron and a univalent positively charged ion is given by

$$E = \frac{-1}{4\pi\varepsilon_0}\frac{e^2}{R}$$

where R is the distance between the ions, ε_0 is the permittivity of free space, and e is the electronic charge. The negative sign is due to the opposing signs of the charges, which makes the interaction attractive in nature. Therefore,

$$E = \frac{-1}{4\pi\times 8.85\times 10^{-12}}\frac{(1.6\times 10^{-19})^2}{2\times 10^{-10}}$$

Simplifying we get

$$E = 1.15\times 10^{-18}\ \text{J}$$

5. Glasses are commonly available solids that are typically isotropic in their properties. This is because glasses are non-crystalline.

6. The four largest allowed wavelengths are 2, 1, 0.67, and 0.5 m.

7. Electrons are confined in atoms and can be confined in nanostructures such as nanoparticles and thin films.

8. The allowed energy values that can be adopted by an electron confined to a dimension "a" are given by

$$E = \frac{\hbar^2 n^2 \pi^2}{2ma^2}$$

9. The wavelength corresponding to this electron is

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{9.11 \times 10^{-31} \times 2} = 3.64 \times 10^{-4}\ \text{m}$$

10. Energy is simply the kinetic energy of the electron,

$$E = \frac{1}{2}mv^2 = \frac{1}{2} \times 9.11 \times 10^{-31} \times 2^2$$
$$= 18.22 \times 10^{-31}\ \text{J}$$

11. The shape of the curve will be the same, however the allowed values from the curve will be more widely spaced.

12. The electrons in an atom are localized to the vicinity of the atom. Based on quantum mechanics, this quantizes their energy. Therefore, the spectra associated with the atoms have only specific values of frequencies associated with them, corresponding to these quantized values of energy.

13. If the walls of the potential well were not infinite, there will be a non-zero probability of finding the electron outside the potential well.

14. Since

$$E = \frac{\hbar^2 n^2 \pi^2}{2ma^2}$$

or

$$E = \frac{h^2 n^2}{8ma^2}$$

Therefore, the lowest allowed values of E are given by

$$E = \frac{(6.626 \times 10^{-34})^2 n^2}{8 \times 9.11 \times 10^{-31} \times (2 \times 10^{-10})^2}$$
$$= 1.5 \times 10^{-18} \times n^2$$

or allowed values of energy are 1.5×10^{-18} J, 6×10^{-18} J, 13.5×10^{-18} J and 24×10^{-18} J for values of $n = 1$, 2, 3 and 4.

15. Since n can only take integer values, n represents a quantum number.

11 Density of States, Fermi Energy, and The Electronic Contribution to Specific Heat at Constant Volume

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Make a plot of the relative spacing of the allowed energy levels of bound and nearly free electrons.
- Define the density of allowed states $D(E)$.
- Define the density of occupied states $N(E)$.
- Derive an expression for $D(E)$ as a function of the energy E.
- Make a plot of $D(E)$ as a function of the energy E.
- Make a plot of $N(E)$ as a function of the energy E.
- Indicate the relationship between Fermi energy and chemical potential of electrons.
- Define and plot the Fermi surface.
- Indicate the significance of the Fermi surface.
- Calculate the Fermi temperature.
- Indicate the significance of the Fermi temperature.
- Derive an expression C_V^e based on the Drude–Sommerfeld model.
- Indicate the manner in which the expression obtained using the Drude–Sommerfeld model corrects the error in the expression obtained using the classical Drude model.

11.1 Introduction

In this chapter, to get a more complete picture of the state of electrons in a solid, we will determine how many states are available at various energy levels, or more specifically, the density of states. This information is analogous to the shape of the vessel that was referred to in Chapter 9. With the filling up of energy levels with electrons, the highest energy level that gets occupied is the Fermi energy. Since electrons at the Fermi energy are in a position to interact with the surroundings of the material, we will examine the significance of these electrons. Finally, we will look at the impact of the Drude–Sommerfeld model on the determination of C_V^e.



11.2 Density of States

The solution to the Schrödinger wave equation showed us that confinement leads to quantization. The smaller the region within which the electron is confined, the more widely spaced are the allowed values of k as well as the corresponding values of E. The larger the region within which the electron is confined, the more closely spaced are the allowed values of k as well as the corresponding values of E. Figure 11.1 shows an illustrative schematic of the allowed values of energy for a bound electron and nearly free electrons.

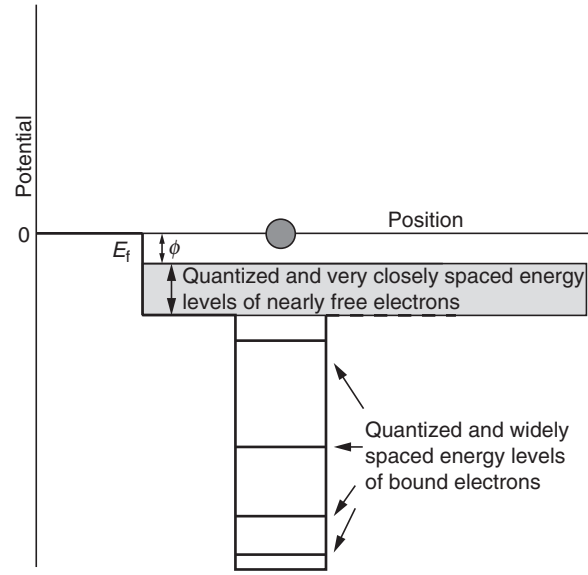


Figure 11.1 Allowed values of energy for nearly free electrons and bound electrons.

The gap between adjacent allowed energy levels for the nearly free electrons is of the order 10^{-10} eV, while between the allowed energy levels for a bound electron is of the order several electron volts to several tens of electron volts. Therefore, the energy levels of nearly free electrons, drawn to the scale of the energy levels of bound electrons, appear continuous even though they are actually discrete. This appearance of continuity arises because there are 10^{10} nearly free electron energy levels occupying the same magnitude in the energy scale that is occupied by two adjacent bound electron energy levels.

Although, we will discuss semiconductors in detail later, it is of interest to note now that the bands are considered to be almost continuous sets of allowed energy values relative to the band gaps that are several electronvolts wide – the reason is same as indicated above.

While we have plotted the energy levels as a function of position in several of the plots so far, we note that

$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2 \pi^2 n^2}{2ma^2}$$

where k is the wave vector and n is the quantum number (since it quantizes the allowed values of energy). These relationships have been obtained in a one-dimensional sense. In three dimensions, we have

$$k^2 = k_x^2 + k_y^2 + k_z^2$$

and

$$n^2 = n_x^2 + n_y^2 + n_z^2$$

Therefore, we can make plots of the system using the x, y, and z components of the k vector or the components of the quantum number. In the language of physics, this is referred to as plotting in k space or plotting in quantum number space, respectively.

We can create plots describing the system using any of the variables that define the system, since the other variables are related to it. The shape of the plot may look different based on the variable chosen, but the information presented will be the same. Based on the information desired, one or the other set of axis, and hence the corresponding "space" will be the more convenient choice. Since

$$E = \frac{\hbar^2 \pi^2}{2ma^2}(n_x^2 + n_y^2 + n_z^2)$$

when energy E is constant,

$$n_x^2 + n_y^2 + n_z^2 = \text{constant}$$

Therefore, all points of constant energy lie on the surface of a sphere in n space.

Given that we have now incorporated the detail of confinement and associated quantization in the model for the material, it is possible to extract further understanding of the workings of materials.

While we made a plot of the Fermi–Dirac distribution, we noted that we did not know how many states were available at a given energy level. The parameter we are looking for $D(E)$ is called the density of allowed states, and is the number of states in the energy range E and $E + dE$.

If we define $N(E)$ as the density of occupied states,

$$N(E) = 2f(E)\ D(E)$$

The factor "2" accounts for the fact that we can have an electron with spin up or spin down in each state.

Stated in words, the above equation means that the density of occupied states is equal to the density of allowed states $D(E)$ multiplied by the probability of occupancy of the states $f(E)$.

It is of interest to obtain an expression for $D(E)$. To obtain this expression in the discussion that follows, we will approach the problem from two different perspectives and obtain two different expressions that will be related to each other. Through this relationship, we will get an expression for $D(E)$.

First, we note that all states of equal energy lie on the surface of a sphere in n space. Since n_x, n_y and n_z are quantum numbers, they can only be integers with positive values. This implies that only the positive octant of a sphere in n space contains the allowed states. Therefore, the total number of states up to an energy value E, which is given by

$$\frac{\text{The volume occupied by the states in } n \text{ space}}{\text{Volume occupied by a single state in } n \text{ space}}$$

becomes one-eighth of the volume of a sphere in n space, divided by 1^3, since the unit volume in n space is defined by a cube with unit dimensions, that is, $n_x = n_y = n_z = 1$.

Therefore, the total number of states up to an energy value E is given by

$$N_T(E) = \frac{(1/8) \times [(4/3)\pi n^3]}{1} = \frac{1}{6}\pi n^3$$

Since

$$E = \frac{\hbar^2 \pi^2 n^2}{2ma^2}$$

we have

$$n = \left(\frac{2ma^2 E}{\hbar^2 \pi^2} \right)^{1/2}$$

Therefore, the total number of states up to an energy value E is given by

$$N_{\mathrm{T}} = \frac{1}{6}\pi \left(\frac{2ma^2}{\hbar^2 \pi^2} \right)^{3/2} E^{3/2}$$

Second, if $D(E)$ is the density of allowed states or the number of states in the energy level E and $E + dE$, then another expression for the total number of states up to an energy value E is given as follows:

$$N_{\mathrm{T}}(E) = \int_0^E D(E)dE$$

or

$$D(E) = \frac{dN_{\mathrm{T}}(E)}{dE}$$

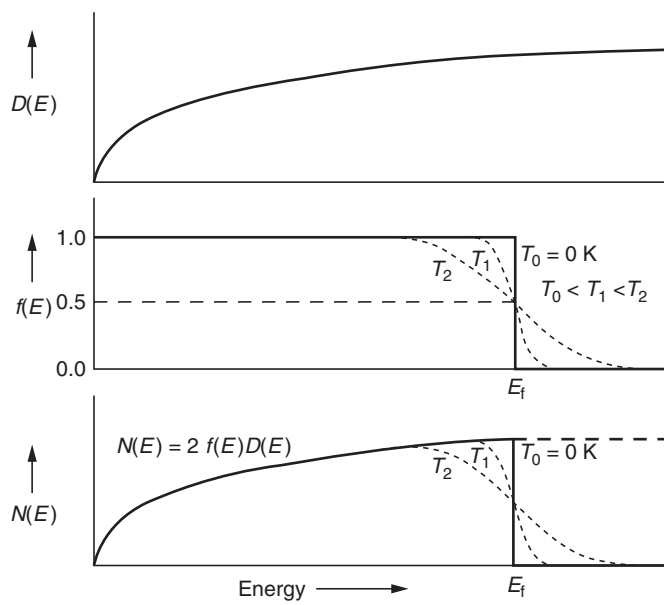


Figure 11.2 Plots of the density of allowed states $D(E)$, the probability of occupancy $f(E)$, and the density of occupied states $N(E)$ as a function of energy and at different temperatures.

Since we independently have arrived at an expression for $N_T(E)$ in our discussion earlier, we can differentiate the same and arrive at the expression for the density of allowed states $D(E)$. Therefore,

$$D(E) = \frac{d}{dE}\left[\frac{1}{6}\pi\left(\frac{2ma^2}{\hbar^2\pi^2}\right)^{3/2} E^{3/2}\right]$$

Simplifying,

$$D(E) = \frac{1}{4}\pi\left(\frac{2ma^2}{\hbar^2\pi^2}\right)^{3/2} E^{1/2}$$

Therefore, $D(E)$ is related to E through a parabolic relationship.

The plots of $D(E)$, $f(E)$, and hence $N(E) = 2\,f(E)D(E)$, as a function of energy are shown in Fig. 11.2 along with their variation with temperature.

Later in this chapter, we will consider the calculations associated with estimating C_V^e, which has only been addressed descriptively so far. Given our present knowledge of the density of occupied states, we can calculate C_V^e consistent with the Drude–Sommerfeld model. We will now look more closely at aspects associated with the Fermi energy.

11.3 Aspects Associated with Electrons Filling Energy Levels



We have noted that the density of occupied states $N(E)$ is the density of allowed states $D(E)$ multiplied by the probability of occupancy of the states $f(E)$. The density of occupied states, therefore, gives a more complete picture of the electrons in a solid.

We also noted that at 0 K when the nearly free electrons in the solid are used for filling up the available energy levels, in accordance with the rules applicable to fermions, the energy levels are filled up starting from the lowest to highest energy level. The energy level that is filled up by the last available nearly free electron is referred to as the Fermi energy E_f.

We found that the energy of the electron relates to its wave vector through the following relationship:

$$E = \frac{\hbar^2 k^2}{2m}$$

where k is the wave vector and is similar to k_x in a one-dimensional case. The wave vector k has the dimension of L^{-1}, since it is related to the inverse of the wavelength.

Since the system can be observed from the perspective of different variables, we can obtain plots of different topologies that convey similar or the same information. Constant energy will appear as a flat surface in some plots that have energy on one of the axes, but with other variables a spherical surface may correspond to constant energy.

We can make a plot using the components of the wave vector k. As indicated earlier, in a one-dimensional case $k = k_x$. In three dimensions, the magnitude of the wave vector will be such that $k^2 = k_x^2 + k_y^2 + k_z^2$.

Since

$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m}(k_x^2 + k_y^2 + k_z^2)$$

Therefore, similar to the quantum number space that we discussed earlier in this chapter, in k space, points corresponding to constant energy will lie on the surface of a sphere.

As we fill up the available states with the nearly free electrons, the electrons having the same energy will be represented by a sphere. Corresponding to electrons of higher energies, there will spheres of larger radii. Corresponding to the electrons with the highest energy in the system, which is the Fermi energy E_f at 0 K, the sphere that will be generated is called the Fermi surface.

Figure 11.3 shows a schematic of the Fermi surface and corresponding plot of density of occupied states. The Fermi energy shows up as a straight line in the density of states plot, since energy is on the x-axis. In the plot in k space, the outermost sphere corresponds to the Fermi energy.

Based on the system we choose, E_f can differ. The E_f can change even within the same system if there is a change of phase that affects the number of free electrons per unit volume.

We have therefore identified two concepts so far: The energy at which we run out of free electrons while filling up the energy levels at 0 K – which is the Fermi energy – and its corresponding surface in k space, the Fermi surface. At the moment these are the only definitions. We will see the significance and use of these concepts a little latter.

While examining the translational kinetic energy of classical particles, we found that

$$\frac{1}{2}mv^2 = \frac{3}{2}k_b T$$

While recognizing that we are not dealing with classical particles, we can still utilize the idea that if the energy of the particle is considered to be thermal energy, then it is related to the temperature T consistent with that thermal energy through the expression $k_b T$. Therefore, corresponding to the

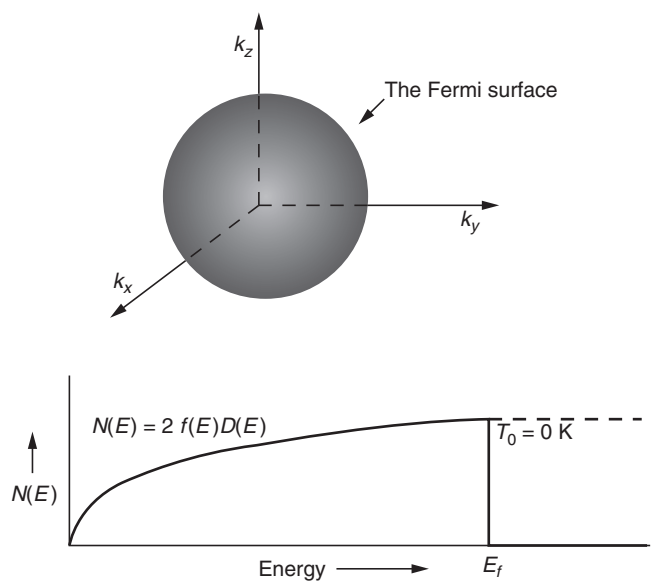


Figure 11.3 Plot of allowed states in k space and the corresponding density of states plot. Fermi energy is represented by a sphere in k space called the Fermi surface, and as a straight line in the density of states plot.

electrons at the Fermi energy E_f, a temperature can be defined, which is called the Fermi temperature T_f, such that

$$E_f = k_b T_f$$

We have so far identified three quantities:

1. Fermi energy E_f.
2. Fermi surface.
3. Fermi temperature T_f.

Let us now examine their significance.

11.4 Fermi Energy

To understand the significance of Fermi energy, let us compare it with temperature, pressure, and chemical potential. In general, we can think of higher temperature and pressure as measures of greater tendency for heat and mass, respectively, to leave a location. Therefore, for example, heat moves from a hot body to a cold body till the temperatures of the two bodies equalize.

Similarly, chemical potential is associated with each specific species in a phase and represents the tendency of the species to leave the phase, and go to phases where it is at a lower chemical potential. Electrons can be considered as one of the species that build up the material. When two materials come in contact with each other, if electrons in one of the materials have higher energy, they will tend to flow into the material where they have lower energy. Since electrons in a material fill up a range of energy levels, which energy should be used for the comparison? Since all the energy levels up to the Fermi energy are filled in the ground state of the material, the most appropriate energy used for the comparison between electrons in two materials is the Fermi energy itself. It is similar to the comparison of the highest level of water in two containers for determining the direction of the flow of water when the two containers are connected. Thus, the Fermi energy of electrons is analogous to the chemical potential μ of the electrons.

$$E_f \approx \mu$$

The equality is quite reasonable at low temperatures, but becomes less accurate at very high temperatures since E_f represents the highest occupied energy level for metals only at 0 K.

The identification of E_f with the chemical potential of electrons is very useful since it helps us predict what will happen when two dissimilar materials are brought in contact. In the electronics industry, junctions between different semiconductors are common and knowledge of the E_f of the two materials assists in determining the direction of natural flow of electrons due to the formation of the junction. Therefore, knowledge of E_f has significant practical applications.

11.5 Fermi Surface

Fermi surface is the surface in k space that corresponds to the Fermi energy. In quantum mechanics, we accept the wave particle duality and associate particles with waves consistent with their momentum. In k space, we are plotting wave vectors $k = 2\pi/\lambda$, where λ is the wavelength corresponding to electrons with specific energy. Therefore, the Fermi surface represents wave vectors of electrons having the Fermi energy E_f. To understand the significance of wave vectors, let us briefly look at the interaction of radiation with matter.

When radiation in the form of X-rays or an electron beam interacts with matter, we see diffraction effects. Figure 11.4 shows a schematic of a typical experiment to record diffraction patterns.

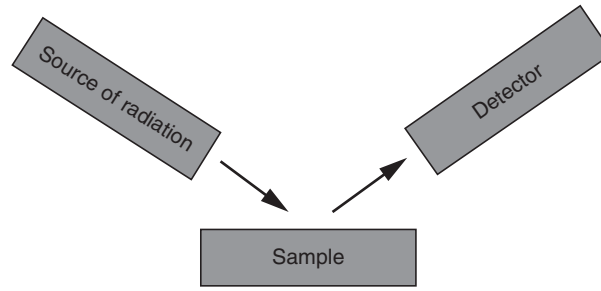


Figure 11.4 A schematic showing how radiation can be made to interact with matter to produce and record diffraction effects in a laboratory setting.

To enable recording diffraction effects, in general we need

1. A source of radiation.
2. Sample with a periodic structure.
3. Wavelength of the radiation to be in a range that enables diffraction effects to be recorded (which is of the order of the spacing of the periodic structure, or less).

However, the important aspect to note is that even though the schematic shows the source of the radiation to be independent of the sample, this is only an experimental convenience. Waves of electrons already within the sample can interact with the periodic structure of the sample and display diffraction effects. Unlike a separate electron beam source in an electron microscope, this is equivalent to an electron beam within the sample.

The Fermi surface represents the wave vectors of the electrons corresponding to the Fermi energy. *k*-space plots this information in inverse length dimensions. We can plot the periodic structure of the material also using inverse length notation on the same plot. This combined plot can help us in identifying diffraction conditions for the nearly free electrons within the material due to their interaction with the periodic structure of the same material. The plot of the periodic structure of the material in inverse length dimensions is referred to as a plot in "reciprocal space," which will be examined in greater detail in Chapter 12 and is a very important tool in understanding the behavior of electrons in solids. For the moment, we will simply summarize that the diffraction of nearly free electrons results in the formation of permitted energy levels and forbidden energy levels, or bands for the free electrons – a more detailed account of this interaction will also appear in Chapter 13.

11.6 Fermi Temperature

The Fermi temperature, T_f, is defined as the temperature corresponding to the Fermi energy such that

$$E_f = k_b T_f$$

With the Fermi energy of the order of a few electron volts, the Fermi temperature works out to be of the order of several thousand Kelvin (approximately 10000 K). How do we reconcile the fact that the material is at room temperature or lower, or even at 0 K, while the Fermi temperature is 10000 K? One way to look at the situation is as follows: When we measure the temperature of a material, we do not typically measure the

temperature of a single atom or electron. What we measure is the average temperature of the material. There is invariably going to be a distribution of energy within the material. In this distribution, an extremely small thermal mass consisting of a very small fraction of the nearly free electrons (which is itself a very small fraction of the total electrons in the system) is at the Fermi energy, and the temperature corresponding to that energy is the relatively high Fermi temperature. Therefore, the "high" Fermi temperature is not inconsistent with the "low" temperature or the solid as a whole.

Let us now use the Fermi temperature to understand how corrections can be made to the estimate of C_V^e.

11.7 Estimating the Electronic Contribution to Specific Heat at Constant Volume



We obtained the following expressions for the total number of allowed states $N_T(E)$ below an energy level E and for the density of available states $D(E)$:

$$N_T(E) = \frac{1}{6}\pi\left(\frac{2ma^2}{\hbar^2\pi^2}\right)^{3/2} E^{3/2}$$

$$D(E) = \frac{1}{4}\pi\left(\frac{2ma^2}{\hbar^2\pi^2}\right)^{3/2} E^{1/2}$$

Since a represents the extent of the system for nearly free electrons, a^3 represents the volume V of the system. Therefore, we can rewrite the above equations as

$$N_T(E) = \frac{V}{6}\pi\left(\frac{2m}{\hbar^2\pi^2}\right)^{3/2} E^{3/2}$$

and

$$D(E) = \frac{V}{4}\pi\left(\frac{2m}{\hbar^2\pi^2}\right)^{3/2} E^{1/2}$$

We have noted earlier that only electrons close to the Fermi energy can gain energy as we raise the temperature. At 0 K, $f(E) = 1$ for energy levels up to E_f. Therefore,

$$N_T(E_f) = \frac{V}{6}\pi\left(\frac{2m}{\hbar^2\pi^2}\right)^{3/2} E_f^{3/2}$$

Rearranging, we have

$$\frac{N_T(E_f)}{V} = \frac{1}{6}\pi\left(\frac{2m}{\hbar^2\pi^2}\right)^{3/2} E_f^{3/2}$$

The above equation shows the total number of allowed states per unit volume up to the Fermi energy. At 0 K, since $f(E) = 1$ for all $E < E_f$, the equation above also represents the total number of nearly free electrons, up to the Fermi energy, per unit volume. Therefore, it represents the number of free electrons per unit volume, which we had previously designated as N_f.

Similarly, the density of available states per unit volume, at the Fermi energy, is given by

$$\frac{D(E_f)}{V} = \frac{1}{4}\pi\left(\frac{2m}{\hbar^2\pi^2}\right)^{3/2} E_f^{1/2}$$

which we can designate as $D_V(E_f)$. Therefore, we find that

$$\frac{D_V(E_f)}{N_f} = \frac{3}{2}\frac{1}{E_f}$$

or

$$D_V(E_f) = \frac{3}{2}\frac{N_f}{E_f}$$

When we raise the temperature of the system from 0 K to a temperature T, on a per electron basis, the energy provided is $k_b T$. Therefore, only electrons within $k_b T$ of the Fermi energy can participate in gaining this energy and move to unoccupied states of higher energies.

On a per unit volume basis, the number of such electrons is given by

$$D_V(E_f) \times k_b T = \frac{3}{2}\frac{N_f}{E_f} k_b T$$

The energy provided to each of those electrons is

$$\frac{3}{2} k_b T$$

The energy possessed by electrons at temperature T that participate in the energy increase process = Number of electrons available × Energy per electron. Therefore,

$$E = \left(\frac{3}{2}\frac{N_f}{E_f} k_b T\right) \times \left(\frac{3}{2} k_b T\right)$$

or

$$E = \frac{9}{4}\frac{N_f}{E_f} k_b^2 T^2$$

Therefore,

$$\frac{dE}{dT} = C_V^e = \frac{9}{2}\frac{N_f}{E_f} k_b^2 T$$

Since

$$E_{\mathrm{f}} = k_{\mathrm{b}} T_{\mathrm{f}}$$

we have

$$C_V^e = \frac{9}{2}\frac{N_{\mathrm{f}} k_{\mathrm{b}}^2 T}{k_{\mathrm{b}} T_{\mathrm{f}}} = \frac{9}{2} N_{\mathrm{f}} k_{\mathrm{b}} \left(\frac{T}{T_{\mathrm{f}}}\right)$$

The above expression for C_V^e is therefore the prediction of the Drude–Sommerfeld model.

The classical Drude model predicted:

$$C_V^e = \frac{3}{2} N_{\mathrm{f}} k_{\mathrm{b}}$$

The two expressions primarily differ in the term of T/T_{f}. With the Fermi temperature T_{f}, of the order of 10000 K, and room temperature T of the order of a few 100 K,

$$\frac{T}{T_{\mathrm{f}}} \oplus 0.01$$

The Drude–Sommerfeld model, therefore, effectively corrects a major shortcoming of the classical Drude model, and hence represents a significant improvement in our efforts to build a model for the properties of solids.

11.8 Further Improving the Model of the Solid

While it is indeed an improvement, the Drude–Sommerfeld model is still only a free electron model. There are no features in the model to enable it explain anisotropy in material properties. The parameter N_{f} – the number of free electrons per unit volume – is the same regardless of direction, therefore, directional variation in properties cannot be explained using this parameter.

The aspect of the material that differentiates between various directions in the material is its crystal structure. We have not accounted for, or incorporated, the crystal structure of the solid in any way in the model for the material.

Thus, far we have treated the wavelike behavior of the nearly free electrons independently, and ignored any interactions between these nearly free electrons and the periodically arranged ionic cores. Even then we have made significant progress in modeling materials, making predictions of material properties and correcting predictions in earlier models. However, to explain anisotropy, we need to understand the interaction between the ionic cores and the wavelike behavior of the nearly free electrons. This is shown in the schematic in Fig. 11.5. In this regard, we will take a two-step process:

1. We will examine the interaction of waves in general with the periodic structure of the ionic cores, which results in scattering of the waves in general, and in specific cases results in the diffraction process.
2. We will then take into account the fact that the "nearly free" electrons in the material are also showing wavelike behavior, and since the diffraction process does not specify the origin of the waves, it will be possible to examine the interaction between the wave-like behavior of the "nearly free" electrons and the ionic cores using the same diffraction phenomenon.

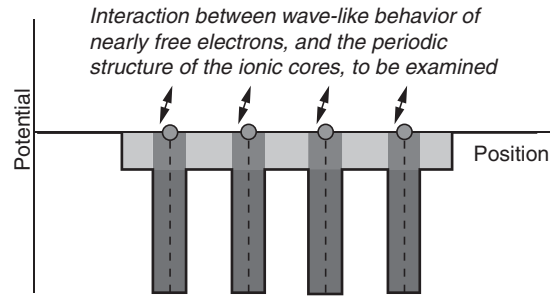


Figure 11.5 A schematic that highlights the interaction between the wavelike behavior of nearly free electrons and the periodic structure of the ionic cores that is necessary to address now to further improve our model for materials.

However, we note that the wave vector k is in reciprocal length units or is a vector in "reciprocal" space. We have typically studied crystal structures using "real" space, which contains distances in ordinary length dimensions and units. To study the interactions between waves and crystal structures, it helps if they are both in the same kind of space and have the same dimensions. In this context, reciprocal space is seen to capture nuances of the diffraction process much more elegantly and effectively than real space. Therefore, as a first step we have to understand what a reciprocal space is, how it is defined, how crystal structures are represented in reciprocal space, and how diffraction is described in reciprocal space. This will be the subject of our discussion in the next chapter. After we have familiarized ourselves with these concepts, we will return to our problem of understanding how waves of nearly free electrons interact with the periodic structure of the material and what is the consequence of the interaction on material properties. Presumably, this will result in the model predicting the presence of anisotropy in material properties.

But first, let us look at reciprocal space in Chapter 12.

Summary

1. Quantized energy levels of nearly free electrons are closely spaced with the gap between adjacent allowed energy levels of the order of 10^{-10} eV.
2. Quantized energy levels of bound electrons are widely spaced with the gap between allowed energy levels of the order of several electron volts to several tens of electron volts.
3. All points of constant energy lie on the surface of a sphere in n space or quantum number space.
4. $N(E)$, the density of occupied states = the density of allowed states $D(E)$ multiplied by the probability of occupancy of the states $f(E)$.
5. The density of allowed states $D(E)$ is given by the expression:

$$D(E) = \frac{1}{4}\pi\left(\frac{2ma^2}{\hbar^2\pi^2}\right)^{3/2} E^{1/2}$$

6. In k space, points corresponding to constant energy lie on the surface of a sphere. The outermost sphere corresponds to the Fermi energy.
7. Corresponding to the electrons at the Fermi energy E_f, a temperature can be defined, which is called the Fermi temperature T_f, such that

$$E_f = k_b T_f$$

With the Fermi energy of the order of a few electron volts, the Fermi temperature works out to be of the order of several thousand Kelvin (approximately 10000 K).

8. Fermi energy of electrons is analogous to the chemical potential μ of the electrons.

9. Fermi surface is the surface in k space that corresponds to the Fermi energy. Its interaction with the periodic structure of the material has implications on the band structure of the material.

10. The density of states per unit volume at the Fermi energy is given by

$$D_{\mathrm{V}}(E_{\mathrm{f}}) = \frac{3}{2}\frac{N_{\mathrm{f}}}{E_{\mathrm{f}}}$$

11. C_V^e as predicted by the Drude–Sommerfeld model is given by

$$C_V^e = \frac{9}{2} N_{\mathrm{f}} k_{\mathrm{b}} \left(\frac{T}{T_{\mathrm{f}}} \right)$$

12. C_V^e as predicted by the classical Drude model predicted:

$$C_V^e = \frac{3}{2} N_{\mathrm{f}} k_{\mathrm{b}}$$

13. Since

$$\frac{T}{T_{\mathrm{f}}} \oplus 0.01$$

the Drude–Sommerfeld model corrects a major shortcoming of the classical Drude model.

Practice Questions

Multiple Choice Questions

1. The bands in a semiconductor consist of
- **(a)** Continuous energy levels
- **(b)** Discrete but closely spaced energy levels
- **(c)** Discrete and widely spaced energy levels
- **(d)** A single energy level

2. Band gap in a metal is
- **(a)** Large compared to the spacing of the allowed energy levels
- **(b)** Equal to that in a semiconducting material
- **(c)** Equal to that in a ceramic material
- **(d)** Zero

3. In quantum number space, the allowed energy levels form a
- **(a)** Sphere
- **(b)** Hemisphere
- **(c)** One fourth of a sphere
- **(d)** One eighth of a sphere

4. Density of allowed states is
- **(a)** Independent of energy
- **(b)** Related to E through a parabolic relationship
- **(c)** Related to E through a cubic relationship
- **(d)** Related to E through an exponential relationship

5. The density of occupied states, at zero Kelvin is
- **(a)** Zero for energies greater than the Fermi energy
- **(b)** Zero for energies less than Fermi energy
- **(c)** Finite non zero value for energies greater than the Fermi energy
- **(d)** Undefined for energies less than the Fermi energy

6. Fermi energy of metals
- **(a)** Is the same for all metals
- **(b)** Will vary from metal to metal
- **(c)** Changes with temperature
- **(d)** Is highest at room temperature

7. Fermi energy of a metal
- **(a)** Is the same for all phases of the metal
- **(b)** Can vary from phase to phase
- **(c)** Depends on the temperature and not on the phase
- **(d)** Is valid only for the room temperature phase

8. Fermi temperature of most metals is of the order of
- **(a)** Few degrees Kelvin
- **(b)** Few tens of degrees of Kelvin

(c) Few hundred of degrees of Kelvin
(d) Few thousands of degrees of Kelvin or more

9. Band structure of a material
(a) Evolves independent of its Fermi energy
(b) Evolves independent of its crystal structure
(c) Depends on its Fermi surface
(d) Depends only on the type of material it is

10. While comparing wave vector space with quantum number space
(a) Surfaces corresponding to constant energy have the same overall shape
(b) Surfaces corresponding to constant energy have different shapes
(c) Constant energy corresponds to a flat surface in both these spaces
(d) Constant energy corresponds to a straight line in both these spaces

11. When temperature is increased, as per the Drude–Sommerfeld model
(a) Electrons within $k_b T$ of Fermi energy can participate in the process of energy increase
(b) Electrons within $k_b T$ of zero energy can participate in the process of energy increase
(c) All electrons can participate in the process of energy increase
(d) No electrons can participate in the process of energy increase

12. The density of states per unit volume at the Fermi energy is
(a) Inversely proportional to the number of free electrons per unit volume
(b) Directly proportional to the Fermi energy
(c) Directly proportional to the number of free electrons per unit volume
(d) Directly proportional to the Fermi temperature

13. In magnitude, Fermi temperature for most metals is approximately of
(a) The same order as room temperature
(b) Two orders higher than room temperature
(c) Two orders lower than room temperature
(d) One order lower than room temperature

14. Compared to the Drude model, the C_V^e as predicted by the Drude–Sommerfeld model is
(a) One order of magnitude less
(b) Two orders of magnitude less
(c) One order of magnitude more
(d) Two orders of magnitude more

15. Waves of nearly free electrons
(a) Do not interact with the ionic cores
(b) Interact only with the waves of free electrons
(c) Interact only with external materials (other than their source)
(d) Interact with the periodic structure of the lattice they travel in

Short Answer Type Questions

1. If the density of allowed states were a constant (independent of energy), what difference will it make to the temperature dependence of the density of occupied states?

2. To what extent must the energy increase so that the density of allowed states doubles?

3. At 0 K at what energy, is the density of allowed states the maximum?

4. How can the density of states in a metal be experimentally measured?

5. In wave vector space, what sort of surface will be defined by the states of same energy?

6. What are some techniques for measuring the Fermi energy?

7. What is the Fermi energy of Fe?

8. What is the Fermi temperature of Fe?

9. What is the Fermi temperature of Cu and how does it compare with the melting point of Cu?

10. What happens when two materials with different Fermi energies are physically in contact with each other?

11. What is the significance of a high Fermi temperature?

12. What is the temperature dependence of the specific heat of ceramic materials at low temperatures?

13. What is the temperature dependence of the electronic contribution to specific heat at low temperatures?

14. Comment on the applicability and limitations of the Dulong–Petit law.

15. Comment on the temperature dependence of specific heat of copper at low temperatures.

Answers

Multiple Choice Questions

1. (b) **5.** (a) **9.** (c) **13.** (b)
2. (d) **6.** (b) **10.** (a) **14.** (b)
3. (d) **7.** (b) **11.** (a) **15.** (d)
4. (b) **8.** (d) **12.** (c)

Short Answer Type Questions

1. The primary difference will be that the energy of the states affected by the increase in temperature will be more symmetric about the Fermi energy.

2. In view of the parabolic relationship between the density of allowed states and energy, the energy must quadruple for the density of allowed states to increase by a factor of 2.

3. In view of its dependence on energy, the density of allowed states is maximum at the Fermi energy at 0 K.

4. Techniques based on photo emission and optical measurements have been used for experimentally measuring the density of states of several metals.

5. In wave vector space, states of the same energy lie on the surface of a sphere. This can be seen from the relations:

$$E = \frac{\hbar^2 k^2}{2m}$$

where

$$k^2 = k_x^2 + k_y^2 + k_z^2$$

Substituting, we get

$$E = \frac{\hbar^2 (k_x^2 + k_y^2 + k_z^2)}{2m}$$

which defines the surface of a sphere.

6. Fermi energy can be measured using thermionic emission and diffusion through a charged barrier.

7. The Fermi energy of Fe is 11.1 eV.

8. Since $E_f = k_b T_f$, the Fermi temperature of Fe is 12.7×10^4 K.

9. The Fermi temperature of Cu is 8.14×10^4 K. In comparison, the melting point of Cu is only 1358 K.

10. When two materials with different Fermi energies come in contact with each other, electrons flow from the material with the higher Fermi energy to the material with the lower Fermi energy till the Fermi energies of the two materials equalize.

11. A high Fermi temperature can be interpreted to mean that an extremely small thermal mass of the few electrons at the Fermi energy has energies consistent with the high Fermi temperature, while a vast majority of the solid is at a much lower energy – may even very close to absolute zero.

12. At low temperatures T, the specific heat of non-metals is proportional to T^3. This is based

on the phonon (lattice vibrations) contributions to the specific heat.

13. At low temperatures, the electronic contribution to the specific heat increases linearly with temperature.

14. The Dulong–Petit law is valid over a wide range of temperatures, but fails at very low temperatures and very high temperatures.

15. As the temperature decreases, the specific heat of copper initially displays the cubic dependence on temperature associated with phonon (lattice vibrations) contribution to the specific heat. At even lower temperatures, the specific heat of copper displays the linear dependence on temperature associated with electronic contributions to the specific heat.

12 The Reciprocal Space

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- State the definition based on which reciprocal space is generated.
- Plot the unit vectors of the reciprocal space based on their definitions.
- Derive the relationship between $\vec{H}_{hkl}$ and the plane ($h\ k\ l$).
- Derive the relationship between $\vec{H}_{hkl}$ and d_{hkl}.
- Derive the equation that represents the condition for diffraction in real space.
- Derive the equation that represents the condition for diffraction in the reciprocal lattice notation.
- Represent the condition for diffraction as a plot in reciprocal space.
- Draw the Ewald sphere in reciprocal space.
- Indicate the significance of the Ewald sphere touching a reciprocal lattice point.
- Derive and interpret the reciprocal lattice vectors corresponding to the real lattice vectors of simple cubic, face-centered cubic (FCC) and body-centered cubic (BCC) lattices.
- Prove that the reciprocal lattice is the Fourier transform of the real lattice.
- Show that $e^{i(\vec{k}\cdot\vec{R})} = 1$ is one way to define a reciprocal lattice corresponding to a real lattice.

12.1 Introduction

Many properties of solid materials stem from the fact that they have periodic internal structures. Electronic properties are no exception. Why do electronic properties of materials vary from one material to another? One may even be tempted to ask, "Is there a difference in the electrons present in different materials leading to the differences in their electronic properties?" Even within the same material why is graphite conducting in one direction and insulating in another?

In response to the above questions, it is relevant to note that the electrons as such are the same in all materials. The difference in electronic properties is therefore not a result of differences in the electrons present in the materials but rather due to "other differences" between the materials. In crystalline materials, the feature that has a significant impact on the electronic properties is the crystal structure of the specific material. It is therefore of interest to understand the periodic structure of crystalline materials, which is the focus of this chapter. While crystal structures are discussed from high school onwards, in this discussion we will revisit some of the concepts and expand our understanding of periodic structures. We will also familiarize ourselves with a concept referred to as "reciprocal space" that is very useful in describing periodic structures. This concept is not very intuitive at first glance but is very powerful in capturing key features of periodic structures, and hence very useful in understanding the impact of periodicity on electronic properties.

12.2 Defining Reciprocal Space



The wave vector $k = 2\pi/\lambda$ has the dimensions L^{-1}. We will now examine how to represent crystallographic information in the same framework as the wave vector information.

Crystal structure has information about crystal directions, axis, unit vectors, which are usually presented in the context of real space where the quantities have the dimensions of L. We will now define a reciprocal space where we will represent the same crystal structure information within a different framework, where the dimensions of the quantities are L^{-1}. This will enable us to more easily relate the crystal structure to the waves of electrons traveling through it, since they are presented in the same framework. Reciprocal space is credited to Ewald whose work in the 1920s laid the groundwork for this concept. In this chapter, we will look at the reciprocal lattice as an independent entity and then link it back to the models we have examined so far.

In real space, we use the unit vectors $\vec{a_1}$, $\vec{a_2}$, and $\vec{a_3}$. We will now define unit vectors for the space that will be called the reciprocal space. It is important to note that the unit vectors for reciprocal space are defined based on our convenience – or rather that they are choices we make. So at first, the selection of the unit vectors of the reciprocal lattice seems arbitrary. However, they are deliberately defined in a particular manner because it then gives the corresponding reciprocal space some useful properties and enables interesting relationships with real space.

We will define the unit vectors $\vec{b_1}$, $\vec{b_2}$, and $\vec{b_3}$ in reciprocal space, which relate to the real space vectors $\vec{a_1}$, $\vec{a_2}$, and $\vec{a_3}$ in a specific manner as shown below.

$$\vec{b_1} = \frac{\vec{a_2} \times \vec{a_3}}{V}$$

$$\vec{b_2} = \frac{\vec{a_3} \times \vec{a_1}}{V}$$

$$\vec{b_3} = \frac{\vec{a_1} \times \vec{a_2}}{V}$$

where the product designated by "□" is the vector cross-product and V is the volume of the unit cell in real space.

It is important to note that the above definition is something that we are enforcing because it leads to useful results later on. The calculations and discussions that follow simply enforce the above definitions.

12.3 Properties of Reciprocal Space

Let us examine the consequence of the above definition. Let us consider a general triclinic cell in real space. In view of it being triclinic, the three crystal unit vectors need not be equal in length, nor do the angles of the unit cell have to be equal. Therefore, the unit cell in real space is defined as

$$\vec{a_1} \neq \vec{a_2} \neq \vec{a_3}$$

and

$$\alpha \uparrow \beta \uparrow \gamma$$

Besides it is important to note that the relation "≠"is used for indicating "not necessarily equal to," implying, for example, that a cubic cell is a subset of the triclinic cell. Figure 12.1 shows a triclinic cell. Since

$$\vec{b_3} = \frac{\vec{a_1} \times \vec{a_2}}{V}$$

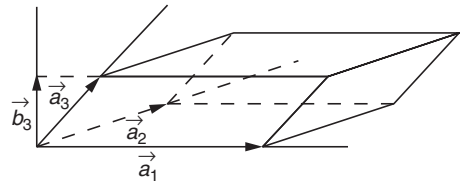


Figure 12.1 A triclinic cell showing the unit vectors $\vec{a}_1$, $\vec{a}_2$, and $\vec{a}_3$. A unit vector of reciprocal space, $\vec{b}_3$, is also shown in the figure to indicate how it relates to the real-space vectors.

The reciprocal lattice vector $\vec{b}_3$ is therefore perpendicular to the real lattice vectors $\vec{a}_1$ and $\vec{a}_2$, and the plane defined by $\vec{a}_1$ and $\vec{a}_2$.

Based on standard vector mathematics, $\vec{a}_1 \times \vec{a}_2$ is the area of the parallelogram at the base of the triclinic cell shown in Fig. 12 1, and is the numerator for the equation for $\vec{b}_3$.

The volume of the triclinic unit cell, which is the denominator for the equation for $\vec{b}_3$, is simply the product of the area of the base of the unit cell, with the height of the unit cell. The magnitude of $\vec{b}_3$ is given by

$$|\vec{b}_3| = \frac{\text{Area of base of unit cell}}{\text{Area of base of unit cell} \times \text{Height of unit cell}}$$

Simplifying, we get

$$|\vec{b}_3| = \frac{1}{\text{Height of unit cell}}$$

Since the height of the unit cell represents the distance between nearest plane parallel to the basal plane, it is essentially the spacing between (0 0 1) planes in the real space. Therefore,

$$|\vec{b}_3| = \frac{1}{d_{001}}$$

Incidentally, since $\vec{b}_3$ is perpendicular to the plane defined by $\vec{a}_1$ and $\vec{a}_2$, the height of the unit cell is simply the projection of $\vec{a}_3$ on $\vec{b}_3$. Similarly,

$$|\vec{b}_1| = \frac{1}{d_{100}}$$

and

$$|\vec{b}_2| = \frac{1}{d_{010}}$$

These are directly a result of how we have defined the relationship between the real-space and reciprocal-space unit vectors. Also, due to the definitions,

$$\vec{b}_3 \cdot \vec{a}_2 = 0$$

$$\vec{b}_3 \cdot \vec{a}_1 = 0$$

and

$$\vec{b}_3 \cdot \vec{a}_3 = \frac{1}{\text{Projection of } \vec{a}_3 \text{ on } \vec{b}_3} \times \text{Projection of } \vec{a}_3 \text{ on } \vec{b}_3 = 1$$

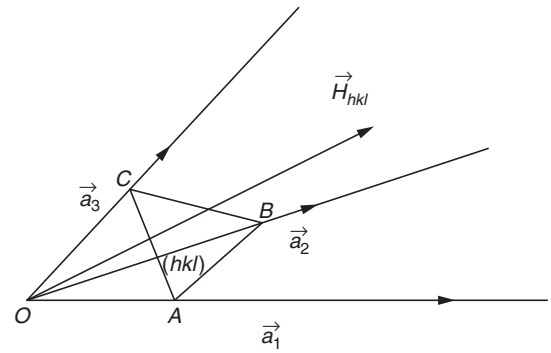


Figure 12.2 The plane ($h\ k\ l$) and a vector designated as $\vec{H}_{hkl}$.

In general,

$$\vec{b}_i \cdot \vec{a}_j = 0, \text{ if } i \neq j$$

$$\vec{b}_i \cdot \vec{a}_j = 1, \text{ if } i = j$$

We have looked at the specific case of the unit vectors going to build the real space and reciprocal space and how they relate to each other. In view of how the reciprocal space is defined, we are able to generalize further. It turns out that if $\vec{H}_{hkl}$ is a vector in reciprocal space, then it is perpendicular to the plane ($h\ k\ l$) of the real space and

$$\left|\vec{H}_{hkl}\right| = \frac{1}{d_{hkl}}$$

Consider Fig. 12.2, which shows the unit vectors of real space, plane ($h\ k\ l$) and a vector designated as $\vec{H}_{hkl}$. Purely based on conventional nomenclature for vectors, we have

$$\vec{H}_{hkl} = h\vec{b}_1 + k\vec{b}_2 + lb_3$$

where h, k, and l are integers. We will not associate any other significance to these integers at this time, except to say that they coincide with the miller indices of the plane ($h\ k\ l$) of real space.

Based on the definition of the plane ($h\ k\ l$), the intercept of this plane with the real-space axis occurs at $\vec{a}_1/h$, $\vec{a}_2/k$, and $\vec{a}_3/l$, and are the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$, and $\overrightarrow{OC}$, respectively. Since

$$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$$

Therefore, rearranging and substituting, we get

$$\overrightarrow{AB} = \frac{\vec{a}_2}{k} - \frac{\vec{a}_1}{h}$$

This implies

$$\vec{H}_{hkl} \cdot \overrightarrow{AB} = (h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3) \cdot \left(\frac{\vec{a}_2}{k} - \frac{\vec{a}_1}{h}\right) = -1 + 1 = 0$$

Therefore, $\vec{H}_{hkl}$ is perpendicular to $\overrightarrow{AB}$. By a similar analysis, it can be shown that $\vec{H}_{hkl}$ is perpendicular to $\overrightarrow{BC}$, as well as to $\overrightarrow{CA}$. Since any two of the vectors $\overrightarrow{AB}$, $\overrightarrow{BC}$, and $\overrightarrow{CA}$ define the plane $A\ B\ C$, which is also

the plane ($h\ k\ l$), we have effectively shown that any reciprocal lattice vector $\vec{H}_{hkl}$ is perpendicular to a real lattice plane whose miller indices match h, k, and l, that is, the real lattice plane ($h\ k\ l$). Therefore, we have now been able to relate the direction of a reciprocal lattice vector to the physical orientation of a real lattice plane.

Let us consider the magnitude of $\vec{H}_{hkl}$ and examine what it means. Consider a unit vector along $\vec{H}_{hkl}$, which we will designate as $\hat{n}$, it is defined by

$$\hat{n} = \frac{\vec{H}_{hkl}}{|\vec{H}_{hkl}|}$$

In real space, the meaning of defining a plane ($h\ k\ l$) as one which intercepts that respective unit cell vectors at $\vec{a}_1/h$, $\vec{a}_2/k$, and $\vec{a}_3/l$ is that one of the nearest parallel planes of this family passes through the origin. Therefore, d_{hkl} is simply the shortest distance between the origin and the plane ($h\ k\ l$) – or the distance of the origin from the plane ($h\ k\ l$) – along the perpendicular to the plane that passes through the origin. Since $\vec{H}_{hkl}$ is perpendicular to the plane ($h\ k\ l$) and passes through the origin, d_{hkl} is simply the magnitude of the projection of any of the vectors $\vec{a}_1/h$, $\vec{a}_2/k$, or $\vec{a}_3/l$ along the direction of $\vec{H}_{hkl}$ or along $\hat{n}$. Therefore,

$$d_{hkl} = \frac{\vec{a}_1}{h} \cdot \hat{n} = \frac{\vec{a}_1}{h} \cdot \frac{(h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3)}{|\vec{H}_{hkl}|} = \frac{1}{|\vec{H}_{hkl}|}$$

We started this discussion with merely an enforcement that the integers h, k, and l, which were Miller indices of a plane in real space, matched the components of a vector in reciprocal space along its axes. However, we now find that due to the manner in which reciprocal space has been defined, the real space plane ($h\ k\ l$) and the reciprocal lattice vector $\vec{H}_{hkl}$ are related in interesting ways. Specifically, we find that

1. $\vec{H}_{hkl}$ is perpendicular to the plane ($h\ k\ l$)
2. $d_{hkl} = 1/|\vec{H}_{hkl}|$

We will now consider the description of diffraction in the context of the reciprocal space.

12.4 Diffraction in Reciprocal Space



We are proceeding with a general discussion on how waves interact with matter. Later, we will discuss how electron waves interact with the periodic structure of the same material in which the electrons exist. In our earlier discussions, we noted that the wave vector $k = 2\pi/\lambda$ has reciprocal length dimensions, and therefore belongs to "reciprocal space." It is therefore of interest to understand how waves interact with a periodic structure depicted using a reciprocal lattice in reciprocal space.

We found that the reciprocal lattice vector $\vec{H}_{hkl}$ is perpendicular to the plane ($h\ k\ l$) and $d_{hkl} = 1/|\vec{H}_{hkl}|$. Let us now look at diffraction as it is represented in real space and as it is represented in reciprocal space.

Figure 12.3 shows how a parallel set of monochromatic waves, which are in phase with each other, interact with planes of atoms. It is important to note that "planes of atoms" is itself a concept we have created for our convenience. In reality, atoms in a crystalline material sit at specific locations or can be thought of as having organized addresses associated with each of them. When viewed along specific directions in space, several atoms appear to reside on specific planes. These planes are then described as planes of atoms. The atoms represent discrete points on the plane and hence the planes are not "solid" planes.

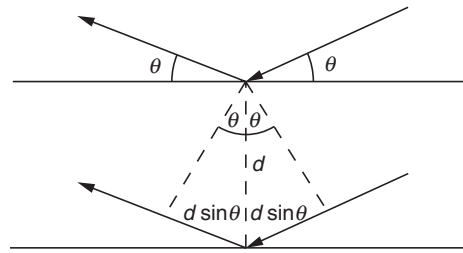


Figure 12.3 Interaction of parallel monochromatic waves, which are in phase, with planes of atoms in a crystalline material.

As shown in Fig. 12.3, the additional path traveled by the second ray is

$$d \sin\theta + d \sin\theta = 2d \sin\theta$$

When this additional distance traveled by the second ray is an integral multiple of the wavelength λ, constructive interference occurs and strong diffraction peaks are observed. Therefore, the condition for diffraction in real space is

$$n\lambda = 2d \sin\theta$$

which is the *Bragg law for diffraction*.

Let us now examine how the condition for diffraction may be indicated using reciprocal space notation. Consider the interaction of two parallel rays with two atoms – one located at the origin O and the other at a lattice point A – as shown in Fig. 12.4. Since $\overrightarrow{OA}$ is a valid real lattice vector since A is a designated lattice point, we can write

$$\overrightarrow{OA} = p\vec{a}_1 + q\vec{a}_2 + r\vec{a}_3$$

where p, q, and r are *integers* (since we define regular lattice points as those which can be arrived at using integral steps of the unit vectors corresponding to the lattice), and $\vec{a}_1$, $\vec{a}_2$, and $\vec{a}_3$ are the unit vectors corresponding to the lattice.

Let us define $\vec{S}_0$ and $\vec{S}$ as unit vectors along the direction of the incident beam and the direction of the diffracted beam, respectively, as shown in Fig. 12.4. If $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are perpendicular to $\vec{S}_0$ and $\vec{S}$, respectively, the path difference between the two rays is

$$|\overrightarrow{BA}| + |\overrightarrow{AC}|$$

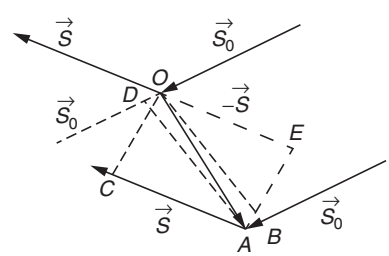


Figure 12.4 The interaction of parallel rays with atoms at two lattice points. One at the origin O and the other at a designated lattice point A.

From Fig. 12.4, this is the same as

$$|\overrightarrow{EO}| + |\overrightarrow{OD}|$$

Since $\overrightarrow{S_0} \cdot \overrightarrow{OA} = \overrightarrow{OD}$ and $-\overrightarrow{S} \cdot \overrightarrow{OA} = \overrightarrow{EO}$, the path difference is given by

$$(\overrightarrow{S_0} - \overrightarrow{S}) \cdot \overrightarrow{OA}$$

or by convention as

$$-(\overrightarrow{S_0} - \overrightarrow{S}) \cdot \overrightarrow{OA}$$

By definition, a path difference of λ equals a phase difference of 2π. Therefore, the above path difference can be proportionally written as a phase difference given by

$$-2\pi \frac{(\overrightarrow{S} - \overrightarrow{S_0})}{\lambda} \overrightarrow{OA}$$

which can be rearranged and written as

$$-2\pi \left(\frac{\overrightarrow{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda} \right) \cdot \overrightarrow{OA}$$

We note that both the vectors $\overrightarrow{S}/\lambda$ and $\overrightarrow{S_0}/\lambda$ have L^{-1} dimensions or reciprocal space dimensions. Therefore, their difference can be written as $h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}$. However, it is important to note that at this point in the derivation we have no information to suggest that the difference results in a valid reciprocal lattice vector, which means that it ends at valid reciprocal lattice point. It could perfectly well end in an empty location in reciprocal space, which does not correspond to a valid reciprocal lattice point. The condition that will indicate if a vector in reciprocal space is a valid reciprocal lattice vector is if it can be written as $h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}$ **and** if h, k, and l turn out to be *integers*. Therefore, even though $(\overrightarrow{S}/\lambda) - (\overrightarrow{S_0}/\lambda)$ can be written as $h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}$, in view of its dimensions being consistent with reciprocal space, at present we have no information to suggest that h, k, and l are integers. Since we can freely select the directions $\overrightarrow{S_0}$ and $\overrightarrow{S}$ and wavelength λ, we can easily pick values for these such that any of h, k, and l do not turn out to be integers.

The phase difference can be written as

$$-2\pi \left(\frac{\overrightarrow{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda} \right) \cdot \overrightarrow{OA} = -2\pi (h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}) \cdot (p\overrightarrow{a_1} + q\overrightarrow{a_2} + r\overrightarrow{a_3}) = -2\pi (hp + kq + lr)$$

where p, q, and r are integers, while h, k, and l need not be integers at this time. However, for constructive interference between the two rays, the phase difference between them should be an integral multiple of 2π. Therefore, for constructive interference, $(hp + kq + lr)$ should be an integer. Since p, q, and r can change as we move to the next lattice point along $\overrightarrow{OA}$, and in principle, constructive interference should still occur since the orientation is the same, we require $hp + kq + lr$ to be an integer regardless of the values p, q, and r. This can only be accomplished with certainty if h, k, and l are also integers. Therefore, the condition for constructive interference and hence diffraction requires that h, k, and l are also integers.

In other words, we now find that diffraction requires $h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}$ to be a valid reciprocal lattice vector, that is, it ends at a valid reciprocal lattice point or starts at the origin and ends at the valid reciprocal lattice point $h\ k\ l$, or can be designated as $\overrightarrow{H}_{hkl}$.

Therefore, the condition for diffraction in reciprocal space is that

$$\frac{\overrightarrow{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda} = \overrightarrow{H}_{hkl}$$

The left-hand side of the equation contains information about the incident beam and diffracted beam – both magnitude as well as direction – and the right-hand side of the equation contains the information about the crystal lattice. This equation lays out the condition for diffraction in the context of reciprocal space.

We have therefore derived the condition for diffraction in real space as well as in reciprocal space. However, we have not stated anything specific on the origin of the waves that are getting diffracted. They could originate from a source external to the sample or even be the waves corresponding to the electrons within the sample itself.

In Chapters 13 and 14, we will discuss the interaction between electron waves that are due to the nearly free electrons within the sample, and the crystal structure of the same sample itself.

We will now examine a pictorial manner in which the condition for diffraction in reciprocal space can be represented and also examine how some common crystal structures get represented in reciprocal space.

12.5 The Ewald Sphere



We have seen that diffraction occurs when in reciprocal space,

$$\frac{\vec{S}}{\lambda} - \frac{\vec{S}_0}{\lambda} = \vec{H}_{hkl}$$

Let us now plot this information. Let us designate O as the origin of reciprocal space, and draw the incident beam to contact the origin. Therefore, $\overrightarrow{AO}$ is the incident beam that has the magnitude $1/\lambda$ as shown in Fig. 12.5 and is therefore $\vec{S}_0/\lambda$. With A as the center (not O), let us draw a circle of radius $1/\lambda$ (we are drawing a circle in 2D, but a more complete picture is that of a sphere in 3D). Consider the vector $\overrightarrow{AB}$ having the magnitude $1/\lambda$, which is same as the incident beam, but has a different direction. To see if diffraction occurs in that direction, let us first designate $\overrightarrow{AB}$ as $\vec{S}/\lambda$. The vector $\overrightarrow{OB}$ is therefore $(\vec{S}/\lambda) - (\vec{S}_0/\lambda)$ and represents the left-hand side of the equation above. All points on the circle in 2D or sphere in 3D represent different directional possibilities of the left-hand side of the equation above.

The equation suggests that diffraction will occur if $\overrightarrow{OB}$ turns out to be a valid $\vec{H}_{hkl}$ for the material interacting with the waves indicated in Fig. 12.5. Therefore, on this same plot we also need to draw the reciprocal lattice corresponding to the material to the same scale. Taken together, it will be possible to identify the directions at which diffraction will occur.

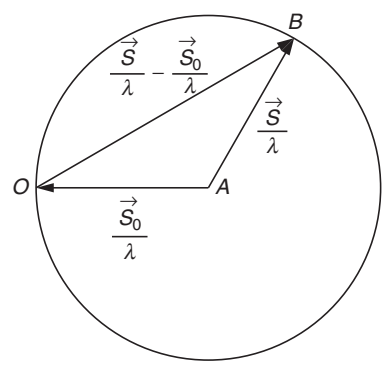


Figure 12.5 Representation of the left-hand side of the equation for diffraction in reciprocal space.

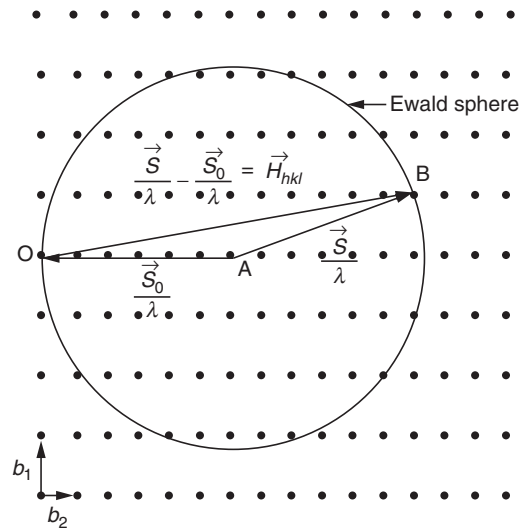


Figure 12.6 Information of the waves as well as the periodic structure of the material in a single plot in reciprocal space. The sphere corresponding to the waves is called the Ewald sphere or sphere of reflection.

Starting from the origin of reciprocal space, let us plot all combinations of $\vec{b_1}$, $\vec{b_2}$, and $\vec{b_3}$, that is, the reciprocal lattice points. All the points obtained in this manner will therefore represent valid $\vec{H}_{hkl}$ vectors or the right-hand side of the equation. Figure 12.6, therefore, represents the information of the waves as well as the periodic structure of the material in a single plot in reciprocal space.

Any direction in which the sphere touches a reciprocal lattice point corresponds to a situation where the entire equation is satisfied simultaneously, that is,

$$\frac{\vec{S}}{\lambda} - \frac{\vec{S_0}}{\lambda} = \vec{H}_{hkl}$$

and therefore the condition for diffraction is satisfied, and diffraction will occur. For all other points in Fig. 12.6, either the right-hand side of the equation is not satisfied (no valid reciprocal lattice point is present) or the left-hand side of the equation is not satisfied (sphere/circle is not present), or both are not satisfied. Therefore, at all such points, diffraction condition is not satisfied and hence diffraction will not occur.

It is important to note that the sphere and lattice can be selected independently – that is, independent of each other. We can choose any wavelength λ to carry out the study – thereby we can select a sphere of our choice. We can also choose any material to study, therefore we can select the reciprocal lattice of our choice. For a given pair of λ and material, the above procedure will show us the directions at which diffraction will occur.

12.6 Crystal Lattices in Reciprocal Space

Let us now look at the reciprocal lattice more closely to understand how structures get represented in it and its implications. The real lattice vectors $\vec{a_1}$, $\vec{a_2}$, and $\vec{a_3}$ give us the reciprocal lattice vectors $\vec{b_1}$, $\vec{b_2}$, and $\vec{b_3}$ based on the relationship of the kind

$$\vec{b_1} = \frac{\vec{a_2} \times \vec{a_3}}{V}$$

A real material may have a specific structure such as simple cubic structure or FCC or BCC or so on for which there will be corresponding vectors $\vec{a_1}$, $\vec{a_2}$, and $\vec{a_3}$. It is of interest to see how such a material will be depicted in reciprocal space, or in other words, what will be the corresponding values of $\vec{b_1}$, $\vec{b_2}$, and $\vec{b_3}$.

Let us consider some specific cases and examine the results obtained when the structure is represented in reciprocal space. Consider a simple cubic structure. The unit vectors will have the same magnitude in all three directions. Therefore,

$$\vec{a_1} = a\hat{x}$$

$$\vec{a_2} = a\hat{y}$$

$$\vec{a_3} = a\hat{z}$$

where $\hat{x}$, $\hat{y}$, and $\hat{z}$ are unit vectors in the x, y, and z directions, respectively. The volume of the unit cell is given by

$$V = \vec{a_1} \cdot (\vec{a_2} \times \vec{a_3}) = ax \cdot (ay \times az) = ax \cdot (a^2 x) = a^3$$

Therefore,

$$\vec{b_1} = \frac{a^2 \hat{x}}{a^3} = \frac{1}{a}\hat{x}$$

Hence $\vec{b_1}$ has the magnitude $1/a$ and still has the direction $\hat{x}$. By symmetry we have

$$\vec{b_2} = \frac{1}{a}\hat{y}$$

$$\vec{b_3} = \frac{1}{a}\hat{z}$$

Therefore, a simple cubic structure of side a in real space is represented by a simple cube in reciprocal space. Only the magnitude of the side has changed.

Let us now look at an FCC structure in real space. By convention, the unit vectors of choice connect the origin to the three face centers. Therefore,

$$\vec{a_1} = \frac{a}{2}(\hat{y} + \hat{z})$$

$$\vec{a_2} = \frac{a}{2}(\hat{x} + \hat{z})$$

$$\vec{a_3} = \frac{a}{2}(\hat{x} + \hat{y})$$

So

$$V = \vec{a_1} \cdot (\vec{a_2} \times \vec{a_3}) = \frac{a^3}{4}$$

and

$$b_1 = \frac{\vec{a_2} \times \vec{a_3}}{V}$$

gives us

$$\vec{b_1} = \frac{1}{a}(-\hat{x} + \hat{y} + \hat{z})$$

Similarly, by symmetry

$$\vec{b_2} = \frac{1}{a}(\hat{x} - \hat{y} + \hat{z})$$

$$\vec{b_3} = \frac{1}{a}(\hat{x} + \hat{y} - \hat{z})$$

A plot of these vectors will result in a BCC structure, where these are vectors that connect the origin to the nearest three body centers. It is important to note that the material has not changed its crystal structure, it just so happens that the representation of a FCC structure in reciprocal space shows the attributes of a BCC structure.

Finally, let us consider a BCC structure in real space. By convention, the unit vectors of choice connect the origin to the nearest body centers:

$$\vec{a_1} = \frac{a}{2}(-\hat{x} + \hat{y} + \hat{z})$$

$$\vec{a} = -(\hat{x} - \hat{y} + \hat{z})$$

$$\vec{a_3} = \frac{a}{2}(\hat{x} + \hat{y} - \hat{z})$$

This gives us the following:

$$V = \vec{a_1} \cdot (\vec{a_2} \times \vec{a_3}) = \frac{a^3}{2}$$

and

$$\vec{b_1} = \frac{\vec{a_2} \times \vec{a_3}}{V} = \frac{1}{a}(\hat{y} + \hat{z})$$

Similarly, by symmetry

$$\vec{b_2} = \frac{1}{a}(\hat{x} + \hat{z})$$

$$\vec{b_3} = \frac{1}{a}(\hat{x} + \hat{y})$$

These represent vectors from the origin to three nearby face centers. Therefore, a body-centered structure in real space is represented by a face-centered structure in reciprocal space.

Therefore, we have seen examples of how a material with a given structure in real space should be represented in reciprocal space. If we combine this information with the Ewald sphere corresponding to the waves being used, we can determine if diffraction will occur. We will use this information to understand how waves corresponding to electrons in a material interact with the crystal structure of the same material.

12.7 Reciprocal Lattice as the Fourier Transform of the Real Lattice

To prove that the reciprocal lattice is the Fourier transform of the real lattice, we will adopt the following four-step approach:

1. Write a mathematical expression that represents a 1D real lattice.
2. Determine the Fourier coefficients of the Fourier series corresponding to this real and periodic lattice.
3. Look at a derivation of the Poisson's summation formula that relates Fourier coefficients of a periodic function to its Fourier transform.
4. Use the Poisson summation formula and the Fourier coefficients of the 1D lattice to determine its Fourier transform.

12.7.1 Mathematical Representation of a 1D Lattice

Consider a 1D lattice as shown in Fig. 12.7. The analysis we will follow can be extended to 2D and 3D lattices as well, by treating each of the axes independently.

First it is necessary to represent the 1D lattice mathematically. To do this, let us consider what a 1D lattice is, to begin with. A 1D lattice implies that there are equally spaced lattice points along a straight line. It also implies that there is nothing in between those points. Therefore, the mathematical representation of a 1D lattice is accomplished by placing what are known as δ functions at each of the lattice points. A δ function, by definition, is valid only at one point and is zero everywhere else. Mathematically, a δ function is defined such that

$$\delta(x) = 0 \text{ for all } x \neq 0$$

In other words, it has a value only at one point, and that is at $x = 0$. At all other points, the δ function itself evaluates to 0. In addition, the δ function is also defined such that

$$\int_{-a}^{+b} \delta(x)dx = 1$$

Note: The region of integration includes the point $x = 0$

To place a δ function at a point other than $x = 0$, say $x = a$, we simply modify the δ function such that its argument is 0 at $x = a$ as opposed to at $x = 0$. In other words, $\delta(x - a)$ implies that a δ function has been placed at the point $x = a$, because at $x = a$, $x - a = 0$, and the δ function becomes the same as our original definition.

Additionally:

$$\int f(x)\delta(x - a) = f(a)$$



Figure 12.7 A 1D lattice of spacing a.

as long as the region of integration includes the point a. The above is true since the δ function is only valid at $x = a$, when $f(x) = f(a)$.

So if we can write a function that places δ functions equally spaced apart along a straight line, we have a mathematical representation of a 1D lattice. An example is as follows:

$$f(x) = \sum_{n=-\infty}^{\infty} \delta(x - na)$$

where n is an integer. This function places δ functions at integral intervals of a. This series of uniformly spaced δ functions, gets referred to as a Dirac comb, in view of the uniformly spaced spikes. Therefore, we now have a mathematical representation of a 1D lattice.

12.7.2 Fourier Coefficients Corresponding to the 1D Lattice

Consider the function

$$f(x) = \sum_{n=-\infty}^{\infty} \delta(x - na)$$

$f(x)$ can be represented by a Fourier series as follows:

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{2\pi i n x/a}$$

where the Fourier coefficients C_n are obtained using

$$C_n = \frac{1}{a} \int_{-a/2}^{a/2} f(x) e^{-2\pi i n x/a} dx$$

Since $f(x)$ is a δ function, by definition the above integral has a value only when $x = 0$ within this time period. And it is simply the value of $e^{-2\pi i n x/a}$ when $x = 0$, which is 1 (see the properties of the δ function indicated earlier in this section). Therefore, the Fourier coefficients are

$$C_n = \frac{1}{a}$$

for all values of n.

12.7.3 Poisson's Summation Formula

A Fourier series consists of a sum of infinite terms each of which has a periodicity of 2π. Therefore, $f(x)$ can be written as

$$f(x) = \sum_{m=-\infty}^{\infty} g(x + 2m\pi)$$

If L is the periodicity of $f(x)$, the Fourier coefficients corresponding to $f(x)$ can be obtained as

$$C_n = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-2\pi i n x/L} dx$$

$$C_n = \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} \int_0^{2\pi} g(x + 2m\pi) e^{-2\pi i n x/L} dx$$

When $x = 0$, $g(x + 2m\pi)$ is $g(2m\pi)$; and when $x = 2\pi$, $g(x + 2m\pi)$ is $g(2\pi + 2m\pi) = g[2\pi(m+1)]$. Therefore, the limits of integration from 0 to 2π can be changed to $2\pi m$ to $2\pi(m + 1)$, while correspondingly changing $g(x + 2m\pi)$ in the integral to $g(x)$. Therefore

$$C_n = \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} \int_{2\pi m}^{2\pi(m+1)} g(x) e^{-2\pi i n x/L} dx$$

Since the integration is between $2\pi m$ and $2\pi(m + 1)$, and the integral is further summed between $m = -\infty$ and $m = \infty$, this is the same as integrating between $-\infty$ and ° . Therefore,

$$C_n = \frac{1}{2\pi} \int_{-\infty}^{\infty} g(x) e^{-2\pi i n x/L} dx$$

The integral in the RHS of this equation is by definition the Fourier transform of $f(x)$ and is designated as $\tilde{f}(k)$. Therefore

$$C_n = \frac{1}{2\pi} \tilde{f}(k)$$

where

$$k = \frac{2\pi n}{L} = \frac{2\pi n}{a}$$

This implies that

$$f(x) = \sum_{m=-\infty}^{\infty} g(x + 2m\pi) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \tilde{f}(k) e^{ikx}$$

The above relation between the Fourier coefficients and the Fourier transform of a function is called the Poisson summation formula.

12.7.4 Fourier Transform of a 1D Lattice

Using the Fourier coefficients of the 1D lattice from Section 12.7.2 and the Poisson summation formula derived in Section 12.7.3, we have

$$C_n = \frac{1}{a} \text{ and } C_n = \frac{1}{2\pi} \tilde{f}(k)$$

Therefore,

$$\tilde{f}(k) = \frac{2\pi}{a}$$

Since $C_n = (1/a)$, for all integral values of n, from $-\infty$ and ° ; $\tilde{f}(k)$ must evaluate to $2\pi/a$ at integral values of n. Since $k = (2\pi n/a)$, this implies that $\tilde{f}(k)$ must have the form

$$\tilde{f}(k) = \frac{2\pi}{a} \sum_{p=-\infty}^{\infty} \delta\left(k - \frac{p2\pi}{a}\right)$$

(the new index p has simply been introduced so as to distinguish it from n used in the relation for k), which is a 1D lattice with spacing $(2\pi/a)$. Therefore, the Fourier transform of a 1D lattice of spacing a is a 1D lattice with reciprocal spacing $(2\pi/a)$. Hence, reciprocal space is described as the Fourier transform of real space.

12.8 Another Way to Define or Describe Reciprocal Space

Consider a plane wave, which has a constant frequency and can therefore be represented using

$$A(x,t) = A_0 \cos(kx - \omega t + \phi)$$

Note: The pairs of variables in the argument k, x and ω, t have been chosen so that their respective products are dimensionless. ϕ is a phase constant. Therefore, k has the dimensions of L^{-1}, and is the wave number $2\pi/\lambda$, corresponding to the wave being considered. "k-space" can be thought of as wave vector space or reciprocal space in view of its dimensions.

Since $e^{i\theta} = \cos\theta + i\sin\theta$; the plane wave can be more generally represented as

$$A(r,t) = A_0 e^{i(\vec{k}\cdot\vec{r} - \omega t + \phi)}$$

which can be expanded to

$$A(r,t) = A_0 \cos(\vec{k}\cdot\vec{r} - \omega t + \phi) + iA_0 \sin(\vec{k}\cdot\vec{r} - \omega t + \phi)$$

The real component of which is the same as the plane wave equation we had written for $A(x,t)$ above.

In the equation for the plane wave, if we focus on the spatial variation of the wave at constant time, we are essentially focussed on only the impact of the kx term (for example) on the amplitude of the wave.

In the complex exponential notation, this implies, we can write the wave as

$$A(r,t) = A_0 e^{i(\vec{k}\cdot\vec{r})}$$

where we have absorbed the effect of the ωt and ϕ terms into the A_0 constant.

Expanding the complex exponential notation we have

$$A(r,t) = A_0 \cos(\vec{k}\cdot\vec{r}) + iA_0 \sin(\vec{k}\cdot\vec{r})$$

Let $\vec{r}$ represent any vector in real space, and let $\vec{R}$ represent a valid real lattice vector, belonging to a lattice. Therefore, $\vec{R}$ begins and ends at a lattice point and is consistent with the periodicity of the lattice.

If the plane wave is compared to this periodic lattice and if it relates to the periodicity of the Real lattice such that the amplitude of the wave is the same when the argument is changed in increments of $\vec{R}$. Mathematically, this periodicity of the plane wave in relation to that of the lattice it is being compared to, can be represented as

$$A_0 e^{i(\vec{k}\cdot\vec{r})} = A_0 e^{i[\vec{k}\cdot(\vec{r}+\vec{R})]}$$

Or, ignoring the constant A_0, we have

$$e^{i(\vec{k}\cdot\vec{r})} = e^{i[\vec{k}\cdot(\vec{r}+\vec{R})]}$$

This implies

$$e^{i(\vec{k}\cdot\vec{r})} = e^{i(\vec{k}\cdot\vec{r})} e^{i(\vec{k}\cdot\vec{R})}$$

Therefore,

$$e^{i(\vec{k}\cdot\vec{R})} = 1$$

As we noted earlier k (or in general the vector $\vec{k}$) has the dimensions of L^{-1}, and corresponds to the wave number $2\pi/\lambda$. Therefore the collection of vectors $\vec{k}$, that relate to the periodicity of a *Real lattice*, but with dimensions of L^{-1}, or inverse of real space dimensions, satisfy the equation:

$$e^{i(\vec{k}\cdot\vec{R})} = 1$$

Such a collection of vectors $\vec{k}$, which have inverse of real space dimensions, and are related to the periodicity of a Real lattice, as described by the equation above, form the Reciprocal lattice corresponding to the lattice. Therefore,

$$e^{i(\vec{k}\cdot\vec{R})} = 1$$

is one way to define a *Reciprocal lattice*, corresponding to a Real lattice.

With $e^{i\theta} = \cos\theta + i\sin\theta$, the equation $e^{i(\vec{k}\cdot\vec{R})} = 1$ implies $\vec{k}\cdot\vec{R} = 2n\pi$. Or, for example,

$$k_x = \frac{2n\pi}{a_x}$$

where n is an integer. This result is consistent with our prior definition of reciprocal space, therefore the definition $e^{i\vec{k}\cdot\vec{R}} = 1$, is often used to indicate that the k space generated is reciprocal to the real space under consideration.

Summary

1. Unit vectors for reciprocal space are defined based on our convenience in order to give the corresponding reciprocal space some useful properties and enable interesting relationships with real space.
2. If $\vec{H}_{hkl}$ is a vector in reciprocal space, then it is perpendicular to the plane (h k l) of the real space.
3. The magnitude of a reciprocal lattice vector is the reciprocal of the d spacing of the corresponding planes.
4. The condition for diffraction in real space, the Bragg law for diffraction is
$$n\lambda = 2d\sin\theta$$
5. The condition for diffraction in reciprocal space is that
$$\frac{\vec{S}}{\lambda} - \frac{\vec{S_0}}{\lambda} = \vec{H}_{hkl}$$
6. A sphere generated by all possible orientations of the incident beam with one end of the incident beam vector as the origin is called the Ewald sphere.
7. Any direction in which the Ewald sphere touches a reciprocal lattice point, the condition for diffraction is satisfied and diffraction will occur.
8. The reciprocal lattice corresponding to a real space simple cubic structure is also a simple cubic structure.
9. The reciprocal lattice corresponding to a real space FCC structure is a BCC structure.
10. The reciprocal lattice corresponding to a real space BCC structure is an FCC structure.
11. Reciprocal lattice is the Fourier transform of the Real lattice.
12. $e^{i(\vec{k}\cdot\vec{R})} = 1$ is one way to define a Reciprocal lattice corresponding to a Real lattice.

Practice Questions

Multiple Choice Questions

1. In the standard definition of the reciprocal lattice, $\vec{b_3}$ is related to the vectors by $\vec{a_1}$ and $\vec{a_2}$, as
- **(a)** $\vec{b_3}$ is perpendicular to both
- **(b)** $\vec{b_3}$ is parallel to both
- **(c)** $\vec{b_3}$ is parallel to $\vec{a_1}$ but perpendicular to $\vec{a_2}$
- **(d)** $\vec{b_3}$ is parallel to $\vec{a_2}$ but perpendicular to $\vec{a_1}$

2. The dimensions of moduli of vectors in reciprocal space are
- **(a)** L
- **(b)** L^{-1}
- **(c)** L^{-2}
- **(d)** L^{0}

3. $\vec{H}_{hkl}$ is, by convention, a compact method to represent:
- **(a)** A vector in real space only
- **(b)** A vector in reciprocal space and a corresponding plane in real space
- **(c)** A vector in real space and a plane in reciprocal space
- **(d)** Vectors in both real space and reciprocal space

4. Inter-planar spacing of real lattice planes are
- **(a)** Proportional to modulus of corresponding reciprocal lattice vectors
- **(b)** Proportional to modulus of corresponding real lattice vectors
- **(c)** Proportional to inverse of modulus of corresponding real lattice vectors
- **(d)** Proportional to inverse of modulus of corresponding reciprocal lattice vectors

5. A vector in reciprocal space is
- **(a)** Perpendicular to the corresponding plane in real space
- **(b)** Perpendicular to the corresponding plane in reciprocal space
- **(c)** Perpendicular to the corresponding vector in real space
- **(d)** Perpendicular to the corresponding vector in reciprocal space

6. Diffraction from polycrystalline samples with large crystallite sizes implies
- **(a)** Constructive interference at Bragg angles
- **(b)** Destructive interference at non-Bragg angles
- **(c)** Constructive higher order interference at the corresponding Bragg angle
- **(d)** All of the above

7. Constructive interference in real space is represented by:
- **(a)** Path difference being an integral multiple of $\lambda/2$
- **(b)** Phase difference being π
- **(c)** Path difference being an integral multiple of λ
- **(d)** Phase difference being $\pi/2$

8. In general, the difference between the incident and diffracted wave vectors in reciprocal space is
- **(a)** Any vector in reciprocal space
- **(b)** A valid reciprocal lattice vector
- **(c)** A valid real lattice vector
- **(d)** A scalar

9. The condition for diffraction in reciprocal space requires that the difference between the incident and diffracted wave vectors be
- **(a)** Any vector in reciprocal space
- **(b)** Any vector in real space
- **(c)** A valid reciprocal lattice vector
- **(d)** A valid real lattice vector

10. Diffraction will always occur if:
- **(a)** Condition for diffraction is satisfied in reciprocal space but not in real space
- **(b)** Condition for diffraction is satisfied in real space but not in reciprocal space
- **(c)** Condition for diffraction is satisfied in both real space as well as in reciprocal space
- **(d)** A crystalline sample is present

11. For real materials, the Ewald sphere represents the condition for diffraction in
- **(a)** One-dimensional space only
- **(b)** Two-dimensional space only

(c) Three-dimensional space only
(d) Real space only

12. In the representation using Ewald sphere for diffraction, the reciprocal lattice depends on
(a) The wavelength of the radiation used
(b) The frequency of the radiation used
(c) The size of the sample used
(d) The crystal structure of the material used

13. With respect to the Ewald sphere and a specific sample, condition for diffraction is satisfied in
(a) All directions within the sphere
(b) Specific directions in which the Ewald sphere touches valid reciprocal lattice points
(c) All valid reciprocal lattice directions
(d) All directions in reciprocal space

14. Reciprocal lattice corresponding to a real lattice is
(a) Always the same as the real lattice
(b) Can be the same as the real lattice
(c) Never the same as the real lattice
(d) Only weakly related to the real lattice

15. Since the reciprocal lattice corresponding to a real space FCC structure is a BCC structure, it means
(a) Crystal structure of the FCC material has changed to BCC
(b) The material can display both BCC as well as FCC structure
(c) The material has become amorphous
(d) The crystal structure of the material remains FCC

Short Answer Type Questions

1. How is the volume of the unit cell defined with respect to the three lattice vectors $\vec{a_1}$, $\vec{a_2}$ and $\vec{a_3}$?

2. What factors decide the size and shape of the reciprocal lattice?

3. What is the planar density of atoms on (111) for Cu?

4. For the (111) plane of Cu, what is the magnitude of the reciprocal lattice vector?

5. How is the reciprocal lattice vector of (111) of Cu different from that of Ag?

6. Using Cu Kα radiation, a sample of CeO_2 is examined. At what angle will a peak of (2 2 0) be visible?

7. What is the significance of the order of a reflection?

8. Why does broadening of diffraction peaks occur when the sample size is small?

9. What is the difference between X-rays used for X-ray diffraction studies and those used in energy dispersive spectroscopy (EDS) studies?

10. In the derivation for the condition for diffraction in reciprocal space, what is the significance of h, k, l being integers?

11. Using the Ewald sphere construction, indicate the differences between typical X-ray diffraction experiments conducted in laboratories and electron diffraction experiments in electron microscopes.

12. What is the significance of rotating a single crystal sample along one axis, from the perspective of the use of an Ewald sphere to identify diffraction?

13. How can the Ewald sphere be used if a range of wavelengths are present?

14. During electron diffraction experiments in electron microscopes, what is the relationship between the data obtained and planes of a zone?

15. How are planes in real space represented in reciprocal space?

Answers

Multiple Choice Questions

1. (a)
2. (b)
3. (b)
4. (d)
5. (a)
6. (d)
7. (c)
8. (a)
9. (c)
10. (c)
11. (c)
12. (d)
13. (b)
14. (b)
15. (d)

Short Answer Type Questions

1. The volume is defined by the vector triple product

$$V = \vec{a_1} \cdot (\vec{a_2} \times \vec{a_3})$$

2. The dimensions of the real lattice determine the dimensions of the reciprocal lattice in an inverse manner, while the symmetry of the real lattice decides the shape of the reciprocal lattice.

3. The (111) plane is a close packed plane. In two dimensions, the atoms appear as circles that touch each other while occupying the vertices of a triangle. If the atomic radius is r, a 60° sector of each circle lies within the triangle. The total area of the circles lying within the triangle is therefore given by

$$3 \times \frac{\pi r^2}{6} = \frac{\pi r^2}{2}$$

The side of the triangle is $2r$, therefore its area is

$$\frac{1}{2} \times 2r \times 2r \times \sin 60° = 2r^2 \sin 60°$$

Therefore, the planar packing fraction is

$$\frac{\pi r^2}{2 \times 2r^2 \sin 60} = \frac{\pi}{4 \times \sin 60} = 0.907 = 90.7\%$$

4. The magnitude of the reciprocal lattice vector is the inverse of the d spacing of the (111) planes. Since the lattice parameter of Cu is 3.61 Å, it is

$$\frac{\sqrt{3}}{3.61} = 0.48 \ \text{(Å)}^{-1}$$

5. The reciprocal lattice vectors of (111) of Cu as well as Ag have the same direction but differ in magnitude based on the spacing of the d_{111} spacing in the respective materials.

6. Cu Kα radiation has a wavelength 1.54 Å. CeO_2 is cubic with a lattice parameter of 5.41 Å. The d spacing of the (2 2 0) plane is given by

$$d = \frac{5.41}{\sqrt{4+4+0}} = 1.91 \text{ Å}$$

Therefore, with Cu Kα radiation, we have from the Bragg law

$$1.54 = 2 \times 1.91 \times \sin\theta$$

Therefore, $\theta = 23.7°$ or $2\theta = 47.4°$.

7. Order of a reflection represents diffraction where the path difference is 2 or more integral multiples of the wavelength.

8. When the sample size is small, there are not enough planes available in the sample to ensure destructive interference at angles other than the Bragg angle, which therefore enables the peaks to display broadening, or finite intensity at off-Bragg angles.

9. In X-ray diffraction, the characteristic X-ray from a target is used for probing a sample, and it provides insight into the crystal structure of the sample and hence the phases present. In EDS studies, the elements in the sample are induced to release their characteristic X-rays and as a result elemental analysis is possible. EDS by itself does not provide phase information.

10. The significance of *h*, *k*, *l* being integers is that they represent a valid lattice point in reciprocal space and hence a valid plane in real space.

11. In the scale of the reciprocal lattice of typical crystals, the Ewald sphere corresponding to the wavelength of typical X-rays used appears as a sphere, since the wavelength used is of the same order of magnitude as the lattice parameter. On the other hand, the wavelength of typical electron beams used in electron diffraction experiments in electron microscopes is two orders of magnitude less than the lattice parameter. As a result, in the scale of the reciprocal lattice of typical crystals, the Ewald sphere corresponding to the electron beam almost looks like a flat plane.

12. Under the conditions described, diffraction will occur from all points that the Ewald sphere can touch when the reciprocal lattice is rotated through the same axis in reciprocal space.

13. In this case, two Ewald spheres will have to be drawn, one corresponding to the lowest wavelength used and the other to the highest wavelength used. All reciprocal lattice points touching either sphere, or lying in the region between the two spheres, will satisfy the condition for diffraction

14. Since the Ewald sphere of the electron beam is almost a plane in the scale of the reciprocal lattice of a crystal, the reciprocal lattice points intersected by the Ewald sphere belong to the planes of a zone with the electron beam direction approximately being the zone axis.

15. Planes in real space are represented as points in reciprocal space.

13 Wigner–Seitz Cell, Brillouin Zones, and The Origin of Bands

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Define a Wigner–Seitz cell.
- Identify the Wigner–Seitz cell corresponding to specific 2D lattices.
- Identify the Wigner–Seitz cell corresponding to simple 3D lattices.
- Define the first Brillouin zone.
- Define and identify Bragg planes.
- Define higher order Brillouin zones.
- Identify the first and higher order Brillouin zones for a square 2D lattice.
- Identify the Wigner–Seitz cell of a BCC lattice.
- Identify the Wigner–Seitz cell of an FCC lattice.
- Identify the first Brillouin zone of an FCC real lattice.
- Identify the first Brillouin zone of a BCC real lattice.
- Identify the significance of a Bragg plane touching one end of a wave vector with respect to satisfying the condition for diffraction in reciprocal space.
- Identify the significance of Brillouin zone boundaries touching wave vectors with respect to satisfying the condition for diffraction in reciprocal space.
- Plot the E versus k relationship of nearly free electrons and the Brillouin zones of a one-dimensional (1D) real lattice on the same plot in "k-space" and indicate the interaction between the two.
- Identify the allowed energy bands and forbidden energy bandgaps based on the interaction between wave vectors corresponding to the nearly free electrons and the Brillouin zones.
- Identify the correlation between the "extended zone representation" and the "flat-band diagram."
- Indicate the significance of the Fermi energy in these diagrams.
- Draw the "repeated zone representation" and "reduced zone representation" diagrams.
- Draw and identify the interaction between a "Fermi circle" and Brillouin zones in 2D.
- Draw and indicate the interaction between a "Fermi sphere" and the first Brillouin zone for an FCC real lattice.

13.1 Introduction

We have discussed diffraction in real space and reciprocal space, and seen examples of what happens when real space structures are represented in reciprocal space. We have also noted that such transformations are carried out primarily for our convenience; the material retains its structure regardless of how we choose to look at it. In this chapter we will look at specific constructs, namely, Wigner–Seitz cells and Brillouin zones, that correspond in specific ways to the crystalline structure of materials. Using these constructs and by examining the interaction of electron waves with these structures, it is possible to understand the origin of the band structure in materials.

13.2 The Wigner–Seitz Cell



Let us now introduce a few new terms that will be used extensively later. The first is "Wigner–Seitz cell." Stated in words, a Wigner–Seitz cell about a lattice point is the region in space that is closest to that lattice point than to any other lattice point. While the definition may not be easy to visualize at first glance, it is actually quite easy to determine. As shown in Fig. 13.1, take any two points in space and draw a straight line joining them. Draw a plane that perpendicularly bisects the line joining these two points. All the points on one side of the perpendicular bisector are by definition closer to the point on the same side of the perpendicular bisector than to the other point. This is illustrated in Fig. 13.1.

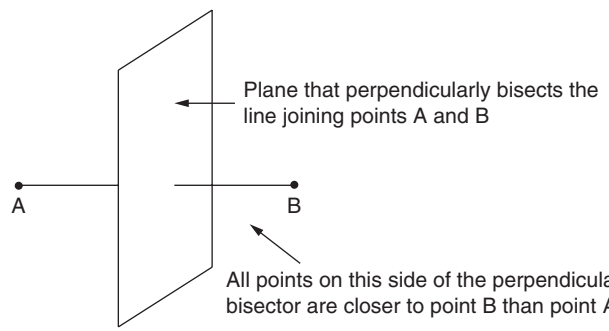


Figure 13.1 Identifying a region closest to a given lattice point than to its neighbor.

A lattice, as we have seen, is an array of points. In the most general case, it is a 3D array of points. To identify the region in space closest to a single lattice point than to any other lattice point, we merely extend the approach we have adopted above. As a first step, we identify all the nearest neighbors of the point being examined. Lines are then drawn from the point to all its nearest neighbors. Perpendicular bisectors are drawn to each of the lines. Once these steps are completed, it will be possible to identify the innermost region bounded by these perpendicular bisectors. The innermost region then consists of all the points in space, which are closest to the lattice point in that region than to any other lattice point. This region, bounded by the perpendicular bisectors, is then the Wigner–Seitz cell about that lattice point.

The procedure described above is shown in Fig. 13.2. Consider a 2D square lattice as drawn in Fig. 13.2(a) and see how its Wigner–Seitz cell is identified.

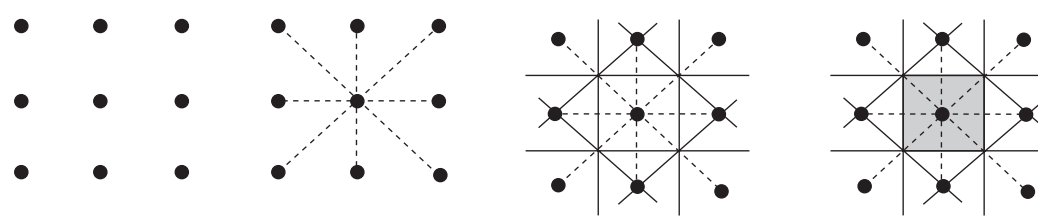


Figure 13.2 The steps to identify the Wigner–Seitz cell about a lattice point in a 2D square lattice.

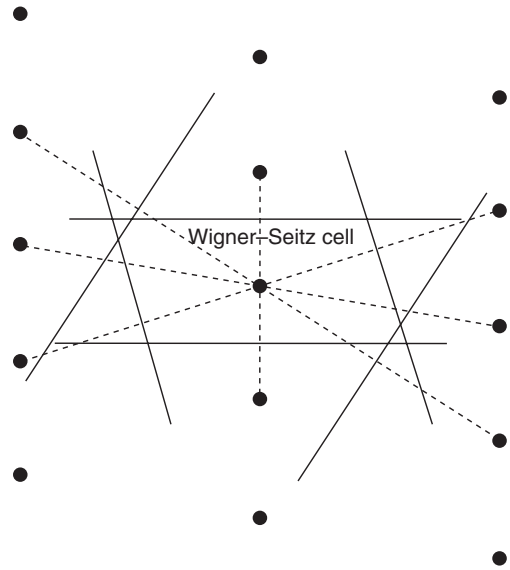


Figure 13.3 A general 2D lattice and its Wigner–Seitz cell. In the most general case, a 2D Wigner–Seitz cell will be a hexagon.

The procedure described above can easily be extended to three dimensions. The primary difference will be that the perpendicular bisectors will now be planes instead of lines. So, for a 3D cubic lattice, the 2D lattice shown in Fig. 13.2 will form a section of the lattice, and the perpendicular bisectors shown in the figure will be planes instead of lines. Additionally, there will be a perpendicular bisector above the plane of the paper and one below the plane of paper corresponding to the lines joining the central point to the lattice points above and below the plane of the paper. The Wigner–Seitz cell for the 3D cubic lattice will therefore be a cube about the lattice point chosen. In the case of the 2D square and 3D cube, the Wigner–Seitz cells also have similar structures, but for other lattices the shape of the lattice and shape of the corresponding Wigner–Seitz cell may not display such immediate equality.

In two dimensions, a square lattice displays a square Wigner–Seitz cell and a rectangular lattice displays a rectangular Wigner–Seitz cell. Consider the lattice shown in Fig. 13.3, which is a more general case of a 2D lattice and its Wigner–Seitz cell.

In the most general case, the 2D Wigner–Seitz cell will be a hexagon. Due to symmetry, in specific cases it reduces to a rectangle or a square based on the lattice chosen, as discussed above.

13.3 The Brillouin Zone



Let us now define a "Brillouin zone." Given that we now understand how a Wigner–Seitz cell is defined, it is easy to understand a Brillouin zone. We will first define and discuss the "first" Brillouin zone and then subsequently discuss the second, third, and higher Brillouin zones. The first Brillouin zone is defined as the Wigner–Seitz primitive cell about a lattice point in *reciprocal space*. In other words, to identify the first Brillouin zone for a lattice, we first draw the reciprocal lattice corresponding to the given lattice, and then identify the Wigner–Seitz cell about a point in that reciprocal lattice. This specific Wigner–Seitz cell is then the first Brilloun zone corresponding to the original real space lattice.

At this stage, it is relevant to note that Wigner–Seitz cell is a more general concept and can be defined both with respect to real space as well as reciprocal space. However, Brillouin zones are defined only with respect to reciprocal space. Further, Brillouin zones have greater significance in terms of the electronic properties of the materials and hence are treated as a concept worthy of independent recognition over and above their relationship to Wigner–Seitz cells.

While describing Wigner–Seitz cells, we considered planes that perpendicularly bisect lines joining a lattice point to its nearest neighbors. In reciprocal space, planes bisecting lines joining reciprocal lattice points have special significance with respect to diffraction phenomena displayed by that lattice. Planes that are perpendicular bisectors of lines joining the origin to lattice points in reciprocal space therefore have a special name – they are referred to as "Bragg planes" – this is the third term that we are defining in this chapter. It is important to recognize that Bragg planes bisect lines joining a reciprocal lattice point chosen as the origin to all the other points in the reciprocal lattice, and are not restricted to just lines joining the reciprocal lattice point to its nearest neighbors. The first Brillouin zone is thus the region bounded by the nearest collection of Bragg planes around a reciprocal lattice point.

To identify the second, third, and higher Brillouin zones, we extend the procedure followed thus far. The second Brillouin zone is identified as the region in reciprocal space that extends beyond the nearest (first) Bragg plane in all directions, but not beyond the very next (second) Bragg plane in each of those directions. Generalizing further, the nth Brillouin zone is all the points between the $(n-1)$th Bragg plane and nth Bragg plane in all directions. In other words, if we start from each of the points in the $(n-1)$th Brillouin zone and continue to move outward from the central point, the nth Brillouin zone consists of all the points that can be reached by crossing just one more Bragg plane.

The process of identifying various Brillouin zones is illustrated in Fig. 13.4. The points in the figure represent reciprocal lattice points (in this case, corresponding to a square real lattice). The lines drawn are various Bragg planes with respect to the central point. The numbers indicate the Brillouin zone to which each region belongs.

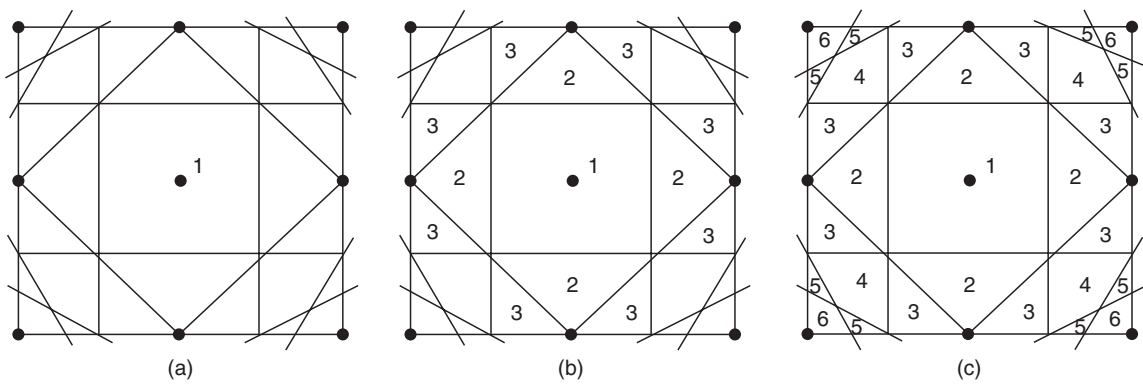


Figure 13.4 Identification of the first and higher order Brillouin zones for a 2D square lattice. Although all the Bragg planes have been drawn in the context of the figure, for clarity in (a) only the first Brillouin zone is identified; in (b) first, second, and third Brillouin zones are identified; and in (c) first second, third, fourth, fifth, and sixth Brillouin zones are identified. All the zones identified are only within the context of the figure drawn, that is, there are other regions that will correspond to the fourth, fifth, and sixth Brillouin zones, which do not show up in the region shown above.

Animation: "Brillouin zones for a square 2D lattice" in DVD.

The first Brillouin zone for a 2D square lattice is a square, which is easy to see in the figures presented above. The second Brillouin zone consists of four parts. As can be seen from Fig. 13.4, if these four parts are cut and rearranged, a square will emerge once again. Or, conceptually, if they are moved by one lattice vector into the first Brillouin zone, they will reassemble into a square. Higher order Brillouin zones get more fragmented and appear at various places in the diagram. However, these fragments too can be reassembled to obtain a square just the way we did with the second Brillouin zone. This process is possible as a direct result of the symmetry of the lattice.

We have so far seen Wigner–Seitz cell, Bragg plane, and Brillouin zones. We have seen examples of these in two dimensions. In three dimensions, the corresponding diagrams can look complicated; however, the concept is the same. Let us now consider 3D versions of the first Brillouin zone and see how it helps us understand the behavior of electrons in solids. Specifically we will look at the interaction of Brillouin zones and wave vectors of nearly free electrons.

As indicated in Chapter12, a simple cubic structure in real space transforms to a simple cubic structure in reciprocal space. Further, the Wigner–Seitz cell for a simple cubic lattice is a simple cube as well. Therefore, the first Brillouin zone for a material that has a simple cubic structure in real space is the Wigner–Seitz cell corresponding to its reciprocal lattice, and is therefore also a cube.

Consider a BCC lattice. It consists of a lattice point at the body center and lattice points at the corners of the cubic lattice. Since the body center is conveniently at the center of the unit cell, we choose it as the central point around which the Wigner–Seitz cell is identified. By treating the body center as the origin, imaginary lines can be drawn to the corner lattice points that are the nearest neighbors of the lattice point at the body center. Perpendicular bisectors to these imaginary lines can be drawn and the region in space bounded by these perpendicular bisectors will then form the Wigner–Seitz cell of the BCC lattice. The Wigner–Seitz cell of a BCC lattice is shown in Fig. 13.5.

We can also consider imaginary lines connecting the body center of the unit cell chosen to its adjacent body centers. The perpendicular bisector to these lines will be the sides of the unit cell since they will be midway between the adjacent body centers. These planes will not cut out any portion from the Wigner–Seitz cell we have identified above. Therefore, Fig. 13.5 correctly identifies the Wigner–Seitz cell for a BCC lattice.

Consider now an FCC lattice. Since lattice points are present at the corners of the cube as well as at the face centers, at first glance it is not possible to identify a convenient central point about which the Wigner–Seitz cell can be defined. However, by taking two adjacent unit cells together, it is possible to identify the shared face center as a common central point about which the Wigner–Seitz cell can be defined. In view of the symmetry of these structures, the choice of origin is only a matter of convenience. The choice of origin is shown in Fig. 13.6.

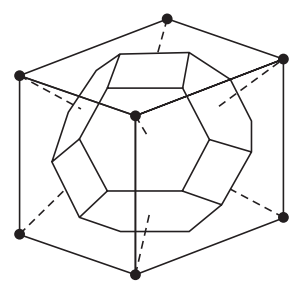


Figure 13.5 Wigner–Seitz cell of a BCC lattice. The central lattice point of the BCC lattice is hidden within the Wigner–Seitz cell.

Animation: "Wigner–Seitz cells for FCC and for BCC lattices" in DVD.

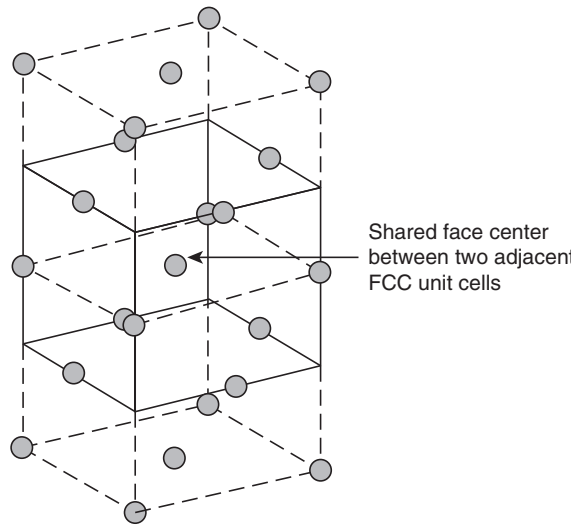


Figure 13.6 Choice of origin for an FCC unit cell for the purpose of identifying its Wigner–Seitz cell. A face center shared between two adjacent FCC unit cells is the origin of choice.
Animation: "Wigner–Seitz cells for FCC and for BCC lattices" in DVD.

The nearest neighbors of the face-centered lattice point are four face centers of the unit cell on top, four face centers of the unit cell below, and four shared corner lattice points of the two unit cells. By drawing imaginary lines to these nearest neighbors and perpendicular bisectors to those lines, the Wigner–Seitz cell of the FCC lattice is identified as shown in Fig. 13.7.

Since a BCC real lattice is represented in reciprocal space by an FCC lattice, as seen in Chapter 12, the Wigner–Seitz cell of the FCC lattice becomes the first Brillouin zone of the BCC real lattice. Similarly, since an FCC real lattice is represented in reciprocal space by a BCC lattice, the Wigner–Seitz cell of the BCC lattice becomes the first Brillouin zone of the FCC real lattice. These structures are therefore conceptually inversely related as indicated in Fig. 13.8.

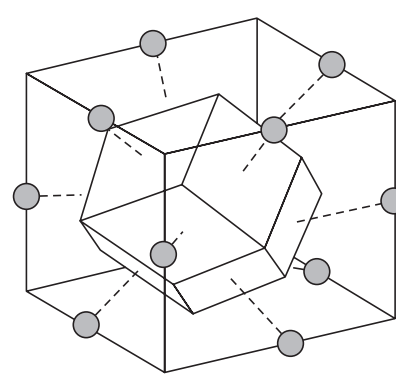


Figure 13.7 Wigner–Seitz cell of an FCC lattice. One shared face-centered lattice point is hidden within the Wigner–Seitz cell.

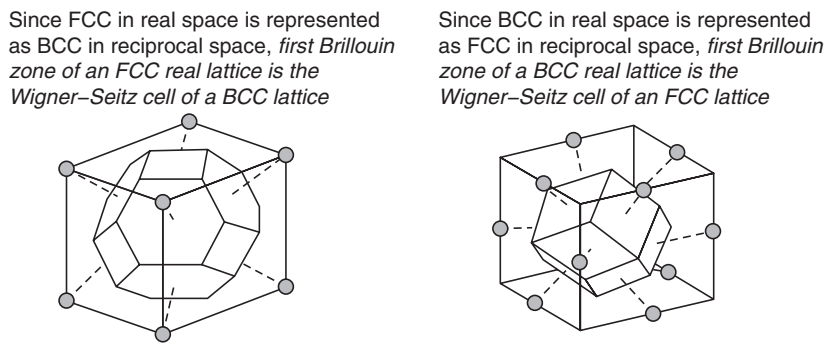


Figure 13.8 The relationship between real lattice structures and their Brillouin zones for FCC and BCC real lattice structures.

We have also noted that in the manner that the Brillouin zones are identified, their boundaries are Bragg planes. Therefore, if specific phenomena occur at Bragg planes, then these phenomena can be expected to occur at the Brillouin zone boundaries as well.

13.4 Significance of Bragg Planes

Let us re-examine the description we have seen for diffraction in reciprocal space and identify the significance of Bragg planes in that description.

In reciprocal space, the condition for diffraction is written as

$$\frac{\vec{S}}{\lambda} - \frac{\vec{S}_0}{\lambda} = \vec{H}_{hkl}$$

A schematic of this relationship is shown in Fig. 13.9.

$$\frac{\vec{S}}{\lambda} - \frac{\vec{S}_0}{\lambda} = \vec{H}_{hkl}$$

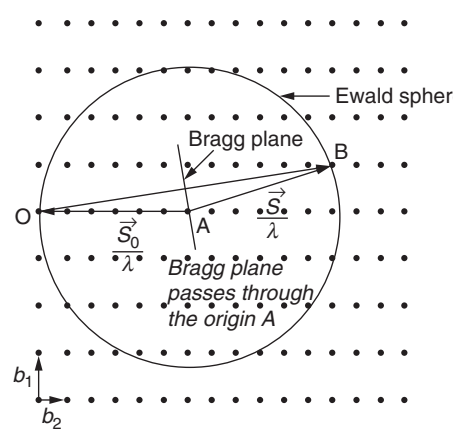


Figure 13.9 Diffraction relationship in reciprocal space showing that the corresponding Bragg plane passes through the origin and therefore touches one end of the wave vector.

If we look at the plot of this relationship, we note that purely due to geometric considerations the perpendicular bisector of $\vec{H}_{hkl}$ will pass through the origin. This occurs because $\vec{H}_{hkl}$ is a chord to the circle drawn. In reciprocal space, we have called these perpendicular bisectors of reciprocal lattice vectors, $\vec{H}_{hkl}$, as Bragg planes. Therefore, Bragg planes of those reciprocal lattice vectors which lie on the surface of the Ewald sphere and hence satisfy the condition for diffraction, pass through the origin of the Ewald sphere, where they touch one end of the wave vector $\vec{S}_0/\lambda$. In other words, whenever a Bragg plane touches one end of a wave vector, the rest of the diagram automatically "falls in place" and the condition for diffraction is satisfied with respect to the Ewald sphere corresponding to that specific wave vector.

As mentioned earlier, the process of diffraction is not impacted by the source of the waves. As long as the condition for diffraction – as indicated by the diffraction equation – is satisfied, diffraction will occur.

13.5 Interaction of Electron Waves with Brillouin Zones

Nearly free electrons in a material are roaming through the material displaying wavelike behavior and we have studied how to associate wave vectors with these electrons based on the momenta they possess. Since they roam through a material that has a periodic structure, like most metals have, the electrons get opportunities to interact with the Brillouin zones associated with the periodicity of these materials. Therefore, if we draw a consolidated plot that shows the wave vectors allowed for the free electrons, and the Brillouin zone boundaries corresponding to the structure of the material, then whenever the wave vectors touch the Brillouin zone boundaries in this consolidated plot, diffraction will occur – the only additional condition being that the wave vectors as well as the Brillouin zones should be drawn to the same scale on that plot.

So far we have discussed about real space and reciprocal space. With a scaling factor of 2π, reciprocal space can be converted to "k-space" as follows.

Consider a 1D lattice of spacing a; the corresponding reciprocal lattice has a spacing $1/a$. Since k-space plots λ as $2\pi/\lambda$, for the scaling to be the same, real lattice spacing of a should be plotted as $2\pi/a$ in k-space, which is dimensionally the same as reciprocal space. It is important to draw our attention to a detail at this stage, since it will directly impact the details in the consolidated plot we are going to draw. The lattice has a spacing a, which is of the order of 10^{-10} m. Therefore, the corresponding k-space spacing of $2\pi/a$ is relatively large. On the other hand, the extent of confinement of the nearly free electrons L is of the order of meters, or the extent of the solid. Therefore, the corresponding allowed wave vectors, $n\pi/L$, are very closely spaced. In fact, we can expect of the order of 10^{10} allowed wave vectors in the scale defined by two adjacent lattice points in k-space. The allowed wave vectors, though discrete, will look continuous on this plot simply as a result of the scale of the plot.

Since the reciprocal lattice vectors are $2\pi/a$ in length, the Bragg planes for this reciprocal lattice occur at

$$\frac{1}{2}\times\frac{2\pi}{a}=\frac{\pi}{a}$$

Therefore, starting at the origin, Brillouin zone boundaries occur with a spacing of π/a. The extent of the Brillouin zones for this lattice are therefore as follows:

First Brillouin zone:

$$-\frac{\pi}{a}\text{ to }+\frac{\pi}{a}$$

Second Brillouin zone:

$$-\frac{2\pi}{a}\text{ to }-\frac{\pi}{a}\text{ and }+\frac{\pi}{a}\text{ to }+\frac{2\pi}{a}$$

***n*th Brillouin zone:**

$$-\frac{n\pi}{a} \text{ to } -\frac{(n-1)\pi}{a} \text{ and } +\frac{(n-1)\pi}{a} \text{ to } +\frac{n\pi}{a}$$

The Brillouin zones of this 1D lattice are shown in Fig. 13.10.

Let us now examine the interaction of the allowed wave vectors and Brillouin zones. This interaction results in forbidden values of energy for the electrons, and hence results in the band structure. We will now plot the reciprocal lattice information as well as the wave vector information on the same plot and examine the interaction between them in a pictorial manner.

We have noted that while the E versus k relationship is the same for free electrons, nearly free electrons – as well as the bound electrons – it is a continuous curve only for free electrons. For nearly free electrons as well as bound electrons, confinement of the electron results in only specific values of k to be permitted, and therefore only the corresponding values of energy are permitted. The resulting plot has discrete points that are laid out in the form of the parabola consistent with

$$E = \frac{\hbar^2 k^2}{2m}$$

To reiterate, we wish to plot the periodicity of the lattice as well as the wave vectors on the same plot. The reciprocal space differs from k-space only in the scaling factor 2π. Therefore, reciprocal lattice information is multiplied by 2π to enable it to be plotted in k-space. Therefore, real lattice vector a, which is plotted in reciprocal space as $1/a$, is now plotted in k-space as $2\pi/a$. We note that a, which of the order of inter-atomic spacing, is of the order of 10^{-10} m, while L, the extent of confinement of the nearly free electrons, is of the

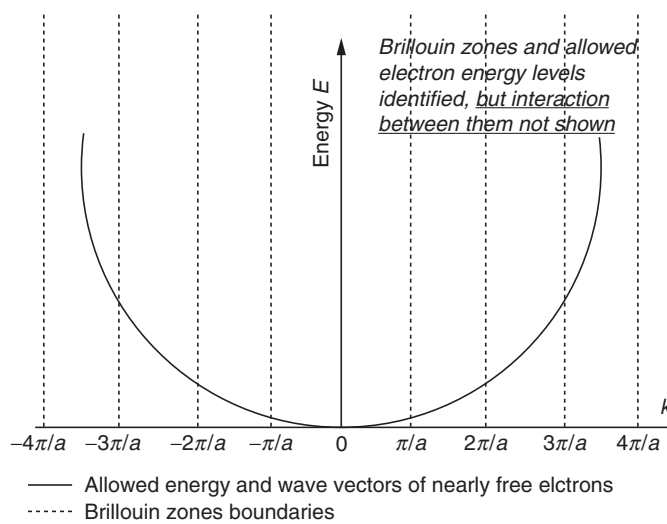


Figure 13.10 Brillouin zones corresponding to a 1D real lattice of spacing a, and wave vectors corresponding to nearly free electrons, drawn to the same scale in k-space. The interaction between the Brillouin zones and electron wave vectors *has not been shown* in this figure. The figure primarily highlights the relative scales of the two pieces of information that have been plotted. The discrete allowed wave vectors of the electrons look continuous when drawn to this scale.

order of meters. Therefore, the reciprocal lattice points plotted in k-space as $2n\pi/a$, where n is an integer, can contain 10^{10} allowed wave vectors of nearly free electrons that are of the form $n\pi/L$ between two adjacent reciprocal lattice points. This is the reason that allowed wave vector plots drawn on the scale of the order of $2n\pi/a$ to look continuous even though they contain discrete points spaced π/L apart.

13.6 The Origin of Bands



We have noted that diffraction occurs when one end of a wave vector touches a Bragg plane. In the context of nearly free electrons, diffraction causes the E versus k relationship to distort in the vicinity of the Bragg planes (or Brillouin zone boundaries). This distortion causes some of the energy levels to become forbidden to the nearly free electrons – which results in the presence of band gaps in the material.

Let us now examine this interaction between wave vectors and the Brillouin zones pictorially and not focus on the exact values of the band gaps that result. We will discuss the examples of 1D, two-dimensional (2D) and three-dimensional (3D) lattices and interaction of wave vectors with the Brillouin zones corresponding to these lattices in k-space. In Chapter 14, we will examine the formation of band gaps in a mathematical manner.

Since a 1D real lattice of spacing a is plotted in k-space as $2\pi/a$, the Brillouin zone boundaries occur in intervals of π/a, that is, at π/a, $2\pi/a$, $3\pi/a$, etc. Figure 13.11 shows the interaction of the E versus k relationship of nearly free electrons with the Brillouin zone boundaries of a 1D lattice of spacing a.

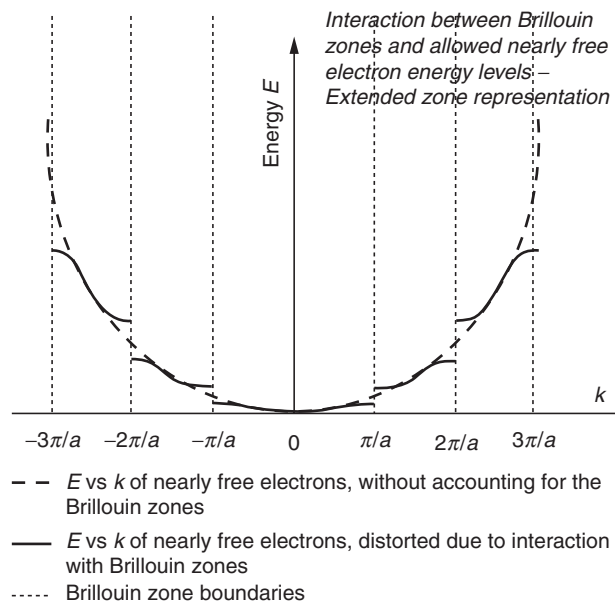


Figure 13.11 Interaction of the E versus k relationship of nearly free electrons with the Brillouin zone boundaries of a 1D lattice of spacing a.

The distortion of the E versus k relationship at the Brillouin zone boundaries and the resulting energy gaps can be described as follows: The traveling waves of nearly free electrons undergo diffraction at the Brillouin boundaries and result in standing waves; as a result energy gaps appear.

The above form of representing the interaction between the allowed energy values and Brillouin zones of the material is called the "extended zone" representation, since the information presented is spread across several lattice points, but has a single origin.

The band gaps described here and the remaining allowed energy levels are, in concept, the same as the band gaps and the allowed bands we are familiar within the band structure of materials. However, in high school texts, bands of allowed energy levels are represented as boxes separated by gaps that represent the band gaps and such a diagram is called a "flat-band diagram." How does the flat-band diagram relate to Fig. 13.11 above? This relationship is shown in Fig. 13.12.

The flat-band diagram is sufficient to explain specific material phenomena, but is insufficient to explain many others. The other representations we are presenting in this chapter, including the extended zone representation and some more that will follow, are much more capable of explaining a variety of material phenomena when compared to the flat-band diagram. We have also noted that at 0 K, the Fermi energy represents the highest occupied energy level in the system, and interaction of electrons with the outside world begins here. Where does this fit into our representations of the allowed energy levels, and how does that relate to the flat-band diagram? Figure 13.13 shows an example of this.

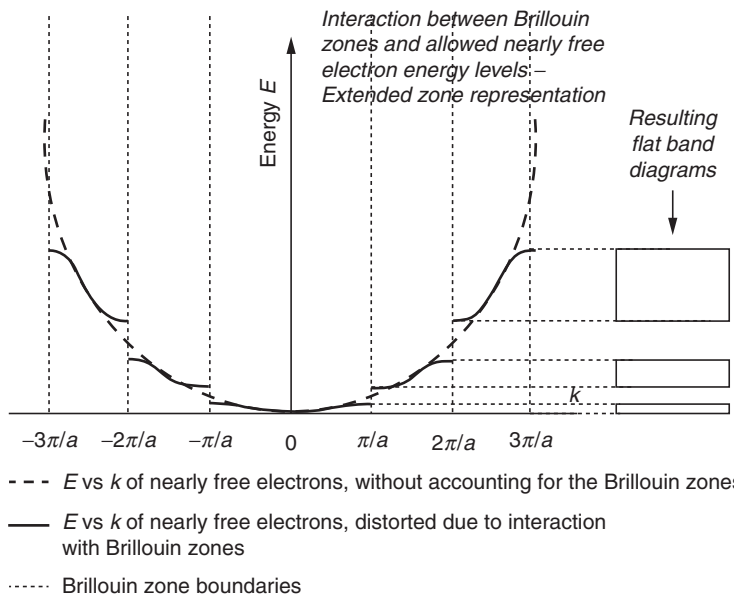


Figure 13.12 Relationship between the extended zone representation and the flat-band structure.

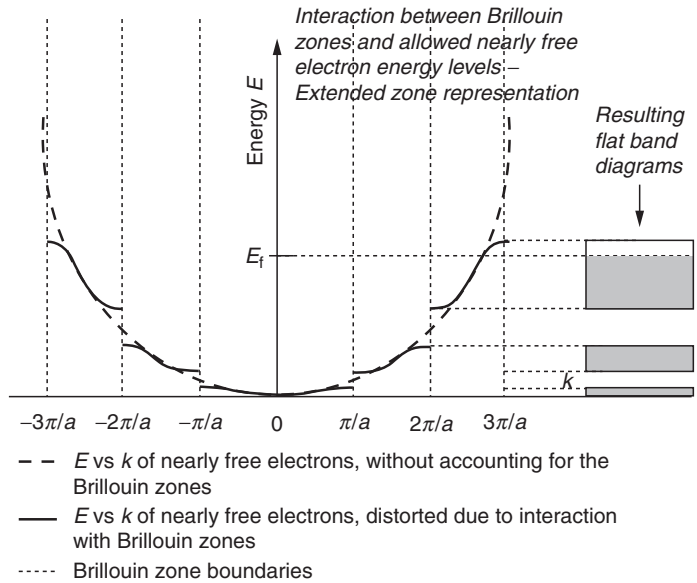


Figure 13.13 The Fermi energy in the extended zone scheme and in the flat-band diagram. The actual value of the Fermi energy will depend on the material chosen and has been arbitrarily fixed in the schematic above, for illustrative purposes.

While the extended zone scheme plots the essential details, it is important to note that the choice of origin of *k*-space is arbitrary. Due to symmetry, each lattice point is the same as every other lattice point. Therefore, the *E* versus *k* diagram can be repeated at each lattice point to give a more complete picture of the situation in the material. This type of a diagram is called a "repeated zone" representation and is shown in Fig. 13.14.

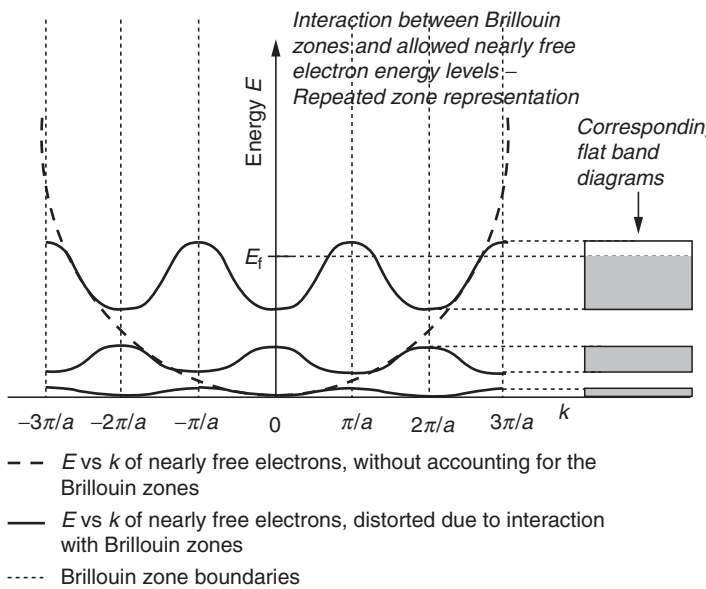


Figure 13.14 The repeated zone representation.

Looking at the repeated zone representation, it can be concluded that the region in the first Brillouin zone between $-\pi/a$ and π/a contains all the information in a compact manner. Therefore, it is often considered sufficient to show all the allowed wave vectors and energy levels within the first Brillouin zone itself. Such a representation is referred to as the "reduced zone" representation and is shown in Fig. 13.15.

In these diagrams, it is the location of the Fermi energy (E_f) and/or the nearest band gap above the Fermi energy, which decides whether the material is metallic, semiconducting, or insulating. In the diagrams drawn here, E_f is in the middle of a band, therefore the material being depicted is metallic.

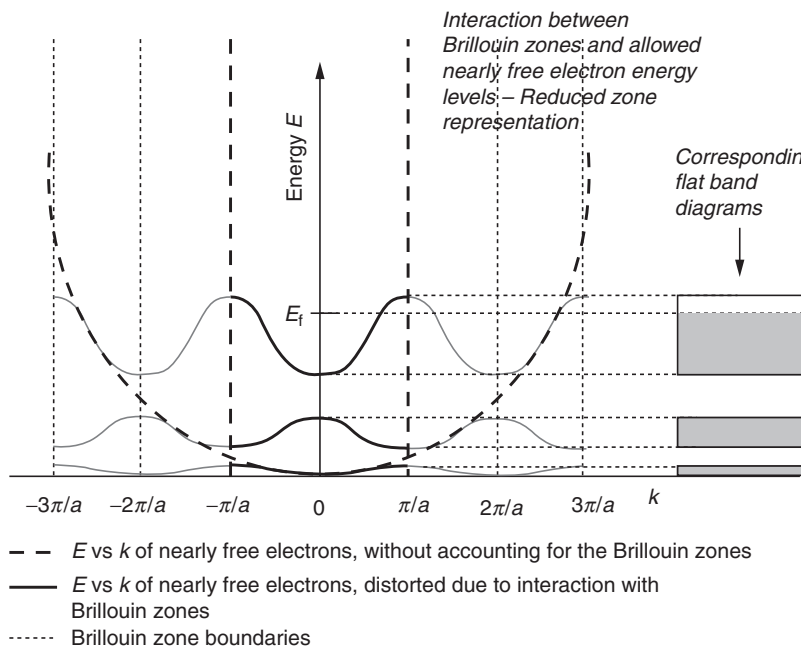


Figure 13.15 The reduced zone representation.

The above diagrams were for a 1D lattice. Let us consider a 2D square lattice and schematically consider its interaction with the wave vectors of nearly free electrons. In 2D *k*-space, electrons of the same energy are represented by a circle. The Fermi energy corresponds to the largest such circle, and we are therefore interested in its interaction with the 2D Brillouin zone boundaries. Figure 13.16 shows the first two Brillouin zones of a 2D lattice and consider four possible values of N_f, resulting in four different values of E_f, and hence circles of corresponding diameters. As long as the circle is far from the Brillouin zone boundary, it does not interact with the boundary and remains undistorted. When the circle gets closer to the boundary, diffraction effects distort it. When the circle is slightly larger than the Brillouin zone, distorted sections of the circle appear inside the first Brillouin zone as well as in the second Brillouin zone.

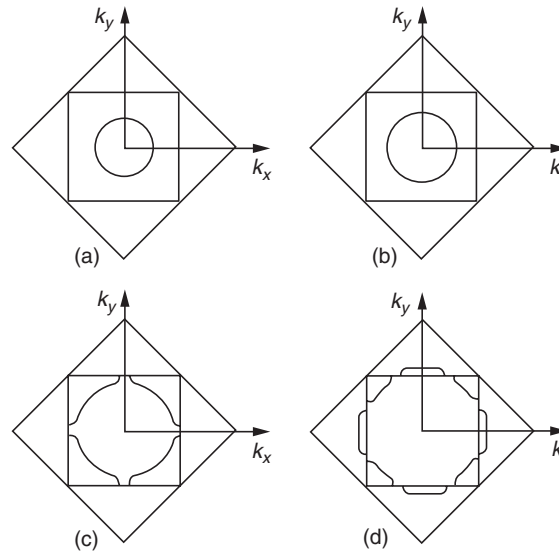


Figure 13.16 Interaction between allowed wave vectors of nearly free electrons and Brillouin zones of a 2D square lattice. The sub-parts (a), (b), (c), and (d) differ in the value of N_f being depicted. N_f in (a) < N_f in (b) < N_f in (c) < N_f in (d). The allowed wave vectors are contained within the first Brillouin zone in sub-parts (a), (b), and (c). In sub-part (d), the allowed wave vectors appear in the first as well as in the second Brillouin zones.

In two dimensions, as well as in three dimensions, it is possible to plot the reduced zone representation in just the manner it was accomplished in the 1D case. The figures get complicated but the concept is the same. By moving higher Brillouin zones by valid lattice vectors in *k*-space, the information corresponding to higher Brillouin zones can be represented in the first Brillouin zone itself. Figure 13.17 shows this for the 2D square lattice.

Finally, we can also consider an example of a material that has an FCC real lattice, and therefore a Brillouin zone that is the Wigner–Seitz cell of a BCC lattice. Spheres will indicate states of the same energy and based on the value of N_f, the largest possible sphere will indicate the E_f, which is the Fermi sphere or Fermi surface. Figure 13.18 shows a case where the sphere corresponding to the Fermi energy just touches the Brillouin zone in specific directions. If N_f were lower, the sphere would not touch the Brillouin zone at any location and if N_f were higher, the sphere would distort and extend into the second zone in specific directions.

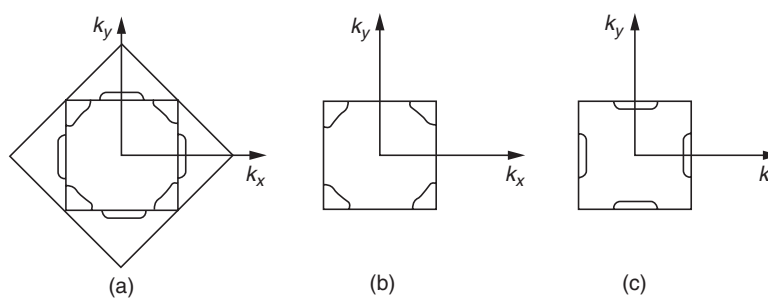


Figure 13.17 (a) Extended zone representation; (b) and (c) reduced zone representation.

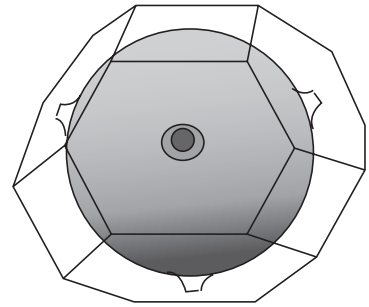


Figure 13.18 The interaction between the Brillouin zone of an FCC real lattice with the Fermi sphere. **Animation: "Brillouin zone–Fermi sphere interaction for FCC lattice" in DVD.**

We have therefore seen the interaction between the Brillouin zones and Fermi surface pictorially and how this results in the band structure of the materials. In Chapter 14, we will look at this interaction in an analytical manner.

Summary

1. A Wigner–Seitz cell, about a lattice point, is the region in space that is closest to that lattice point than to any other lattice point.
2. In the most general case, the 2D Wigner–Seitz cell will be a hexagon. Due to symmetry, in specific cases it reduces to a rectangle or a square based on the lattice chosen.
3. The first Brillouin zone is defined as the Wigner–Seitz primitive cell about a lattice point in reciprocal space.
4. Planes that are perpendicular bisectors of lines joining the origin to lattice points in reciprocal space are referred to as "Bragg planes."
5. The nth Brillouin zone is all the points between the $(n-1)$th Bragg plane and the nth Bragg plane in all directions.
6. The first Brillouin zone for a material that has a simple cubic structure in real space is the Wigner–Seitz cell corresponding to its reciprocal lattice, and is therefore also a cube.
7. Since a BCC real lattice is represented in reciprocal space by an FCC lattice, the Wigner–Seitz cell of the FCC lattice is the first Brillouin zone of the BCC real lattice.
8. Since an FCC real lattice is represented in reciprocal space by a BCC lattice, the Wigner–Seitz cell of the BCC lattice is the first Brillouin zone of the FCC real lattice.
9. With respect to the Ewald sphere corresponding to that specific wave vector, whenever a Bragg plane touches one end of a wave vector, the condition for diffraction is satisfied.
10. If we draw a consolidated plot that shows the wave vectors allowed for the free electrons, and the Brillouin zone boundaries corresponding to the structure of the material, to the same scale, then in this consolidated plot whenever the wave vectors touch the Brillouin zone boundaries, diffraction will occur.
11. Diffraction causes the E versus k relationship to distort in the vicinity of the Bragg planes (or Brillouin zone boundaries) for nearly free electrons. This distortion causes some of the energy levels to become forbidden to the nearly free electrons – which results in the presence of band gaps in the material.
12. The traveling waves of nearly free electrons undergo diffraction at the Brillouin zone

boundaries and result in standing waves – as a result energy gaps appear.

13. Bands are more commonly represented using flat-band diagrams, which mask important features related to k-space.
14. Bands can be more appropriately represented using E versus k diagrams in the extended zone scheme, repeated zone scheme, and the reduced zone scheme. These can be easily related to the corresponding flat-band diagram.
15. The location of the Fermi energy (E_f) and/or the nearest band gap above the Fermi energy decide whether the material is metallic, semiconducting or insulating.
16. For a given shape and size of a Brillouin zone, the value of N_f decides whether or not the Fermi surface will distort and hence the resultant band structure.

Practice Questions

Multiple Choice Questions

1. For a lattice corresponding to a real material, the Wigner–Seitz cell
   (a) Is a finite region in space
   (b) Is a half the size of the sample of the real material
   (c) Is quarter the size of the sample of the real material
   (d) Is an infinite region in space
2. Regions outside the Wigner–Seitz cell about a central point
   (a) Are closer to the central point than to any other point
   (b) Are closer to other points than to the central point
   (c) Belong to the Wigner–Seitz cell
   (d) Can also be placed in the Wigner–Seitz cell by appropriate compression
3. In 3D space, walls of the Brillouin zones are always
   (a) Bragg planes
   (b) Perpendicular to each other
   (c) Parallel to each other
   (d) Curved in shape
4. In 2D a Wigner–Seitz cell cannot have the following shape:
   (a) Square **(b)** Rectangle
   (c) Hexagon **(d)** Circle
5. Brillouin zones are
   (a) Defined in real space
   (b) Defined only when diffraction occurs
   (c) Defined in reciprocal space
   (d) Defined with respect to the Ewald sphere
6. For FCC lattice and BCC lattice, the Brillouin zones are Wigner–Seitz cells of
   (a) BCC and FCC, respectively
   (b) FCC and BCC, respectively
   (c) Simple cubic
   (d) Hexagonal
7. Brillouin zone boundaries are made of
   (a) Bragg planes
   (b) Ewald sphere
   (c) Convex surfaces
   (d) Concave surfaces
8. Condition for diffraction is satisfied when wave vector
   (a) Is equal to a reciprocal lattice vector
   (b) Is equal to a real lattice vector
   (c) Touches a Bragg plane
   (d) Is twice the length of a reciprocal lattice vector
9. For a 1D lattice of spacing a, successive Brillouin zone boundaries occur with a spacing of
   (a) $4\pi/a$ **(b)** π/a
   (c) $\pi/2a$ **(d)** $\pi/4a$
10. In comparison to the scale of the reciprocal lattice, the allowed wave vectors of nearly free electrons in a metal are
    (a) Spaced similarly apart as the reciprocal lattice
    (b) Spaced twice as apart as the reciprocal lattice

(c) Spaced very widely apart
(d) Spaced very closely, appearing almost continuous.

11. Given a Fermi surface and a Brillouin zone
(a) They have to necessarily interact
(b) They have to necessarily not interact
(c) Interaction depends on their relative sizes
(d) Interaction occurs only if both are curved surfaces

12. At Brillouin zone boundaries waves of electrons become
(a) Traveling waves
(b) Standing waves
(c) Square waves
(d) Triangular waves

13. When bands are represented using flat-band diagrams
(a) They do not show the band gap
(b) They do not show the valence band
(c) The show the band gap but not the extent of the band gap
(d) The show the extent of the band gap but do not show its variation in *k*-space

14. Whether the material is metallic, semiconducting or insulating depends
(a) Only on location of Brillouin zones
(b) Only on location of Brillouin zone boundaries
(c) Only on the Fermi energy
(d) On the position of the Fermi energy relative to the Brillouin zone boundaries

15. The flat band diagram can be related to
(a) Extended zone scheme representation
(b) Repeated zone scheme representation
(c) Reduced zone scheme representation
(d) All of the above

Short Answer Type Questions

1. What is symmetry?
2. What are Bravais lattices?
3. What are crystal systems?
4. What is a crystal structure?
5. Starting from a general 2D lattice that has a six-sided first Brillouin zone, how does the first Brillouin zone of a rectangular lattice become a four-sided figure?
6. Why must the reciprocal lattice and wave vectors be drawn to the same scale?
7. Identify the Wigner–Seitz cell for a primitive hexagonal lattice.
8. Why are Wigner–Seitz cells also called primitive cells?
9. What is the spacing between allowed energy levels within a band in Ge?
10. What is the band gap in Ge?
11. Why does diffraction result in forbidden energy levels?
12. What happens at the Brillouin zone boundaries that are significantly inside the Fermi surface?
13. Using the *E* versus *k* diagram for a 1D lattice, what can be said about the extent of bands as a function of energy?
14. Using the *E* versus *k* diagram for a 1D lattice, what can be said about the extent of band gaps as a function of energy?
15. Comment on the effective mass of the electrons at the Brillouin zone boundaries.

Answers

Multiple Choice Questions

1. (a)
2. (b)
3. (a)
4. (d)
5. (c)
6. (a)
7. (a)
8. (c)
9. (b)
10. (d)
11. (c)
12. (b)
13. (d)
14. (d)
15. (d)

Short Answer Type Questions

1. Although symmetry is a concept that many of us intuitively recognize, it is surprisingly difficult to put down in words. In general, the idea of symmetry is that if an object is taken and subject to specific operations such as rotation about an axis by a specific amount of degrees, and if in the process the object in its new orientation overlays exactly with what the object was before the operations were performed, then it is said to display symmetry relative to that operation.

2. There are 14 Bravais lattices in 3D space, and these indicate symmetry of lattice points including points that represent specific centers such as base center, face center and in center, in addition to a primitive lattice that only has lattice points in the corners.

3. There are seven crystal systems based on point groups. Point groups themselves represent symmetry.

4. The crystal structure of a material includes details specific to that material such as its lattice parameters in addition to the symmetry displayed by the material.

5. As the angles in the general 2D lattice approach 90°, two sides of the six-sided first Brillouin zone move toward the corner and progressively become smaller. In the limiting case of the lattice becoming rectangular, the two sides pass through the corner exactly and hence only four sides remain to define the first Brillouin zone.

6. Since diffraction phenomena are shown geometrically using the wave vectors and the reciprocal lattice, they have to necessarily be drawn to the same scale.

7. The Wigner–Seitz cell for a primitive hexagonal lattice is a hexagonally shaped prism. In any plane of lattice points, six vertical faces separate the region closest to a given lattice point from its six nearest neighbors. Two horizontal planes separate the region closest to a given lattice point from the lattice point directly above it and the lattice point directly below it. The result is the hexagonally shaped prism.

8. A Wigner–Seitz primitive cell highlights the information that there is one lattice point per unit cell.

9. The spacing between allowed energy levels in Ge is of the order of 10^{-10} eV.

10. The band gap of Ge is 0.67 eV.

11. Diffraction results in standing waves as opposed to traveling waves. This results in the corresponding energies not to be favorable for the electrons. Hence, a band gap appears.

12. At Brillouin zone boundaries that are significantly inside the Fermi surface, diffraction of the waves occurs and results in gaps in energy that are forbidden for the electrons. However, it is the energy gap that is in the vicinity of the Fermi surface, which is identified as the "band gap" associated with the sample.

13. In the E versus k diagram for a 1D lattice, the extent of the bands increases as a function of energy since the zone boundaries are at regular linear intervals, while the E versus k curve is increasing in a parabolic manner.

14. Similar to the answer above, for a 1D lattice, the extent of the band gaps increase as a function of energy, since the zone boundaries are at regular linear intervals, while the E versus k curve is increasing in a parabolic manner.

15. Since the electrons undergo diffraction at the Brillouin zone boundaries resulting in band gaps, the electrons behave as though they have infinite mass.

14 Bands, Band Gaps, Free Electron Approximation, and Tight Binding Approximation

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Draw and indicate the characteristics of the 1D lattice for which the calculation of allowed and forbidden energy levels is to be carried out.
- Solve the Schrödinger wave equation in one dimension in regions where $V = 0$ and $V = V_0$.
- Write the equations that represent the boundary conditions present in the system, including the impact of Bloch's condition for periodicity.
- Simplify the resulting determinant to arrive at an equation.
- Interpret the equation to identify the allowed energy levels and forbidden energy band gaps.
- Draw the flat-band structure corresponding to insulators, semiconductors, and metals.
- Differentiate between the starting positions and approaches taken by the "free electron approximation" as opposed to the "tight binding approximation."
- Draw energy versus inter-atomic separation diagram to illustrate the tight binding approximation approach.
- Draw the flat-band diagrams corresponding to the energy versus inter-atomic separation diagram.
- Draw and indicate the variation of the band structure of a material as a function of pressure applied on the material.
- Indicate the types of materials that are more easily understood using the "free electron approximation," and those that are more easily understood using the "tight binding approximation."

14.1 Introduction

In Chapter 13, we looked at the origin of bands pictorially. In this chapter we will look at an analytical approach to describing the origin of bands. We will subject the wave vectors of electrons to the periodic potential of the lattice and study the interaction that occurs. Based on this analysis, it can be seen that specific energy levels are allowed, which form the allowed energy bands of the material. It will also be seen that specific energy levels become forbidden for the material, resulting in band gaps in the material.

14.2 Analytical Approach to The Band Structure – Free Electron Approximation



Let us now examine the origin of band gaps in an analytical manner. This approach is credited to Kramers and Kronig, and Penney and is shown in Fig. 14.1.

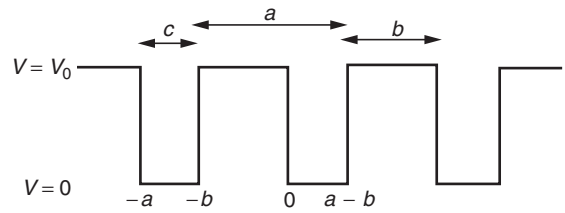


Figure 14.1 The dimensions of the potential wells, and the potentials involved, for the purpose of determining the band gaps corresponding to a solid.

The Schrödinger wave equation in one dimension is as follows:

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+V(x)\right)\psi(x)=E\psi(x)$$

This equation is solved for regions where $V=0$ and $V=V_0$, and the results are compared. In the equations that follow, it is important to note the subscripts to keep track of the potential in the region where the calculation is being done. As indicated above, typically at the boundary, on one side $V=0$ and on the other side $V=V_0$. So calculations in these two regions will be compared with each other in many of the equations that follow.

When $V=0$,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}=E\psi(x)$$

or

$$\frac{d^2\psi(x)}{dx^2}=\frac{-2mE}{\hbar^2}\psi(x)$$

which can be solved by examination and we get the following solution:

$$\psi(x)=A\exp(i\beta x)+B\exp(-i\beta x)$$

where

$$\beta=\frac{\sqrt{2mE}}{\hbar}$$

and A and B are constants that need to be determined. Similarly, when $V=V_0$,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+V_0\psi(x)=E\psi(x)$$

or

$$\frac{d^2\psi(x)}{dx^2}=\frac{2m(V_0-E)}{\hbar^2}\psi(x)$$

which can be solved by examination and we get the following solution:

$$\psi(x) = C\exp(\alpha x) + D\exp(-\alpha x)$$

where

$$\alpha = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

and C and D are the constants that need to be determined. For the solutions to be reasonable we require $\psi(x)$ to be continuous. Therefore at $x = 0$ we require

$$\psi_0(0) = \psi_V(0)$$

and

$$\frac{d\psi_0(0)}{dx} = \frac{d\psi_V(0)}{dx}$$

Periodicity has been mathematically expressed as follows by Bloch:

$$\psi(x+a) = \psi(x)\exp(ika)$$

Therefore comparing the two positions $x = -b$ and $x = a - b$, which are related by the periodicity of the lattice, we have

$$\psi_0(a-b) = \psi_V(-b)\exp(ika)$$

or

$$\psi_V(-b) = \exp(-ika)\psi_0(a-b)$$

and

$$\frac{d\psi_V(-b)}{dx} = \exp(-ika)\frac{d\psi_0(a-b)}{dx}$$

Therefore, we have the following four equations:

$$\psi_0(0) = \psi_V(0)$$

$$\frac{d\psi_0(0)}{dx} = \frac{d\psi_V(0)}{dx}$$

$$\psi_V(-b) = \exp(-ika)\psi_0(a-b)$$

$$\frac{d\psi_V(-b)}{dx} = \exp(-ika)\frac{d\psi_0(a-b)}{dx}$$

Based on the solutions at $V = 0$ and $V = V_0$, the above four equations result in the following:

$$A + B = C + D$$
$$i\beta(A - B) = \alpha(C - D)$$
$$\exp(-ika)[A\exp\{i\beta(a-b)\} + B\exp\{-i\beta(a-b)\}] = C\exp(-\alpha b) + D\exp(\alpha b)$$
$$i\beta[\exp(-ika)][A\exp\{i\beta(a-b)\} - B\exp\{-i\beta(a-b)\}] = \alpha C\exp(-\alpha b) - \alpha D\exp(\alpha b)$$

Therefore, we have four equations in *A*, *B*, *C,* and *D*. These can be solved if the determinant corresponding to their coefficients is equated to zero, that is,

$$\begin{vmatrix} 1 & 1 & -1 & -1 \\ i\beta & -i\beta & -\alpha & \alpha \\ \exp(-ika)\exp i\beta(a-b) & \exp(-ika)\exp[-i\beta(a-b)] & -\exp(-\alpha b) & -\exp(\alpha b) \\ i\beta\exp(-ika)\exp i\beta(a-b) & -i\beta\exp(-ika)\exp[-i\beta(a-b)] & -\alpha\exp(-\alpha b) & \alpha\exp(\alpha b) \end{vmatrix}$$

The determinant above should evaluate or equate to zero. Since

$$\cosh x = \frac{1}{2}(e^x + e^{-x})$$

and

$$\sinh x = \frac{1}{2}(e^x - e^{-x})$$

We can rewrite the above determinant and designate the columns as C1, C2, C3, and C4.

C1 C2 C3 C4

$$\begin{vmatrix} 1 & 1 & -1 & -1 \\ i\beta & -i\beta & -\alpha & \alpha \\ \exp(-ika)\exp i\beta(a-b) & \exp(-ika)\exp[-i\beta(a-b)] & -\exp(-\alpha b) & -\exp(\alpha b) \\ i\beta\exp(-ika)\exp i\beta(a-b) & -i\beta\exp(-ika)\exp[-i\beta(a-b)] & -\alpha\exp(-\alpha b) & \alpha\exp(\alpha b) \end{vmatrix}$$

If we designate the columns of the determinant as C1, C2, C3, and C4 and first carry out the operations C1 − C2, C4 − C3, and follow that with the operation C3 + C2, we get

$$\begin{vmatrix} 0 & 1 & 0 & 0 \\ 2i\beta & -i\beta & -\alpha - i\beta & 2\alpha \\ 2i\exp(-ika)\sin\beta(a-b) & \exp(-ika)\exp[-i\beta(a-b)] & \exp(-ika)\exp[-i\beta(a-b)] - \exp(-\alpha b) & -2\sinh(\alpha b) \\ 2i\beta\exp(-ika)\cos\beta(a-b) & -i\beta\exp(-ika)\exp[-i\beta(a-b)] & -i\beta\exp(-ika)\exp[-i\beta(a-b)] - \alpha\exp(-\alpha b) & +2\alpha\cosh\alpha b \end{vmatrix}$$

Expanding the determinant using row 1, or obtaining its Laplace expansion, we see that the determinant evaluates to $A_{12}C_{12}$, where C_{12} is the cofactor of A_{12}. Since A_{12} is 1, we can simplify our problem to evaluating C_{12} which is

$$(-1)^{1+2}\begin{vmatrix} 2i\beta & -\alpha - i\beta & 2\alpha \\ 2i\exp(-ika)\sin\beta(a-b) & \exp(-ika)\exp[-i\beta(a-b)] - \exp(-\alpha b) & -2\sinh\alpha b \\ 2i\beta\exp(-ika)\cos\beta(a-b) & -i\beta\exp(-ika)\exp[-i\beta(a-b)] - \alpha\exp(-\alpha b) & 2\alpha\cosh\alpha b \end{vmatrix}$$

Since the determinant must evaluate to zero, we can ignore the constant $(-1)^3$ as well as the multiplicative factor 2 in column 1 as well as in column 3. Therefore, the determinant now becomes

$$\begin{vmatrix} i\beta & -\alpha - i\beta & \alpha \\ i\exp(-ika)\sin\beta(a-b) & \exp(-ika)\exp[-i\beta(a-b)] - \exp(-\alpha b) & -\sinh\alpha b \\ i\beta\exp(-ika)\cos\beta(a-b) & -i\beta\exp(-ika)\exp[-i\beta(a-b)] - \alpha\exp(-\alpha b) & \alpha\cosh\alpha b \end{vmatrix}$$

Adding columns 1 and 3 to column 2, we get

$$\begin{vmatrix} i\beta & 0 & \alpha \\ i\exp(-ika)\sin\beta(a-b) & \exp(-ika)\cos\beta(a-b) - \cos\alpha b & -\sinh\alpha b \\ i\beta\exp(-ika)\cos\beta(a-b) & -\beta\exp(-ika)\sin\beta(a-b) + \alpha\sinh\alpha b & \alpha\cosh\alpha b \end{vmatrix}$$

We can now evaluate this determinant using the Laplace expansion, $A_{11}C_{11} + A_{13}C_{13}$, and set it equal to 0. Expanding using the first row, the determinant evaluates to

$$\begin{aligned} &i\beta[\exp(-ika)\cos\beta(a-b)\alpha\cosh\alpha b - \alpha\cosh^2\alpha b - \beta\exp(-ika)\sin\beta(a-b)\sinh\alpha b + \alpha\sinh^2\alpha\beta] \\ &+\alpha[-i\beta\exp(-2ika)\sin^2\beta(a-b) + i\exp(-ika)\sin\beta(a-b)\alpha\sinh\alpha b - i\beta\exp(-2ika)\cos^2\beta(a-b) \\ &+i\beta\exp(-ika)\cos\beta(a-b)\cosh\alpha b] \end{aligned}$$

$[\text{since } \sinh^2\alpha b - \cosh^2\alpha b = -1]$. Therefore, determinant is given by

$$\begin{aligned} &i\alpha\beta\exp(-ika)\cos\beta(a-b)\cosh\alpha b - i\alpha\beta - i\beta^2\exp(-ika)\sin\beta(a-b)\sinh\alpha b - i\alpha\beta\exp(-2ika) \\ &+i\alpha^2\exp(-ika)\sin\beta(a-b)\sinh\alpha b + i\alpha\beta\exp(-ika)\cos\beta(a-b)\cosh\alpha b \end{aligned}$$

After removing the common constant $i\exp(-ika)$, we get determinant to evaluate to

$$\begin{aligned} &\alpha\beta\cos\beta(a-b)\cosh\alpha b - \alpha\beta\exp(ika) - \beta^2\sin\beta(a-b)\sinh\alpha b - \alpha\beta\exp(-ika) \\ &+\alpha^2\sin\beta(a-b)\sin h\alpha b + \alpha\beta\cos\beta(a-b)\cosh\alpha b \end{aligned}$$

Since the determinant evaluates to zero, after rearranging to group common terms we get

$$2\alpha\beta\cos\beta(a-b)\cosh\alpha b + (\alpha^2 - \beta^2)\sin\beta(a-b)\sinh\alpha b - \alpha\beta[\exp(ika) + \exp(-ika)] = 0$$

Dividing throughout by $2\alpha\beta$, we get

$$\cos\beta(a-b)\cosh\alpha b + \left(\frac{\alpha^2-\beta^2}{2\alpha\beta}\right)\sin\beta(a-b)\sinh\alpha b - \left[\frac{\exp(ika)+\exp(-ika)}{2}\right] = 0$$

Since

$$\frac{e^{ika}+e^{-ika}}{2} = \cos ka$$

therefore,

$$\frac{\alpha^2-\beta^2}{2\alpha\beta}\sin\beta(a-b)\sinh\alpha b + \cos\beta(a-b)\cosh\alpha b = \cos ka$$

On examining the parameters at the left-hand side of the equation, we find

$$\alpha = \frac{\sqrt{2m(V_0-E)}}{\hbar}$$

and

$$\beta = \frac{\sqrt{2mE}}{\hbar}$$

For both α and β, the parameter that can be varied is E, the energy of the electron. The other parameters, a and b are constant once a crystal structure is chosen, and specific atoms are chosen to occupy the lattice sites. So effectively, once a material is chosen, only E can be varied to obtain various values for the expression on the left-hand side of the equation.

While the right-hand side of the equation is restricted to assume values in the window ±1, the left-hand side of the equation can evaluate to values outside this window. This situation is interpreted to mean that those energies for which the left-hand side of the equation evaluates to values outside the ±1 window are energies that are forbidden for the system.

Once specific details of the system are known, we therefore see how it is possible to evaluate the allowed energy bands and forbidden energy band gaps of a system.

14.3 Band Structure and Types of Materials



We have discussed the origin of energy bands in solids both pictorially as well as analytically using the Schrödinger wave equation. While the flat-band diagrams that can be drawn corresponding to the band structure of the solid do have some utility, the E versus k relationship is able to explain material phenomena that cannot be explained using the flat-band diagrams. In Chapter 15, we will explore this capability of the E versus k diagram in greater detail.

We will now consider the utility of band diagrams and also examine two different starting points from which band structure can be explained. Consider the band structures shown in Fig. 14.2.

The band structure of an insulator consists of a fully filled valence band and an empty conduction band at higher energy. The band gap is typically greater than 2 eV. The band gap represents the minimum

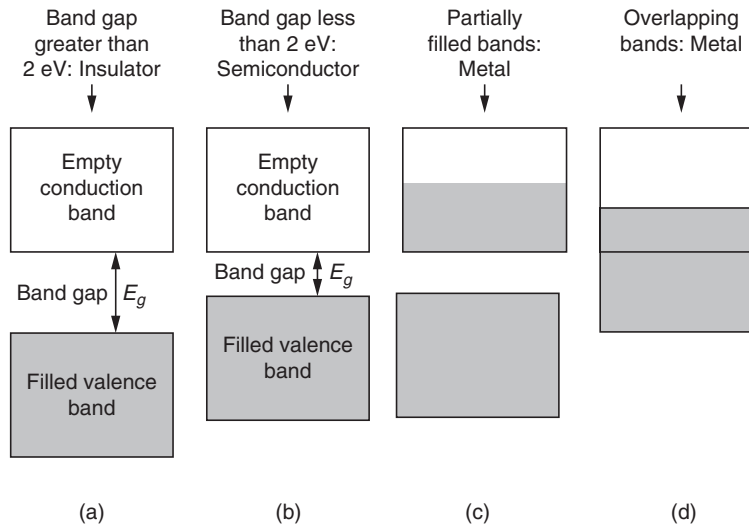


Figure 14.2 Band structure: (a) Band structure of an insulator; (b) band structure of a semiconductor; (c) and (d) band structures associated with a metal.

amount of energy that is required to enable electrons to move from the valence band to the conduction band. Once in the conduction band, the electrons are able to participate in the process of conduction.

The band structure of a semiconductor is similar to an insulator, except that the band gap is typically less than 2 eV.

Metallic systems have two possible band structures. Either they have a half-filled band as the highest energy band or they have an empty band and a full band overlapping in energy. In either case, there are vacant energy levels immediately above the occupied energy levels and hence electrons very easily find vacant states and participate in electronic conduction.

It is important to note that, by convention, the Fermi energy for insulators and intrinsic semiconductors is assumed to be in the middle of the band gap, half way between the valence band and the conduction band. This is counter-intuitive since by definition a bandgap implies that there are no allowed energy levels in that range of energy. However, this is the convention used since it helps explain material behavior. For extrinsic semiconductors, which are doped, the Fermi energy of the parent material moves up or down in the band gap, and now coincides with energy levels of the dopants.

In our analysis through all the earlier chapters, we have adopted a picture of the solid that says that there are ionic cores at fixed lattice locations and that there is a free electron gas enveloping these ionic cores. In other words, we have assumed that the solid already exists and the ionic cores are "tightly bound" to their lattice locations while the *electrons are "free"* to run through the extent of the solid. This is called the "free electron approximation."

14.4 Band Structure – The Tight Binding Approximation

There is another approach to modeling materials which starts from a diametrically opposite position. In this approach, we do not have a solid to begin with. Instead the atoms are initially independent and are brought together to build the solid. All the electrons are bound to their respective individual atoms initially. In this

case, the atoms are free initially, while the *electrons are tightly bound to begin with*. In view of the focus on the electronic properties of the materials, this approach is referred to as the "tight binding approximation" – highlighting the status of the electrons at the start of the model.

Figure 14.3 shows how the tight binding approximation builds the band structure of the solid. The tight binding approximation considers the solid as follows: When the atoms are far apart, all the bound electrons associated with each atom have fixed energy levels occupied by them. Assuming that we are building the solid using atoms of the same element, the energy levels occupied by the electrons in each atom will be identical. As we bring the atoms closer to each other to form the solid, as long as the inter-atomic separation is large the electrons will still maintain their original energy levels. When the atoms get close enough that the outer shell electrons begin to overlap with each other, the energy levels of these outer shell electrons are forced to split into new energy levels above and below the energy level of the same electrons when they belonged to individual well-separated atoms. This splitting of energy levels occurs because electrons obey Pauli's exclusion principle (just as the individual energy levels could only hold limited numbers of electrons each in the free electron approximation as well). Initially only the outer shell electrons overlap, therefore only their levels split and the inner shell electrons still maintain their individual atom-based energy levels. If the inter-atomic separation keeps decreasing even further, progressively more of the inner shell electron levels will overlap and hence also split. At each energy level, the level will split to enough new energy levels so as to accommodate the electrons present in the original level for all the atoms in the solid taken together. So, for example, if 100 atoms come together and there is one electron in the outer shell occupying one state, the solid will split this energy level to a hundred energy levels in the vicinity of this original level to accommodate those 100 outer shell electrons corresponding to the combined solid.

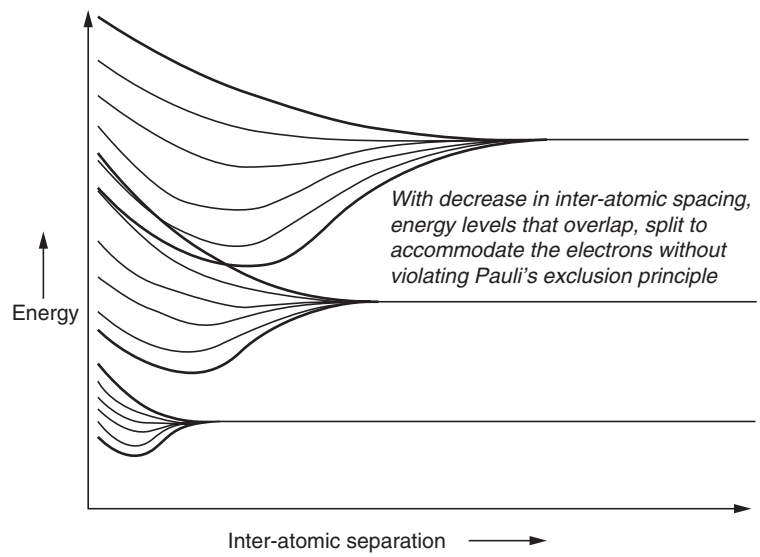


Figure 14.3 An illustration of the tight binding approximation to explain the properties of solids.

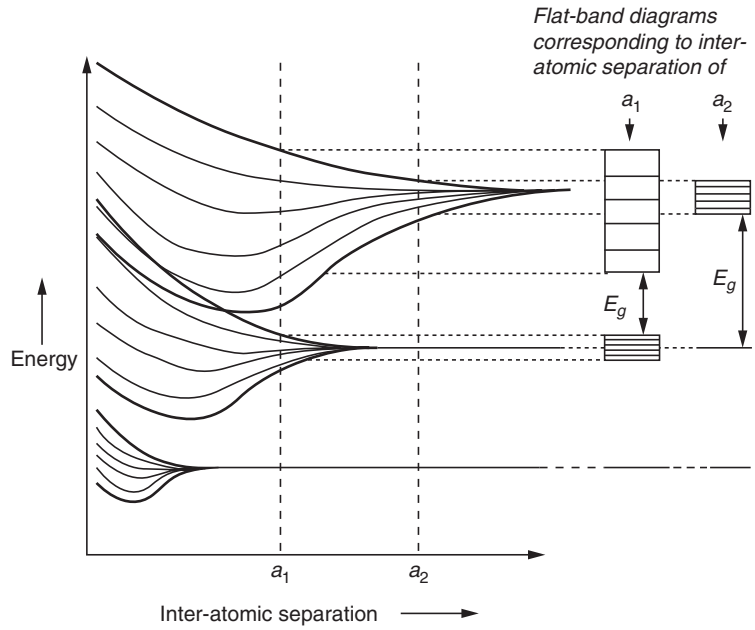


Figure 14.4 Different flat-band structures corresponding to different inter-atomic spacings chosen in a material. The existence of the bands, width of each of the bands as well as the extent of the band gaps change based on our selection of the equilibrium inter-atomic spacing.
Animation: "Inter-atomic separation and band structure" in DVD.

Figure 14.3 shows what all are the possibilities when we vary the inter-atomic spacing. In reality a given material will have a specific inter-atomic spacing, which is the equilibrium spacing of that material under the experimental conditions experienced. Corresponding to this equilibrium spacing, a specific band structure consistent with the tight binding approximation will be displayed by the material. Figure 14.4 shows the different flat-band structures that a material might display based on the choice of the equilibrium inter-atomic spacing.

In this figure it is assumed that six atoms are being brought together, therefore each individual level splits into six levels when the energy levels overlap.

14.5 Effect of Pressure on Band Structure

The information in Fig. 14.4 is of interest because we can experimentally manipulate the equilibrium inter-atomic spacing observed in the material. We can reduce the spacing by applying pressure on the material or increase the spacing by applying a tensile stress on the material, which is less than the yield point of the material. Therefore, with the same material we can get different band structures, and that is of technological interest.

Since the inter-atomic spacing, and hence band structure, can be changed by applying pressure, Fig. 14.5 shows the variation of the extent of the band and the extent of the band gap as a function of pressure.

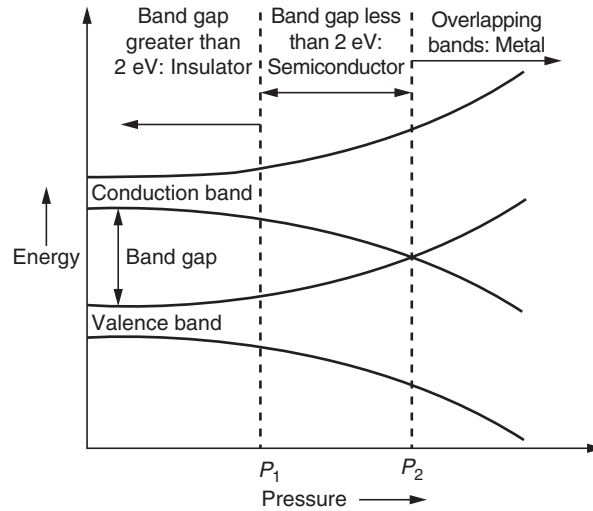


Figure 14.5 Impact of pressure on the band structure of the material.
Animation: "Changes in band structure due to pressure" in DVD.

Figure 14.5 shows that when the pressure applied is less than P_1, the material has a large band gap and is hence an insulator. When $P_1 < P < P_2$, the band gap is less than 2 eV and the material behaves like a semiconductor. When $P > P_2$, the bands overlap and the material behaves like a metal.

We must note the two aspects associated with Fig. 14.5. The figure makes us aware that material properties are sensitive to pressure, and this includes electronic properties such as the band structure. Therefore, while the material may be synthesized and tested in the laboratory under one set of conditions, if it used in another set of conditions, its band structure and hence behavior, may be very different. While we tend to be more aware of the sensitivity of material properties to changes in temperature, we often take pressure being equal to 1 atm, as granted. Depending on the end use, which could be in space where the pressure is very low – essentially vacuum, or on other planets where the pressure can be high, or in deep seas where also the pressure can be very high, the pressure experienced by the material can definitely be significantly different from the ambient 1 atm. So, the effect of pressure should not be summarily ignored.

At the same time, we must also note that the sensitivity of material properties to changes in pressure is rather weak. In other words, the pressure will typically have to change over several orders of magnitude to cause significant impact on material properties. For example, consistent with Fig. 14.5, hydrogen gas is predicted to become a solid that is metallic at high pressures. But the pressure at which this change to metallic hydrogen is expected to occur, is of the order of a million atmospheres, or six orders of magnitude higher than the pressure experienced under ambient conditions. Therefore, we are often not terribly in error in ignoring the effects of pressure, especially on solids.

14.6 Comparison of Free Electron and Tight Binding Approximations

We have therefore discussed two approaches: The free electron approximation and tight binding approximation. In principle, the two approaches must give the same results since they are modeling the same materials. As it turns out, largely they are consistent with each other and with the experimental data. However, in view of the starting points of these two approaches, the free electron approximation lends itself more easily

to the treatment of metallic systems where the state of the material is consistent with the picture that the free electron approximation tries to address. The tight binding approximation is typically more consistent with the state of the material in the case of insulators, so it is better suited for modeling insulators.

We therefore note that in the free electron approximation, Brillouin zone boundaries are an important factor in determining the band structure. Since the Brillouin zone boundaries occur at different distances along different directions in crystalline solids, the free electron approximation is able to indicate the cause for anisotropy in the properties of crystalline solids.

In the tight binding approximation, the inter-atomic spacing is a critical parameter in determining the band structure of the material. Since the inter-atomic spacing in crystalline solids varies based on the crystallographic direction chosen, the tight binding approximation is also able to explain the cause for anisotropy in the properties of crystalline solids.

Therefore, both the approaches can explain anisotropy.

14.7 Conventions Regarding Representing Band Diagrams

Band diagrams are often represented in 2D even though the associated phenomena occur in 3D. To enable representation in 2D, band information along specific high symmetry directions of the crystal are determined and plotted together in a 2D plot. Directions of high symmetry are shown along the X-axis, and the corresponding allowed energy levels plotted along the vertical Y-axis. In order to compress the 3D information into 2D, specific high symmetry directions are only shown in these plots.

By convention, the centre of the Brillouin zone is designated as Γ and points as well as lines inside the Brillouin zone are designated using upper-case Greek alphabets, while points on the surface of the Brillouin zone are designated using upper-case Roman alphabets. Band diagrams indicate these points and directions along the X-axis, while plotting energy along the Y-axis. The allowed E vs. k relations are therefore displayed on the plot. Table 14.1 indicates **a few** of the high symmetry directions for the 14 Bravais Lattices. Texts on group theory have elaborate discussions on this and vastly more comprehensive tables. Several research publications also highlight specific symmetry directions.

Table 14.1 A few of the high symmetry directions for the 14 Bravais Lattices

S. No.	*Real Lattice*	*A Few Points and Directions of High Symmetry in the Brillouin Zone of Each Lattice, and their Designation in Band Diagrams. (Starting point, ending point, direction, and designation)*	*Examples of Elements with the Structure*
1.	Cubic Simple	000 to $0\ \frac{1}{2}\ 0$; [010]; Γ Δ X	α-Po
		000 to $\frac{1}{2}\ \frac{1}{2}\ \frac{1}{2}$; [111]; Γ Λ R	
		000 to $\frac{1}{2}\ \frac{1}{2}\ 0$; [110]; Γ Σ M	
2.	Cubic Face Centred	000 to $\frac{1}{2}\ 0\ \frac{1}{2}$; [101]; Γ Δ X	γ-Fe, Si, GaAs
		000 to $\frac{1}{2}\ \frac{1}{2}\ \frac{1}{2}$; [111]; Γ Λ L	

(*Continued*)

Table 14.1 (Continued)

S. No.	*Real Lattice*	*A Few Points and Directions of High Symmetry in the Brillouin Zone of Each Lattice, and their Designation in Band Diagrams. (Starting point, ending point , direction, and designation)*		*Examples of Elements with the Structure*
3.	Cubic Body Centred	000 to $\frac{1}{2}\ \frac{\bar{1}}{2}\ \frac{1}{2}$;	$[1\bar{1}1]$; Γ Δ H	$\alpha-$Fe
		000 to $\frac{1}{4}\ \frac{1}{4}\ \frac{1}{4}$;	[111]; Γ Λ P	
		000 to $00\frac{1}{2}$;	[001]; Γ Σ N	
4.	Hexagonal	000 to $0\frac{1}{2}0$;	[010]; Γ Σ M	C (Graphite)
		000 to $00\frac{1}{2}$;	[001]; Γ Δ A	
5.	Trigonal	000 to $0\frac{1}{2}\ \frac{\bar{1}}{2}$;	$[01\bar{1}]$; Γ Σ F	Sm, $\alpha-$B
		000 to $\frac{1}{2}\ \frac{1}{2}\ \frac{1}{2}$;	[111]; Γ Λ Z	
6.	Tetragonal Simple	000 to $0\frac{1}{2}0$;	[010]; Γ Δ X	$BaTiO_3$
		000 to $00\frac{1}{2}$;	[001]; Γ Λ Z	
		000 to $\frac{1}{2}\ \frac{1}{2}0$;	[110]; Γ Σ M	
7.	Tetragonal Body Centred	000 to $00\frac{1}{2}$;	[001]; Γ Δ X	In
		000 to $\frac{\bar{1}}{2}\ \frac{1}{2}\ \frac{1}{2}$;	$[\bar{1}11]$; Γ Σ Z	
8.	Orthorhombic Simple	000 to $0\frac{1}{2}0$;	[010]; Γ Σ X	Np
		000 to $\frac{1}{2}00$;	[100]; Γ Δ Y	
		000 to $00\frac{1}{2}$;	[001]; Γ Λ Z	

(*Continued*)

Table 14.1 (Continued)

S. No.	*Real Lattice*	*A Few Points and Directions of High Symmetry in the Brillouin Zone of Each Lattice, and their Designation in Band Diagrams. (Starting point, ending point , direction, and designation)*		*Examples of Elements with the Structure*
9.	Orthorhombic Body Centred	000 to $\frac{1}{2}\ \frac{\bar{1}}{2}\ \frac{1}{2}$;	[1 $\bar{1}$ 1]; Γ Σ X	Li_2NiO_2
10.	Orthorhombic Base Centred	000 to $\frac{1}{2}\ \frac{1}{2}$ 0;	[110]; Γ Σ Y	Ga
		000 to 00$\frac{1}{2}$;	[001]; Γ Λ Z	
11.	Orthorhombic Face Centred	000 to 0$\frac{1}{2}\ \frac{\bar{1}}{2}$;	[01 $\bar{1}$]; Γ Δ Y	S
		000 to $\frac{1}{2}\ \frac{1}{2}$ 0;	[110]; Γ Λ Z	
12.	Monoclinic Simple	000 to 00$\frac{1}{2}$;	[001]; Γ Λ Z	Se
13.	Monoclinic Base Centred	000 to 0$\frac{1}{2}\ \frac{1}{2}$;	[011]; Γ Λ Z	Bi
14.	Triclinic	000 to 00$\frac{1}{2}$;	[001]; Γ G	P, $CdSiO_3$

In Chapter 15, we will discuss semiconductors and examine how the theories we have developed so far help us understand the behavior of semiconductors.

Summary

1. An analytical approach to determine allowed bands and forbidden band gaps, as proposed by Kramer, has been discussed.
2. The approach results in an equation that shows acceptable behavior when the energy E has certain values. These are the energies corresponding to the allowed bands.
3. For some values of the energy E, the equation becomes unacceptable with the left-hand side exceeding values allowed by the right-hand side. These values of energy are interpreted to correspond to the forbidden band gaps in the system.
4. The band structure of an insulator has a band gap greater than 2 eV. The band structure of a semiconductor is similar to that of an insulator, except that the band gap is typically less than 2 eV. Metallic systems have two possible band structures – either they have a half-filled band as the highest energy band or an empty band and a full band overlapping in energy.
5. The Fermi energy for insulators and intrinsic semiconductors is in the middle of the band gap, half way between the valence band and conduction band.

6. The free electron approximation and tight binding approximation are models for solids that start from diametrically opposite starting points. The free electron model starts with the solid already present and containing a free electron cloud, while the tight binding model starts with individual atoms and their tightly bound electrons and builds the solid by bringing those individual atoms together.
7. In the tight binding model, as the atoms come closer, splitting of overlapping energy levels occurs because electrons obey Pauli's exclusion principle.
8. Band structure depends on the equilibrium inter-atomic spacing.
9. Inter-atomic spacing, and hence band structure can be changed by applying pressure, however the sensitivity of material properties to changes in pressure is rather weak.
10. Free electron approximation lends itself more easily to the treatment of metallic systems.
11. The tight binding approximation is better suited for modeling insulators.

Practice Questions

Multiple Choice Questions

1. In a crystalline material, potential wells
   (a) Are infinitely deep
   (b) Are finitely deep
   (c) Can be ignored
   (d) Do not exist
2. In a crystalline material potential wells are
   (a) Uniformly spaced, with the same spacing in all crystal directions
   (b) Uniformly spaced, with possibly a different spacing for each different crystal direction
   (c) Non uniformly spaced in any given direction
   (d) Uniformly spaced and arranged in a circular manner around an origin
3. In an amorphous material, potential wells are
   (a) Approximately uniformly spaced, with approximately same spacing regardless of direction
   (b) Exactly uniformly spaced, with possibly a different spacing for each different direction
   (c) Uniformly spaced and arranged in a circular manner around an origin
   (d) Uniformly spaced and arranged in a rectangular manner around an origin
4. When different metals are compared
   (a) The depth of their potential wells can differ
   (b) The spacing of their potential wells can differ
   (c) Width of their potential wells can differ
   (d) All of the above
5. In the analytical approach used in this chapter, an expression equating to cos *ka* is obtained. Based on the value of this expression it is interpreted that
   (a) Bands exist when the expression evaluates to within ±1
   (b) Band gaps exist when the expression evaluates to within ±1
   (c) Bands exist when the expression evaluates to outside of ±1
   (d) Band gaps exist when the expression evaluates to 0
6. ZrO_2 has a band gap of 5 eV. It is a
   (a) Metal **(b)** Semiconductor
   (c) Insulator **(d)** Metallic Alloy
7. Fermi energy for an intrinsic semiconductors is
   (a) At the top of the valence band
   (b) At the bottom of the conduction band
   (c) Half way between the valence band and conduction band
   (d) Undefined

8. In the tight binding approximation, as a starting point
- **(a)** Electrons are free and atoms are bound to their crystallographic locations
- **(b)** Atoms are free and electrons are bound to their respective atoms
- **(c)** Both atoms as well as electrons are free
- **(d)** Both atoms as well as electrons are bound

9. In the free electron approximation, as a starting point
- **(a)** Electrons are free and atoms are bound to their crystallographic locations
- **(b)** Atoms are free and electrons are bound to their respective atoms
- **(c)** Both atoms as well as electrons are free
- **(d)** Both atoms as well as electrons are bound

10. On increasing pressure
- **(a)** Metals will tend to become semiconductors
- **(b)** Semiconductors will tend to become insulators
- **(c)** Metals will tend to become insulators
- **(d)** Semiconductors will tend to become metals

Short Answer Type Questions

1. What are the dimensions and shapes of the potential wells?

2. How do potential wells differ between materials with different forms of bonding?

3. Comment on the probability of finding the electron in regions outside the potential well.

4. What happens if the potential well is made narrow and deep?

5. What is the necessary condition for a set of simultaneous equations to result in a non-trivial solution?

6. What is strained silicon?

7. Do all the energy levels split into bands in the tight binding approximation?

8. How does the band structure of graphite support electronic conductivity?

9. Do glasses have a band structure?

10. What can be expected of the optical properties of insulators when subject to increasing pressures?

Answers

Multiple Choice Questions

1. (b)	**4.** (d)	**7.** (c)	**9.** (a)
2. (b)	**5.** (a)	**8.** (b)	**10.** (d)
3. (a)	**6.** (c)		

Short Answer Type Questions

1. Potential wells in reality get narrower and deeper as the ionic core is approached. Square potential wells are an approximation of the same.

2. Materials with stronger bond will have deeper potential wells, while those with weak bonds will have shallow potential wells.

3. Since the potential wells are finite, there is a non-zero probability of finding the electrons in regions outside the potential well, but this rapidly decays further away from the potential well.

4. The band gaps will widen.

5. For a set of simultaneous equations to result in a non-trivial solution, the determinant of the coefficients should evaluate to zero.
6. Strained silicon is stretched to change the interatomic spacing, which changes its band gap.
7. No, only energy levels that overlap split into bands in the tight binding approximation.
8. Graphite has an overlapping indirect band gap.
9. Yes, glasses and amorphous solids may also display band structure.
10. Since the band gap decreases with increasing pressure, clear transparent insulators can be expected to become colored and eventually opaque with increasing pressure.

15 Material Phenomena Explained using Theories Developed

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Draw the flat-band diagram of a semiconductor.
- Define and give examples of intrinsic elemental semiconductors.
- Define and give examples of intrinsic compound semiconductors.
- Define and give examples of extrinsic semiconductors.
- Write and utilize the equation for the conductivity of a semiconductor.
- Draw the flat band diagram and identify the location of the Fermi energy for intrinsic, *p*-type extrinsic, and *n*-type extrinsic semiconductors.
- Schematically plot the charge carrier concentration of a semiconductor as a function of temperature, for intrinsic and extrinsic semiconductors, and explain the features seen in the plot.
- Indicate how semiconductors interact with light incident on them.
- Indicate the difference between a flat-band diagram and an E versus k diagram.
- Differentiate between direct band-gap semiconductors and indirect band-gap semiconductors and explain the significance of the difference with respect to potential use of the semiconductors.
- Describe the features of phonons.
- Distinguish between optically active and acoustically active phonons.
- Describe the role played by phonons in optical properties of indirect band gap semiconductors.
- Determine the magnetic field generated by a coil carrying electricity.
- Write and utilize the equation for magnetic induction and other common magnetic parameters.
- Distinguish between ferromagnetic materials, paramagnetic materials and diamagnetic materials on the basis of χ.
- Explain the phenomenon of magnetism on the basis of E versus $N(E)$ curve.
- Describe the phenomenon of electron compounds or Hume Rothery phases and give examples.
- Explain the formation of electron compounds using $N(E)$ versus E diagrams.

15.1 Introduction

In Chapter 14, we have looked at the free-electron approximation and the tight-binding approximation. We found that both these approaches are able to explain anisotropy in material properties. They are, therefore, quite complete as models for materials. In this chapter we will examine the utility of the theories we have developed in explaining details of various material properties and phenomena such as the optical properties of semiconductors, the role of phonons in semiconductors, magnetism, and formation of electron compounds or Hume Rothery phases.

15.2 Semiconductors



In this chapter, we will look at semiconductors and we will also see the utility of the E versus k diagrams, especially the ability of the E versus k diagrams to explain material phenomena better than the flat-band diagrams. We will begin by looking at semiconductors in general.

Semiconductors are a class of materials that display significantly less conductivity compared to metals. However, their conductivity can be controlled and manipulated in many technically interesting ways and hence useful devices can be made using these materials. They typically have a filled valence band and an empty conduction band with a band gap of less than 2 eV.

In semiconductors, with the availability of adequate amounts of energy, electrons will make transition from the filled valence band to the empty conduction band. When such a transition occurs, the electron leaves behind an empty state in the valence band. Such empty states, which originally contained electrons, are called "holes" and these are also treated as charge carriers with a positive charge, numerically equivalent to that of electronic charge. Conceptually, the situation in semiconductors can be thought of as follows: For current to exist, electrons must move and for electrons to move they must find empty sites adjacent to them to which they can then move. When the valence band is full, the electrons in the valence band do not have vacant sites within the valence band to move to. At the same time, the conduction band is empty and there are no electrons available in the conduction band to support conduction as well. Therefore, no electronic conductivity is possible. If sufficient energy, in the form of thermal energy or light, is provided, the energy enables a few electrons to make transition from the valence band to the conduction band. The conduction band now contains a few electrons and a large number of vacant sites. The electrons are able to move in this situation and are able to support current. Similarly, the large number of electrons that remain in the valence band also have a few vacant sites or holes that have been left behind by the electrons that transitioned to the conduction band. Therefore, the electrons in the valence band can also move in response to an electric field and support current. It turns out that it is easier to follow the conduction process in the valence band by focusing on the movement of the small number of holes, rather than the large number of electrons. Electrons moving in one direction are the same as the holes moving in the opposite direction. This approach of focusing on electrons in the conduction band and holes in the valence band is, therefore, commonly used in the discussion of semiconductors. This same approach will be used in the discussion here as well.

15.3 Types of Semiconductors

Semiconductors are broadly of two types, intrinsic semiconductors and extrinsic semiconductors.

15.3.1 Intrinsic Semiconductors

These are materials which do not have any deliberately added dopants and are made as pure as possible. Their conductivity is a function of temperature only. They could be elemental or compound semiconductors. Elemental semiconductors are Group IVA elements of the Periodic Table, such as Si and Ge. Compound semiconductors consist of pairs of elements on either side of Group IVA. Elements of Group IIIA and Group VA, such as GaAs and InSb can be combined in a 1:1 ratio to synthesize compound intrinsic semiconductors. These are referred to as III–V compounds. Similarly, we can combine elements which are even further apart in the Periodic Table, on either side of Group IVA, such as Group IIB and Group VIA, and synthesize compound intrinsic semiconductors. These are called II–VI compounds and CdS is an example of such a compound. However, it is not possible to go to groups further and further away from Group IVA and expect combination of elements to result in semiconductors. As the elements

get more widely separated in the Periodic Table, the bonds between the elements tend to become more ionic and less covalent. In ionic bonds, the electrons are significantly localized to specific atoms and they do not display semiconducting behavior. Incidentally, it is worth noting that semiconductors can be discussed and described using bonds in the material as well as in the band structure of the material. The description chosen is a matter of convenience and essentially identical discussion will result in both cases.

15.3.2 Extrinsic Semiconductors

These are primarily Group IVA elements with tiny quantities of other elements deliberately added as dopants. Based on the dopant chosen, there is typically either a slight excess of electrons available as charge carriers, in which case the material is called an "*n*-type semiconductor," or there is a slight excess of holes available as charge carriers, in which case the material is called a "*p*-type semiconductor." Doping a IVA element with a VA element results in an *n*-type semiconductor, while doping the IVA element with a IIIA element, results in a *p*-type semiconductor.

The Ohm's law states that

$$J = \sigma\varepsilon$$

Since the flux of charge equals the product of the concentration of charge carriers, times the charge of the charge carrier, times the drift velocity corresponding to the charge carrier, we have

$$J = \sigma\varepsilon = n|e|v_{\mathrm{d}}$$

where v_{d} is the drift velocity of the electrons. Rearranging, we have

$$\sigma = n|e|\left(\frac{v_{\mathrm{d}}}{\varepsilon}\right)$$

Since $(v_{\mathrm{d}}/\varepsilon) = \mu_{\mathrm{e}}$ or the mobility of the electrons, we have

$$\sigma = n|e|\mu_{\mathrm{e}}$$

In an intrinsic semiconductor, for each electron in the conduction band, there is a hole in the valence band. Due to the vacant sites available, both the bands are able to support the conduction process. Electrons moving in one direction are the same as holes moving in the opposite direction. Therefore

$$\sigma = n|e|\mu_{\mathrm{e}} + p|e|\mu_{\mathrm{h}}$$

where p is the concentration of the holes in the valence band and μ_{h} is the mobility of the holes.

In intrinsic semiconductors, $n = p$, since each electron in the conduction band has left behind a hole in the valence band. This is, therefore, called the intrinsic carrier concentration, n_{i}. Therefore, for intrinsic semiconductors

$$n = p = n_{\mathrm{i}}$$

and

$$\sigma = n_{\mathrm{i}}|e|(\mu_{\mathrm{e}} + \mu_{\mathrm{h}})$$

In general, $\mu_e > \mu_h$. This can be understood as follows: From the perspective of electrons, the electrons in the conduction band have a large number of vacant sites to move to and are therefore able to move more easily in this "open space." Therefore, electrons in the conduction band are more mobile, resulting in a larger μ_e. Relatively speaking, the electrons in the valence band are situated in an extremely "crowded space" with a large number of other electrons and very limited vacant sites or holes to move into. Therefore, the electrons in the valence band are less mobile, and since we use holes to describe the charge transport in the valence band, this results in a smaller μ_h.

In the case of extrinsic semiconductors, at room temperature, the charge carriers are dominated by the contributions of the dopants in the system. Therefore, for an *n*-type semiconductor $n \gg p$ which results in

$$\sigma \approx n|e|\mu_e$$

Similarly, for a *p*-type semiconductor, at room temperature

$$\sigma \approx p|e|\mu_h$$

The reason the dopants dominate the extrinsic semiconductor behavior is that doping shifts the Fermi energy of the material to a value corresponding to the dopant level. Electron transfer to or from this level and the adjacent band becomes very easy or requires a lot less energy compared to the energy required to transfer electrons from the valence band to the conduction band. Intrinsic behavior is not precluded in an extrinsic semiconductor; it is just that it is negligible at room temperature and has an impact only at relatively high temperatures. The Fermi energies of intrinsic semiconductors, *n*-type semiconductors and *p*-type semiconductors, are shown in Fig. 15.1.

The position of the Fermi energy in semiconductors is an important piece of information because when two semiconductors are brought in physical contact with each other, the value of the Fermi energy of the two materials decides the direction in which the electrons will flow to try and equalize the Fermi energies.

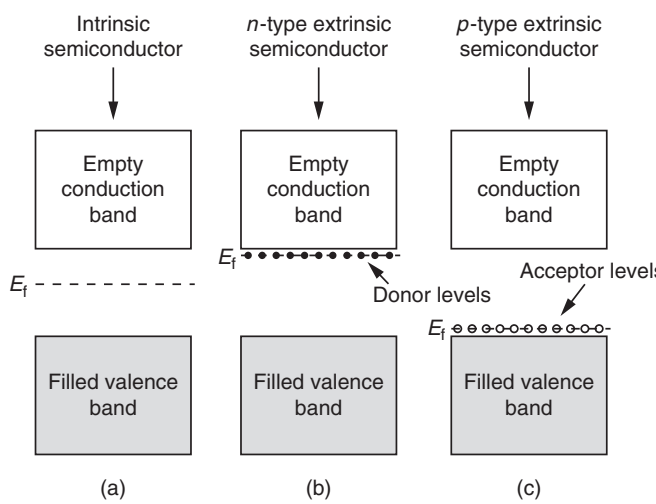


Figure 15.1 Fermi energy of (a) intrinsic, (b) *n*-type, and (c) *p*-type semiconductors.

15.4 Effect of Temperature on Semiconductors

As noted earlier, conductivity depends on the charge carrier concentration, the value of the charge and the mobility of the charge. This is true regardless of whether the conductor is metallic or semiconducting. However, we see that metals show a positive thermal coefficient for resistivity, that is, resistance increases with temperature, whereas for semiconductors the resistance decreases with temperature. Why does this occur?

The answer lies in the fact that for metals, the charge carrier concentration is largely constant as the temperature is increased. But the mobility of the electrons decreases on increasing temperature due to increased probability of collisions, due to the higher velocities of the electrons.

Therefore, in the equation

$$\sigma = N_f |e| \mu_e$$

As the temperature increases, the only parameter affected is μ_e, which decreases with increase in temperature. Therefore, the conductivity decreases (or resistivity increases) with increase in temperature.

For semiconductors, both the charge carrier concentration as well as mobility of the charge carriers is impacted by changes in temperature. The charge carrier concentration increases with the temperature and more than compensates for the decrease in the charge carrier mobility. Hence, for semiconductors, the conductivity increases with temperature or the resistivity decreases with temperature.

The behavior of charge carrier concentration as a function of temperature for the semiconductors is summarized in Fig. 15.2.

For all semiconductors, at very low temperatures, the charge carriers do not have enough energy to occupy vacant sites and hence the charge carrier concentration is very small. For extrinsic semiconductors, as the temperature increases, the dopant levels contribute to the charge carrier concentration, even at relatively low temperatures, since the energy required to transfer the electrons to the conduction band from the donor level, for example, is very small. This contribution to the charge carrier concentration is fixed

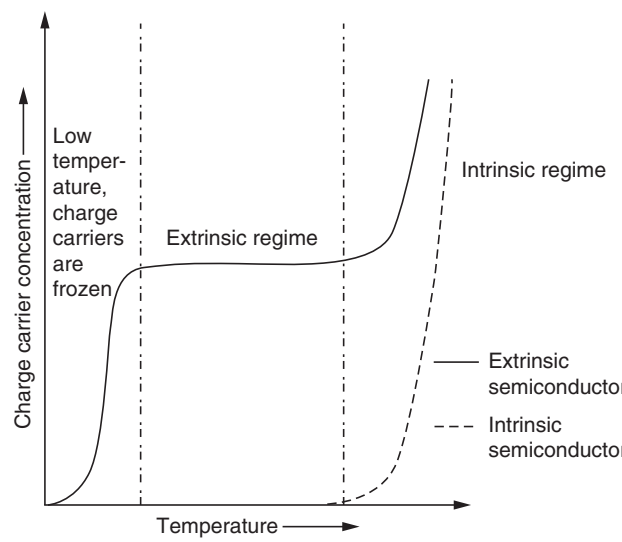


Figure 15.2 Schematic of charge carrier concentration for semiconductors as a function of temperature.

over a wide range of temperatures since the dopant concentration is fixed and the energy required is small. Only at much higher temperatures, the intrinsic contributions begin to take effect with transitions from the host material's valence band to the conduction band across the band gap, which requires a relatively large amount of energy. For intrinsic semiconductors, this last contribution is the only one that is present. The intrinsic contribution increases with further increase in temperature, since more electrons can make the transition at increased temperatures.

15.5 Optical Properties of Semiconductors

Having examined the basics of semiconductor behavior, we can now try to understand the optical properties of materials. In general, when a material has a band gap of, E_g, it can only absorb radiation that has enough energy to enable transition across the band gap. Therefore, the minimum frequency, ν_{min}, of the radiation that it can absorb is such that:

$$E_g = h\nu_{min}$$

The material will not absorb frequencies lower than this and will therefore be transparent to such radiations. This is also the reason that metals are opaque. In metals, there is a continuous range of energy levels almost immediately above the Fermi energy. Therefore, even for very small frequencies, there are energy levels that can accept electrons gaining that small quantity of energy.

From the above discussion, we understand how materials absorb radiation. However, even after exceeding the threshold, ν_{min}, it turns out that some materials with relatively low band gaps are quite ineffective in absorbing radiation, while other materials absorb more of the radiation incident on them. Why does this happen? Some process in the material impacts the probability of the absorption process and this information is not captured by the ν_{min} information alone.

The information that is not captured by just stating the band gap is that in some cases the highest occupied energy level and the next lowest unoccupied energy level are not in the same location in *k*-space. This implies that the electron not only has to gain energy greater than that corresponding to the band gap, but it also has to move to a different location in *k*-space to complete the transition. For an electron to gain energy when light falls on the material, it needs to interact with a photon of the appropriate energy. Since the photons are randomly striking the material, there is a probability associated with a given electron encountering a photon and gaining energy from it. If the next lowest unoccupied energy level is in the same location in *k*-space, then interaction with the photon is all that is required for the electron to absorb the radiation and go to the conduction band. Materials which have the highest occupied energy level and the next lowest unoccupied energy level in the same location in *k*-space are called "direct band gap" semiconductors. On the other hand, if the next lowest unoccupied energy level is at another location in *k*-space, the electron that has gained energy from the photon must also encounter an appropriate lattice vibration to transport it to that location in *k*-space. Lattice vibrations are referred to as Phonons, and this encounter with an appropriate phonon, also has a probability associated with it, which is relatively low. The overall probability of such a material absorbing radiation is the product of the probability of the electron interacting with a photon of sufficiently high energy and the probability of the electron then encountering an appropriate phonon. This product of probabilities results in a relatively low overall probability for the process. Therefore, such materials are ineffective in absorbing radiation. Materials which have the highest occupied energy level and the next lowest unoccupied energy level at different locations in *k*-space are called "indirect band gap" semiconductors. Direct band gap materials are therefore preferred for optical and opto-electronic applications over indirect band gap materials.

15.6 The Utility of *E* versus *k* Diagrams

The flat-band diagrams do not distinguish between the direct band gap materials and the indirect band gap materials, while the *E* versus *k* diagrams do indicate the difference. Therefore, the *E* versus *k* diagram is much more capable in explaining material properties relative to flat-band diagrams. The difference between direct band gap and indirect band gap semiconductors is highlighted in Fig. 15.3.

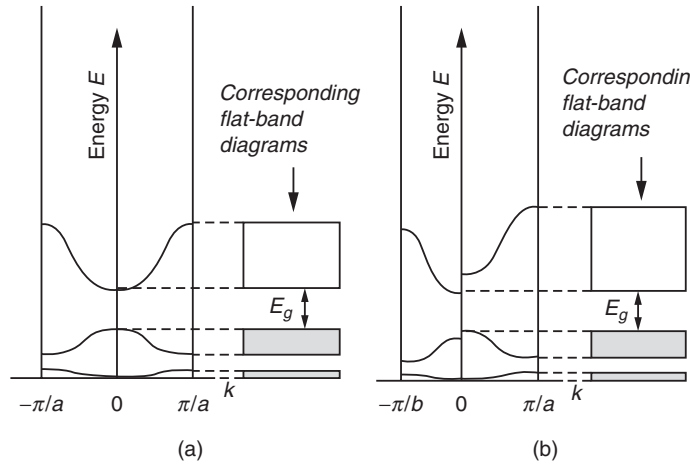


Figure 15.3 (a) Direct band gap semiconductor and (b) indirect band gap semiconductor.

Therefore, Fig. 15.3 highlights the utility of the *E* versus *k* diagrams over and above that of the flat-band diagrams in the context of optical properties of semiconductors.

15.7 Phonons

In this course, we have so far focused largely on the electrons in the materials and ignored the atoms or the ionic cores. Atoms can vibrate about their mean positions and can gain or release energy, or in other words, participate in the energy transaction process. Atoms vibrating about their mean positions result in waves of lattice vibrations that can travel through the solid at v_s, the speed of sound.

The form of the wave that travels through the solid can be arbitrary, but it is possible to write it down as a sum of waves that are well-defined and sinusoidal, as obtained using a Fourier series. Therefore, each wave can be identified with specific wavelength and frequency.

Waves in matter or waves of lattice vibrations are called phonons. Taken in its entirety, the specific heat of a solid has a significant contribution from phonons and this represents the atomic contribution to the specific heat; so far we have only looked at the nearly free electron contribution to specific heat. One of the successes of quantum mechanics is that the phonons too behave consistent with quantum mechanical rules. For phonons, in just the manner we saw for photons,

$$E = h\nu$$

In other words, the energy of phonons is also quantized. One major difference between photons and phonons is that the former travel at the speed of light, while the latter travel at the speed of sound. Therefore, for phonons:

$$\nu = \frac{v_s}{\lambda}$$

where v_s, as we indicated earlier, is the speed of sound in that material.

Phonons can therefore be thought of as lattice wave equivalent of photons. In a solid of length L and inter-atomic spacing d, the maximum wavelength that can be supported is $2L$, in which case the neighboring atoms are displaced in the same direction from their mean positions and the minimum wavelength that can be supported is $2d$. To support specific wavelengths in between these two limits, some adjacent atoms will move in the opposite directions.

In case the solid has a two-atom basis, and if the two atoms are ionically bonded, then due to the specific wavelengths where the atoms move in opposite directions, an electromagnetic wave is generated. In this case, the phonons are able to interact with incident electromagnetic radiation of corresponding frequencies. The typical range of such frequencies places them in the infrared region of the electromagnetic spectrum.

Phonons that can interact with electromagnetic radiation are referred to as "optically active" phonons. The rest of the phonons are referred to as "acoustically active" phonons.

As indicated earlier in this chapter, phonons become necessary to enable electron transitions in indirect band gap semiconductors, such as silicon (Si), but are not required to enable transitions in direct band gap semiconductors, such as GaAs. Therefore, direct band gap semiconductors are preferred for opto-electronic materials which depend on electrical as well as optical properties for their functioning. The functioning of indirect band gap semiconductors is highlighted in Fig. 15.4.

In just the manner in which we looked at the statistical behavior of electrons and found that they are Fermions, photons and phonons, follow a statistical behavior that is credited to Satyendra Nath Bose and Albert Einstein and is called the Bose–Einstein statistics. Photons and phonons are therefore referred to as Bosons.

Let us now consider the use of the theories developed, in the context of explaining phenomena related to magnetism.

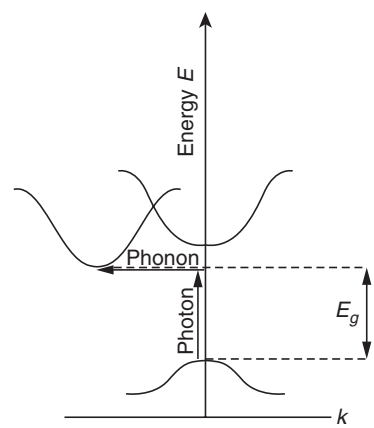


Figure 15.4 The photon followed by phonon process required for an indirect band gap semiconductor to absorb incident light.

15.8 Magnetism



Magnetic properties are commonly used in a variety of technologies. Audio speakers and motors are common examples of technologies that use magnets. More exotic technologies that use magnetism are the magnetic resonance imaging (MRI) scanners used in the medical field.

Magnetism is a phenomenon that has been known and used long before any understanding was developed on the science behind the phenomenon. Magnetism is observed in the form of an attractive force that some materials exert on some others that is in addition to any other forms of attraction, such as electrostatic attraction. While originally it was discovered as existing in some materials, as obtained in nature, it was later discovered that moving electrical charges create a magnetic field. This discovery led to the manufacture of electromagnets where coils of current carrying conductors generate magnetic fields that are then used to serve specific purposes. Industrially, the use of electricity to generate magnetism is the most commonly observed usage of magnetism, since it enables considerable control on the phenomenon.

The common equations associated with magnetism are as follows:

1. An externally generated magnetic field using a coil carrying electricity is indicated by

$$H = \frac{NI}{L}$$

where N is the number of turns of the coil, I is the current flowing through it, and L is the length of the coil.

2. An externally generated magnetic field can induce a magnetic field in another material which is placed within the coil and this is given by

$$B = \mu H$$

where μ is the permeability of the material.

3. In case the coil is in vacuum, the field in the vacuum is given by

$$B = \mu_0 H$$

where μ_0 is the permeability of vacuum.

4. The ratio

$$\frac{\mu}{\mu_0} = \mu_r$$

is called relative permeability and is therefore without units.

5. The response of a sample to the externally generated magnetic field can be thought of as a response that is in addition to the response of vacuum to the same field. This additional response, which is the extent of internal reinforcement or opposition to the applied field, is referred to as magnetization, denoted by M. Therefore, the response of the material can be written as

$$B = \mu H = \mu_0 H + \mu_0 M$$

6. And since the magnetization M is itself generated in response to the external field, it can be related to the external field through

$$M = \chi H$$

where χ is called the susceptibility of the material. Therefore,

$$B = \mu_0 H + \mu_0 M = \mu_0 H + \mu_0 \chi H$$

Therefore,

$$\mu H = \mu_0 H (1 + \chi)$$

and

$$\frac{\mu}{\mu_0} = \mu_r = 1 + \chi$$

or

$$\chi = \mu_r - 1$$

7. Since μ_r can be less than or greater than 1 (since some materials will oppose applied magnetic fields while some others may augment the applied magnetic field), χ can be positive or negative.

Figure 15.5 shows the response of the specific types of materials to externally applied magnetic fields. Diamagnetic materials weakly oppose the applied field ($\mu_r < 1$, χ is a small negative number), paramagnetic materials weakly augment the applied field ($\mu_r > 1$, χ is a small positive number), while ferromagnetic materials strongly augment the applied field ($\mu_r \gg 1$, χ is a large positive number).

In the theories for materials we have developed so far, we noted that the density of occupied states, as a function of energy, has a profile as shown in Fig. 15.6.

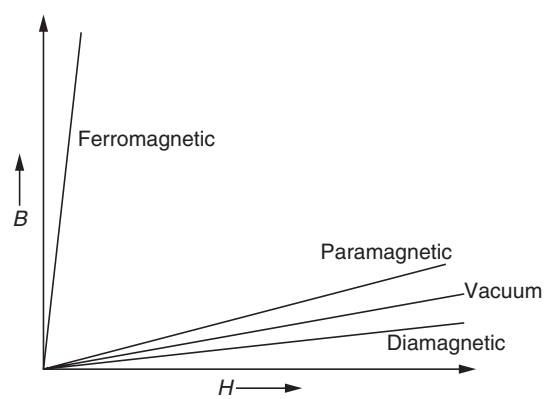


Figure 15.5 Schematic of the response of specific materials to an externally applied magnetic field.

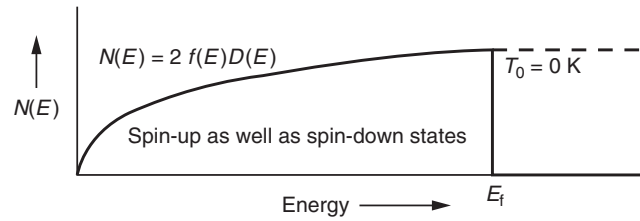


Figure 15.6 Density of occupied states as a function of energy. Both spin-up as well as spin-down states are included in the figure together.

The occupied states consist of electrons that are spin-up and spin-down. The spin of the electrons contribute to the magnetic behavior of the material. The density of occupied states of the material can therefore be redrawn as shown in Fig. 15.7, where the axes have been interchanged. The electrons with spin-down appear on one side of the figure, while the electrons with spin-up, appear on the other side of the figure.

As an aside, it is important to note that although the form of Fig. 15.7 above is similar to the E versus k diagram, they are not the same.

When a ferromagnetic material experiences an externally applied magnetic field, the states with spin aligned in the direction of the applied magnetic field attain lower energies, while the states with spin opposed to the applied magnetic field attain a higher energy state. Assuming that the applied magnetic field is aligned with the spin-up direction of the electrons, temporarily, the situation can be thought of as shown in Fig. 15.8(a). Since the Fermi energy of the system has to be a uniform value for the system, electrons move from the states with spin opposed to the magnetic field to states with spin aligned with magnetic field, till the Fermi energy reaches a uniform value, as shown in Fig. 15.8(b).

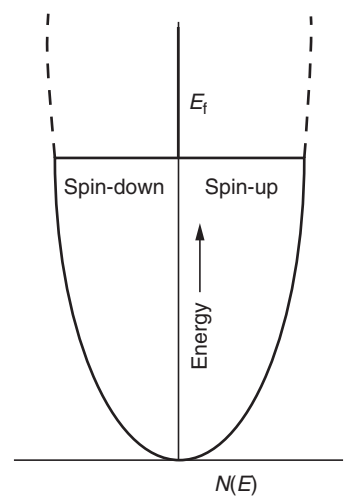


Figure 15.7 Density of occupied states as a function of energy with the electrons having spin-up and electrons having spin-down being shown separately. For the sake of clarity, the axis have been interchanged with respect to the earlier figure (Fig. 15.6) and therefore the figure is rotated by 90°, counter clockwise, with respect to the earlier figure (Fig. 15.6).

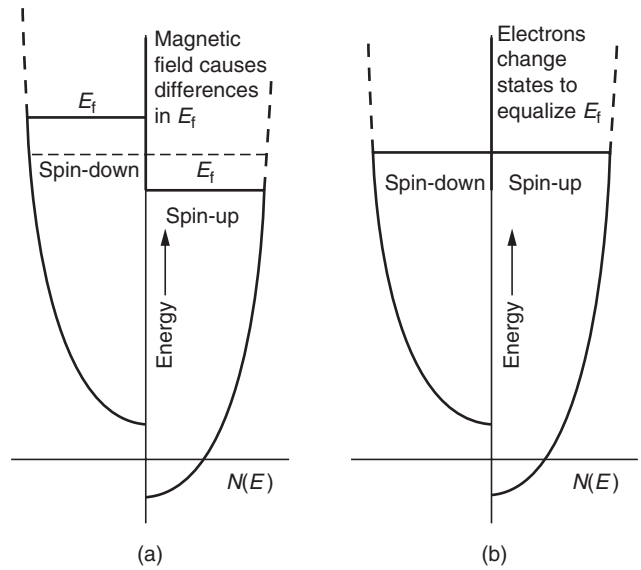


Figure 15.8 Response of a ferromagnetic material to an externally applied magnetic field: (a) Intermediate state where the states with spin aligned in the direction of the applied magnetic field attain lower energies, while the states with spin opposed to the applied magnetic field attain a higher energy. (b) The final state of the system after electrons change states to attain a uniform Fermi energy for the system.

This increase in the number of electrons with spin aligned favorably with the applied magnetic field explains the magnetization behavior of materials, *M*. The theories we have developed thus far are, therefore, able to explain magnetic behavior of materials.

Let us now look at one more instances where the theories developed so far in this course help us get additional insight into the properties of materials. Let us look at electron compounds, where a phase change is explained using the theories we have developed in this course.

15.9 Electron Compounds



The success of the theories we have developed so far can be gauged from the number of experimentally observed phenomena they are able to explain. In particular, we have concentrated on the theories that apply to solid metallic systems, which contain considerable amounts of free electrons. Therefore, the theories developed are found to be generally effective in explaining properties that are dependent on the nearly free electrons. Some of these properties are the more commonly observed properties, such as electrical conductivity, thermal conductivity, optical properties, and magnetic properties. However, there are also other phenomena in materials that are not so commonly encountered by us in day-to-day life and it is of interest to see if these too can be explained using the theories developed so far. Let us now look at a specific experimental phenomenon that is observed in some of the binary phase diagrams in silver (Ag)- and copper (Cu)-based systems.

Both Ag and Cu have the FCC crystal structure. The terminal solid solutions in these systems also have the FCC crystal structure. It is observed that on adding an alloying element which forms a substitutional solid solution with the solvent, up to a certain composition, the structure remains FCC. After this, the

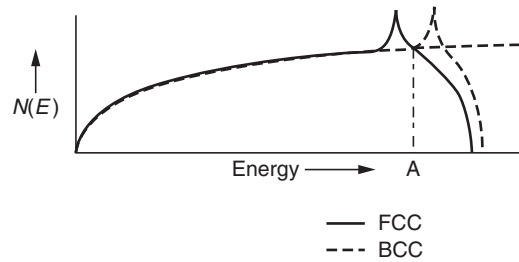


Figure 15.9 The density of occupied states as a function of energy for the FCC structure, as well as for the BCC structure. This schematic representation can be assumed to apply for both Ag and Cu, since they are FCC in the elemental state and change to BCC when alloyed appropriately.

structure changes to BCC. The exact composition at which the structure changes from FCC to BCC varies, depending on the solvent atom.

These binary systems were examined to determine the reason for the phase change. A natural valency could be associated with each of the elements used as the solute. Based on the valency of the solvent and the solute and the molar ratio of the solid solution, it is possible to assign an "e/a" ratio, which refers to the average number of valence electrons per atom in the alloy. Hume Rothery discovered that regardless of the alloying element used, the phase change occurs at a fixed value of the e/a ratio. Since the different solute atoms could have different valencies, the same e/a ratio occurs at different compositions in each of the different systems studied. These compounds which occur as a result of the valence electron to atom ratio are called "electron compounds" or "Hume Rothery phases." While the experimental observation was useful, it still does not, by itself, explain why the phase change must occur at those specific e/a ratios. It is of interest to see if the theories we have developed are able to explain the formation of electron compounds.

Consider the plot of the density of available states as a function of energy, as shown in Fig. 15.9. In the case of nearly free electrons in a solid at room temperature, the free electron parabola is distorted at the Brillouin zone boundaries. Further, due to the effect of the temperature, the highest occupied energy levels are spread out in energy and do not occur at a single specific value as they do at the E_f at 0 K.

Figure 15.9 shows the density of occupied states as a function of energy for the FCC structure as well as for the BCC structure. For both Ag and Cu, this schematic representation can be assumed to apply since they are FCC in the elemental state and change to BCC when alloyed appropriately.

Near the Brillouin zone boundaries, with increasing energy, the density of occupied states increases as a function of energy and then decreases sharply. From Fig. 15.9, it can be seen that since the Brillouin zone boundary of the BCC structure occurs at a larger value of energy, the distortion of the free electron parabola occurs at a higher level of energy in the BCC structure, compared to the FCC structure. As can be seen from Fig. 15.9, at energy levels greater than an energy level identified on the energy axis by the point "*A*," the $N(E)$ for the FCC structure is less than that of the BCC structure and continues to decrease, while the $N(E)$ for the BCC structure continues to increase for some more values of energy. Therefore, if the number of electrons in the system is such that the E_f is greater than that corresponding to the point "*A*," the nearly free electrons in the system can be accommodated in lower energy states if the structure is BCC, than if it is FCC. Since there are a large number of nearly free electrons in the system, the difference in energy is significant enough to cause the structure to change, once a threshold level of electrons is exceeded. This concentration is expressed in the form of the nearly free electrons to atom ratio and when this e/a ratio exceeds 1.4,

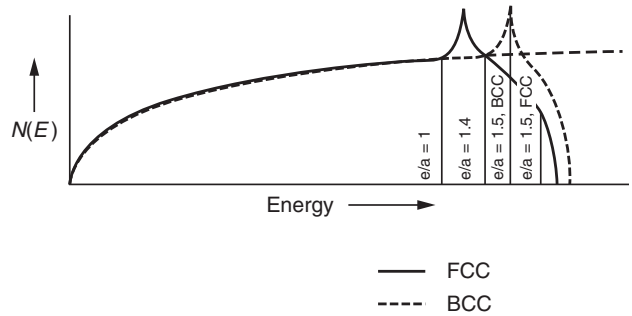


Figure 15.10 The effect of different e/a ratios on the highest energy level occupied in the FCC structure and in the BCC structure. At e/a ratios greater than 1.4, electrons are held at lower energy levels in the BCC structure in comparison to the FCC structure; an example of e/a = 1.5 is shown for illustrative purposes.

the phase change occurs. Figure 15.10 shows the situation for e/a ratios of 1, 1.4, and 1.5. At an e/a ratio of 1.5, the electrons are held at lower energy levels in the BCC structure compared to the FCC structure. Hence, the phase change occurs.

In this chapter, we have seen several material phenomena explained by the theories discussed so far. In Chapter 16, we will look at superconductivity as a phenomenon, which also involves another Boson and we will derive the Bose–Einstein statistics.

Summary

1. Semiconductors are a class of materials that display significantly less conductivity compared to metals. However, their conductivity can be controlled and manipulated in many technically interesting ways and hence useful devices can be made using these materials.
2. In semiconductors, with the availability of adequate amounts of energy, electrons will transition from the filled valence band to the empty conduction band and leave behind empty states in the valence band called "holes." These holes are also charge carriers with a positive charge, numerically equivalent to that of electronic charge.
3. Semiconducting materials which have do not have any deliberately added dopants and are made as pure as possible, are called intrinsic semiconductors. Their conductivity is a function of temperature only.
4. Intrinsic semiconductors can be elemental or compound semiconductors. Elemental semiconductors are Group IVA elements. Compound semiconductors consist of pairs of elements on either side of Group IVA.
5. Extrinsic semiconductors are primarily Group IVA elements with tiny quantities of other elements deliberately added as dopants. Their conductivity is dominated by the concentration of dopants added.
6. Doping a IVA element with a VA element results in an *n*-type semiconductor, while doping the IVA element with a IIIA element, results in a *p*-type semiconductor.
7. For semiconductors:

$$\sigma = n|e|\mu_e + p|e|\mu_h$$

8. In general, $\mu_e > \mu_h$.

9. Doping shifts the Fermi energy of the material to a value corresponding to the dopant level.

10. For metals, only mobility of charge carriers is affected by the temperature, whereas for semiconductors, both the charge carrier concentration and the mobility of the charge carriers are impacted by the changes in temperature. Hence, for metals, resistivity increases with temperature, while that of the semiconductors decreases with increase in temperature.

11. A material with a band gap of E_g can only absorb radiation that has enough energy to enable transition across the band gap. Therefore, the minimum frequency, ν_{min}, of radiation that it can absorb is such that:

$$E_g = h\nu_{min}$$

The material will not absorb frequencies lower than this and will therefore be transparent to such radiations.

12. Materials which have the highest occupied energy level and the next lowest unoccupied energy level in the same location in *k*-space are called "direct band gap" semiconductors. These are better suited for optical and opto-electronic applications.

13. Materials which have the highest occupied energy level and the next lowest unoccupied energy level at different locations in *k*-space are called "indirect band gap" semiconductors.

14. The flat-band diagrams do not distinguish between the direct band gap materials and the indirect band gap materials, while the *E* versus *k* diagrams do indicate the difference and hence have a greater utility.

15. Waves in matter or waves of lattice vibrations are called phonons. Phonons can be thought of as lattice wave equivalent of photons. Phonons behave consistent with quantum mechanical rules, but travel at the speed of sound, unlike photons which travel at the speed of light.

16. Phonons in ionically bonded solids, with ions moving in opposite directions, result in electromagnetic waves in the infrared (IR) region and can interact with electromagnetic radiation.

17. Phonons that can interact with electromagnetic radiation are referred to as "optically active" phonons. The rest of the phonons are referred to as "acoustically active" phonons.

18. Phonons become necessary to enable electron transitions in indirect band gap semiconductors.

19. Phonons and photons are Bosons.

20. Diamagnetic materials weakly oppose the applied field ($\mu_r < 1$, χ is a small negative number), paramagnetic materials weakly augment the applied field ($\mu_r > 1$, χ is a small positive number), while ferromagnetic materials strongly augment the applied field ($\mu_r \gg 1$, χ is a large positive number).

21. When a ferromagnetic material experiences an externally applied magnetic field, the states with spin aligned in the direction of the applied magnetic field attain lower energies, while the states with spin opposed to the applied magnetic field attain higher energies. Since the Fermi energy of the system has to be a uniform value for the system, electrons move from the states with spin opposed to the magnetic field to states with spin aligned with magnetic field, till the Fermi energy reaches a uniform value. This increase in the number of electrons with the spin aligned favorably with the applied magnetic field, explains the magnetization behavior of materials, *M*.

22. In some of the binary phase diagrams in Ag- and Cu-based systems, regardless of the substitutional alloying element used, the structure changes from FCC to BCC at a fixed value of the e/a ratio. These phases are called Hume Rothery phases.

23. Due to the shapes of the respective $N(E)$ versus E curves, beyond specific e/a ratios, the electrons are held at lower energy levels in the BCC structure compared to the FCC structure. Hence, the phase change occurs at these e/a ratios.

Practice Questions

Multiple Choice Questions

1. At room temperature, the conductivity of intrinsic semiconductors is
- **(a)** Marginally less than that of metals
- **(b)** Significantly less than that of metals
- **(c)** Marginally less than that of insulators
- **(d)** Significantly less than that of insulators

2. Intrinsic semiconductors consist of
- **(a)** Only single elements
- **(b)** Can be compound semiconductors
- **(c)** Can have small amounts of dopants
- **(d)** Can have large amounts of dopants

3. For metals, with increase in temperature
- **(a)** Only mobility of charge carriers is affected
- **(b)** Mobility of charge carriers a well as number density of charge carriers is affected
- **(c)** Number density of ionic cores is affected
- **(d)** Type of charge carrier is changed

4. For an intrinsic semiconductor, with increase in temperature
- **(a)** Only mobility of charge carriers is affected
- **(b)** Mobility of charge carriers a well as number density of charge carriers is affected
- **(c)** Number density of ionic cores is affected
- **(d)** Type of charge carrier is changed

5. In an indirect band gap semiconductor
- **(a)** Probability of an electron encountering a photon followed by a phonon is low
- **(b)** Probability of an electron encountering a photon followed by a phonon is high
- **(c)** All electrons are continuously encountering photons
- **(d)** All electrons are continuously encountering phonons

6. The strength of an externally generated magnetic field using a coil carrying electricity
- **(a)** Increase with current in the coil
- **(b)** Increases with the number of turns of the coil
- **(c)** Increases with decrease in length of the coil
- **(d)** All of the above

7. In comparison to paramagnetic materials, diamagnetic materials
- **(a)** Augment the externally applied magnetic field twice as strongly
- **(b)** Augment the externally applied field just as strongly
- **(c)** Oppose the externally applied magnetic field weakly
- **(d)** Do not interact with the externally applied magnetic field at all

8. When a ferromagnetic material experiences an externally applied magnetic field
- **(a)** Electron states with spin aligned in the direction of the applied magnetic field attain lower energies
- **(b)** Electron states with spin opposed to the applied magnetic field attain lower energies
- **(c)** Electron states with spin aligned in the direction of the applied magnetic field attain higher energies
- **(d)** Electron states with spin aligned in the direction of the applied magnetic field are unaffected in their energy

9. When a ferromagnetic material experiences an externally-applied magnetic field
- **(a)** Electrons are unaffected
- **(b)** Electrons move to equalize Fermi energy
- **(c)** Electrons become photo electrons
- **(d)** Electrons move to conduction band

10. If χ is a large positive number for a material, it is
- **(a)** Diamagnetic
- **(b)** Paramagnetic
- **(c)** Non-magnetic
- **(d)** Ferromagnetic

11. e/a ratio is the critical parameter for Hume Rothery phases because
- **(a)** It indirectly compares the Fermi surface to the Brillouin zone
- **(b)** It represents changes in conductivity
- **(c)** It represents the bound electrons in the system
- **(d)** It represents the band gap in the system

12. Hypothetically, Hume Rothery phases are unlikely to exist if
- **(a)** Brillouin zones are small
- **(b)** Brillouin zones are large
- **(c)** There is no distortion in the density of occupied states as a function of energy curve
- **(d)** There is distortion in the density of occupied states as a function of energy curve

13. Phonons travel at
- **(a)** The speed of sound
- **(b)** The speed of light
- **(c)** Higher than the speed of sound but less than the speed of light
- **(d)** Lower than the speed of sound

14. With respect to electromagnetic radiation,
- **(a)** All phonons can interact with electromagnetic radiation
- **(b)** Some phonons can interact with electromagnetic radiation
- **(c)** Phonons can never interact with electromagnetic radiation
- **(d)** Phonons are electromagnetic radiation

15. Electronic transition in GaAs require
- **(a)** Phonons alone
- **(b)** Photons necessarily followed by phonons
- **(c)** Photons alone
- **(d)** Phonons necessarily followed by photons

Short Answer Type Questions

1. Distinguish between the optical properties of typical metals and typical ceramic materials.

2. Can ceramic materials also be opaque?

3. What gives material a color?

4. Give some examples of direct band gap semiconductors.

5. Give some examples of indirect band gap semiconductors.

6. What is the source of earth's magnetic field?

7. What is the magnetic field intensity generated using a coil, 1-m long, with 10 turns per meter, carrying a current of 1 A

8. What is the principle behind the operation of MRI equipment?

9. Name a technologically important diamagnet.

10. Name a commonly available ferrimagnetic material.

11. Why are electron compounds primarily observed only in metallic systems?

12. What fraction of the specific heat of a material can be attributed to phonons?

13. Are indirect band gap semiconductors incapable of absorbing light?

14. What is the operating principle of a light emitting diode (LED)?

15. How are LEDs of different colors made?

Answers

Multiple Choice Questions

1. (b)	**5.** (a)	**9.** (b)	**13.** (a)
2. (b)	**6.** (d)	**10.** (d)	**14.** (b)
3. (a)	**7.** (c)	**11.** (a)	**15.** (c)
4. (b)	**8.** (a)	**12.** (c)	

Short Answer Type Questions

1. Metals are opaque, while single crystals of typical ceramic materials are transparent.
2. Ceramic materials can also be opaque, even though their band gap is consistent with transparent behavior. The presence of grain boundaries and fine porosity can cause the samples to be opaque due to the dispersion of light incident on the sample.
3. Materials display a color based on the wavelengths transmitted through the sample, in case they are transparent. Based on the band gap of the material, only wavelengths less than a threshold value will be absorbed. The transmitted wavelengths give the material its color.
4. GaSb and GaAs are examples of direct band gap semiconductors.
5. Si and Ge are examples of indirect band gap semiconductors.
6. The source of earth's magnetic field is the motion of the molten iron-based alloys in its interior.
7. The magnetic field intensity, H, is given by $\frac{10 \times 1}{1} = 10$ ampere turns per meter.
8. MRI imaging is based on the use of a powerful magnet which causes the magnetic moment of molecules to align with the field. The magnetic field is then oscillated to force the molecules to follow the oscillation. The oscillation of the magnetic moments is then detected and interpreted.
9. Superconductors such as Yittrium Barium Copper Oxide are technologically important perfect diamagnets.
10. Magnetite, Fe_3O_4, is a commonly available ferrimagnetic material.
11. Only in metallic systems, the number of free electrons and the contribution of these electrons to the energy of the system is significant enough to result in lower energy configurations being dictated by the energies of the electrons. Hence, electrons compounds are observed in metallic systems.
12. In insulating materials, most of the specific heat at low temperatures can be attributed to phonons.
13. No, indirect band gap semiconductors can also absorb light, they just do so inefficiently.
14. In LEDs, electrons and holes recombine to release light. This is the operating principle of the LED.
15. Based on the band gap of the LED material, LEDs of different colors can be produced.

16 Superconductivity and The Bose–Einstein Statistics

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Describe the phenomenon of superconductivity using a ρ versus T plot.
- Draw and explain a *T-H-J* plot, indicating the realm of superconductivity.
- Distinguish between type-I and type-II superconductors.
- Describe Meissner effect and indicate its significance.
- Explain the features associated with Cooper pairs.
- Explain the BCS theory.
- Indicate features of Bosons.
- Write the properties of the particles for which the Bose–Einstein statistical distribution is derived.
- Write the general expression for the number of microstates corresponding to a single macrostate for Bosons.
- Use the Lagrange method of undetermined multipliers to derive the expression for the Bose–Einstein statistical distribution.

16.1 Introduction

Superconductivity is a phenomena that has attracted considerable theoretical examination since it seems quite extreme from common place experience. In this chapter, we look at the phenomenon of superconductivity. It is being discussed as a separate topic, because superconductivity is significantly different from "exceptionally good" normal conductivity in the sense that the behavior of electrons in superconductors is very different from that in normal conductors, as will be explained later in this chapter. We will look at the phenomenon and examine the theory that explains some instances of superconductivity. We will also look at the statistical behavior of bosons, which form an essential part of the superconducting state.

16.2 Superconductivity



In general, in normal metallic conductors, as the temperature is lowered, the resistivity of the metal decreases and levels-off at some finite value at 0 K. In a few materials, it was discovered that on lowering the temperature, up to a point the behavior was similar in that the resistivity kept gradually decreasing, but then on crossing a specific value of temperature, the conductivity abruptly dropped to virtually zero. These materials, where the resistivity abruptly drops to zero when the temperature goes below a critical value, are called superconductors. The difference in behavior of superconductors and normal conductors, with respect to variation in temperature, is shown in Fig. 16.1.

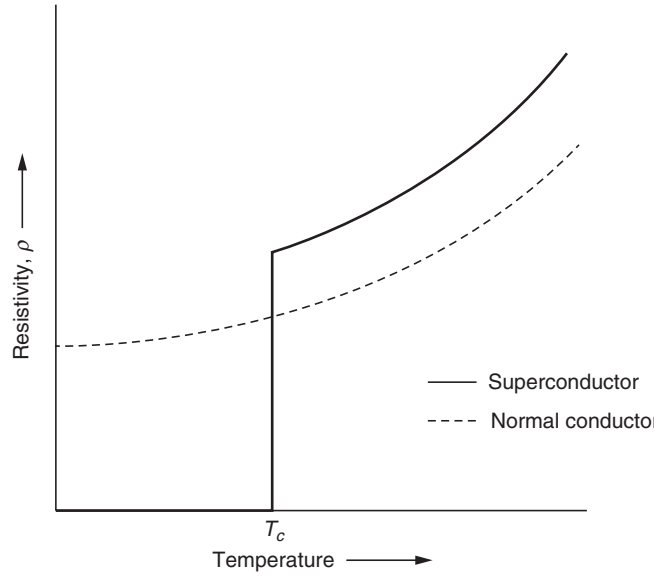


Figure 16.1 Difference in behavior of superconductors and normal conductors, with respect to variation in temperature.

The phenomenon of superconductivity was discovered in 1911 by H. Kammerlingh Onnes in the element mercury. To date, superconductivity has been demonstrated only at temperatures significantly below 0°C, in fact, at least temperatures close to –180°C continue to be required to demonstrate superconductivity. Superconductors can broadly be classified into "high temperature superconductors" and "low temperature superconductors;" however, the phrase "high temperature superconductor" is strictly a relative term, the "high temperature" of –180°C is only in relation to the low temperature of nearly –270°C that is required to demonstrate superconductivity in certain other materials.

Several metals, alloys, and intermetallic compounds show superconductivity at or below 10 K. This is a temperature that is typically attained using liquid helium. In 1986, materials were discovered that showed superconductivity at temperatures that were around –180°C, temperatures that could be attained using liquid N_2. It is interesting to note that these high-temperature superconductors are actually ceramic materials and are very poor conductors of electricity at room temperature.

Investigations into the phenomenon of superconductivity showed that superconductors responded to the conditions they were subject to in interesting ways. It was found that when the current passing through the superconductor was increased, above a certain critical current density, designated as J_c, superconductivity ceased to exist. Similarly, when an external magnetic field was imposed on the superconductor, above a critical value of the external field, designated as H_c, superconductivity ceased to exist. As already pointed out, above a critical temperature, superconductivity ceased as well and this critical temperature is designated as T_c. Combined action of more than one of these influences resulted in a breakdown of the superconductivity state even more easily. Figure 16.2 summarizes the response of a superconductor to the current density flowing through it, the temperature it is subject to, and the magnetic field it is placed in. The material stays a superconductor for all conditions between the origin of the plot and the surface defined in the figure.

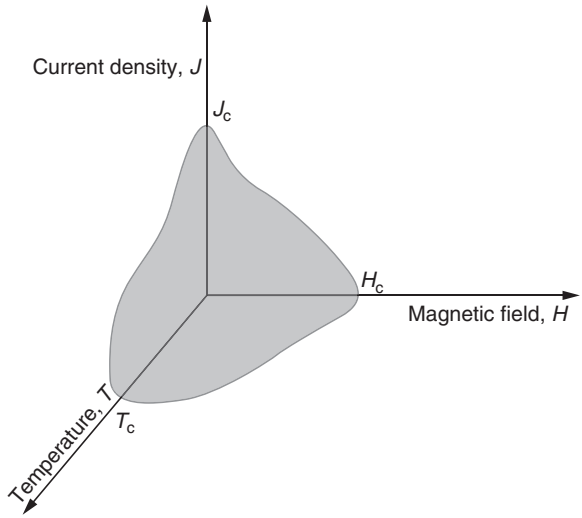


Figure 16.2 Impact of current density, temperature, and magnetic field on superconductivity. The material stays a superconductor for all conditions between the origin of the plot and the surface defined in the figure.

16.3 The Meissner Effect

Superconductors were observed to exclude magnetic fields, an effect called the Meissner effect, after the person who discovered it. This turned out to be a very important observation since this behavior was different from that of a regular metallic conductor. The Meissner effect indicated that superconductivity could not be merely thought of as better or improved conductivity; a new mechanism had to be proposed. The Meissner effect indicated that magnetism of materials opposed what superconductivity required of materials.

16.4 The BCS Theory

The theory that explained low-temperature superconductivity is credited to Bardeen, Cooper, and Schrieffer and is hence called the BCS theory. They theorized that the electrons in superconductors operate in pairs, called cooper pairs, which have opposite spin and opposite k vectors. As a result, the pair of particles operate as though they have zero spin and have no net wave vector. Particles that have integer spins belong to the class of particles called bosons, just as photons and phonons that we discussed in Chapter 15. In the case of superconductors, electrons, which are normally fermions, pair up and behave like bosons. In view of the zero-wave vector of the cooper pairs, they do not suffer from the typical scattering effects that normal electrons experience.

Cooper pairs can have a significant distance between them, of the order of several nanometers, and still maintain the interaction between them. This is accomplished using lattice waves or phonons. In other words, in this situation, one boson (a cooper pair) uses another boson (a phonon) to sustain itself. The formation of a cooper pair results in a small decrease in energy and at least this small amount of energy must be provided to breakdown the superconducting state.

These ideas of how the cooper pairs enable superconductivity, directly address the implications of Meissner effect which showed that magnetism opposes superconductivity. Since magnetism requires spins of electrons to align with each other, the cooper pairs having opposite spins explain the conflict with magnetism.

Based on the interaction with magnetism, superconductors are classified into two types, called type-I and type-II superconductors. Type-I superconductors completely exclude any applied magnetic fields in the superconducting state called the S-state and abruptly become normal conductors or attain the normal N-state above H_c. In the N-state, the magnetic field completely penetrates the sample. These superconductors are usually the low-temperature superconductors.

Type-II superconductors completely exclude the magnetic field up to an applied field H_{c1}, at which point the material develops two regions within it, regions that remain superconducting and others that are normal. In other words, the S-state and the N-state coexist. Above a higher level of applied magnetic field H_{c2} the material becomes entirely normal. This type of superconductor can usually handle much higher magnetic fields compared to type-I superconductors.

It is relevant to note that it takes only 5–10 K of thermal energy to breakdown the low-temperature superconducting state. The stability of the cooper pairs is consistent with this energy. Therefore, the BCS theory is unable to explain the high-temperature superconductivity, since the cooper pairs are not believed to be capable of surviving at those high temperatures.

Superconductors face challenges in commercialization. They require low temperature to operate, which is expensive to generate and sustain. Further, high-temperature superconductors are ceramic materials which are not easy to process and can be brittle. Even with these challenges, there are special applications where superconductors have made their mark. These include:

1. Magnetic resonance imaging (MRI) scans, which are increasingly commonly used in the medical field.
2. Superconducting electromagnets in particle accelerators. In particle accelerators, such as the one in CERN (The European Organization for Nuclear Research), very powerful magnets are required to accelerate particle beams. This is accomplished using electromagnets with high amounts of current running through the coils. With normal conductors, the heat generated would be so high that the conductors would likely melt after a short duration of operation. By using superconductors it is possible to sustain high currents and, hence, very strong magnetic fields, without the associated heat.

Interestingly, one of the primary purposes of the particle accelerator at CERN is to look for a particle called the Higgs Boson. This is therefore an instance where two bosons, phonons and cooper pairs, are used to look for a third boson, the Higgs Boson.

In summary, we have seen the basic features associated with superconductivity. The dream is to have a room temperature superconductor, something that may or may not happen. The BCS theory has not been able to explain the high temperature superconductivity; therefore, it is likely that a more fundamental theory may appear that augments or replaces the BCS theory.

In view of our repeated encounters with bosons, we will now look at the Bose–Einstein statistics, which describes the statistical behavior of bosons and also look at a very unique prediction based on these statistics.

16.5 The Bose–Einstein Statistics



What can we say about the statistics displayed by particles that show quantum mechanical behavior, but do not obey the Pauli's exclusion principle? What are examples of such particles? Particles that have integer spin, such as photons, several atoms, and even pairs of electrons under some conditions, fall under this category. The statistical behavior of these particles was described by Satyendra Nath Bose and Albert Einstein and goes by their names as Bose–Einstein statistics. Particles obeying these statistics are called bosons.

Exemption from Pauli's exclusion principle implies that any number of particles can occupy the same state at the same time. In the derivation that follows, we are examining particles that are identical, indistinguishable, and exempt from Pauli's exclusion principle.

Similar to the analysis in Chapters 7 and 9, let us consider a system of particles that are allowed to occupy energy levels ε_1, ε_2, ε_3, ε_4, …, ε_i. Let us assume that there are s_i number of allowed states at ε_i and that the total number of particles at ε_i is n_i. The values of s_i and ε_i are fixed for the system, since we have defined or identified a system with those values. The value of n_i is however variable and it is our intention to find out what will be the equilibrium values of n_i, at each of the ε_i, given the various constraints we are placing on the system.

The possibility of placing any number of particles in a single state impacts our analysis of the system significantly. In fact, we can even have all of the particles in the same state at a given energy level. In other words, we are free to arrange a collection of states and a collection of particles in any manner possible with almost no other restrictions, excepting one important restriction, the particles cannot exist independent of the states. What does this mean? Let us look at a more general example and see how we need to approach this situation mathematically. Assume we have "X" objects of one kind and "Y" objects of another kind. The number of different ways in which these can be arranged is given by

$$\frac{(X+Y)!}{X!Y!}$$

The situation we are faced with is somewhat similar, but with one important difference. In the above example, we can have all of the "X" objects followed by all of the "Y" objects, as one of the arrangements. However, in the case we are considering, we are excluding the possibility that we can have an arrangement where we have all of the particles followed by all of the states, since that would imply that there are no particles in any of the states and all the particles are outside all of the states. In Fig. 16.3, if the boxes represent states and "*" represents particles that can occupy these states, we cannot have the situation shown below:

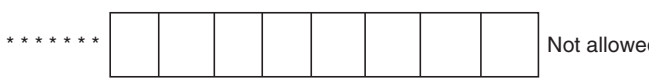


Figure 16.3 An arrangement of particles and states that is not permitted since it implies that all of the particles are outside all of the states.

Therefore, the mathematics leads to acceptable results only if we conceptually treat the situation as one where we have n_i particles to be placed in $(s_i - 1)$ states. The implication being that if we have an arrangement where we have all the n_i particles followed by all the $(s_i - 1)$ states, all the particles can be thought of as being in the first state. Similarly, if we have an arrangement where we have all of the $(s_i - 1)$ states followed by all the n_i particles, it would imply that all the particles are in the last state. Alternately, if the s_i states are thought of as a large box with $(s_i - 1)$ partitions, similar to Fig. 16.3, all arrangements obtained from rearranging $(s_i - 1)$ partitions and n_i particles, are acceptable. The use of $(s_i - 1)$ partitions ensures that particles are never "outside" all permissible states. The s_i states have $(s_i - 1)$ partitions between them.

Mathematically, all the acceptable conditions can therefore be expressed using

$$\frac{(n_i + s_i - 1)!}{n_i!(s_i - 1)!}$$

The above expression is for a single energy level ε_i. If we denote this as $\Omega(n_i)$, then

$$\Omega(n_i) = \frac{(n_i + s_i - 1)!}{n_i!(s_i - 1)!}$$

Considering all the possible energy levels and the number of particles and states at these levels, the total number of microstates in which the system can exist while still at the same macrostate, is given by

$$\omega(\overline{N}) = \Omega(n_1) \times \Omega(n_2) \times \Omega(n_3) \ldots = \prod_i \Omega(n_i) = \prod_i \left[\frac{(n_i + s_i - 1)!}{n_i!(s_i - 1)!} \right]$$

Since $n_i + s_i \gg 1$, we can reasonably approximate the above to

$$\omega(\overline{N}) = \prod_i \left[\frac{(n_i + s_i)!}{n_i!(s_i - 1)!} \right]$$

Our intent is to find the probability of occupancy of a state s_i, at the energy level ε_i, when the system is at equilibrium. This is given by the expression (n_i/s_i). Based on the approach of statistical mechanics, the equilibrium state corresponds to the most probable state of the system. In particular, statistical mechanics assumes that the most probable state swamps the probability of all of the other states combined. Therefore, similar to the approach we have adopted in Chapters 7 and 9, we will proceed to determine the most probable state, which will correspond to the state where $\omega(\overline{N})$ is maximum, that is, it is the state which has the maximum possible microstates. In the process of maximizing $\omega(\overline{N})$, we will obtain an expression for (n_i/s_i) when $\omega(\overline{N})$ is maximum. This is the answer we seek.

Also, as indicated earlier, since the function $\omega(\overline{N})$ and the function $\ln \omega(\overline{N})$ either both increase or both decrease, their maxima occur at the same point $\overline{N_0}$. Therefore, instead of maximizing $\omega(\overline{N})$, we can choose to maximize $\ln \omega(\overline{N})$, which is mathematically easier to handle and arrive at the same desired result, $\overline{N_0}$.

$$\ln \omega(\overline{N}) = \sum_i \ln(n_i + s_i)! - \sum_i \ln n_i! - \sum_i \ln(s_i - 1)!$$

Using Stirling's approximation, which states that for large X

$$\ln X! = X \ln X - X$$

we get

$$\ln \omega(\overline{N}) = \sum_i \{(n_i + s_i)\ln(n_i + s_i) - (n_i + s_i) - (n_i \ln n_i - n_i) - [(s_i - 1)\ln(s_i - 1) - (s_i - 1)]\}$$

Maximizing $\ln \omega(\overline{N})$ implies, $\partial \ln \omega(\overline{N}) = 0$. Since s_i is constant and only n_i can be varied,

$$\partial \ln \omega(\overline{N}) = 0 = \sum_i [\ln(n_i + s_i) - \ln n_i] \, \partial n_i = 0$$

In addition, particle conservation and energy conservation give us the following additional equations, respectively:

$$\sum_i \partial n_i = 0$$

$$\sum_i \varepsilon_i \partial n_i = 0$$

Applying the Lagrange method of undetermined multipliers, discussed in greater detail in Chapter 7, we obtain

$$-\ln\left(\frac{s_i + n_i}{n_i}\right) + \alpha + \beta\varepsilon_i = 0$$

Note: The negative sign in the first term in the equation above is inserted by convention to make the final results easier to interpret. Since the equations leading up to the above are all equal to zero, introduction of a negative sign before summing up is acceptable.

Simplifying further, we get

$$\frac{s_i}{n_i} + 1 = e^{\alpha + \beta\varepsilon_i}$$

Therefore,

$$\frac{n_i}{s_i} = \frac{1}{e^{\alpha + \beta\varepsilon_i} - 1} = f_{\text{BE}}$$

16.6 The Bose–Einstein Condensate

The Bose–Einstein distribution predicts that at very low temperatures, very nearly absolute zero, all particles in the system will condense into a single state, called the Bose–Einstein condensate. This was a prediction made around the year 1924. At that time, the experimental facilities available were not capable of attaining the low temperatures required to test this prediction. While it is not very difficult to attain temperatures of the order of 4 K using liquid helium, getting to lower temperatures is a significant experimental challenge. It took almost 70 years, from the time of the prediction, for experimental facilities to evolve to the stage where it was possible to demonstrate the formation of this condensate. In 1995, Eric Cornell and Carl Weiman demonstrated the formation of the condensate using a small collection of rubidium (Rb) atoms, which they cooled to a few hundred nK. For demonstrating the formation of the Bose–Einstein condensate, they were awarded the Nobel Prize in Physics for the year 2001 along with Wolfgang Ketterle.

Incidentally, the mean background temperature of the universe is of the order of 2.7 K. Therefore, temperatures below this are not expected to occur naturally anywhere in the universe. Seen from that perspective, to attain an experimental condition outside the natural ability of the universe is itself a remarkable accomplishment.

The prediction of the Bose–Einstein condensate is an example of the power of theoretical studies. Surprising predictions can be made of unknown states of matter and it could take years to decades to test the prediction.

Summary

1. Superconductivity is significantly different from "exceptionally good" normal conductivity.
2. Materials, where the resistivity abruptly drops to zero when the temperature goes below a critical value, are called superconductors.
3. Several metals, alloys, and intermetallic compounds show superconductivity at or below 10 K.
4. Some materials show superconductivity at temperatures around 100 K. These high temperature superconductors are actually ceramic materials and are very poor conductors of electricity at room temperature.
5. The superconducting state ceases to exist above a certain critical current density, designated as J_c; as well as above a critical value of an externally applied magnetic field, designated as, H_c; and above a critical temperature, designated as T_c.
6. Superconductors exclude magnetic fields, an effect called the Meissner effect. The Meissner effect indicates that magnetism of materials opposes what superconductivity requires of materials.
7. Based on the interaction with magnetism, superconductors are classified into two types, called type-I superconductors, which displays abrupt transition between the S–N states, and type-II superconductors, which show gradual transition between the S–N states.
8. The BCS theory suggested that the electrons in superconductors operate in pairs, called cooper pairs, which have opposite spin and opposite k vectors. As a result, the pair of particles operates as though they have zero spin and have no net wave vector.
9. Cooper pairs are bosons.
10. BCS theory is unable to explain high-temperature superconductivity.
11. Particles that have integer spin, such as photons, several atoms and even pairs of electrons under some conditions, obey the Bose–Einstein statistics and are called bosons.
12. Since bosons have integer spin, they are exempted from the Pauli's exclusion principle. This implies that any number of particles can occupy the same state at the same time.
13. The Bose–Einstein distribution is given by

$$f_{\text{BE}} = \frac{1}{e^{\alpha+\beta\varepsilon_i} - 1}$$

14. The Bose–Einstein distribution predicts that at very low temperatures, very nearly absolute zero, all particles in the system will condense into a single state, called the Bose–Einstein condensate.

Practice Questions

Multiple Choice Questions

1. When temperature is decreased, typical metals
   (a) Become superconductors
   (b) Become insulators
   (c) Reach a low non-zero resistance
   (d) Reach a high non-zero resistance
2. Cooper pairs are
   (a) Classical particles
   (b) Bosons
   (c) Fermions
   (d) Electron pairs with same spin
3. Meissner effect indicates
   (a) Electronic conductivity opposes superconductivity
   (b) Thermal conductivity opposes superconductivity
   (c) Photo electrons affect superconductivity
   (d) Magnetism opposes superconductivity
4. High temperature superconductivity is observed at around
   (a) 1 K **(c)** 100 K
   (b) 10 K **(d)** 1000 K

5. In type-II superconductors
   (a) Only S-state can exist
   (b) Only N-state can exist
   (c) Neither S-state nor N-state can exist
   (d) Both S-state and N-state can coexist
6. An example of a boson is
   (a) Electron
   (b) All ions
   (c) Photons
   (d) All molecules
7. Pairs of electrons are
   (a) Always bosons
   (b) Bosons only under some experimental conditions
   (c) Bosons when subject to high magnetic fields
   (d) Bosons when subject to high temperatures
8. In the Bose–Einstein condensate
   (a) Several particles occupy one state
   (b) One particle occupies two states
   (c) One particle occupies several sates
   (d) Several particles occupy two states
9. Bosons are
   (a) Subject to Pauli's exclusion principle always
   (b) Subject to Pauli's exclusion principle sometimes
   (c) Exempt from Pauli's exclusion principle always
   (d) Exempt from Pauli's exclusion principle sometimes
10. In the calculation of the Bose–Einstein statistics
    (a) All the particles can be outside of all of the states
    (b) Some of the particles can be outside of all of the states
    (c) All the particles must be in one state
    (d) Each of the particles must be in at least one of the states

Short Answer Type Questions

1. What is the strength of the magnetic fields generated using superconductors at CERN?
2. What is the strength of the magnetic field in a typical MRI scanner in a hospital?
3. What are some of the safety issues associated with MRI scanners in a hospital?
4. What is the magnetic field of the earth?
5. What is the amount of heat generated by the high currents in the superconducting coils at CERN?
6. Describe some of the challenges faced in creating the Bose–Einstein condensate.
7. Indicate some materials where the Bose–Einstein condensate has been demonstrated.
8. What is the criterion for an atom to qualify as a Boson?
9. Indicate some atoms which qualify as Bosons.
10. Indicate a potential use for the Bose–Einstein condensate.

Answers

Multiple Choice Questions

1. (c)
2. (b)
3. (d)
4. (c)
5. (d)
6. (c)
7. (b)
8. (a)
9. (c)
10. (d)

Short Answer Type Questions

1. CERN is believed to use magnets with strength of the order of 8 T.
2. MRI scanners use magnets with strength of the order of 3 T.
3. MRI scanners use very powerful magnets, therefore significant care must be taken not to bring ferromagnetic objects to the vicinity of MRI scanners. Such objects can move very quickly and dangerously due to the strong magnetic attraction.
4. The magnetic field of the earth is between 25 and 65 μT.
5. Negligible heat is generated even though the currents are high, since the resistance is virtually non-existent.
6. The major challenge in creating the Bose–Einstein condensate is the ability to attain extremely low temperatures of the order of nK.
7. Bose–Einstein condensate has been demonstrated using Rb atoms
8. For an atom to qualify as a Boson, it must have integer spin, that is, the sum of protons, electrons and neutrons must be an even number.
9. Rb and He.
10. Superfluidity demonstrated by liquid helium is related to Bose–Einstein condensation and, in this state, it demonstrates zero viscosity, which can have several potential applications including supporting the operation of superconductors.

17 Physics of Nano-Scale Materials

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Indicate why nano-scale materials are of interest to study.
- Define an exciton.
- Indicate the similarities and the differences between an exciton and a hydrogen atom.
- Indicate the parameters that affect the Bohr radius of an exciton.
- Indicate what is meant by confinement of an exciton.
- Define and differentiate between bulk materials, quantum wells, quantum wires and quantum dots.
- State examples of tailoring or tuning of the band gap of materials using quantum confinement effects.

17.1 Introduction



Nano-scale materials or nanomaterials are materials where the particle or crystal size is in the scale of a few nanometers. In recent years, there has been considerable interest in the area of nanomaterials in view of the interesting ways in which their properties can be manipulated. In general, most material properties change as a function of size. The trend in the size dependence of a property may be different in the macroscopic state, when compared to that in the nano regime. Trends in the size dependence of properties may get exaggerated or even reversed when the size scale decreases to a few nanometers. It is therefore of interest to examine nanomaterials, since it becomes possible to obtain vastly different properties while the chemical composition is the same.

It is necessary to note that, while the particle size or crystal size being a few nanometers, in some ways qualifies the material as a nanomaterial, for each property and each material, interesting size dependence of properties may become apparent only below a specific length. In other words, there is no universal size below which all nanomaterials show interesting trends in properties. In some cases, the properties may become interesting below a few tens of nanometers and, in some other cases, the properties may become interesting only when the size drops even further to below a few nanometers. So, if the material simply has a particle size or crystal size of a few nm, but its properties are exactly the same as that of a bulk sample, then there is no interest in this nanomaterial.

In this chapter, we will look at the impact of nano-scale materials on their electrical and optical properties.

17.2 The Exciton

Let us first examine what is the size scale that is of interest for nanomaterials to show electrical and optical properties that are different from the bulk properties of the same material. We have noted earlier that, in intrinsic semiconducting materials, optical absorption occurs by transition of an electron from the valence band to the conduction band. This transfer of an electron to the conduction band creates a hole in the valence band, as shown in Fig. 17.1.

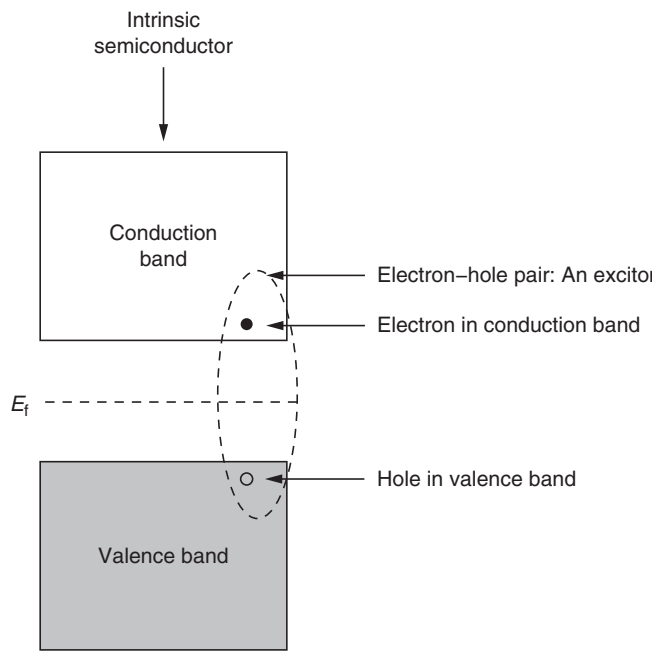


Figure 17.1 Transfer of an electron to the conduction band and the resultant formation of a hole in the valence band of an intrinsic semiconductor, due to the absorption of a photon of appropriate energy. The electron–hole pair exert attractive forces on each other and behave like a combined entity, called the exciton, which has some similarities to the hydrogen atom in view of the number of particles involved and their charges.



Animation: "Exciton formation" in DVD.

Since the electron and the hole have opposite charges, they exert an attractive force on each other and maintain connectivity with each other. This electron–hole pair, which operates as an associated pair of particles, is called an "exciton." The attractive force between them makes them more stable than they would be as independent particles.

The exciton has similarities to a hydrogen atom in the sense that it contains two entities of opposite charges, just like the hydrogen atom. Therefore, the exciton can be treated similar to a hydrogen atom and, just like the Bohr radius for a hydrogen atom, it is possible to define an exciton Bohr radius. While the Bohr radius of the hydrogen atom is of the order of an angstrom, the exciton Bohr radius can be several nanometers; in this respect the exciton differs from the hydrogen atom. The exact value of the exciton Bohr radius

varies from material to material and is dependent on the dielectric constant of the material which impacts the interaction to some degree.

For nanomaterials selected for their optical or electronic properties, the size scale of interest is therefore the exciton Bohr radius for that specific material. This value could vary from system to system and could be specific values such as 3 nm, 8 nm, etc. If the nanomaterial is synthesized in this size scale, we begin to see nano-scale effects in the electrical and optical properties of the material.

17.3 Confining the Exciton

"Confinement" of the exciton, by making the particle size smaller, impacts the band gap displayed by the material. By making the size of the particles smaller and making the confinement more severe, the band gap increases. By synthesizing particles with a narrow size distribution, but with size in the nanometer range, it is possible to "tune" the band gap of the material. It is therefore experimentally possible to get several samples of the same material such that the band gap is different in each sample. This is particularly useful for creating devices without having to use dissimilar materials, thereby eliminating diffusion, corrosion and sealing issues associated with dissimilar material contact.

With regard to synthesizing materials in different size scales, we have a few specific possibilities, as summarized in Fig. 17.2.

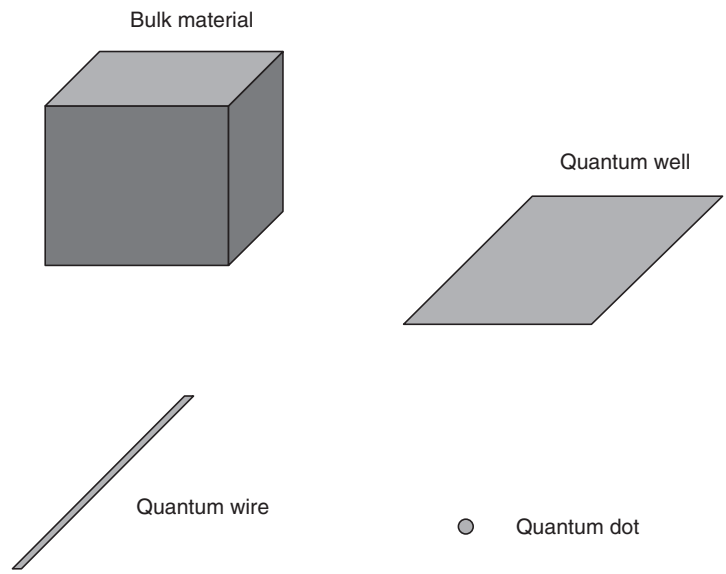


Figure 17.2 Schematic showing bulk material, a quantum well, a quantum wire, and a quantum dot.
Animation: "Quantum confinement" in DVD.

The possibilities with respect to synthesizing materials in different size scales are as follows:

1. **Bulk material:** No confinement effects observed.
2. **Quantum well:** In this case, a very thin layer of the material is synthesized with the thickness of the layer in the range of the exciton Bohr radius. This implies that the exciton is confined in one dimension and free in two other dimensions.

3. **Quantum wire:** A very narrow wire of the material qualifies as a quantum wire, where the confinement effects are observed in two dimensions, but the exciton is free in one dimension.
4. **Quantum dot:** The material has a very small size in all three dimensions and the exciton is therefore confined in all three dimensions.

Materials synthesized in the nano-scale have very high surface area and are therefore very reactive with their environments. It is, therefore, a challenge to stabilize these particles. Such stability is usually brought about by trapping the material in host matrices such as polymers. The particles can be made to grow within the structure of the polymer and by selecting the polymer and processing conditions, specific narrow-size range, nanoparticles can be synthesized.

Examples where these nano-scale effects have been effectively demonstrated are in the Cadmium Sulfide (CdS) system and in the Lead Sulfide (PbS) system. The bulk band gap of CdS system is in the visible region of the electromagnetic spectrum. By confinement effects, the band gap can be tailored or tuned to increase into the ultraviolet (UV) range of the spectrum. Similarly, the bulk band gap of the PbS system is in the infrared (IR) region of the spectrum and, by introducing confinement effects, the band gap can increase into the visible region of the spectrum and further increase into the UV region of the spectrum.

On an aesthetic note, the above ability to manipulate the band gap implies that with the same chemical composition it is possible to obtain samples with a wide range of colors, across the entire visible spectrum.

A technologically important application of this nano-scale effect on the optical properties of the material is that it becomes possible to synthesize a variety of different materials that can be used for solar cells. Solar radiation has a significant amount of energy in the IR region of the spectrum. Therefore, having materials that absorb significantly in this wavelength region of the electromagnetic spectrum, makes them excellent for use in developing solar cells. By utilizing band gap tuning discussed here, it becomes possible to have a larger variety of materials capable of being used for solar cell application, than is possible by merely looking at the bulk band gap of the materials.

Summary

1. It is of interest to examine nanomaterials since it is possible to obtain vastly different properties with the same chemical composition by making the material in the nano-scale.
2. Electron–hole pairs in semiconductors exert attractive forces on each other and behave like a combined entity, which is called an exciton. An exciton has some similarities to the hydrogen atom in view of the number of particles involved and their charges.
3. While the Bohr radius of the hydrogen atom is of the order of an angstrom, the exciton Bohr radius can be several nanometers and can vary from material to material.
4. "Confinement" of the exciton, by making the particle size of the order of the exciton Bohr radius, impacts the band gap displayed by the material and enables tuning of the band gap.
5. Quantum wells, quantum wires and quantum dots confine the exciton in 1, 2, and 3 dimensions respectively.
6. Materials synthesized in the nano-scale have very high surface area and are therefore very reactive with their environments and it is a challenge to stabilize these particles.

Practice Questions

Multiple Choice Questions

1. Properties of nanomaterials differ from those of macroscopic materials primarily due to the contribution of
   (a) Composition
   (b) Crystal structure
   (c) Orientation
   (d) Surface effects
2. Interesting properties displayed by nanomaterials occur
   (a) At the same size for all properties
   (b) At specific sizes for specific properties
   (c) At the same size for all materials
   (d) Only below 5 nm
3. An exciton is similar to a hydrogen atom in that
   (a) It has the same number of particles and charges as the hydrogen atom
   (b) It has only the same number of particles as the hydrogen atom
   (c) It has only the same types of charges as the hydrogen atom
   (d) It has the same mass as the hydrogen atom
4. The exciton Bohr radius
   (a) Is the same as that of the hydrogen atom
   (b) Is different from that of the hydrogen atom but is the same for all materials
   (c) Varies from material to material
   (d) Is defined only for the individual atoms
5. An exciton can be confined in
   (a) 1D
   (b) 2D
   (c) 3D
   (d) All of the above

Short Answer Type Questions

1. Why is it difficult to stably maintain nano-sized particles under open atmospheric conditions?
2. How is the thermodynamic treatment of materials impacted by the materials being nano-sized?
3. Give an example of quantum confined semiconducting materials.
4. What happens to the ferromagnetic behavior in nano-scale materials?
5. Give an example of a quantum well.

Answers

Multiple Choice Questions

1. (d)
2. (b)
3. (a)
4. (c)
5. (d)

Short Answer Type Questions

1. Nano materials are highly reactive since they have a large fraction of atoms at the surface and, hence, have a large fraction of unsaturated bonds. They will react easily with any other species they can access to lower their energy. Hence, it is difficult to maintain them stably under open atmospheric conditions.

2. When nano-sized materials are considered, the surface energy contribution to enthalpy is large due to the large surface areas present. As a result, the thermodynamic stability of phases can be significantly affected by making nano-sized particles of the materials.
3. ZnO particles trapped in polymer matrices can result in the formation of semiconducting quantum dots.
4. At nano-dimensions, ferromagnetic materials display a behavior called superparamagnetism. Under these conditions, magnetization randomly flips under the influence of temperature. The typical time between flips is called Neel relaxation time.

   Average magnetization is zero, if the time to measure is much longer than Neel relaxation time. An applied external magnetic field will magnetize the nanoparticle, like a paramagnet, but the magnetic susceptibility is much higher.
5. Thin layers of GaAs sandwiched between two layers of another material of higher band gap will result in the formation of a quantum well.

Bibliography

This book, an introductory text by its nature, covers topics that have already been covered to varying degrees in various books, courses, and articles. The unique contributions of such a book then are the choice of topics, sequence of topics, style and clarity of presentation, depth of discussion, and the effective use of learning tools to assist the reader. A very important contribution of the book is also how well it gauges the needs of a potential reader and how well it enables the reader to learn efficiently. The reader should feel that the time spent reading the book was time well spent and that the reader's knowledge and confidence in the topics discussed was significantly improved by reading the book. The present book aims to do all of the above. This book includes videos to provide students as well as instructors with a classroom experience. It also includes animations for a more visual experience of various concepts.

Over the years the author has had the opportunity to learn concepts from several teachers, courses, textbooks, research articles, popular science books, and popular science articles. It is impossible to individually acknowledge each and every one of them. At the same time specific books have been very useful in the learning process and these are gratefully acknowledged here:

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Index

A
absolute temperature, 43
absolute zero Kelvin, 114
absorptivity, 10
"acoustically active" phonons, 220
$2\pi/a$ in k-space, 186
allowed values of energy for free electrons and bound electrons, 144
alternating current conductivity measurement, 28–34
  amplitude of the signal, 30
  complex number notation for, 30
  four-probe measurement of the conductivity, 29
  impedance analysis, 33–34
  response of a pure resistor, 29
  schematic of, 28–29
amplitude of the wave, 173
anisotropy, 126–128
anode, 24
Avogadro number, 43, 44

B
band diagrams, 207
bands of Brillouin zones, 188–193
  analytical approach to band structure, 197–202
  band diagrams, 207
band structure, 9, 202–203
  of an insulator, 203
  associated with a metal, 203
  effect of pressure on, 205–206
  of metallic systems, 203
  of a semiconductor, 203
  tight binding approximation, 203–205
Bardeen, Cooper, and Schrieffer (BCS) theory, 233–234
blackbody radiation, 90, 95
  Einstein's equation, 96
  Max Planck's study of, 95
  spectral distribution of, 91
body-centered cubic (BCC) lattices, 168–169, 225–226
  body-centered structure in real space, 169
  Wigner–Seitz cell of, 192
Bohr, Neils, 97–98
Bohr radius for a hydrogen atom, 242–243
Boltzmann constant, 44, 81, 113, 114
bonding energy, 7
Born, Max, 97
Born interpretation, 97
Bose–Einstein condensate, 237
Bose–Einstein statistics, 220, 234–237
Bragg law for diffraction, 164
Bragg planes, 185–186
Bravais lattices, symmetry directions for, 207–209
Brillouin zone–Fermi sphere interaction for FCC lattice, 193
Brillouin zones, 179, 181–184
  boundaries, 186
  corresponding to a 1D real lattice of spacing, 187
  of 2D square lattice, 191–192
  interaction of electron waves with, 186–188
  nth, 187
  origin of bands, 188–193
bulk material, 243

C
Cadmium Sulfide (CdS) system, 244
carbon, 11
cathode, 24
charge carriers, 24–25, 53
chemical engineering, 74
chemical properties, 6–8
classical mechanics, central concepts of, 88
  Maxwell–Boltzmann statistics, use of, 88
classical particles, 86
classical physics, 86
coercivity, 10
complex exponential notation, 173
Compton-modified scattering, 90
conditional probability, 108–109
conduction, 55
confinement, 132–134
  of nearly free electrons and bound electrons, 138
  quantum mechanical approach, 134–138
"confinement" of the exciton, 243–244
conjugate variables, 98
constructive interference, 165
contact resistance, 26
convection, 55–56
corpuscles, 96
corrosion, 9
Coulombs per square meter per second, 54
crystal lattices in reciprocal space, 167–170
crystalline materials, 159
crystalline solids
  anisotropy of, 126–128
  periodic potential in, 128–132
cube, volume of, 38
  volume of atom associated with cube, 38
Curie temperature, 10
current density, 50, 54

D
dark energy, 90
dark matter, 90
de Broglie, Louise, 96–97
density of states, 143–147
  energy level for, 151–153
dielectric constant, 9
diffraction
  in real space, 166
  of reciprocal space, 163–166
direct current conductivity measurement, 25–27, 28
  four-probe measurement of the electrical conductivity, 27
  of the potential drop, 26–27
  response of a pure resistor to, 30
  two-probe measurement of electrical conductivity, 26
discrete nature of atomic and molecular spectra, 90
1D real lattice
  Fourier coefficients corresponding to, 171
  Fourier transform of a 1D lattice of spacing, 172
  mathematical representation of, 170
drift velocity, 51, 54
Drude model, 49, 61, 78–79
  analysis of the predictions of, 86
  density of states, energy level for, 152
  differences between Drude–Sommerfeld model and, 101
Drude–Sommerfeld model, 100–101, 116, 125–126, 147, 153
  differences between classical Drude model and, 101
  features of, 101
2D square lattice, 192
ductility, 6

E
electrical conductivity, 8, 23, 49–55
electrical properties, 6, 8–9
electrolyte, 24
electromagnetic waves, 56
electron affinity, 7
electron collisions, 41
  change in momentum, 42
  mean free time between collisions, 41
electron compounds, 224–226
electronic circuits, 25
electronic contribution to specific heat at constant volume, 151–153
electron mobility, 9
electrons, 24, 25
electrons filling energy levels, 147–149
empty conduction band, 214
energy increment, 94
energy possessed by electrons at temperature *T*, 152
engineer, technical troubleshooting challenges faced by, 1
equilibrium inter-atomic spacing, 205
equilibrium of electrons, 41
E vs k diagrams, 214, 219
Ewald sphere, 166–167, 170
exciton, 242–244
  Bohr radius, 243
  "confinement" of, 243–244
experimental data, 2
extended zone representation, 189, 192
extrinsic semiconductors, 215–216

F
face-centered cubic (FCC) lattice, 168–169
  Brillouin zone–Fermi sphere interaction for, 193
face-centered cubic (FCC) structure, 39–40, 225–226
  crystal structure, 224–225
Fermi–Dirac distribution, 113, 114–117, 145
  comparison with Maxwell–Boltzmann distribution, 117–119
  features of, 114–117
Fermi–Dirac statistics, 89, 100–101, 107–113
Fermi energy, 115, 116, 149, 152, 190
fermions, 100, 111
Fermi surface, 149–150
Fermi temperature, 150–151, 151, 153
ferromagnetism, 222–223
filled valence band, 214
final momentum, 42
first Brillouin zone, 186
flat-band structure, 189
flux of heat, 57
Fourier coefficients corresponding to the 1D lattice, 171
Fourier's law of heat conduction, 57, 59
Fourier transform
  of a 1D lattice of spacing, 172
  of the real lattice, 170–172
Fourier transform of a 1D lattice of spacing, 172
Fourier transforms, 98
four-particle system, statistical mechanics of, 77–79
  observations, 77–78
  possible microstates, 77
free electron approximation, 197–202, 203
free electron cloud, 37
free electron gas, 37–38
  metallic solid containing, 37–38
  with respect to ideal gas, 40
free electron model, 126
free electrons per cubic meter, 40
free electron theory for metals, 41
δ function, 170–171

G
graphite, 11, 91, 127–128
  layered structure of, 127

H
hard magnetic materials, 10
hardness, 7

heat exchangers, 60–61
heat shrink material, use of, 16–17
Heisenberg, 98
holes, 24, 214
Horgan, John, 90

I
ideal gas-free electron gas comparison, 40
identical and indistinguishable particles, 111
impedance analysis
AC, 33–34
infrared (IR) radiation, 90–91
initial momentum, 42
integers, 164–165
interaction of electron waves with Brillouin zones, 186–188
inter-atomic spacing, 205
inter-ionic distance, 18, 20
intrinsic semiconductors, 214–215
inverse of conductivity, 50
ionic conductivity, 28
ionization energy, 7

K
kinetic theory of gases, 41–44
Kirchoff, 91
k space, 174
k-space, 173, 186, 188, 190

L
Lagrange method of undetermined multipliers, 80, 113
Laplace expansion, 201
lattice wave equivalent of photons, 220
Lead Sulfide (PbS) system, 244
localization, 99

M
macrostate, 75
magnetic properties, 6, 10
magnetism, 221–224
common equations associated with, 221–222
material, modeling a, 2–3
material properties, 2, 126
chemical, 6–8
in chemistry books, 11
correlations between properties, 5–6
electrical, 6, 8–9
magnetic, 6, 10
mechanical, 6–7
optical, 6, 10–11
in physics books, 11
thermal, 6, 9
material specialist, 2
Maxwell–Boltzmann distribution, 86
as a function of temperature, 86
Maxwell-Boltzmann distribution8, 81
Maxwell-Boltzmann Statistics, 73, 79–81
Maxwell–Boltzmann statistics, 86, 101
mean free time between collisions, 41
mechanical properties, 6–7
Meissner effect, 233
metal, definition of, 5
metallurgical engineering, 74
microstate, definition of, 75
microstate of the system, 108–110
modeling materials, 153–154
modulus of elasticity, 6
MRI (magnetic resonance imaging) equipment, 10

N
NaCl crystal, 17–19
energy involved in formation of, 19
nano-scale materials or nanomaterials, 241
exciton, 242–244
natural convection, 56
"nearly free" electrons, 153
negatively charged free electron cloud, 38
neutrons, 88
Newton's first law, 52
Newton's idea of particles of light, 96
Newton's laws, 86
*n*th Brillouin zone, 187
n-type semi-conductivity, 25
Nyquist plot, 32

O
occupancy of a state at energy level, 108
Ohm's law, 25, 49
one-dimensional wave vector, 135
"optically active" phonons, 220
optical properties, 6, 10–11
nano-scale effect on, 244
of semiconductors, 218, 219

P
packing fraction for solids, 38–40
parallel monochromatic waves, 163–164
paramagnetic materials, 222
partition function, 81
Pauli, Wolfgang, 99
Pauli's exclusion principle, 18, 99, 100, 107, 111, 115, 204
periodicity of the lattice, 199
permeability of the material, 221
permeability of vacuum, 221
phase diagrams in metallurgical engineering, 74
phonons, 219–220
photoelectric effect, 90, 95–96, 96
photons, 96, 220
Planck, Max, 90, 93–94
study of blackbody radiation, 95
Planck equation, 95
planes of atoms, 163
p-n-p transistor, 24
Poisson summation formula, 171–172
polycrystal, 127
pressure, effect on band structure, 205–206
protons, 88
p-type semi-conductivity, 25

Q
quanta, 94
quantization, 132–134, 143
  quantum mechanical approach, 134–138
quantum dot, 244
quantum mechanics, 97, 100
  brief history of, 90–100
  central concepts of, 88–89
  definition of, 88
  "identical but indistinguishable," context of, 88
quantum number space, 144
quantum well, 243
quantum wire, 244

R
radiation, 56
Rayleigh–Jean equation, 95
Rayleigh–Jean law, 93
real lattice, 173–174
reciprocal lattice, 170–172, 174
reciprocal lattice point, 167
reciprocal space, 154, 159
  crystal lattices in, 167–170
  definition of, 160, 173–174
  diffraction of, 163–166
  properties of, 160–163
reduced zone representation, 191, 192
reflectivity, 10
refractive index, 11
relative permeability, 221
relativity, theory of, 96
remanence, 10
repeated zone representation, 190
resilience, 7
resistivity, 50

S
Schrödinger, Erwin, 97
Schrödinger wave equation, 97, 135, 143, 198
scientist, challenges with materials, 2
second Brillouin zone, 186
semiconductors, 214
  effect of temperature on, 217–218
  extrinsic, 215–216
  intrinsic, 214–215
  optical properties of, 218–219
single crystal, 127
solid, model of, 153–154
solids, packing fraction in, 38–40
Sommerfeld, Arnold, 100
specific heat, 9, 86
sphere of reflection, 166–167
splitting of energy levels, 204
standard temperature and pressure (STP), 40
statistical mechanics
  of systems with large collections of particles, 73–75
  of system with four particles, 77–79
  of system with three particles, 76–77
  of system with two particles, 75–76
Stefan–Boltzmann law, 92, 93
Sterling's approximation, 79
Stirling's approximation, 112
sub-atomic particles, 88
superconductivity, 34, 231–233
superconductors, 221
susceptibility of the material, 222
symmetry directions for Bravais lattices, 207–209

T
technical troubleshooting challenges, 1
Teflon, 23
tensile strength, 6
thermal conductivity, 9, 55–61
  heat transferred per unit time per unit cross-sectional area, 57
  of metallic conductors, 60
  numerical value of, 60
thermal expansion, 9, 15
  model of, 17–21
thermal expansion/contraction, 15–17
thermal properties, 6, 9
three-particle system, statistical mechanics of, 76–77
  possible microstates, 76
tight binding approximation, 203–205
  comparison of free electron and, 206–207
tight-binding approximation, 213
time-dependent Schrödinger wave equation, 97
time-independent Schrödinger wave equation, 97
toughness, 7
transmissivity, 10
two-particle system, statistical mechanics of, 75–76
  possible microstates, 75–76

U
ultraviolet catastrophe, 93
uncertainty principle, 98, 99
units for conductivity, 25
universal gas constant, 43

V
valence band, 214
volume of the cube, 38

W
Watt per square meter per meter, 92
wave function, 99
wave–particle duality, 97
Wein's displacement law, 92
Wigner–Seitz cell of a BCC lattice, 192
Wigner–Seitz cells, 179, 180–181

Y
yield strength, 6

About the Book

This book "Physics of Materials", and the associated DVD, examine essential concepts in Solid-State Physics. The DVD, containing 40 lectures of approximately an hour each, provides the student with the classroom experience. Several animations are also included in the DVD which enable a more visual experience to understanding the concepts discussed.

In both the book as well as the DVD, a significant effort has been put in to present the material at the right level of detail such that it can be accessed and utilized by undergraduate and postgraduate students with a wide range of backgrounds. A conscious effort has been made to ensure that the level of detail used in the presentation enables students to stay with the content throughout. Therefore, in the book as well as in the DVD, derivations are made as complete as possible so that the student has the satisfaction of seeing in one reading how important concepts are developed. Completeness and clarity are given priority. In view of the topics discussed as well as the manner of coverage of these topics, a very wide cross-section of students as well as professionals can benefit from this book and the associated DVD.

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